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***In situ* observations of an ionospheric critical velocity experiment**

Swenson, Charles Merrill, Ph.D.

Cornell University, 1992

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300 N. Zeeb Rd.
Ann Arbor, MI 48106

IN SITU OBSERVATIONS OF AN IONOSPHERIC CRITICAL
VELOCITY EXPERIMENT

A Dissertation
Presented to the Faculty of the Graduate School
of Cornell University
in Partial Fulfillment of the Requirements for the Degree of
Doctor of Philosophy

by
Charles Merrill Swenson
January 1992

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Biographical Sketch

Charles M Swenson was born July 8, 1962 in Lansing Michigan into a wonderful family. He graduated from Logan Senior High School in 1980 with a number of good friends. After high school he attended Utah State University for a year, then took one and a half years off to serve a mission for the Church of Jesus Christ of Latter Day Saints. After his mission, Charles returned to college at Utah State and completed a dual major in electrical engineering and physics in 1985. He stayed an additional year at Utah State University to work on a masters of science in electrical engineering, and completed all requirements in 1989. Charles began his doctoral work at Cornell University in August of 1986 under Dr. Michael Kelley. His most important accomplishment while at Cornell was marrying Melanie Kay Ashworth in 1989. After leaving Cornell University Charles will be returning to Utah State University where he will be assuming the role of an Assistant Professor in the electrical engineering department.

For my wife Melanie,
whom I shure Love.

My spelling is Wobbly.
It's good spelling
but it Wobbles
and the letters
get in the
wrong places.

— A. A. Milne, *Winnie the Pooh*

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and the other engineers and technicians at NASA for their work. Phil Ward from NASA did an outstanding job of figuring out where all the payload pieces went and his contribution was crucial. Thanks also to David Detwiler and his commanding presence who told those in charge of the proceedings to increase the launch elevation about 5° just before launch when there was no time left for argument. They did and CRIT II just barely reached the required altitude.

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Chapter 1

Introduction

‘O Oysters, come and walk with us!’

The Walrus did beseech.

‘A pleasant walk, a pleasant talk,

Along the briny beach ...’

—*Lewis Carroll, The Walrus and the Carpenter, Through the Looking-Glass and What Alice Found There*

Consider a cloud of gas, a material such as air or water vapor, or any other material that can be made into a vapor with sufficient heat. Suppose that there are no big particles or dust in the gas, but that every type of particle is broken down to its smallest parts, atoms and molecules. Now, imagine that this gas is constantly irradiated by ultra violet light and x-rays from the sun or some other star. Such radiation can knock a negatively charged electron free from its atom causing it to be positively charged. When an atom or molecule loses an electron, it has become ionized and is called an ion, and if enough of the atoms or molecules in our hypothetical gas become ionized, then the gas is called a plasma (no relation to that part of our blood with the same name). This is what occurs very high in our atmosphere as it extends out into space in what is called the ionosphere. This process also occurs between the planets, between the stars, and between the

galaxies. In fact, a plasma is the most common state of matter in the universe.

A plasma acts differently than a regular gas because of charged electrons and ions. The charged particles can *feel* each others presence from a long distance away, as common experience with static electricity or the attraction of one magnet to another show. In a regular gas, the particles only *detect* each other when they actually touch, but the electromagnetic forces within a plasma cloud allow it to act and react in ways that are unfamiliar to our every day experiences with gases. This makes studying the plasma clouds surrounding the planets, stars, and galaxies interesting; even though there are few particles and collisions are rare, they still influence each other and exhibit collective effects.

1.1 Neutral Streams in Space Plasma

What would happen if a cloud of regular, unionized gas collided with a cloud of magnetized plasma in space? Would they just pass ghostlike through each other or would something else happen? Some theories do exist, but what would actually happen? These are the questions for an experimental space plasma physicist. The neutral cloud of gas might originate from a satellite in orbit firing its thrusters or a space station dumping waste gases and materials. This cloud of neutral gas would be moving with a high velocity relative to the space plasma because of its orbital velocity and would look more like a jet or stream in the reference frame of the plasma. Streams of neutral gas also occur naturally in space. Examples include the nucleus of a comet when it is close to the sun and giving off gases or volcanic eruptions on small moons which shoot gases into space, like Jupiter's moon Io. Colliding gas and plasma clouds might also occur on much larger scales, such as in the nebula between and around stars.

The most interesting effects will occur when some of the particles of the neutral stream become ionized and join the plasma cloud. There are several known processes that would contribute to ionizing the neutral stream. The first, photoionization, is a well understood process where high energy photons from the sun or another star knock electrons out of an atom. However, if the nearest star is far away or if the gas is in the shadow of some body the effectiveness of this process will be negligible. Then collisions of the neutral particles with ions, electrons, or other neutrals, although rare, will become the dominate source of ionization if the relative velocity is large enough. One kind of collision process, called charge exchange, can occur when an electron from a colliding neutral atom is taken by the resting ion. The two particles exchange the electron and the neutral atom becomes a fast ion in the plasma frame. Another ionization process, called charge stripping, also results in a free electron and a fast ion. The electron is stripped free of the neutral atom when it collides with another neutral atom that is at rest.

The interaction of such newborn fast ions with the surrounding plasma is thought to cause a variety of interesting effects. Waves will be generated as the newborn ions shoot through the plasma communicating their presence by electromagnetic forces. Electric fields, currents and magnetic fields will be generated until the streaming motion of the newborn ions is stopped and they join in the random motion of the plasma particles. Depending on the strength of these plasma waves, currents, and electromagnetic fields, they may affect the functioning of spacecraft or the flow of gases around and between stars and planets.

These are a few of the reasons for studying the collision of a neutral gas with a plasma. In addition, if these perturbations are strong enough some of the plasma electrons may be accelerated to the point that they can act as a source of ionization as well. Since such a process creates new born fast ions the, perturbations and the

ionization may be sustained in a burning like process. This new physics is the subject of the present thesis.

1.2 The Critical Ionization Velocity Effect

The interaction of a stream of neutral gas and a magnetized plasma was a topic Hannes Alfvén considered as he pondered the banded structure of our solar system [Alfvén, 1954; Alfvén and Arrhenius, 1975; Alfvén, 1981]. He proposed that if the solar system accreted out of a neutral cloud, eventually during the course of events leading to stellar formation, streaming neutral gas will interact with a magnetized plasma. The plasma might be created during the formation of the protostar, or by photoionization from other stars. This plasma environment threaded with the magnetic field of the star or protostar would form a barrier through which the rest of the neutral cloud must pass as it fell towards the central body. Alfvén argued that such conditions might explain, by a process he outlined, the different compositions of the planets and their locations. He also suggested this same process would differentiate the elements in moon systems forming around central planets, such as the systems of Jupiter, Saturn and Uranus. He recognized that once any part of a neutral cloud became ionized and magnetized it could no longer fall towards the central body. The magnetic field of the central body would support it, in a sense, against the gravitational pull.

Using simple energy balance principles Alfvén hypothesized that some process might occur to ionize the neutral gas when it had fallen through a gravitational potential equal to its ionization potential. The mechanism was left completely unspecified. Because elements within the homogeneous nebula have different ionization potentials, this provides a mechanism for separation of elements during

the condensation of a large neutral cloud into stars and planets. Alfvén presented Fig. 1.1 to show gravitational potential energy bands which the planets and moons fall into for our solar system.

Later, Alfvén's hypothesis for the initiation of the anomalous ionization mechanism was generalized and restated as a balance between kinetic energy and ionization potential. The postulate is formulated in the reference frame of the magnetized plasma. In words, Alfvén's hypothesis is that if the velocity of a streaming neutral gas perpendicular to the ambient magnetic field is such that its kinetic energy is greater than its ionization potential, some collective process of the plasma and neutrals will result in an energy transfer and ionization of the neutrals. The velocity at which this balance occurs, V_c , is called the critical velocity and is defined as

$$V_c = \sqrt{\frac{2e\phi_n}{m_n}} \quad (1.1)$$

where ϕ_n is the ionization potential of the neutral species, m_n is the mass of a neutral particle, and e is the charge of an electron. Since Alfvén's original proposal, considerable effort has been spent to validate this hypothesis. The ionization phenomena which Alfvén described has been labeled either critical velocity or more properly critical ionization velocity (CIV). It is a process by which ultimately the streaming neutrals are the source of free energy enabling their own ionization. Viewed this way, it is clear that (1.1) is an optimistic minimum velocity at which this reaction would occur, requiring 100% efficiency of the process. CIV can be likened to a burning process or a discharge process in which the ambient magnetized plasma acts as a catalyst, promoting a reaction which would not otherwise occur.

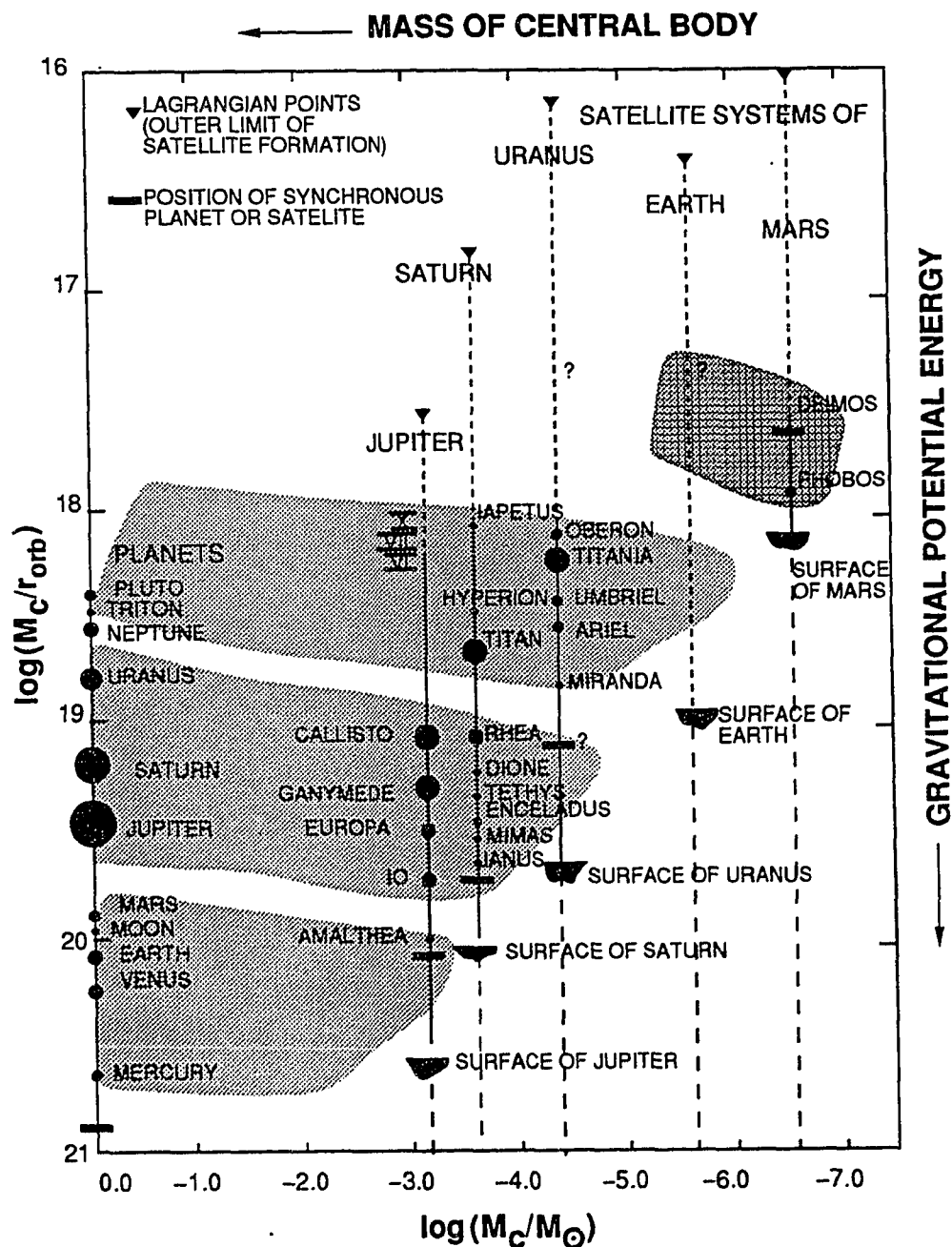


Figure 1.1: Structure of the solar system in terms of the mass of the central bodies and the gravitational potential energy of the bodies orbiting around them. After *Alfvén and Arrhenius* [1975].

1.2.1 Simple CIV Theory

There are two creditable processes by which the neutral particles can be ionized in a CIV interaction. The first is by neutral-ion collisions and the second is by electron-neutral collisions. The first process cannot be reconciled with the type of efficiencies required by (1.1) nor with existing laboratory experiments. This can be seen by considering simple inelastic collisions and conservation of momentum. For the case of equal masses, at most $1/2$ of the kinetic energy of the neutral atom can be released for ionization. This point has been demonstrated for impact experiments where the plasma and neutral gas are different. These experiments exhibit ionization although the center-of-mass energy is quite small (e.g. H^+ plasma colliding with Argon gas, which is an extreme case). Furthermore, the "anomalous" CIV process is expected in parameter regimes where the ion neutral collision rate is insignificant, and indeed rapid ionization occurs in laboratory experiments under such conditions [Mattoo and Venkataramani, 1980]. In other words, ionization is observed for center of mass energies well under the ionization potential and in regimes where the ion-neutral collision frequency is so low that the efficiency of the ionizing process is negligibly small.

The second candidate involves collisions of electrons with neutrals. For this process to efficiently ionize, electrons must be energized or heated since the energy of a collision at the critical velocity, $\frac{1}{2}m_e V_c^2$, is quite small due to the small electron mass. Thus at some point in an electron heating process, the kinetic energy of the neutrals must somehow be transferred to the electron population. The energy of the electrons could then be transferred back to the neutrals in the form of ionizing collisions. The mechanism which achieves the energy transfer is called the CIV microprocess and proceeds from the collective nature of a magnetized plasma. An energy flow diagram for this process is presented in Fig. 1.2. Energy is lost during

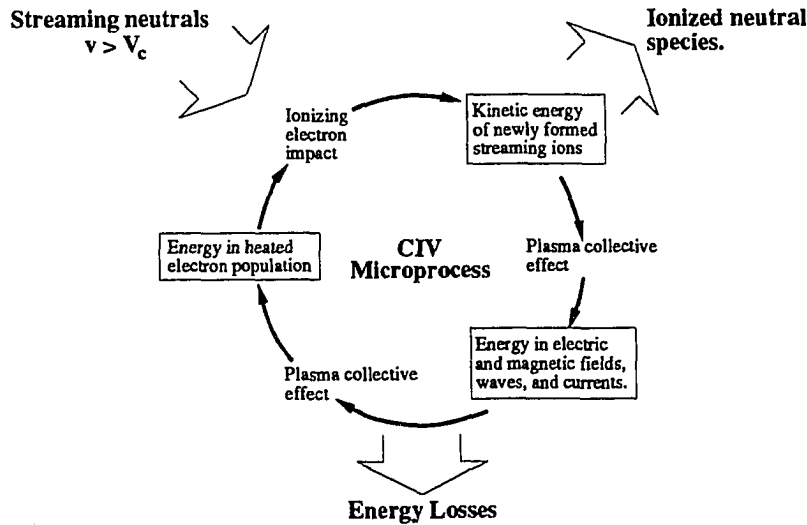


Figure 1.2: Energy flow for the CIV microprocess.

the microprocess in every step, by hot electrons escaping before an ionizing impact, excitation of neutrals and ions, or inefficiencies of the collective process draining energy from newborn ions. These loss mechanisms have been formally incorporated into equation (1.1) with the addition of a factor $\eta < 1$. The factor η has been defined differently by different authors, but the most usual definition is the fraction of the neutral energy which goes to the electrons. This is done because the electron impact ionization process in Fig. 1.2 is well known [Torbert, 1990]. For the remainder of this simplified discussion of CIV we will define η as the fraction of the neutral's kinetic energy which ends up in ionized neutrals. This will allow us to develop a simplified model based on conservation of energy and momentum for the CIV process. Using this definition of η , the threshold velocity for neutrals above which CIV will actually occur, V_i , is thus greater than V_c and given by

$$V_i = \sqrt{\frac{2e\phi_n}{\eta m_n}} \quad (1.2)$$

Some initial ionization of the neutral stream, called seed ionization in the literature,

is required to start the microprocess. If the losses in Fig. 1.2 are small enough, then the seed ionization will result in accelerated electrons and in more ionization until the process avalanches into a full CIV reaction. Seed ionization is provided by the collisional process described in section 1.4.

A second consideration in the CIV process, after energy arguments, is conservation of momentum. For every ionizing electron impact, the background plasma must absorb the momentum of the resulting ion to release available kinetic energy. If the background plasma is thin and the neutral stream is dense, then it may be impossible for the background ionosphere to absorb the momentum of even a fraction of the neutrals thus limiting the ionization that can be produced by CIV. In the laboratory the walls of the device offer an essentially infinite sink for momentum but in space this requirement poses yet another limit on the CIV reaction, referred to in the literature as a mass loading limit. This momentum exchange process was considered by *Haerendel* [1982] in which he proposed that shear Alfvén waves could transport momentum of the ionizing neutrals to a large volume of ionospheric plasma as pictured in Fig. 1.3. Hearendel proposed that a dynamo current, $\vec{j}_D$, within the ionization region would slow the newly created plasma through a $\vec{j}_D \times \vec{B}_0$ force while the polarization current within the emitted wave, $\vec{j}_p$, would produce a $\vec{j}_p \times \vec{B}_0$ force to accelerate the ambient plasma in the direction of the stream and thus absorb the momentum referred to above. Continuity of the j_D and j_p currents is maintained by field aligned currents, $j_{||}$, in shear Alfvén wave as illustrated in Fig. 1.3. The CIV process effectively results in an inelastic collision between the ionizing neutrals and the ionospheric plasma. Conservation of momentum for this inelastic collision is stated as

$$\bar{m}_1 v_1 = (\bar{m}_1 + \bar{m}_2) v_3 \quad (1.3)$$

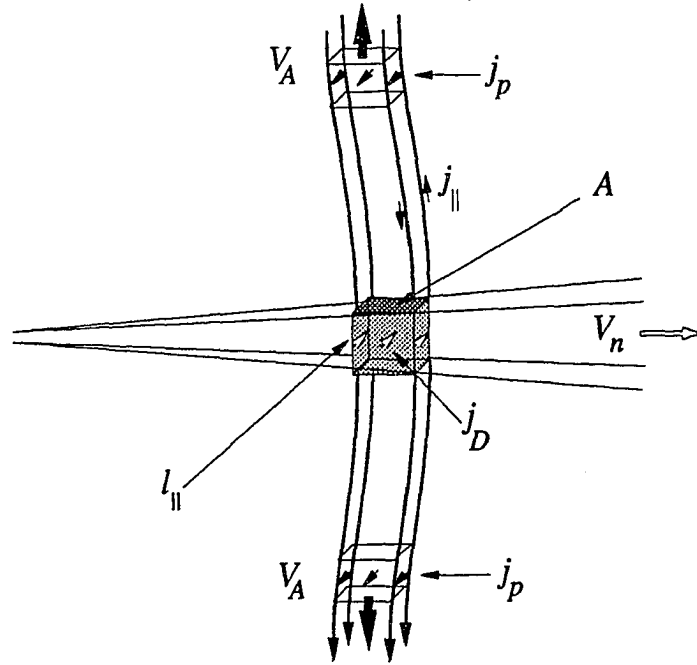


Figure 1.3: Momentum coupling between an ionizing stream and the ambient ionosphere via Alfvén waves. Bending of the magnetic lines of force is caused by the mass injection.

where $\bar{m}_1$ is the total mass of ionized neutrals in the interaction volume, v_1 is the velocity of the neutral particles just before they are ionized, $\bar{m}_2$ is the mass of the interacting ionosphere, and v_3 is resulting velocity of the ionosphere and new ions after the inelastic collision. If a CIV reaction occurs then the transfer of kinetic energy to internal energy must be greater than the ionization energy to allow for loss. This can be expressed as

$$\bar{m}_1 v_1^2 - (\bar{m}_1 + \bar{m}_2) v_3^2 = \frac{\bar{m}_1 V_c^2}{\eta} \quad (1.4)$$

where η is the efficiency of the CIV reaction as defined in (1.2). Equations (1.3) and (1.4) can be solved for $\bar{m}_1$, the total mass of neutrals that can become ionized

given conservation of momentum and energy.

$$\bar{m}_1 = \bar{m}_2 \left(\eta \frac{v_1^2}{V_c^2} - 1 \right) \quad (1.5)$$

This equation inherently requires $\sqrt{\eta}v_1 > V_c$ to be consistent with the energy conservation expressed in (1.2). The neutral mass which becomes ionized can also be expressed in terms of the ionization rate of the neutrals.

$$\bar{m}_1 = m_n \dot{n} t A l_{\parallel} \quad (1.6)$$

where m_n is the mass of a neutral particle, $\dot{n}$ is the ionization rate of neutrals, and $A l_{\parallel}$ is the volume in which the reaction is occurring. The mass of the ionosphere that can interact in the process is calculated, as a function of time, from the volume swept out by two Alfvén waves traveling away from the interaction region along the flux tube (Fig. 1.3).

$$\bar{m}_2 = 2\beta n_o m_i V_a t A \quad (1.7)$$

where $n_o m_i = \rho$ is the mass density of the ionosphere, V_a is the Alfvén velocity, and β is a factor to incorporate momentum absorbing effects of the plasma surrounding the flux tube of cross section A . *Haerendel* [1982] takes β to be 2 and calculates a quantity he calls the mass loading factor, λ_m , which is the ratio of $\bar{m}_1$ and $\bar{m}_2$.

$$\lambda_m = \frac{m_n \dot{n} l_{\parallel}}{4\rho V_a} \quad (1.8)$$

Using the expressions (1.6) and (1.7) in equation (1.5) we arrive at an expression for the ionization rate, $\dot{n}$.

$$\dot{n} = 2\beta \frac{m_i}{m_n} \frac{n_o V_a}{l_{\parallel}} \left(\eta \frac{v_1^2}{V_c^2} - 1 \right) \quad (1.9)$$

This equation, (1.9), contains only the energy and momentum constraints on the process. The energy released by the neutrals is immediately available for ionization in this model. For example, time constraints for the heating of electrons and finite ionization time have not been included. This simplified model is therefore an upper limit to the ionization rate for a given η . A more complete model/computer simulation taking into account momentum, energy, and time constraints has been developed by *Torbert* [1990], but has not yet been applied to the data presented here.

1.2.2 Review of Existing CIV Theories

Two different mechanisms have been proposed for the CIV microprocess. The first, called the homogeneous model, builds on a work by *Sherman* [1969] and *Raadu* [1978]. The second model proposed by *Piel et al.* [1978] has been labeled the ionizing front model. Both models originate from laboratory experience with CIV and are only concerned with the transfer of kinetic energy from newly ionized neutrals to electrons. Initially, both models ignored the conservation of momentum problem and details of how required electric fields and current systems are setup and maintained. *Haerendel* [1982] considered the momentum problem as he applied these models to CIV experiments in space plasmas, previously discussed, but details are still unspecified. The two model are compared schematically in Fig. 1.4. Although the models invoke different methods for driving a plasma instability, both models suggest that lower hybrid waves are the dominant mode excited. These electrostatic lower hybrid waves heat electrons through wave particle interactions. Both theories require some seed ionization to initiate the process, that is, some classical collisions which create a few fast ions.

The homogeneous model is named because density gradients are not considered

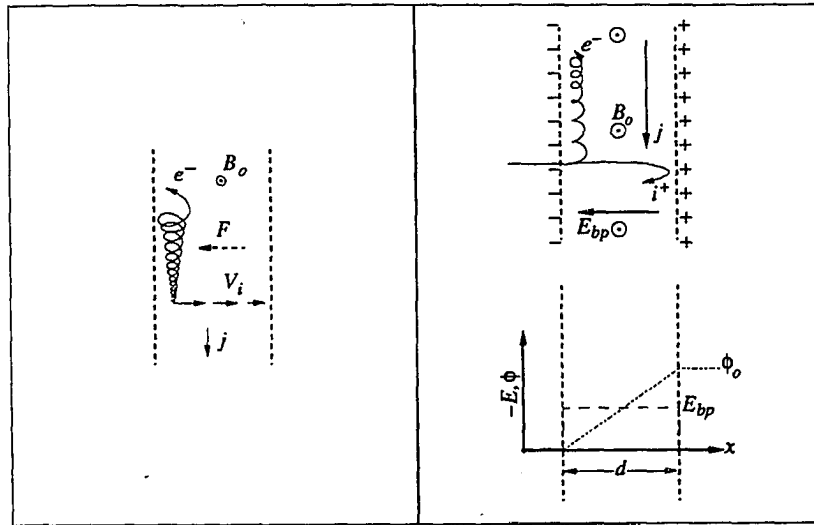


Figure 1.4: (left) Homogeneous model as proposed by *Sherman* [1969]. (right) Ionizing front model as proposed by *Piel et al.* [1978].

and the plasma is assumed to be homogeneous. The electrons in the homogeneous model are heated as a result of the modified two stream instability which is driven by the relative motion between magnetized electrons and the beam of newly ionized neutrals. The efficiency of the process, η in equation (1.2), has been estimated to be in the range of 0.25 to 0.67 [*Galeev and Chabibrachmanov*, 1983; *Raadu*, 1978]. The slowing force, F , which acts on ions created out of the neutral stream is unspecified, but a $\vec{F} \times \vec{B}_o$ force is required to drive the current j in Fig. 1.4. This current is in the direction of the dynamo current, j_D , of Fig. 1.3 and affects the momentum coupling to the ionosphere. Although unspecified, a cross-stream self polarization electric field (E_{csp}) in the $-\vec{v}_n \times \vec{B}_o$ direction is required to launch Alfvén waves along the field. The $j_p \times \vec{B}_o$ force in the Alfvén wave front accelerates the background plasma in the direction of the neutral stream, while the current, j_D , within the stream slows the newborn ions as previously discussed in connection with Fig. 1.3. The cross-stream self polarization electric field must take on a value that is consistent with the currents required in the Alfvén wave for momentum exchange. If we assume

the electric field is constant along the flux tube, then the electric field is given by a simple formula [Brenning *et al.*, 1991a].

$$E_{csp} = -V_n \times B \frac{\lambda_m}{1 + \lambda_m} \quad (1.10)$$

where V_n is the velocity of the neutral stream and the quantity λ_m is defined in (1.8).

Piel *et al.* [1978] proposed the ionizing front model for CIV experiments in rotating plasma devices. A polarization electric field is set up by an ion density gradient which is created by new ions shooting out in front of their magnetized electrons. The width of this front is less than a ion gyro radius but greater than an electron gyro radius (Fig. 1.4). The potential, ϕ_o , in the front is such that in a steady state it will stop the forward motion of the newborn ions.

$$e\phi_o > 1/2mv_n^2 \quad (1.11)$$

This polarization field is antiparallel to the beam direction and is denoted by E_{bp} . It drives an electron Hall current in the $E_{bp} \times \vec{B}_o$ direction which is across the neutral stream. Lower hybrid waves are excited by the electron current across the neutral stream due to the modified two stream instability. This is one of the major differences between the two models. An electron current induces a two stream instability in this ionizing front model while an ion beam drives the two stream instability in the homogeneous model. This electron current must close across the magnetic field in the ionosphere to meet the momentum transfer requirements. The ionizing front model, in its present form, does not include this process [Brenning *et al.*, 1991a].

Table 1.1: Properties of Barium [*CRC Handbook*, 1984].

Atomic number	56
Atomic weight	137.33 amu
First ionization potential	5.211 eV
Second ionization potential	10.004 eV
Critical ionization velocity, V_c	2.6962×10^3 m/s
Specific gravity	3.5
Melting point	725° C
Boiling point	1640° C

1.3 A Barium Stream in the Ionosphere

Laboratory results, when applied to space, are subject to the criticism that the chamber walls are an infinite sink for momentum. In space this momentum must be accommodated by radiation of an Alfvén wave as discussed above in a process which is not completely understood. So the fact that CIV occurs in the lab is no guarantee that it will work in space.

To study this problem a series of experiments were carried out in the period 1983-89 using sounding rockets for access to space. The primary material of choice for these experiments is barium. Barium is a metallic element of the rare earth group discovered by Sir Humphrey Davy in 1808. It is soft and silvery white like lead but oxidizes easily. Properties of barium are presented in Table 1.1. Barium has the third lowest ionization potential, after sodium and potassium and is not nearly as dangerous to work with, making it ideal for a CIV experiment. With high explosives, barium can be accelerated and vaporized to form a stream. This procedure is discussed in Section 2.2.1, and details of the experiments reported on in this thesis are given in subsequent chapters.

There are two components to the ionosphere, the plasma component and the neutral gas component. The neutral density is greater than the plasma density by

Table 1.2: Typical mid latitude F peak (400 km) characteristics.

Plasma density (day)	$5 \times 10^5 \text{ cm}^{-3}$
Plasma density (night)	$1 \times 10^5 \text{ cm}^{-3}$
Neutral density (sunspot maximum)	$4.7 \times 10^8 \text{ cm}^{-3}$
Neutral density (sunspot minimum)	$2.4 \times 10^6 \text{ cm}^{-3}$
Oxygen Ions	98%
Hydrogen Ions	2%
Temperature, $T_e \approx T_i$	1100° K

several orders of magnitude to at least 1000 km in altitude. The highest plasma density is found in the 300 to 600 km range which is termed the F-region of the ionosphere. Characteristics of the F region vary with solar cycle, solar activity, and time of day. Surprisingly, even the neutral density in this region can change by several orders of magnitude over a solar cycle or within a few days of an intense magnetic storm at a given height. Some parameters for the mid-latitude ionosphere in the F region at the peak plasma density, at about 400 km, are presented in Table 1.2. The height choice for an ionospheric CIV experiment is a compromise between the desire for low collisionality, high electron density, and existing rocket lift capability.

1.4 Seed Ionization Processes

Barium atoms will rapidly photoionize if exposed to wavelengths less than 3265 Å, for example, in the presence of solar ultra violet light. The time constant for ionization in full sunlight is only about 28 s, so the entire neutral stream would rapidly ionize if exposed [Stenbaek-Nielsen *et al.*, 1990a]. If the neutral barium stream is not exposed to solar UV the dominant ionization process is not as clearly understood. If conditions are appropriate, a CIV avalanche might be ignited, otherwise, collisional processes will produce the only ionization. Since in any case a few seed

ions must exist to start a CIV discharge, we need to discuss what is known about the collisional processes.

Charge exchange and charge stripping are expected to be the dominant collisional ionization processes in the ionosphere. Such collisions occur between barium and all trace components of the ionosphere, but we will concentrate on the reactions which are expected to dominate. Production of barium ions by the charge stripping reaction with neutral oxygen,



is possible only if the interaction energy is greater than 5.51 eV (the ionization energy for barium). The subscripts fast and slow refer to the velocity of the species relative to the lab reference frame which we define to be that of the ambient plasma. If the oxygen is at rest then the barium velocity must be greater than 8.4 km/s for stripping to be energetically possible. The cross section for this process is estimated to be $9 \times 10^{-22} \text{ m}^2$ [Stenbaek-Nielsen *et al.*, 1990b]. Charge exchange of barium with oxygen ions,



at energies near the ionization potential for barium is possible because of energetic resonances between the two species. These resonances occur when the energy of an internal state of the atom losing the charge combined with the energy required for ionization equals the energy available from recombination with the interacting atom [G. Swenson *et al.*, 1991]. This type of process can result in relatively large cross sections that depend strongly on the interaction energy, as shown in detailed calculations for charge exchange of barium with barium ions published by Sramek [1980].

Ion production by stripping or charge exchange can be simply calculated with the assumption that the reacting species in the ionosphere do not become extinct during the process. Extinction, when one of the reactants becomes totally or nearly consumed, is more of a concern for ionospheric oxygen ions than the other species. For this calculation we assume the ionospheric species is stationary in the laboratory reference frame. The rate at which ions are produced from the neutral barium stream is given by

$$\frac{dn_{Ba^+}}{dt} = n_{Ba} n_r v_{Ba} \sigma_r \quad (1.14)$$

where n_r is the density of the reacting species (oxygen or oxygen ions), v_{Ba} is the velocity of the barium stream and σ_r is the cross section for the process. The neutral, n_{Ba} , and ionized, n_{Ba^+} , barium density are related to the initial density of the barium stream.

$$n_{Ba^+} + n_{Ba} = n_{Ba \text{ stream}} \quad (1.15)$$

If the cross section, the barium density $n_{Ba \text{ stream}}$, and velocity v_{Ba} are taken to be constant then the barium ion density can be simply found.

$$n_{Ba^+} = n_{Ba \text{ stream}} (1 - e^{-n_r \sigma_r v_{Ba} t}) \quad (1.16)$$

If $n_{Ba \text{ stream}}$ and v_{Ba} vary with time and σ_r is a function of the interaction energy, then integration of (1.14) and (1.15) is not quite as straight forward. In principal, the ionization resulting from the charge exchange or charge stripping process can be calculated. If the quantities are known by measurement, then these equations can be numerically integrated to estimate ionization by these processes. In practice, the cross sections for the process (1.12) and (1.13) have not been measured. Furthermore, the theoretical calculations for these processes becomes complex for interaction energies less than 100 eV [Sramek, 1980]. Various estimates for these

Table 1.3: Estimated cross sections for charge exchange and charge stripping.

Charge Exchange: $\text{Ba}_{\text{fast}} + \text{O}_{\text{slow}}^+ \rightarrow \text{Ba}_{\text{fast}}^+ + \text{O}_{\text{slow}}$	
σ	Notes
$1 \times 10^{-20} \text{m}^2$	<i>Personal communication, R. Olson, 1991; [Olson, 1972]</i>
$1 \times 10^{-19} \text{m}^2$	<i>[Brenning, 1982]</i>
$2 \times 10^{-19} \text{m}^2$	<i>[Torbert, 1991]</i>
$1 \times 10^{-18} \text{m}^2$	<i>[G. Swenson et al., 1991]</i>
Charge Stripping: $\text{Ba}_{\text{fast}} + \text{O}_{\text{slow}} \rightarrow \text{Ba}_{\text{fast}}^+ + \text{O}_{\text{slow}} + e^-$	
σ	Notes
$9 \times 10^{-22} \text{m}^2$	<i>[Brenning et al., 1991a]</i>
$1 \times 10^{-23} \text{m}^2$	<i>Personal communication, G. Swenson, 1991</i>

cross sections are compiled in Table 1.3. Unfortunately the estimates for the charge exchange cross section vary by more than two orders of magnitude. This has become an important question for ionospheric CIV experiments using barium. Depending on the value used, data either confirm CIV (*Stenbaek-Nielsen et al. [1990a]*) or were contaminated by resonant charge exchange *G. Swenson et al. [1991]*! A detailed calculation of charge exchange cross sections after the manner of *Sramek [1980]* would require effort on the order of a Ph.D thesis (*personal communication R. Olson, 1991*). The best solution would be a laboratory measurement as called for by *G. Swenson et al. [1991]*. We try here to cope with these discrepancies, particularly with regards to the charge exchange, but find the situation problematical to say the least. We tend to believe the result quoted by Torbert rather than *G. Swenson et al. [1991]* since they are based on extrapolation of a Ca-O^+ cross section. The *G. Swenson et al. [1991]* cross-sections are somewhat controversial and meant to make the point that we do not really understand the barium oxygen reactions.

1.5 Thesis Overview

CIV is perhaps of cosmic importance playing a role in the shaping of planetary, star, or even galactic systems. It is proposed as a fundamental reaction between a magnetized plasma and a neutral gas. Laboratory experiments have confirmed CIV, but this is not proof that CIV actually occurs in nature. The theories for CIV have concentrated on the energy balance and energy flow of the process, and efficiencies have been calculated with the assumption that lower hybrid waves are the heart of the process. The associated electric fields and current systems which would be required are largely unspecified. The next steps in the search for CIV have been tests in space and tests in space with *in situ* diagnostics. In this thesis we report on a set of CIV experiments conducted in the ionosphere which made *in situ* measurements.

In the following chapters we will review the CRIT I and CRIT II experiment and observations. The CRIT II experiment will be more heavily emphasized since data are available from a greater variety of instruments and the over all data quality is superior to CRIT I. In Chapter 2 we will review the results of previous CIV experiments in the ionosphere and outline the goals and methods of the CRIT experiments. The deployment of the payloads and descriptions of the instruments are also presented in this chapter. In Chapter 3 we present an overview of the data. Since an enormous amount of data has been collected, we use CRIT II data as a guide and summarize all of the other data in Appendix B. In Chapter 4 we focus closely on the plasma density observations of CRIT II. Possible explanations for the observed density enhancement will be considered, and the discussion will conclude with an argument that these plasma enhancements are signatures of CIV, the first unambiguous *in situ* detection of this process in space. In Chapter 5 we discuss the magnetic and electric field fluctuations accompanying the density enhancements of

Chapter 4. We will examine aspects of the data that are possible signatures of CIV, that are related to the various current theories for the CIV micro-process, and that show the data set is internally consistent. We conclude that a different instability than lower hybrid waves is observed in the data. In the final chapter, we summarize the previous chapters and discuss areas of the CRIT data set needing further analysis. Problems related to conducting ionospheric tests for CIV and making meaningful measurements during such tests will be discussed as will future research possibilities.

Chapter 2

CRIT I & II Experiments

The objective of the CRIT experiments was to make meaningful measurements of the interaction between a streaming neutral gas and a magnetized plasma. It was expected that ions would be created out of the neutral stream, and the *in situ* measurements were aimed primarily at this interaction. In particular, we wanted to study the energy and momentum transfer of these newborn ions with the surrounding plasma. We expected these processes to be characterized by plasma waves, plasma density variations, and particle heating. The experiment design and instrument choices reflected these goals.

Another goal was to initiate and study a CIV type reaction within a neutral stream in space. Laboratory experiments, documenting the critical velocity effect, have been carried out over the years and reviewed in the scientific literature [Danielsson, 1973; Möbius *et al.*, 1979]. These experiments make Alfvén's predictions about the CIV process quite credible. Since there are fundamental differences between laboratory and space experiments, such as containing walls, a natural extension of the laboratory experiments is to test for CIV in space. It was expected that if a CIV reaction were initiated within the neutral stream, a significant per-

centage of the neutrals would become ionized, and this alone would distinguish the phenomenon from competing ionization processes. The newly produced ion species would be easily detectable from the ground by optical observations of their characteristic spectral qualities. Further confirmation of such a reaction would come from the *in situ* measurements, particularly from electron energy and plasma density measurements. Broadly stated, this objective was to determine the process or processes causing ionization and to distinguish predictable charge stripping and charge exchange processes from CIV. For the purpose of initiating a CIV reaction, we chose a neutral species, barium, with a low ionization potential and created a stream with a velocity perpendicular to the geomagnetic field that exceeded the predicted CIV threshold.

2.1 Previous Experiments

The CRIT I and CRIT II experiments were part of a series of dedicated experimental tests in space involving streaming neutral gases and investigations of CIV. These include Porcupine [Haerendel, 1982], Star of Lima [Wescott *et al.*, 1986a; Kelley *et al.*, 1986], Star of Condor [Wescott *et al.*, 1986b; Torbert and Newell, 1986], and SR1990 [Wescott *et al.*, 1990]. In addition some “accidental” experiments were performed in previous 1972, and 1982. One of the fundamental objectives of the CRIT I and CRIT II experiments was to resolve conflicting results of previous space based CIV experiments. Recent reviews of both the laboratory and space based CIV experiments have been presented in the literature [Newell, 1985; Torbert, 1990]. A summary of these experiments is presented in Table 2.1, after the manner of Torbert [1990] but updated and expanded with data from CRIT I and CRIT II added. References to papers on individual experiments can be found

Table 2.1: Summary of Space based chemical release experiments that have been examined for Alfvén's Critical Ionization Velocity effect and their results as determined from ground based optical observations.

Experiment	Year	Plasma Density	$\vec{B}_0$ Angle	Type	Prompt Ionization Total	CIV
Chachalaca [†]	1972	?	10 – 15°	30° Cone	≈5%	≈5%
Buaro ^{†*}	1976	1×10^6	90°	30° Cone	-	0%
Porcupine [†]	1979	6×10^5	28°	Cone	6-7%	16-18%
Bubble Machine II ^{†*}	1982	?	-	Radial	0%	0%
Star of Lima [†]	1983	2×10^4	90°	Cone	0.05%	0%
Star of Condor [†]	1983	2×10^4	-	Radial	0.004%	0.0017%
George Orwell [†]	1984	?	-	Radial	0%	0%
SR90 ^{†*}	1986	1.5×10^4	48°	Cone	0.34%	0.17%
CRIT I [†]	1986	3.4×10^4	47°	Cone	0.02%	0.01%
CRIT II [†]	1989	5.4×10^5	57°	Cone	4%	1.67%

[†] - Barium [‡] - Strontium * - In solar UV

in the bibliography. Table 2.1 is an attempt to summarize different experimental conditions for the events and allow comparison of the results for the different reporting methods. The ionization percentages presented in this table come from ground based optical observations. The two percentages given for prompt ionization within the neutral stream are calculated as follows. The entries in the Total column are calculated from the ratio of the total number of ions produced to the total number of neutrals initially introduced. The entries in the CIV column are calculated from the ratio of ions produced to the number of neutrals that are energetically able to contribute to a CIV reaction (those with $V_{\perp} \geq V_c$) and neglecting ions produced by known processes. Two kinds of neutral streams have been created for these ionospheric tests. A radial release approximately creates a flat disk of neutral gas that expands from its center into the ionosphere. The second method creates a directed stream of barium that expands as a cone shaped beam. The resulting clouds differ in the rate of expansion and in the number of neutrals with similar velocities perpendicular to the geomagnetic field. Two neutral species, barium and strontium, have been used in these tests because the critical velocities for these two

elements, 2.7 and 3.5 km/s, are among the lowest. Of all the previous experiments, only Porcupine created significant ionization attributed to CIV [Haerendel, 1982]. Null results occurred mostly from radial releases and those using strontium. The only other conical barium release before the CRIT experiments was Star of Lima. The Porcupine experiment had the highest background density, a parameter now thought to significantly effect yield of CIV. The CRIT I and CRIT II experiments were planed to maximize, as far as possible, parameters that would contribute to a CIV reaction, and to probe the differences between positive and negative results of previous experiments.

Several experiments in Table 2.1 have included *in situ* measurements of the chemical release. The Porcupine experiment included several instrumented payloads, but they were positioned neither inside the stream nor on a geomagnetic field line passing through the stream. The data obtained were of little value for understanding the process occurring within the stream [*private communication Nils Brenning*, 1990], and only a small part of the data has so far been published in the scientific literature. An earlier experiment by *Koons and Pongratz* [1981], named Buaro, made measurements inside a photoionizing barium stream of electric and magnetic fields and energy distributions of charged particles. The electric field measurements were made with a single dipole electric field detector inside the barium stream. This experiment was not an attempt to study the ionization mechanism within the stream since it occurred in solar UV, but was designed to investigate the interaction of a dense barium plasma stream and the ionosphere. The Star of Lima experiment included electric field and energetic particle measurements within the barium stream [*Kelley et al.*, 1986]. The electric field measurements were also made with a single probe set. One of the objectives of the CRIT experiments was to improve on these previous *in situ* measurements with a more complete instrument

complement. It was hoped that additional data might help in the understanding of these previous measurements.

2.2 Experiment Overview

Both the CRIT I and CRIT II experiments were carried out with Black Brant X sounding rockets launched from Wallops Island, Virginia (see Fig. 2.1). Each three stage rocket boosted to approximately 400 km a payload which dispersed along the way into four separate modules. Two of the modules contained high explosive chemical canisters which when detonated produced the high velocity barium streams for the experiment. The other two modules were instrumented diagnostic packages called the main and subpayloads. The relative location of the chemical canisters, main, and subpayloads was a critical item in the design of the experiment. The objective was to place the main payload in the path of the neutral barium stream and the subpayload outside of the stream but on the same field line as the main payload. The chemical canisters were to be pointed back at the main payload at detonation, thus creating a fast neutral stream that would sweep past the payload. The first chemical release was to be about 2 km and the second about 4 km from the main payload, providing two measurements with different neutral stream densities as the beam expands through the ionospheric background. The chemical canisters and instrumented payloads were to be placed below the ultraviolet terminator, in the earth's shadow, to prevent photoionization. The dispersal directions of the various payload subsections was determined by an inertial attitude control system. It was hoped to orient the payload at specific times to planned directions with at least a 5° accuracy. Poor performance of the attitude control systems during both the CRIT I and CRIT II experiments resulted in deployments that were not planned,

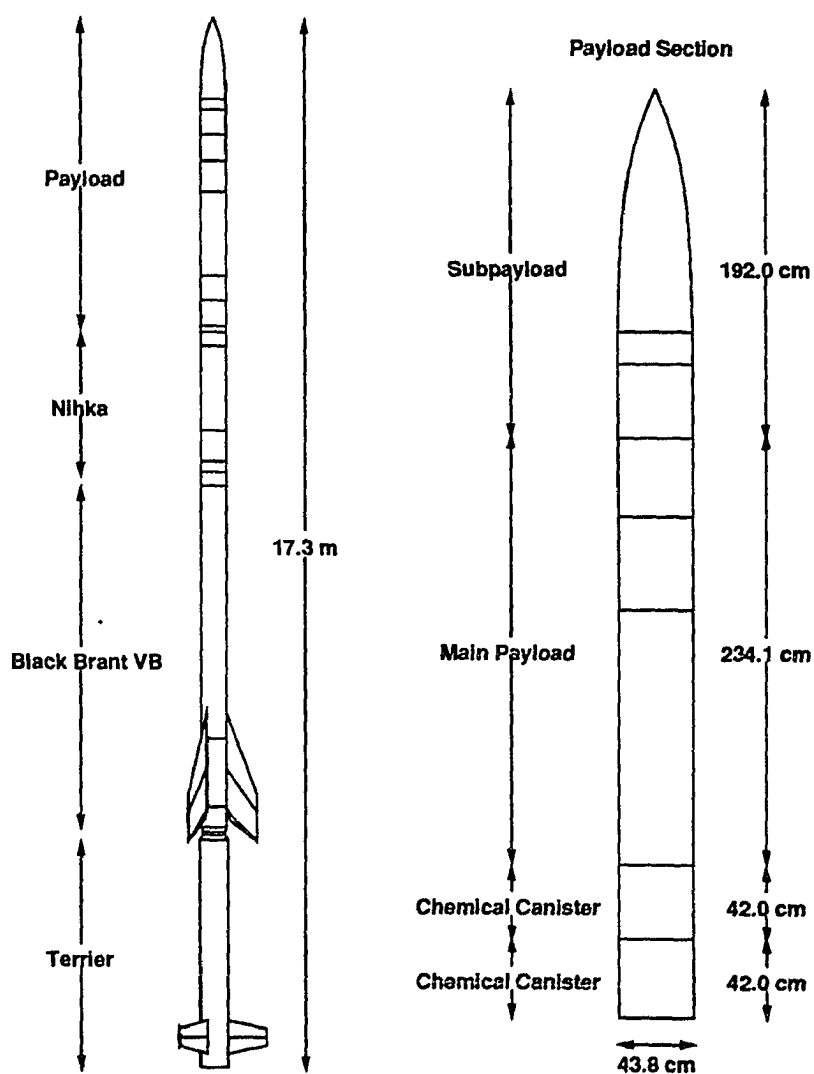


Figure 2.1: General drawing of the Black Brant X sounding used for CRIT II (NASA rocket 35.019) and an enlarged view of the payload section. The CRIT I and CRIT II rockets and payloads were very similar.

but suitable for most of the experimental objectives.

Deployment Sequence

The payload stack and intended deployment sequence of both CRIT I and CRIT II were very similar. A general drawing of the payload deployment sequence is presented in Fig. 2.2. The attitude control system, which controlled the deployment, was part of the main payload section. After about 70 s of flight, at 67 km altitude, the atmospheric forces on the payload were reduced to the point that deployment could begin. After successful firing and detachment of the motors the nose cone was ejected, exposing the subpayload. For CRIT I this occurred after third stage had fired, while for CRIT II this was done before the third stage had fired to achieve the required altitude by shedding excess weight. The payload stack was decoupled from the spent third stage motor casing and oriented nearly anti-parallel to the local geomagnetic field. Booms were deployed on the subpayload and the spin rate of the entire stack adjusted. At 147 seconds into the flight the subpayload was separated by an air spring. The intention was to position the subpayload exactly along the same field line as the main payload about 210 seconds later at the time of the first release. The attitude of the payload with respect to $\vec{B}_0$ was adjusted to 45° from the geomagnetic field, and one of the chemical canisters was ejected by air springs. This maneuver occurred about 178 seconds into the flight, and this attitude was maintained by the attitude control system through the remainder of the mission. About 6 seconds later came the ejection of the other chemical canister, which is the first one to be detonated. Booms on the main payload were deployed and the spin rate of the payload adjusted. The first chemical canister was detonated about 350 seconds into the flight, just before apogee, by a command radioed from the main payload. The second chemical cannister was detonated about 85 seconds later, just after apogee.

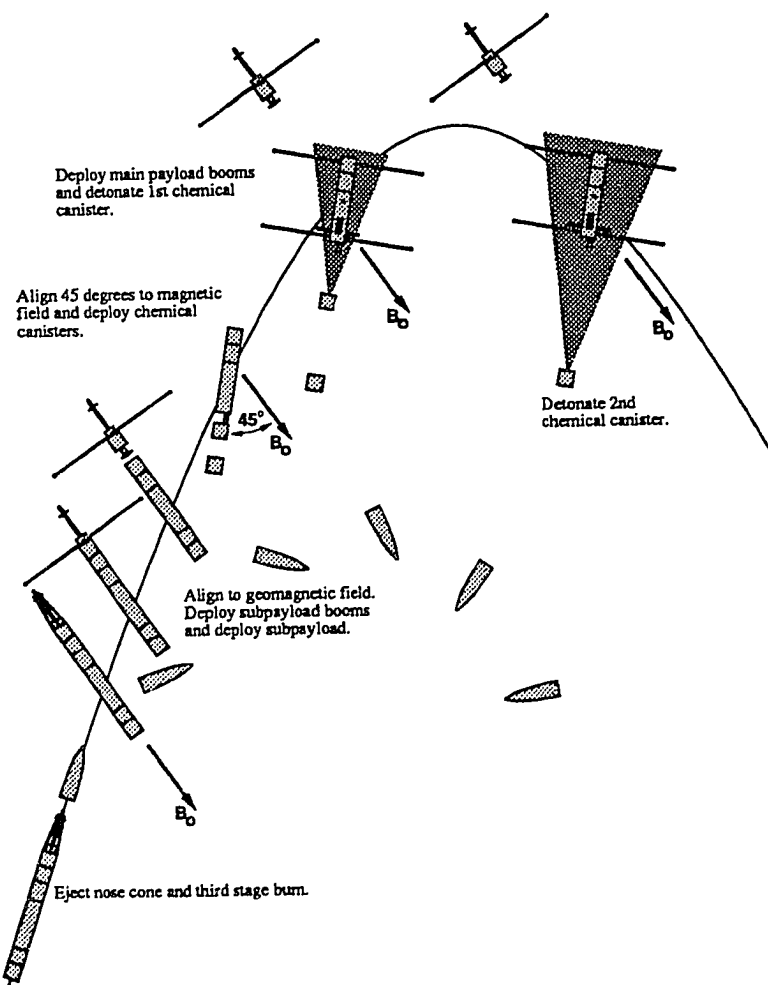


Figure 2.2: The subpayload and chemical cannister deployment sequence for the CRIT II experiment.

Deployment Verification Methods

Knowing exactly where the various payload pieces went is crucial to any interpretation of the measurements, so redundant information was gathered about the deployment during the flight. From this flight information, coupled with the mass and moments of inertia of each module, an attempt has been made to determine the resulting deployment. Fully determining the deployment requires knowledge of the attitude and location of each part of the payload relative to each other, the earth, and the geomagnetic field. The solution arrived at should be internally consistent with the measurements, orbital mechanics, and rigid body dynamics of the system. Indeed, this aspect of the experiment analysis has consumed considerable time of people at NASA, Cornell University, and University of Alaska, much more time in fact than had been anticipated.

The primary positional information was gathered from radio beacons located on the main and subpayloads. Ground based receivers use these beacons to determine location and velocity from which a ballistic trajectory for the payload can be calculated. The CRIT II subpayload used the TRADAT system which is based on the Doppler shift technique of the transmitter instead of a radio beacon, which would have created magnetic disturbances on that payload, but similar information was obtained. In addition, other radars were used to "skin track" the various objects. Skin tracking is not always reliable since extraneous objects, a spent booster or nose cone, are sometimes tracked. The location of the chemical canisters could not be determined by any of these methods since they were too small to be skin tracked and did not carry beacons or telemetry systems for reasons of weight control. Their location was determined by knowing the attitude of the main payload, their separation velocity and mass, and calculation of their position by orbital mechanics.

The separation velocity for each part of the payload was determined by several

methods where possible. The expected separation velocities were calculated from the pressure in the air springs, the separation mechanism, and the mass of the sections. The actual velocities were measured by pull strings attached to each body where possible. Unfortunately, pull strings were not used on the CRIT II chemical cannisters, these were removed just before flight for fear they might induce coning in the chemical cannisters during ejection. These methods give the magnitude of the separation velocity and must be combined with directional measurements to become useful in calculating the three-dimensional deployment. Lastly, the velocities were calculated from deviations in the ballistic trajectories of the main and subpayloads. To use this method the main payload must be observed for some period of time before and after an event. Comparing the two trajectories and knowing the mass of the objects, the separation velocity and direction can be calculated for each body. This last method is the most precise, with the added advantage of determining the vector separation velocity. This technique also very precisely defines the attitude of both bodies at the instant of separation. Unfortunately, due to the relatively short period of time between chemical cannister ejections, it could not be used to determine the individual separation velocities of the chemical cannisters.

The primary source of attitude information was to be provided by the gyro, part of the attitude control system. Several rate gyros, measuring different axes, made up the gyro package. For the purposes of this discussion, it is easier to imagine the gyro package as a single multi-gimbal gyro like those sold in many toy stores. The gyro is only effective for determining attitude if the reference direction is maintained. This direction is established when the gyro is spun up and allowed to freely move in its gimbals. A set of Euler angles are computed (pitch, yaw, roll), based on the reference direction, to describe the motion of the payload during the flight. If some anomalous event causes the gyro's reference direction to move during the flight,

then all attitude data from that time on is invalid. The only way to salvage the remaining measurements would be to determine the new reference direction by some other means. It is impossible to distinguish between motion of the payload and a drifting reference by examination of gyro data alone. For this reason it is important to have other ways to either fully or partially determine the payload's attitude at some time during the flight, such as at the instant of separation of the main and subpayloads as discussed in the previous paragraph. Other sources of attitude information for the CRIT experiments come from magnetometer measurements, electric field instruments, and radar tracking. Magnetic field models (IGRF and DGRF) are used in conjunction with the magnetometer measurements to partially determine attitude. Radar tracking data was used during the boost phase to verify the attitude gyro for the early portion of the flight since, when the rocket was still in the atmosphere, the attitude of the payload must be along the flight path.

2.2.1 Barium Release

The chemical canisters, provided by the University of Alaska, were of the shaped-charge design, and separated from the main payload by air springs. The coning of the chemical cannisters induced by the separation was expected to be less than 3° . Each release was successfully detonated by radio command from the main payload, but a backup timer would have detonated the canisters if the local radio link had failed. The cannisters consisted of a receiver, backup timer, batteries, and a single charge. The charge was aimed forward along the direction of the cannister spin axis and consisted of 7.4 kg of 75/25 Octol high explosives with a tapered 30° conical liner of barium metal. The design was identical to the shape-charge used for the SR1990 experiment [Wescott *et al.*, 1990] except barium was substituted for strontium. About 900 g of barium was contained in each shaped charge but only

15% of the metal is expected to be vaporized, producing approximately 6×10^{23} particles. Because the releases were made in darkness, the data were not ideal for optically diagnosing the neutral jet directly. Hence all estimates of the performance of the barium releases are based on previous releases made in sunlight.

Neutral Distribution

The SR1990 strontium release, made in full solar UV, was optically observed under near ideal conditions and "...obtained the best alkali metal shaped-charged neutral distribution as a function of radial angle from the detonation point and as a differential number of neutral per unit velocity" [Wescott *et al.*, 1990]. From the limited data obtained during the CRIT I releases *Stenbaek-Nielsen et al.* [1990a] conclude that "...observations of the major features (tip and peaks) in the barium jet are consistent with the SR1990 release to better than a few tenths of a km/s". Based on these data we present Fig. 2.3 as the estimated differential velocity distribution we shall use for all releases in the CRIT experiments. The angular distribution is nearly Gaussian and the half maximum cone angle is 6.6° for velocities above 7.0 km/s, $10-11^\circ$ for velocities between 7.0 km/s and 3.5 km/s, and $21-24^\circ$ for the lowest velocities (*personal communication, H.C. Stenbaek-Nielsen, 1990*). The ordinate of Fig. 2.3 is based on the assumption that 15% of the of barium liner was vaporized. This percentage is based on laboratory tests of similar devices where the yield of vaporized barium had this value. We point this out because several calculations which will be presented in this and following chapters are based on the data of Fig. 2.3.

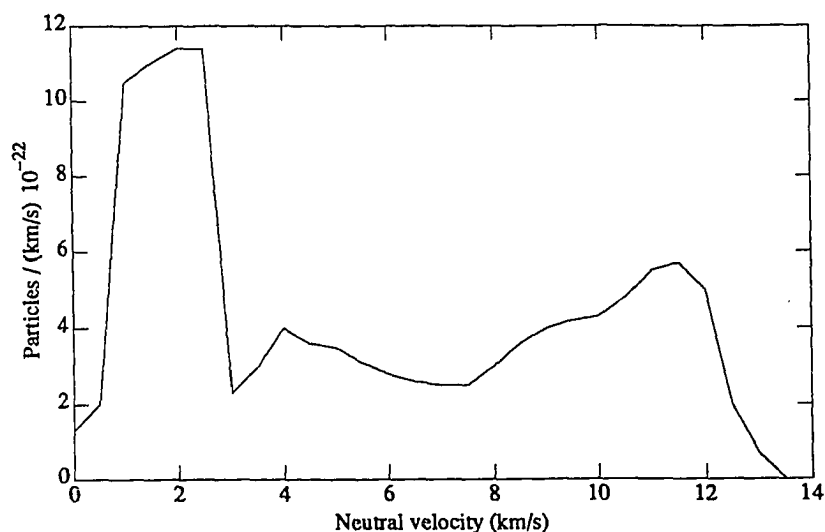


Figure 2.3: Differential velocity distribution of the neutral barium stream as estimated from the SR1990 strontium release [Wescott *et al.*, 1990]. The ordinate scale is derived from the assumption that 15% of the barium liner is vaporized. After [Stenbaek-Nielsen *et al.*, 1990a].

Neutral Stream Calculations

The time evolution of the neutral barium density at the payload is an important parameter for interpreting the *in situ* observations of the CRIT experiments. The neutral stream is the ultimate source of free energy driving the observed density and field fluctuations. In addition binary collisional interactions of the neutral stream with background neutral gas and plasma are of interest. The question must be answered as to whether either of these binary collisional processes are significant contributors to the observations.

Several papers have been published with models to calculate synthetic [Wescott *et al.*, 1990; Stenbaek-Nielsen *et al.*, 1990b] images to compare with optical results. In these images, absolute densities of neutrals or ions clouds are not made; instead, the model is calibrated to measured data at some location and time. The purpose of these models has been to explore ion production rates as a function of time and relative density. For comparisons with *in situ* measurements, absolute models

are needed. Using data compiled from optical measurements presented in section 2.2.1 we can calculate the time history of the stream density at the location of a payload. The angular distribution of the neutral jet was observed to be nearly Gaussian [Wescott *et al.*, 1990]. This distribution can be expressed as

$$n(r, t, \theta) = f(r, t)e^{-a\theta^2} \quad (2.1)$$

where θ is the angle between the burst direction and a radial vector, $a = \ln(2)/\theta_0^2$, θ_0 is the half maximum half angle of the neutral stream expressed in radians and $f(r, t)$ is a normalizing function, to be determined, depending on distance and time. The number of particles in a thin spherical surface dr wide at distance r is given by

$$dN = 2\pi r^2 f(r, t) dr \int_0^\pi e^{-a\theta^2} \sin(\theta) d\theta \quad (2.2)$$

If $a \gtrsim 4$ then the exponential function will decrease fast enough to allow us to approximate $\sin(\theta)$ with θ and to replace the upper limit of the integration with ∞ . If this approximation is made then (2.2) reduces to

$$dN = \frac{\pi r^2 f(r, t)}{a} dr \quad (2.3)$$

This approximation becomes very good if the half density maximum angle is less than 75° , a value much larger than the barium beams used for the experiments. The quantity dN can also be calculated from $F(v)$, the differential velocity distribution of the neutral barium stream, plotted in Fig. 2.3. At time t the number of barium neutrals in a shell dr wide at a distance r from the release is given by

$$dN = F\left(\frac{r}{t}\right) \frac{dr}{t} \quad (2.4)$$

Combining (2.3) and (2.4) gives an expression for $f(r, t)$. Using this result in (2.1), the expression for the neutral density at distance r and θ radians from the release direction at time t is found to be

$$n(r, t, \theta) = \frac{\ln(2)}{\pi \theta_0^2} \frac{F\left(\frac{r}{t}\right) e^{\frac{-\theta^2 \ln(2)}{\theta_0^2}}}{t r^2} \quad (2.5)$$

The neutral density falls off as the square of the distance from the release resulting in a rapidly expanding barium cloud. Equation (2.5) assumes θ_0 is not a function of velocity but in reality the beam widens with lower velocity (*personal communication, H.C. Stenbaek-Nielsen, 1990*).

The calculated time evolution of the neutral barium density at the location of the main payloads for both chemical releases of CRIT I and CRIT II is presented in Fig. 2.4. These plots include the effects of beam widening for the lower velocity neutrals. The neutral density rise time is about 20 ms for the near releases and about 50 ms for the farther releases. The peak neutral density at the payload occurs for beam particles with velocities of 11.5 km/s. The density tapers off by an order of magnitude in about 0.2 seconds.

The neutral barium stream will affect the background ionosphere through collisions. The neutral stream velocity, V_b , is much larger than the thermal speed of oxygen ions in the ionosphere. Therefore, the ion-neutral collision frequency may be computed with the simplifying assumption that the ionospheric ions are a stationary target. This gives neutral density, n , by

$$\nu = n \sigma V_b \quad (2.6)$$

where V_b is the velocity of the barium beam and σ is the cross section for momentum transfer. In the limit of treating the particles as hard spheres the cross section for

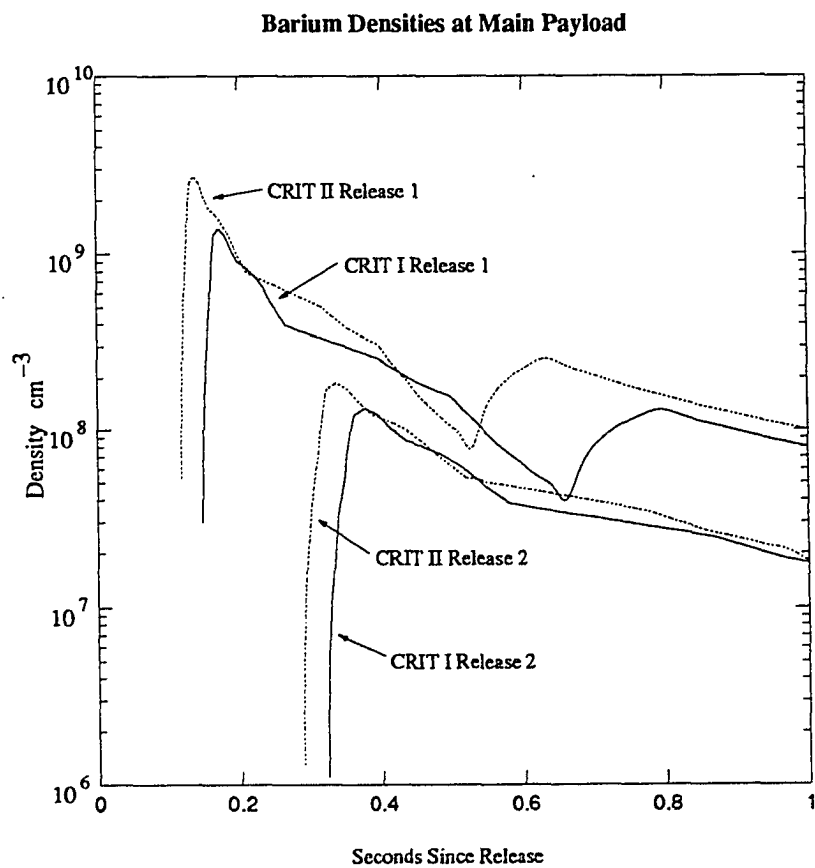


Figure 2.4: Barium density at main payload for each of the four releases of the CRIT I and CRIT II experiments.

Table 2.2: Estimated cross sections for momentum exchange of barium with ionospheric components in the hard sphere limit.

Cross Section for Momentum Exchange (m^2)			
	Ba	O	H
Ba	7.0×10^{-20}	8.3×10^{-21}	3.7×10^{-22}
O	7.1×10^{-20}	1.8×10^{-20}	
H	5.1×10^{-20}		3.2×10^{-21}

momentum exchange from grazing collisions can be calculated as

$$\sigma = 2/3\pi(r_1 + r_2)^2 \frac{m_2}{m_1 + m_2} \quad (2.7)$$

where r_1 , r_2 , m_1 , and m_2 are the effective radii and masses of the particles. Tables of atomic radii are published in the scientific literature ([Kittel, 1976]) and can be used for this approximate formula. Cross sections for momentum exchange of barium with various ionospheric components are presented in Table 2.2.

The ion-neutral collision frequency, (2.6), can be calculated using the cross sections in the previous table and the effect of the neutral stream on the ionosphere through collision can be estimated. The collision frequencies are plotted for collisions of the neutral barium stream with the singly ionized species of hydrogen, oxygen, and barium in Fig. 2.5. Oxygen forms the bulk of the natural plasma background and are of primary interest. The curve for barium ions is nearly identical to that for oxygen ions and is only valid to the extent that barium ions are formed from the neutral stream and stopped with respect to the ionospheric ions. The gyro frequencies for these species are also plotted for reference. If the collision frequency is greater than the gyro frequency then ionized particles are essentially being dragged with the neutral stream through collisions and interaction is classified as collisional. This situation would occur for background oxygen ions less than 0.9 km from the release. At the location of the main payload the neutral stream

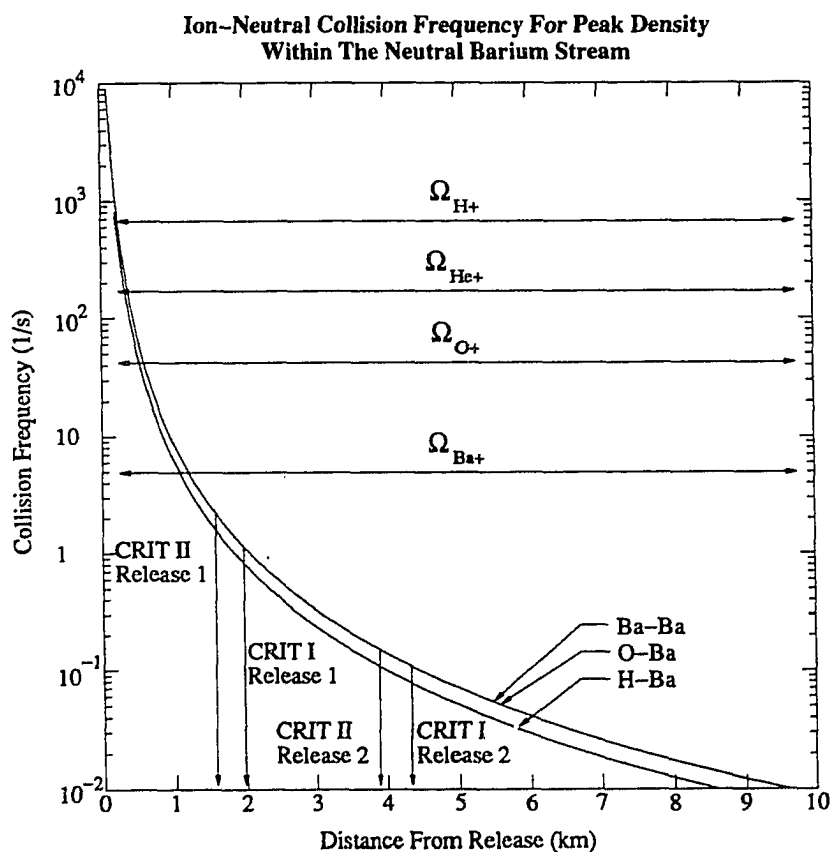


Figure 2.5: The peak ion-neutral collision frequencies for the neutral barium stream with ions of oxygen, hydrogen, and barium are plotted versus distance from release point. The location of the main payloads and gyro frequencies of interest are also noted.

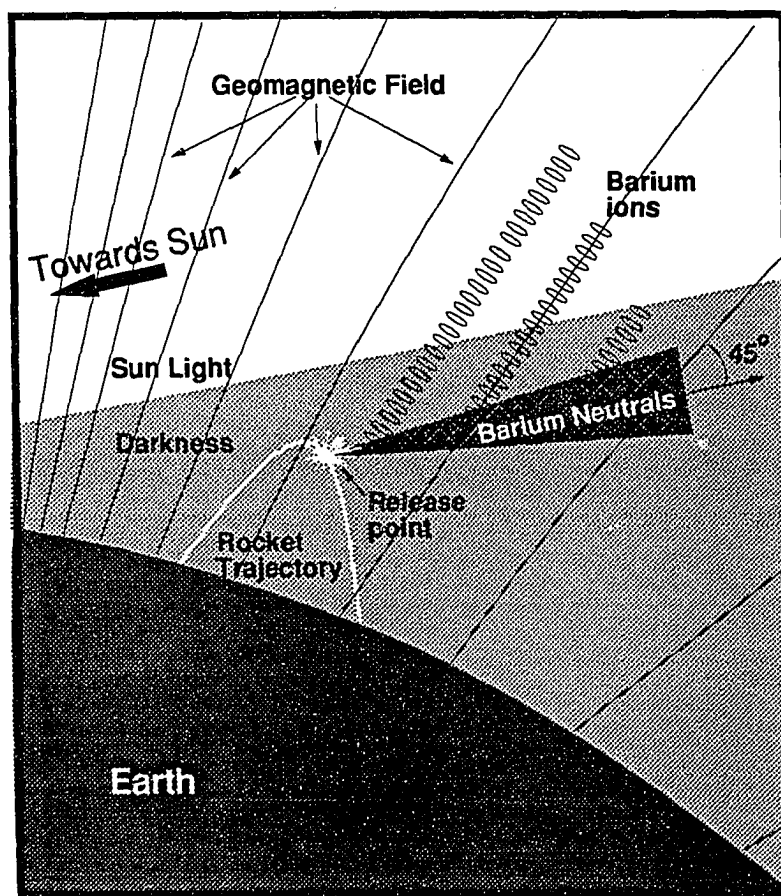


Figure 2.6: Diagram of the barium release direction relative to the magnetic field and the solar UV terminator for the CRIT experiments. This configuration enables detection of barium ions as they enter sunlight and prevents solar UV from ionizing the barium neutrals.

was clearly collisionless with the background ionosphere for all releases. Collisional effects of the neutral barium stream on the ionosphere will be treated in more detail in Chapter 4.

2.2.2 Optical Experiment

The configuration of the CRIT II experiment for optical measurements is presented in Fig. 2.6. The neutral barium stream was directed such that it did not enter sunlight and was $\approx 45^\circ$ to the geomagnetic field. Any ionized neutral barium be-

comes magnetized and spirals along the geomagnetic field into sunlight where it can be optically detected. The distance from the release to the terminator was about 100 km and the release was done after sunset so the terminator is effectively traveling upward. Three optical sites, located at Wallops Island, Virginia, Duck, North Carolina, and Cape May, New Jersey, were used to observe the release. The experiments described in Table 2.1 were done with slight variations to this arrangement. The CRIT I experiment was launched just before dawn so the terminator was falling in altitude and the whole experiment would eventually be illuminated. The results of the optical measurements for CRIT I and CRIT II have been published [Stenbaek-Nielsen *et al.*, 1990a; Stenbaek-Nielsen *et al.*, 1990b]. The motivation behind the optical experiments was to detect ions produced by the critical velocity effect. The results of the CRIT I and CRIT II optical experiments are summarized in Table 2.1.

2.2.3 Main Payload Experiment

The instrumentation on the CRIT I and CRIT II payloads was virtually identical except for some implementation details. Both payloads made three axis electric field and magnetic field measurements. The energetic particle measurements were made for both electrons and ions. The CRIT II experiment also included a photometer at two separate wavelengths. The electric field detectors were provided by Cornell University and the particle detectors by University of New Hampshire. For the purposes of this thesis the largest differences between the CRIT I and CRIT II experiments were the quality of the magnetometer experiment, the placement of the electric field detectors, and the success of the telemetry systems.

Two approaches were taken with the telemetry system on the main payload. A conventional FM-FM system was used where the effective sampling rate is con-

strained by the real time bandwidth of the downlink. The second method made use of a PCM system integrated with a computer and a large solid state memory used as a buffer. Instrumentation on the main payload was rapidly sampled during the barium release, and data were stored in memory for later transmission at the lower rate of the telemetry system. This onboard computer and telemetry system was provided by the University of New Hampshire. The burst storage mode of the PCM system was activated at the same time the command to detonate the barium shaped charges was transmitted by the main payload. The buffer was completely emptied by transmission to ground before the second chemical cannister was detonated. During the CRIT I flight this burst storage mode of the PCM system failed, resulting in the loss of PCM data at the times of the chemical releases. Consequently, only data from the lower performance FM-FM telemetry system was available for analysis. There were no energetic particle measurements from the main payload during CRIT I for the same reasons. The times of these detonation commands provide convenient reference points from which to define a time base. We call this time base, referenced from the start of the burst storage mode, burst time for each of the releases. Light from the chemical canister detonation was detected by the photometers located on the main payload at 0.013 seconds burst time for each of the CRIT II releases and lasted about 10 ms. These data give the best measurement of when the barium jet was actually produced. The times the PCM telemetry system entered burst mode for the various releases are presented in Table 2.3.

Electric Field Detector

The electric field detector on the CRIT I main payload consisted of three sets of dipole antennas that measured 375 cm tip to tip with 3.4-cm-diameter spherical sensors at the ends, as pictured in Fig. 2.7. Two boom sets were positioned or-

Table 2.3: Time of entering telemetry burst mode and sending detonate commands for CRIT I and CRIT II.

Barium Release	Universal Time	Flight Time
CRIT I Release 1	7h 45m 58.507s	358.507
CRIT I Release 2	7h 47m 23.632s	443.632
CRIT II Release 1	1h 47m 58.262024s	348.262024
CRIT II Release 2	1h 49m 30.811035s	440.811035

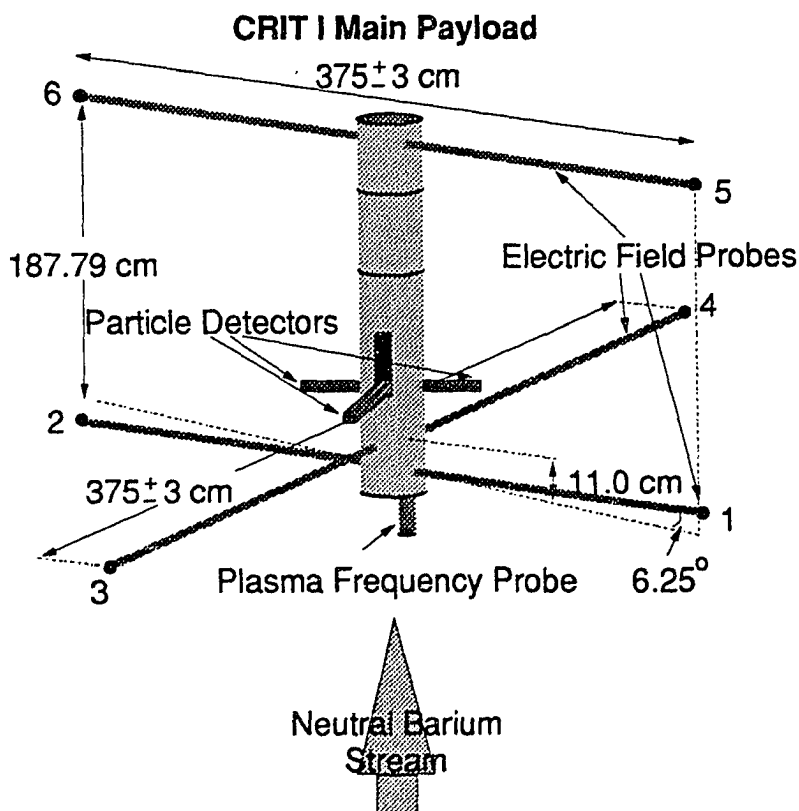


Figure 2.7: Diagram of the CRIT I main payload, NASA flight 35.014. The X-axis (top dead center) of the payload was aligned with electric field detector 1.

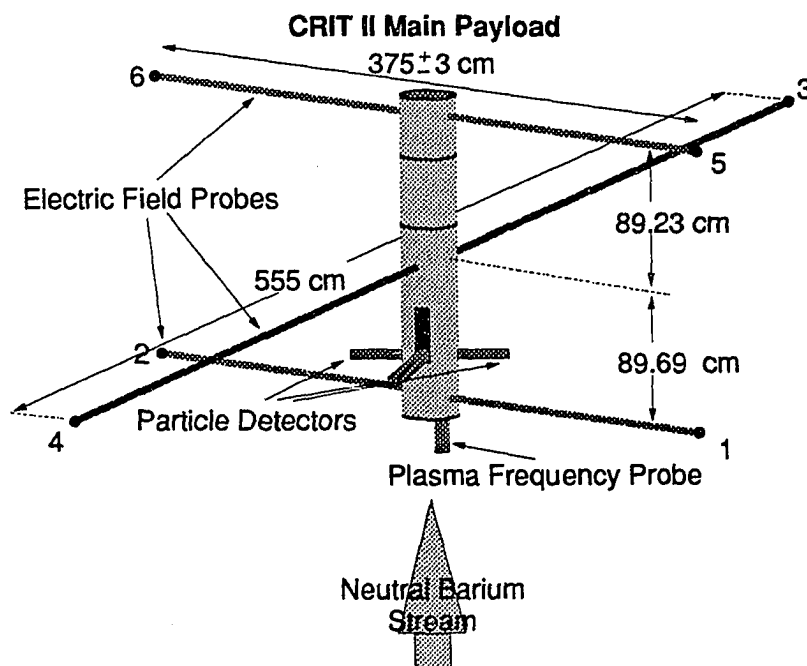


Figure 2.8: Diagram of the CRIT II main payload, NASA flight 35.019. The X-axis (top dead center) of the payload was aligned with electric field detector 1.

thogonally at the end of the main payload closest to the burst. The third set was positioned 1.87 m back and rotated 6.25° about the axis. This configuration provided three linearly independent measurements: two orthogonal measurements across the barium stream and one aligned with the neutral barium and perpendicular to the other two. The frequency response of the instrument was from dc to at least 11.9 kHz in each component. A complete list of CRIT I electric field measurements along with bandwidths and gains is given in Appendix C.

The electric field probe on the main payload of CRIT II consisted of three sets of extendable dipole antennas from which eleven independent measurements were made. A general idea of how the booms were deployed can be obtained from the illustration of the main payload in Fig. 2.8. The three dipole antenna pairs were equally spaced along the main payload with the center booms being deployed orthogonal to the plane of the other sets to a tip-to-tip length of ≈ 555 cm. The

boom sets at either end were deployed in the same plane to a tip-to-tip length of ≈ 375 cm. The sensing elements were 3.4-cm-diameter spheres. A complete list of CRIT II electric field measurements is given in Appendix C.

The electric field dipole antennas on the main payload were configured such that combinations of measurements can be used as an interferometer. With appropriate assumptions, techniques can be used to obtain some information about the phase velocity of the structures being observed. One disadvantage of this boom placement is associated with measurements of short wave length phenomena. In this case various field measurements will not share common phase centers. The phase delay between the various measurements must be considered when trying to determine the vector direction of the field. These subjects are considered in more detail in Chapter 5.

Magnetometer

The aspect magnetometer was not intended as a scientific instrument for CRIT I but was provided for redundant attitude determination. Only two measurements of the magnetic field were made: one along the barium stream direction, that is along the spin axis of the main payload, and the other perpendicular to it. The main payload rotated about the spin axis at about 2 Hz. The frequency response of the magnetometer system is essentially unknown since no attempt was made to calibrate the device for time-varying fields. The outputs were recorded on an FM/FM subcarrier oscillator with information bandwidths of the order of 15 Hz. Such systems can be pushed to a limiting frequency response a factor of 5 higher by using a modulation index of 1 for the ground-based discriminator. However, such a technique creates considerable ambiguity in both the amplitude and the phase of the signal.

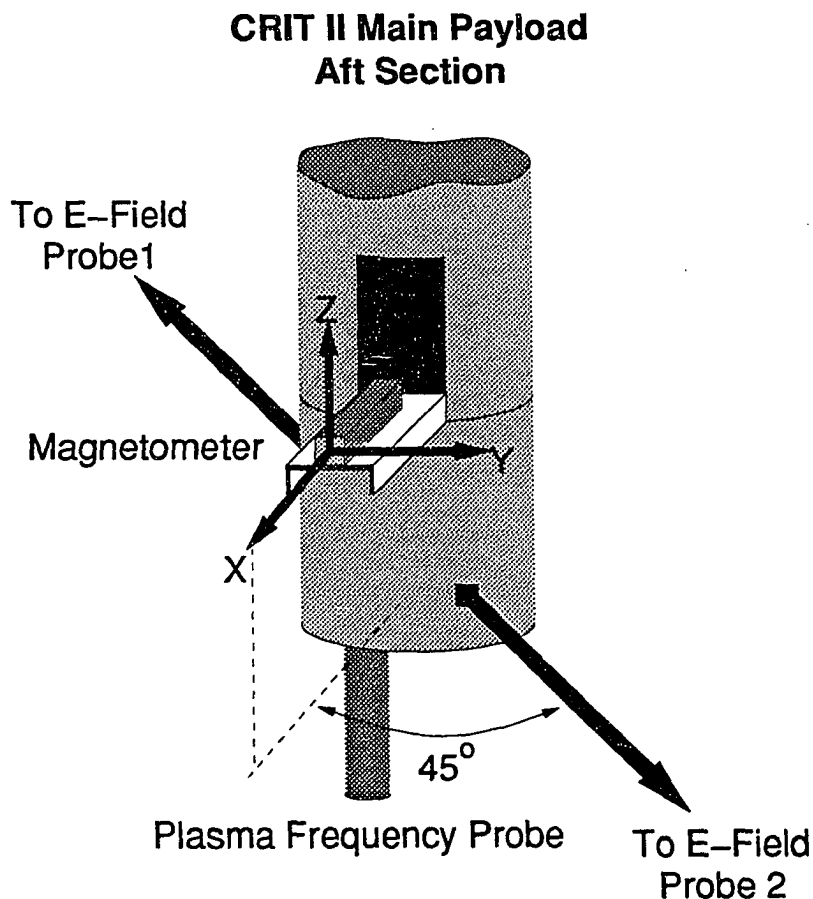


Figure 2.9: Subsection of the CRIT II main payload and measurement coordinate system of the magnetometer. The angle between the x-axis of the magnetometer and the boom to sensor 2 was 45°.

A calibrated, three-axis-fluxgate magnetometer was included with the CRIT II instrumentation. This magnetometer was used to measure perturbations of the geomagnetic field due to the ionizing neutral beam. To achieve this a high pass filter was used to remove the spin signal resulting from the steady state geomagnetic field. The resulting signal was then amplified and sent to the PCM telemetry system. The magnetometer was deployed outside of the payload skin on a short boom identical to those used for the particle detectors (Fig. 2.9).

Plasma Frequency Probe

The plasma density measurements at the main payload were made with a plasma frequency probe, PFP, which measures the upper-hybrid frequency from which absolute plasma density can be determined [*Baker et al.*, 1966; *C. Swenson*, 1989; *Jensen and Baker*, 1990]. The PFP is particularly well suited for this kind of experiment since it is insensitive to changes in vehicle potential and vehicle spin. The sensing antenna was mounted on the aft end of the main payload just off of the spin axis as pictured in Fig. 2.9.

The low density threshold for the CRIT I instrument was $3.3 \times 10^4 \text{ cm}^{-3}$ at the second burst location. Above this density the instrument will lock, and density can be measured. Near this lower sensitivity threshold the quantization of detection is about $4.0 \times 10^2 \text{ cm}^{-3}$ or about 1% of the background plasma. The sampling period was 0.17 ms and constant throughout the flight.

The CRIT II instrument, also flown on the main payload, had two sampling rates, 1.5 kHz and 12 kHz. The higher rate was used during the burst mode of the PCM system, and the lower rate was used at all other times. The data to be presented in Chapter 3 were taken in this faster mode and all further discussion of the CRIT II plasma frequency probe will refer to this fast mode. The PFP had a low frequency threshold just below the local electron cyclotron frequency at the altitude of the releases. The low density threshold is therefore undefined since the probe theory breaks down when the plasma frequency is less than the electron cyclotron frequency, but in practice the low density limit was about $1.0 \times 10^4 \text{ cm}^{-3}$. The resolution of the instrument at the ambient density was $2.33 \times 10^3 \text{ cm}^{-3}$ or about 0.54% of the background plasma density.

Photometers

Two solid state photon detectors measured the light emissions with 10 nm wide filters centered at 550 nm and 633 nm, corresponding to expected emissions from barium and oxygen neutrals respectively [Torbert *et al.*, 1991]. They were positioned such that their field of view was along the payload spin axis towards the chemical release. Only the CRIT II experiment was equipped with these detectors.

2.2.4 Subpayloads

Two telemetry links were used on both the CRIT I and CRIT II subpayloads. One link was an FM-FM telemetry system, and the other was a PCM system. A burst storage mode was not part of the system. Therefore, the number of instruments and measurements had to be limited so the sampling rate would be great enough to make meaningful measurements. Particle detectors were provided by the University of New Hampshire.

The CRIT I subpayload was instrumented to measure energetic particles and electric fields in two dimensions. Two dipole probes were to be deployed in a plane perpendicular to the payload's spin axis (Fig. 2.10). The subpayload magnetometer was provided for vehicle attitude determination and was not sensitive enough for scientific purposes. Due to the placement error of the whole subpayload and the deployment problems of the electric field booms, discussed in Appendix A, the data gathered from the CRIT I subpayload was not as useful as was hoped.

Because of the failures with the CRIT I subpayload, the CRIT II subpayload was completely redesigned (Fig. 2.11). The instrumentation was improved and a three axis scientific magnetometer included. The electric field instrument was provided by Cornell University and consisted of two dipole probes orthogonal to each other with a fifth probe extending perpendicular to the plane of the other two probes.

CRIT I Subpayload

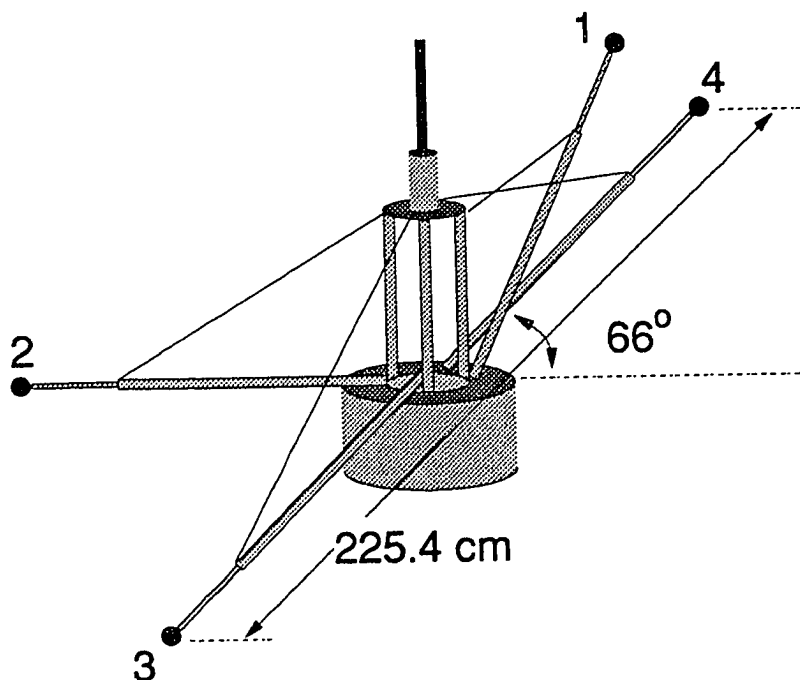


Figure 2.10: Diagram of the CRIT I subpayload in flight.

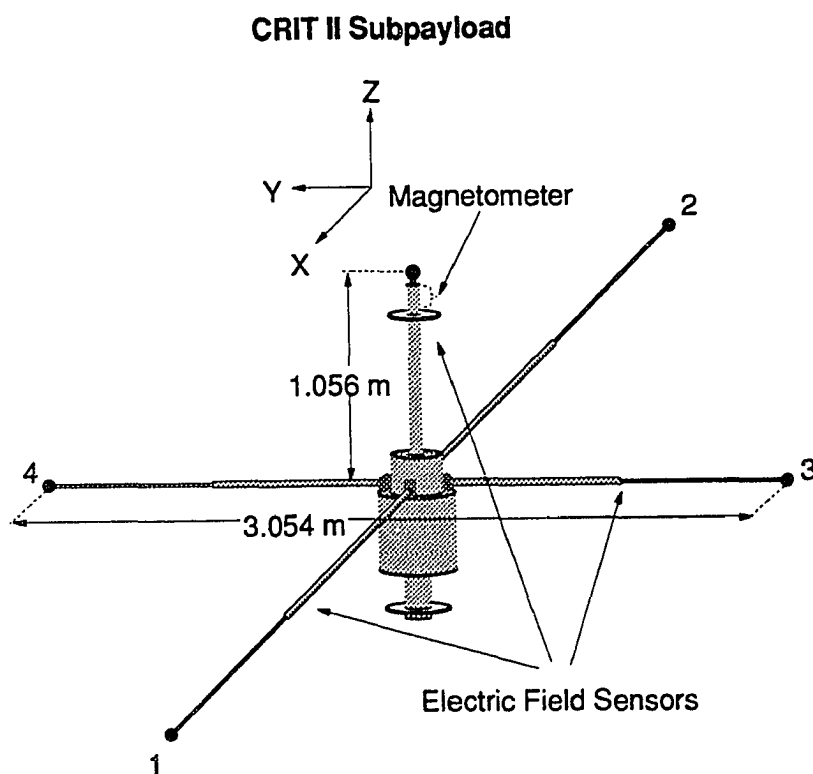


Figure 2.11: Diagram of the CRIT II subpayload.

All five 7.62-cm-diameter, spherical sensors were graphite coated. This probe configuration provides three linearly independent, but nonorthogonal, measurements of the electric field from which, with suitable processing, the entire electric field can be determined. The magnetometer, provided by the Danish Space Research Institute, was a three axis measurements with about 2 nT resolution.

Chapter 3

Data Overview

'The time has come', the Walrus said,
 'To talk of many things:
Of Shoes—and ships—and sealing-wax—
 Of cabbages—and kings—
And why the sea is boiling hot—
 And whether pigs have wings.'
—*Lewis Carroll, The Walrus and the Carpenter, Through the Looking-Glass and What Alice Found There*

Taken as a whole, an enormous and bewildering amount of data has been accumulated during the four chemical releases on CRIT I and CRIT II. We have chosen to summarize all the observations in an appendix, rather than subject the reader to a description in the main body to the thesis, and to concentrate in this chapter on the most exciting results of CRIT II. This experiment provides the first clear indication of CIV in space and is clearly well worth the concentrated effort. In the remainder of this introductory section we briefly overview the ionospheric conditions during the two experiments.

Results from these experiments can be combined and treated as different parameter regimes of the same study since each of the four barium release was expected to produce identical neutral barium streams. Broadly speaking, three parameters,

Table 3.1: Parameters for CRIT I and CRIT II experiments.

Ambient Ionosphere	CRIT I	CRIT II
Magnetic Field	0.44×10^4 T	0.43×10^4 T
Neutral density (400 km)	4×10^7 cm ⁻³	1.7×10^8 cm ⁻³
Plasma density (400 km)	3.3×10^4 cm ⁻³	5.5×10^5 cm ⁻³
Alfvén Velocity (400 km)	1000 km/s	300 km/s
Oxygen lower hybrid frequency	$\omega_{lh}/2\pi = 6.3$ kHz	$\omega_{lh}/2\pi = 6.9$ kHz
Barium Streams	CRIT I	CRIT II
Main payload to first release	1.987 km	1.585 km
Main payload to second release	4.344 km	3.881 km
Injection Angle first release	47.6°	57.0°
Injection Angle second release	47.8°	44.4°
Extent along $\vec{B}_0$ first release	600 m	490 m
Extent along $\vec{B}_0$ second release	1330 m	1200 m
Barium gyro radius (10 km/s)	329 m	
Barium energy (10 km/s)	70 eV	

distance from release, ionospheric plasma density, and neutral background density were different for the four releases. The releases either occurred close (1.4 or 1.3 km) or far (3.1 or 2.7 km) from the main payload, thereby varying the neutral barium density at the measurement site; the releases occurred in a low (3.3×10^4 cm⁻³ for the CRIT I releases) or high (5.5×10^5 cm⁻³ for the CRIT II releases) plasma density ionosphere; and the releases occurred in a low (4×10^7 cm⁻³ for CRIT I) or high (1.7×10^8 cm⁻³) neutral background density. A table summarizing the ionospheric composition for the CRIT I and CRIT II releases is presented in Table 3.1. Each data set includes measurements at two locations, those of the main and sub-payloads. Since there are multiple instruments on each payload and multiple data channels on most instruments, this represents a very large data set. A complete presentation of this data would be wearisome for both the author and the reader. Therefore, we concentrate this overview on the first release of CRIT II, occasionally comparing to the other releases. As noted above a particular release was chosen

Table 3.2: Publications of CRIT I and CRIT II data to date.

CRIT I
"Electrodynamic Interaction Between the CRIT I Ionized Barium Streams and the Ambient Ionosphere", [<i>Brenning et al.</i> , 1989 <i>a</i>]
"Critical Ionization Velocity Interaction in the CRIT I Rocket Experiment", [<i>Brenning et al.</i> , 1989 <i>b</i>]
"Electric and Magnetic Field Measurements Inside a High Velocity Neutral Beam Undergoing Ionization", [<i>Kelley et al.</i> , 1991]
"Interpretation of the Electric Fields Measured in an Ionospheric Critical Ionization Velocity Experiment", [<i>Brenning et al.</i> , 1991 <i>a</i>]
"Barium Swarm: an Ionospheric Alternating Current Generator In CRIT I", [<i>Brenning et al.</i> , 1991 <i>b</i>]
CRIT II
"CRIT II Electric, Magnetic, and Density Measurements Within an Ionizing Neutral Stream", [<i>C. Swenson et al.</i> , 1990]

because of the quality of the measurements and the evidence of CIV effects in the data. Figures presenting a more complete data set for the four releases, including main and subpayload measurements, are presented in Appendix B, and the reader is encouraged to intercompare this at his or her leisure. In addition, a number of publications of these data already exist in the literature and are summarized in Table 3.2.

3.1 CRIT II Main Payload Results

We begin our discussion with data originating from the main payload. This payload was positioned within the path of the neutral barium stream and made measurements of the plasma density, energetic particles, electric fields and magnetic fields.

3.1.1 Plasma Density Measurements

The plasma density as observed on the main payload of CRIT II during the first release is presented in the top panel of Fig. 3.1. The middle panel is the neutral barium density calculated from the model discussed in Section 2.2.1. The neutral barium velocity is also noted in the central panel and Alfvén's critical velocity for barium, V_c , is marked along with the time at which the perpendicular component of the neutral velocity equals V_c . The bottom panel shows the output of the photometer which had an optical window centered at 550 nm with a 10 nm width. The time axis for all three panels is referenced to the beginning of the burst data storage mode of the telemetry system or 348.262 seconds after launch. This mode was activated upon the transmission of the command to detonate the barium shaped charge.

First, consider the lower panel in Fig. 3.1. The photodetector was supposed to be pointed back at the release (the shaped charge was pointed away from the earth), but because of errors in the attitude control system it was 15.8° off of a direct view. The light resulting from the chemical canister detonation can be observed as the first peak starting at 0.013 seconds and lasting about 10 ms. This peak is relatively small because the detonation was observed in the very wings of the detector response. That is, since the half-angle field of view of the detector was 6.5° , the detonation flash was much brighter than indicated. During this 10 ms the neutral barium stream is produced by the explosion. Apparently delays in transmitting and decoding the detonate command cause the actual release to occur about 13 ms after the command was given. The bulk of the neutral barium is calculated to arrive at the main payload, see center panel, almost coincident with the large flash occurring at 0.133 s. Although not yet completely understood, it has been suggested by N. Brenning (private communications) that this flash results

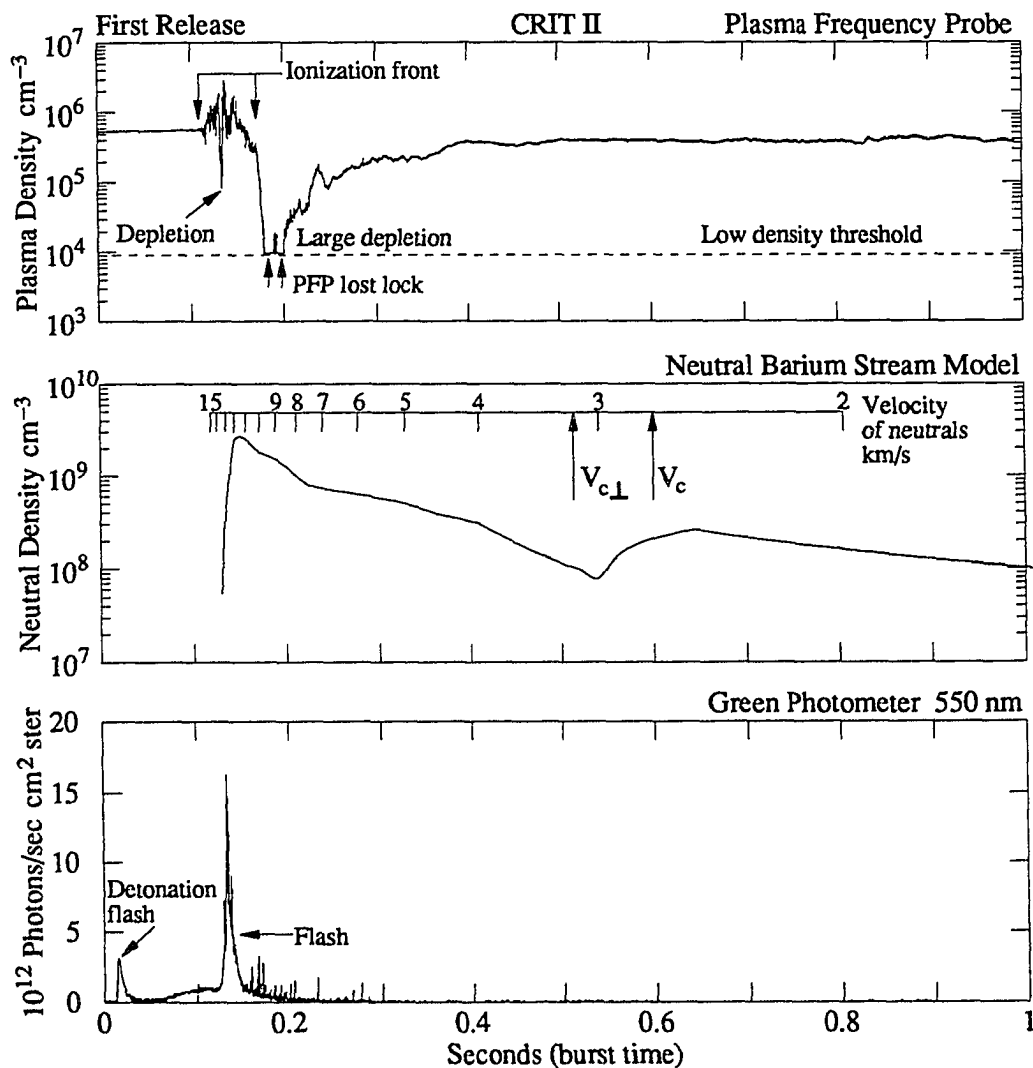


Figure 3.1: A comparison of plasma density measurements, neutral barium density calculations, and photometer observations during the first release of CRIT II.

from hot electron excitation of water vapor, sputtered by neutral barium collisions with the payload surface. Further discussion of the photometer measurements are presented by *Torbert et al.* [1991].

Plasma density is plotted in the top panel of Fig. 3.1. The plasma content is seen to increase in a complicated manner from a background density of $5.4 \times 10^5 \text{ cm}^{-3}$ to a peak of about $2.8 \times 10^6 \text{ cm}^{-3}$, to drop below $1.0 \times 10^4 \text{ cm}^{-3}$ in a large depletion region and then slowly recover. Within the deep depletion the plasma density dropped below the threshold of detection of the instrument. Similar measurements were obtained from the Langmuir probe, but these are not presented since the PFP measurements are more accurate. The initial disturbances in plasma density (0.112 s) correspond to the arrival of 16 km/s material. This is faster than the 13.5 km/s maximum velocity predicted by the strontium based model velocity distribution in the center panel. It has been proposed that collisionally accelerated neutral oxygen, unusually fast neutral barium, or both are arriving at the payload at those times, but there is another possibility. Inaccuracies in determining the distance of the chemical release from the main payload would affect the fitting of the barium model to the data. If the barium release occurred 180 m closer than estimated, the barium neutral model would coincide with the density fluctuations and other hypothesis would not be needed. This explanation, which the author favors, will be discussed in greater detail in Section 4.1. The end of the region of increased density, around 0.175 s, corresponds to a velocity of 9.7 km/s. The time period from 0.112 to 0.175 seconds defines a region we call the ionization front since the largest density increases and field fluctuations occur between these times. This terminology should not be construed as implying that the theoretical ionization front model is necessarily supported by the data. There is every reason to believe that this active region rides with the fastest neutral barium, implying a 768 meter

spatial extent along the stream at 0.135 s. For comparison, the gyro radius of barium ions created at this time is approximately 360 meters. The rise time of the spiky density features (≈ 1 ms) are well within the bandwidth of the plasma frequency probe and must be considered as a real physical effect. Likewise the small depletion which coincides with the peak of the flash detected by the photometer is a real effect. This ionization front will be considered in greater detail in Chapters 4 and 5.

The depletion region, after 0.18 s, occurs well after the passage of the ionization front, and has a minimum less than $9 \times 10^3 \text{ cm}^{-3}$. The formation of this 'hole' in the ionosphere may be due to the effect of heavy barium ions which force out the lighter ionospheric oxygen ions. In such a model the creation of barium ions creates a local increase in the plasma density. The ambipolar diffusion electric field pulls the lighter ions out along the magnetic field. The barium ions then leave the region because of their large gyro radii, and a depleted plasma 'hole' is left behind which slowly recovers on the ion acoustic time scale. This phenomena has been observed to occur in computer simulations of chemical releases [Schunk and Szuszczewicz, 1991]. There may well be other explanations for the hole. Our main concern in any case is the plasma enhancement. We have not made a detailed study of the various hole production processes, even though they are certainly interesting.

The second burst of CRIT II had the same pattern as the first burst; a plasma density increase accompanying the fastest barium followed by a density depletion. To some extent the differences can be explained by the $1/r^2$ expansion of the barium stream. The volume occupied by the neutral stream and thus new born ions, was approximately 6 times larger than at the time of the first release. The velocity differential within the neutral stream causes the region of fastest barium to expand in the stream direction as it moves away from the release site. This scaling can be

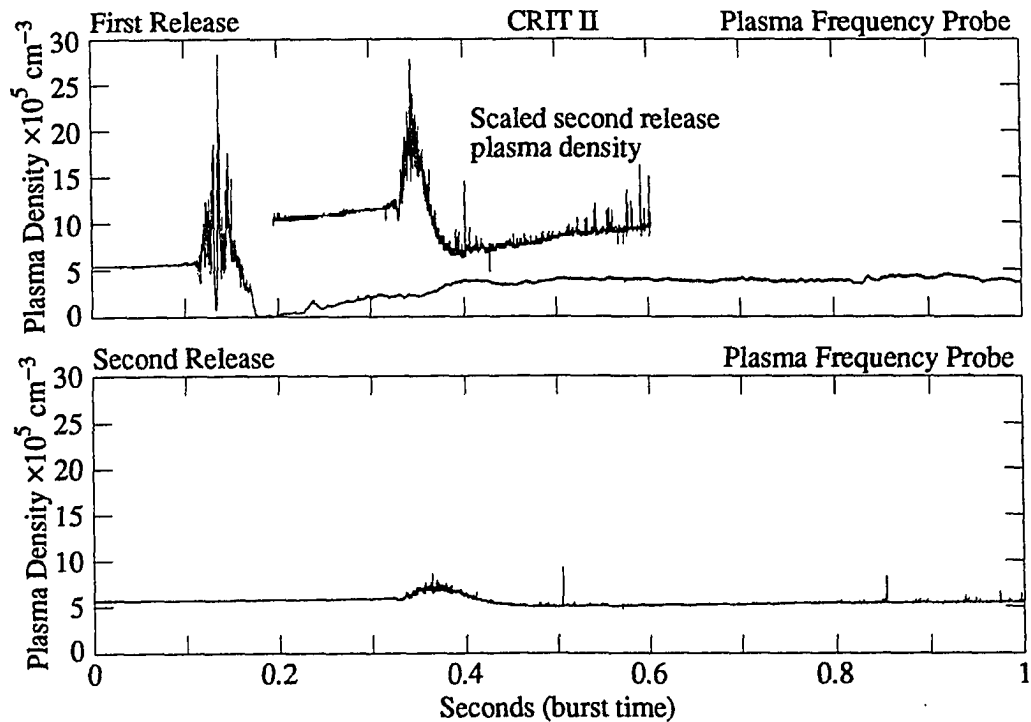


Figure 3.2: CRIT II first and second release plasma density measurements are plotted on the same linear scale. Scaled second release data is inset into the first release plot for comparison.

unfolded as shown in Fig. 3.2. The first and second release plasma density measurements are compared on the same linear scale. The second burst measurements are plotted as a insert to the first burst measurements in the top panel after scaling according to the predicted neutral density. The qualitative agreement between the scaled second burst data and the first burst suggests the process creating the enhancements is equally active at the two measurement locations, provided that ionization rate scales almost linearly with neutral barium density. The density depletion of the scaled second burst data is not as deep as in the first burst.

The ambient plasma density was below the lower threshold of the plasma frequency probe during both CRIT I releases. More importantly, no plasma density increases were detected even though the electric and magnetic field signatures were pronounced (*Kelley et al. [1991]*). The density remained below the detector thresh-

old of roughly 10^4 cm^{-3} except for a brief time during the second release, but well after the passage of the front. Later in the flight the ambient plasma density was greater and the instrument tracked it demonstrating that the instrument was working. Although not completely understood, it is thought that the large background plasma density is related to the higher yield of barium ions on CRIT II and the Porcupine experiment.

3.1.2 Energetic Particle Data

The University of New Hampshire provided detectors for both ions and electrons on the main payload of CRIT II. Time-energy spectrograms of two look directions of the ion electrostatic analyzer are shown in Fig. 3.3 and Fig. 3.4. Gray-scale versions of these plots can be found in Appendix B. Assignment of look direction for the ion detector channels has proven to be problematical. The look directions associated with these channels could not be tested pre-flight with this instrument and the expected directions have, somewhat embarrassingly, been inconsistent with the measurements. The field of view associated Fig. 3.3 is now thought to be along the direction of the neutral barium due to the clear energy dispersion of the ion flux, a dispersion in agreement with the neutral beam model. This type of ion detector cannot distinguish between different ion species, but the highest particle flux band, the red band in the top panel, starting at 100 eV and dropping to 10 eV over 0.3 seconds, is almost certainly caused by barium ionized near the payload. For example, the energy of the flux band agrees with the kinetic energy of the neutral barium at the main payload as a function of time. This continuous ionization cuts off when the neutral barium velocity drops below 3.8 km/s. During the passage of the ionization front, ions with this well defined 'time of flight' energy do not appear. Instead, oxygen or barium ions are scattered in energy above, but mostly

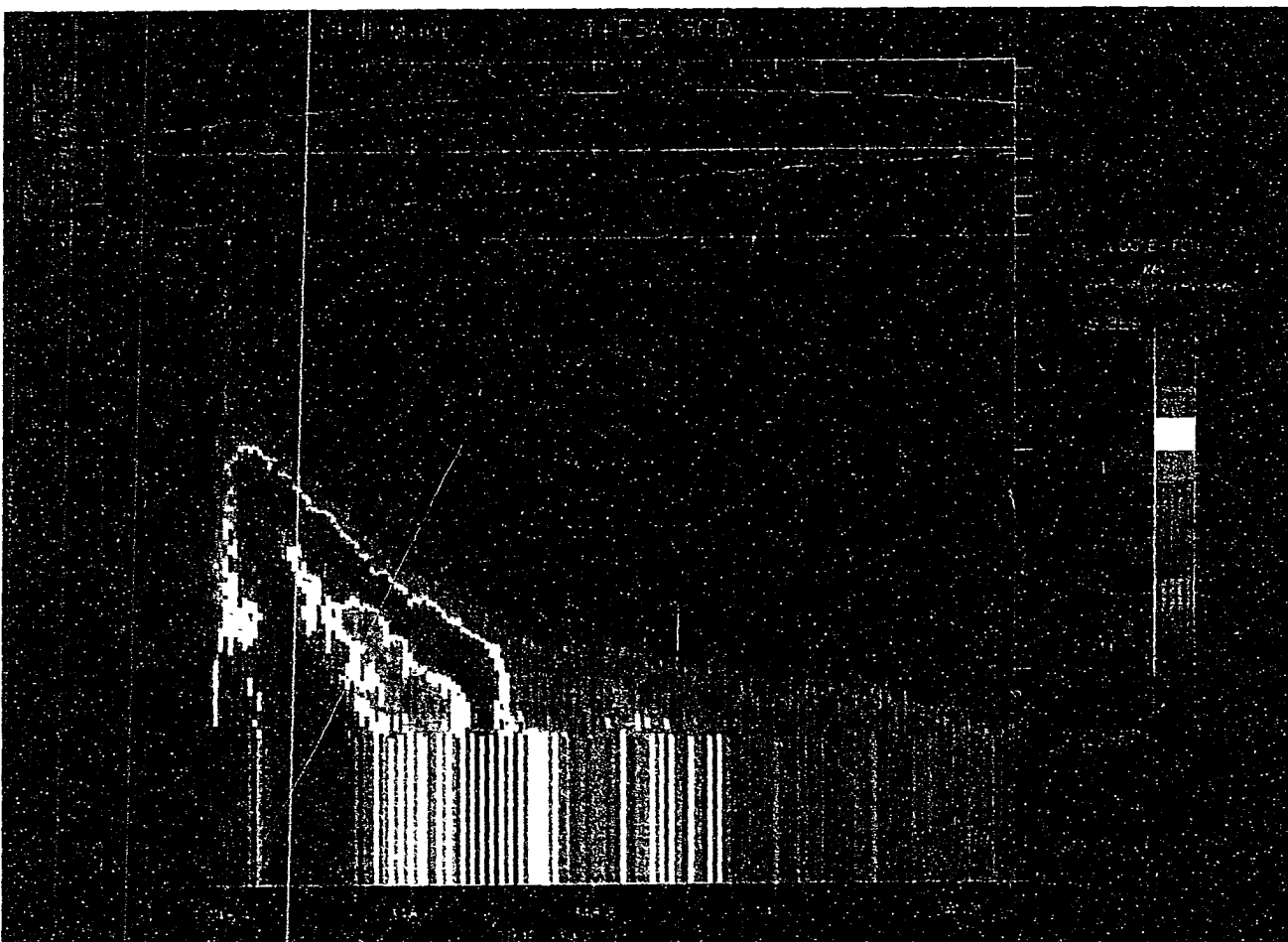


Figure 3.3: Time-energy spectrogram of ions at the CRIT II main payload during the first release. The look direction is back towards the release and nearly along the neutral stream direction.

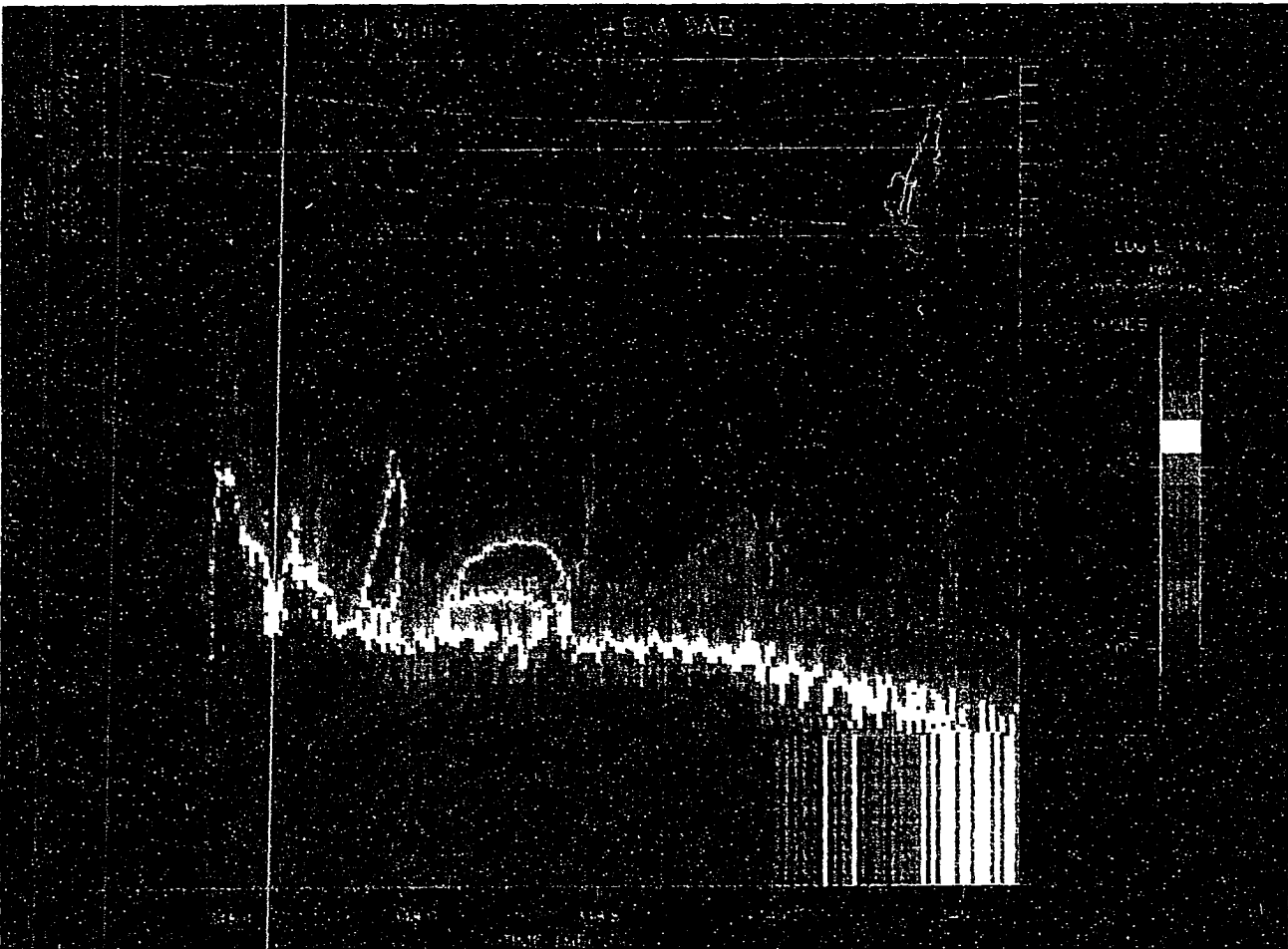


Figure 3.4: Time-energy spectrogram of ions at the CRIT II main payload during the first release. The look direction is nearly perpendicular to the geomagnetic field.

below 100 eV. This scattering presumably results from strong electric fields within the ionization front and is a signature of the energy transfer.

The field of view associated with Fig. 3.4 is thought to be nearly perpendicular to the geomagnetic field. Further evidence that barium ions have been scattered in energy and direction can be found in Fig. 3.4. Two distinctive patterns repeat themselves. The first pattern shows up at 0.3 s and the second at about 0.45 s. Each reappearance is about 0.2 s apart or one gyro period for barium. Manifestly, these are bunches of barium ions scattered such that they have little velocity parallel to the magnetic field and are trapped near the main payload. The barium ions in the first cloud are phase bunched, in time, within $\approx 1/8$ of a barium gyro period. The ions within the second cloud were created or accelerated over $\approx 1/2$ of a barium gyro period. The observed particle fluxes become weaker as the clouds expand and as the payload moves from the original interaction region. We believe the first cloud must have been created rapidly and within the same volume, the ionization front, to be bunched in phase in this manner. This clearly distinguishes the turbulent ionization process within the front from that process continuously creating ions after its passage. The second cloud may have been created through an acceleration process after the passage of the ionization front. This possibility will be further discussed in Section 3.1.4. A more detailed discussion of the CRIT II ion detector data is presented by *Torbert et al.* [1991].

A spectrogram of electron energy fluxes from one channel of the charged particle analyzer is presented in Fig. 3.5. The major heating of electrons occurs with the arrival of the neutral barium stream and the passage of the ionization front. During the ionization front electrons are heated to the 10 eV range and above with energy fluxes exceeding 5×10^{10} keV cm⁻² s⁻¹ ster⁻¹. This is several hundred times that of a large aurora [*Torbert et al.*, 1991]. The fluxes became so large that the detector



Figure 3.5: Time-energy spectrogram of electrons at the CRIT II main payload during the first release.

saturated (blue streaks in the most intense region) around the time of the flash. *Torbert et al.* [1991] estimates that the peak ionization rate of neutral barium by these hot electrons within the ionization front is over $1 \times 10^6 \text{ cm}^{-3} \text{ s}^{-1}$.

Particle measurements during the second burst of CRIT II are qualitatively similar; there are signatures of the ionization front, barium ions trapped in the magnetic field, and heated electrons. The electrons are most strongly heated in the 10 eV range, but electron fluxes are observed at all energies. Due to a failure of the telemetry system all CRIT I particle data were lost, except for the retarding potential analyzer on the subpayload, so these represent the only direct comparisons we have.

3.1.3 Electric Field

Electric field data are presented in Fig. 3.6. The Nyquist frequency for these dc coupled measurements is 12 kHz, but the data were averaged down to 1.5 kHz for plotting on this time scale. Higher resolution data are presented in Chapter 5. This averaging does not affect the amplitude of the dominant waves in any detectable way in these plots. Due to the previously discussed attitude control failure, the spin axis of the main payload was 15.8 degrees off alignment with the neutral stream. We will consider this misalignment of the measurement coordinates small and ignore it for this qualitative discussion. The top two panels contain electric field measurements made nearly perpendicular to the neutral stream and the bottom panel shows the component parallel to the stream. Confusion about coordinate systems is quite possible since we will be using several in this thesis. We have decided to label each figure with the appropriate coordinate system and to provide figures to explain each when they first appear. In the payload coordinate system, the perpendicular axes are rotated approximately 40 degrees from the $\vec{V}_{Ba} \times \vec{B}_0$ direction where $\vec{V}_{Ba}$

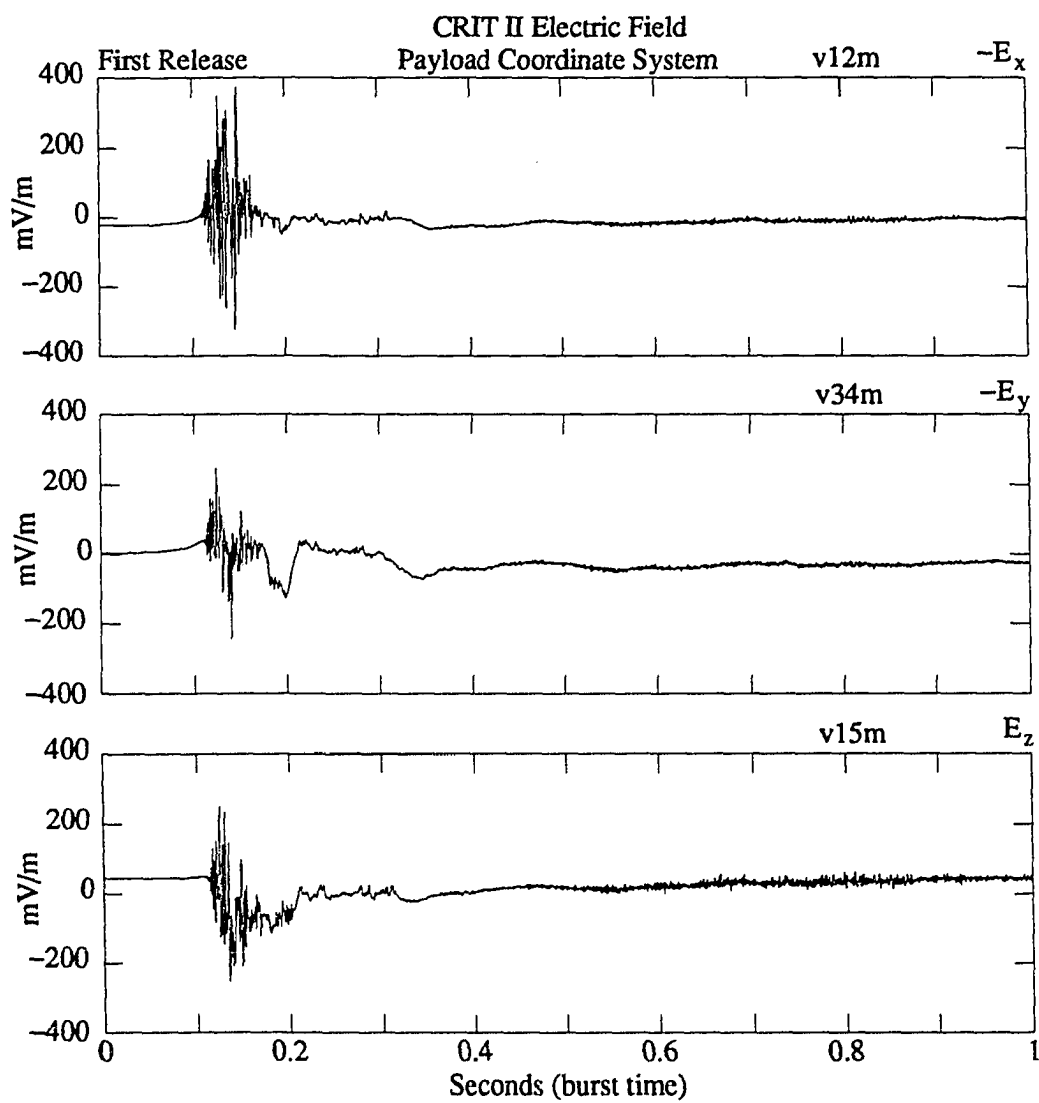


Figure 3.6: CRIT II, first release, electric field measurements, (see Fig. 3.7).

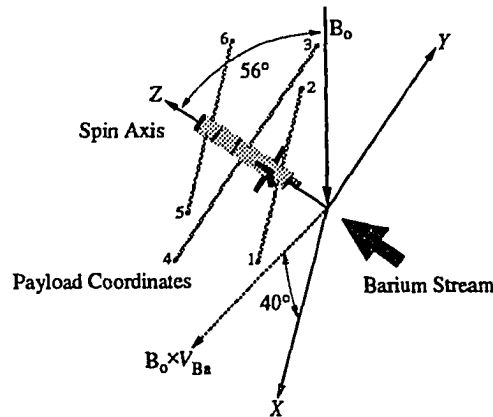


Figure 3.7: The payload coordinate system for the first release of CRIT II.

is the velocity vector of the barium stream and $\vec{B}_0$ is the geomagnetic field vector. This is illustrated in Fig. 3.7. This coordinate system is slowly rotating about the payload spin axis at 0.2 Hz.

DC Fields

Vary large quasi-dc electric fields, ≈ 900 mV/m, were observed and reported by *Kelley et al.* [1991] within the CRIT I ionization front. This data is also presented in Appendix B. The term quasi-dc refers to an electric field which varies on the comparatively slow time scale of the neutral barium stream, that is, on times scales longer than about 20 ms. Summarizing the CRIT I observations, we found that the data were consistent with a positive space charge leading a negative space charge within the ionization front. The electric field was directed back along the neutral stream in general agreement with the ionizing front model of *Piel et al.* [1978], where the space charge results from barium ions, with their large gyro radii, shooting out ahead of magnetized electrons. A large component (order 500 mV/m) of the field was found to be parallel to the geomagnetic field. It appeared as if the classical high conductivity along $\vec{B}_0$ did not short out this parallel electric field. One of the objectives of the CRIT II campaign was to confirm this very surprising

measurement. For the CRIT I electric field experiment only three orthogonal dc measurements are available for analysis, whereas for CRIT II, six independent dc measurements with various sensor spacings are available. With this more complete set of measurements boom length effects could be assessed and redundant measurements could be compared to determine if the geomagnetic aligned electric field is a real effect or just a measurement anomaly.

A detailed analysis of all of the dc electric field measurements of CRIT II indicates the quasi-dc data during the passage of the ionization front cannot be interpreted as physical electric fields. Instead, the physical fields are corrupted by a combination of changes in work function of the various sensor surfaces and variations of floating potential with respect to the plasma. The rate at which neutral barium vapor forms a monolayer over the electric field sensor, and thus causes changes in the work function, is relatively slow and would most likely be uniform over all of the sensors. During the entire 0.1 s duration of the ionization front only $\approx 6\%$ the sensor surface should be coated, but apparently this is enough to shift the contact potential by ≈ 1.0 V. However, only 15% of the barium is actually vaporized during the release, so it is also possible that very small droplets of barium were created and one or more of these droplets impacted a sensor. This could cause an imbalance between sensors and also lead to the type of spurious signals that were detected. When the neutral barium arrives at the CRIT II main payload the various dc measurements shift by up to a volt and then slowly recover over the next 15 seconds. The direction of the jump depends on which sensors are most effected by the barium and are independent of the boom length or the direction of the barium stream. The quasi-dc electric field data, or the fluctuations with frequencies less than about 50 Hz, are only corrupted during the ionization front (approximately 0.1 to 0.25 s). That is, the measurements are consistent internally after this time if

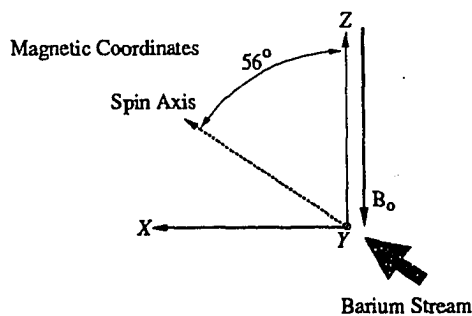


Figure 3.8: Geomagnetic aligned coordinate system for electric field measurements at the main payload.

we assume a slowly varying (≈ 5 Hz) dc bias over the subsequent 15 s. Analysis of the CRIT II data casts doubt on the validity of the CRIT I quasi-dc measurements and suggests that it may be very difficult to make dc electric field measurements inside a chemical release.

With the slow dc bias subtracted and the higher frequencies removed by filtering, the CRIT II electric field data can be transformed to a geomagnetic aligned coordinate system using the attitude information from the gyro and the magnetometer. We can proceed to interpret the quasi-dc data for times well after the peak in the neutral barium or after about 0.3 s. Low frequencies imply long wavelengths, especially if the fields are moving with the barium stream, and small differences in the location of the measurement phase centers of each channel can be neglected. The geomagnetic aligned coordinate system is constructed such that the $\hat{x}$ axis is in the same plane as the magnetic field and the neutral barium stream. The $\hat{z}$ axis is anti-parallel to the geomagnetic field as shown in Fig. 3.8.

Filtered electric field data from the first release transformed to a geomagnetic aligned coordinate system is presented in Fig. 3.9 for the first release. Data during the passage of the ionization front are within the shaded portion of each panel and should be neglected. After the ionization front several oscillations on the time scale of the barium gyro period are observed. These oscillations are mainly in the E_x and

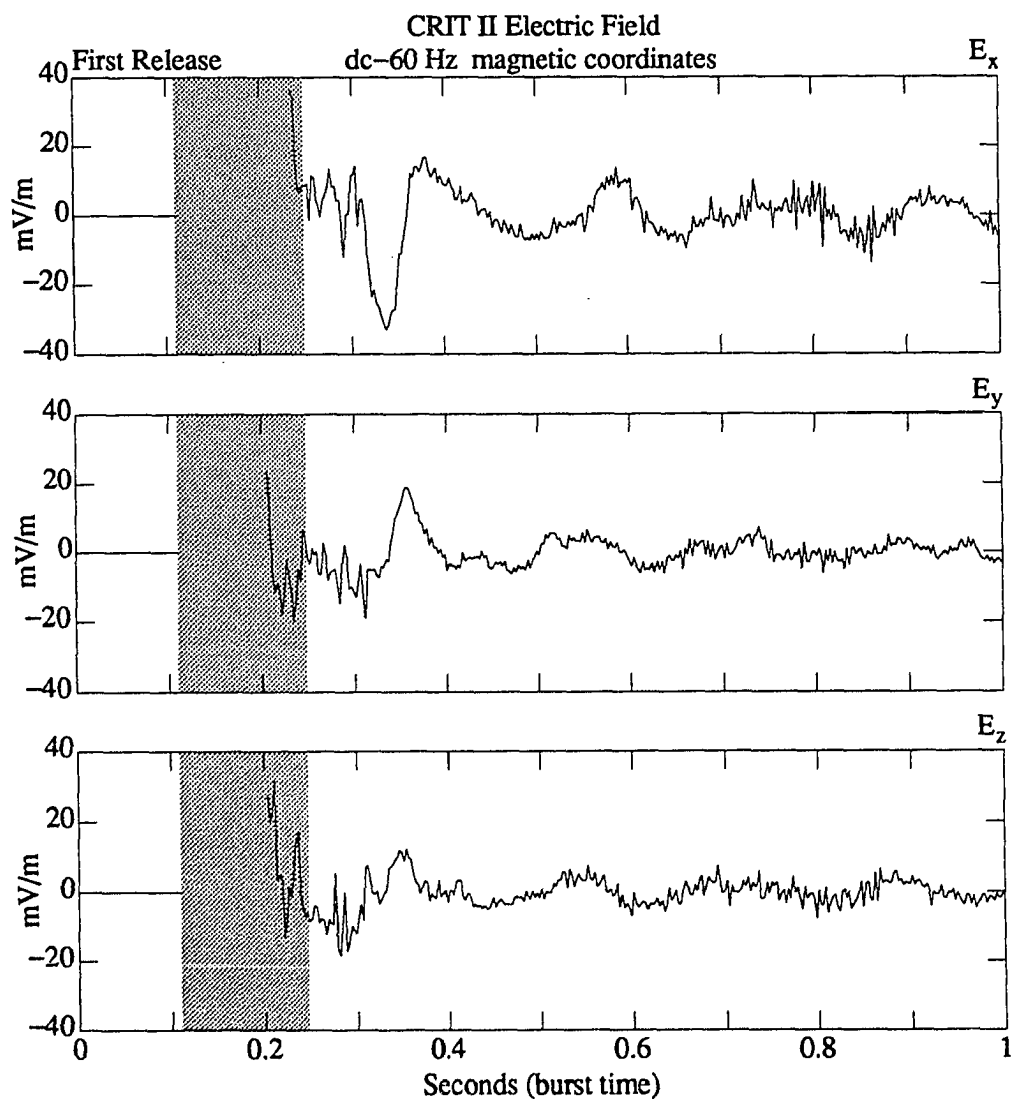


Figure 3.9: CRIT II, first release, electric field measurements in magnetic coordinate system (see Fig. 3.8).

E_y direction as predicted by the barium swarm model of *Brenning et al.* [1991b], which will be discussed in greater detail when we present the magnetometer data containing fluctuations correlated to the electric field data. The small fluctuations in the E_z channel may result from imprecision in the transformation from the measurement coordinate to these magnetic coordinates due to imprecise attitude data since we do not expect that parallel electric field components could be sustained after the passage of the ionization front.

Quasi-dc measurements within the ionization front are corrupted as previously discussed, unless a fortuitous cancellation occurs, but they put an upper limit on the size of the quasi-dc fields. The actual data are a combination of real electric field plus changes in the instrument bias. The largest, quasi-dc measurements for the first release are on the order of 150 mV/m. If the first release quasi-dc field was directed so as to slow barium ions perpendicular to the geomagnetic field and we assume it was distributed in a half sinusoid with a peak value of 150 mV/m, then the integrated potential across a barium gyro radius (≈ 360 m) is 27 V. The potential perpendicular to $\vec{B}_0$ along the stream as predicted by the ionization front model of *Piel et al.* [1978] (1.11) is > 89 V for 13.5 km/s barium. It thus seems that the quasi-dc fields could not be the only cause of slowed and scattered barium ions. The waves within the ionization front have sizable fields, and, although the fields are not maintained over an extended spatial distance, they may contribute to the slowing and scattering of barium ions.

Filtered electric field data for the second release are presented in Fig. 3.10. Again data in the shaded region are in doubt. Clean signatures at the barium gyro period are evident with $\vec{E}$ perpendicular to $\vec{B}_0$ within the attitude and measurement uncertainties. At this distance of about 3.8 km the spurious effects are clearly much smaller and an upper limit of 20 mV/m in the ionizing front seems quite defensible.

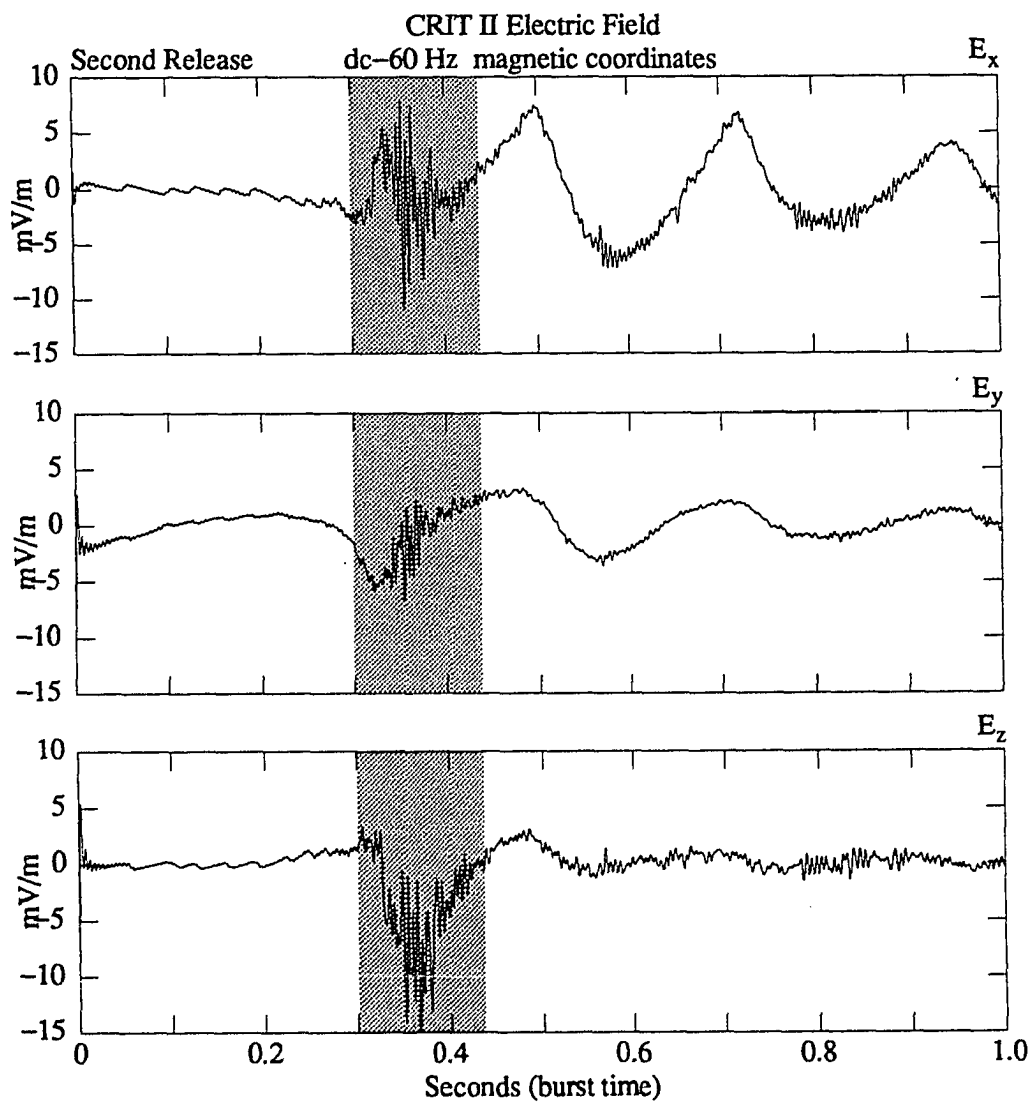


Figure 3.10: CRIT II, second release, electric field measurements in magnetic coordinate system.

Such a limit, which corresponds to about 400 m/s for an $\vec{E} \times \vec{B}_0$ velocity precludes the possibility that an ion beam was somehow generated at the explosion point and propagated across the magnetic field to the main payload. To generate this cross field velocity, the associated electric field would have to be 500 mV/m. In other words, although the data are partially corrupted, we can conclude that the ions detected in the second release could not have been created near the release point and merely transported across $\vec{B}_0$. Although less certain, we feel that similar arguments apply to the first release. Some caution is in order since the gyro radius for barium ions is large. They can travel with the neutral stream for ≈ 300 m before they are significantly turned out of the stream by the geomagnetic field. Still, this implies that the density enhancement observed by the PFP is created near the payload. Processes which could have created the density enhancement will be discussed in detail in Chapter 4

Wave Fields

First, consider the general wave activity in Fig. 3.6. The most intense waves are coincident with the ionization front, and strong wave activity is present in each of the channels. Weaker wave activity continues after the passage of the ionization front through the duration of the plots. The waves within the ionization front will be discussed in detail in Chapter 5. Briefly, the character of the waves changes significantly during the passage of the the ionization front. Initially the waves are narrow banded ($\delta f/f \approx 0.1$) and seem to grow with about a 0.3 ms time constant ($\gamma \approx 0.1\omega_0$ where ω_0 is the observed average wave frequency and γ is the growth rate). Gradually, over about 10 periods the frequency of the most intense waves dropped from 1.6 kHz to about 300 Hz. These narrow band waves peaked at ~ 400 mV/m, an amplitude at which the process apparently saturated. Subse-

quently, the 'saturation' wave pattern changed character, exhibiting less coherent structure and lower amplitudes after the peak, or 'flash', observed in the photometer data.

Fig. 3.11 shows the power spectrum and the associated time series of the electric field measured parallel to $\vec{V}_{Ba}$ during the passage of the ionization front. Here we have utilized the full 12 kHz frequency response of the measurement. The highest spectral intensities observed during the passage of the front are in narrow bands around 50 and 300 Hz. For reference the cyclotron frequencies of oxygen, hydrogen, and helium are noted as are the lower hybrid frequencies for a barium plasma (2362 Hz) and an oxygen plasma (6914 Hz). The frequency of the observed waves does not seem to be associated with any of these frequencies. Since lower hybrid waves traditionally had been thought of as the mechanism for heating the electrons in the CIV microprocess, as discussed in Section 1.2.2, we have endeavored to hypothesize a wave phase velocity consistent with a lower hybrid mode which would have a Doppler shifted frequency equal to the observed value of about 300 Hz. As discussed in Chapter 5, where the modified two stream instability and lower hybrid waves are discussed in detail, we have not been able to reconcile the observed frequencies with the lower hybrid mode. These measurements thus indicate that other instabilities must be considered. The absence of a clean control of the physics by lower hybrid turbulence is in agreement with other ionospheric *in situ* CIV observations which also found intense signals only well below the lower hybrid range [Kelley *et al.*, 1986,1991] and with laboratory observations [Nickenig and Piel, 1987]. In the latter the Doppler shift idea has been offered as an explanation for the discrepancy.

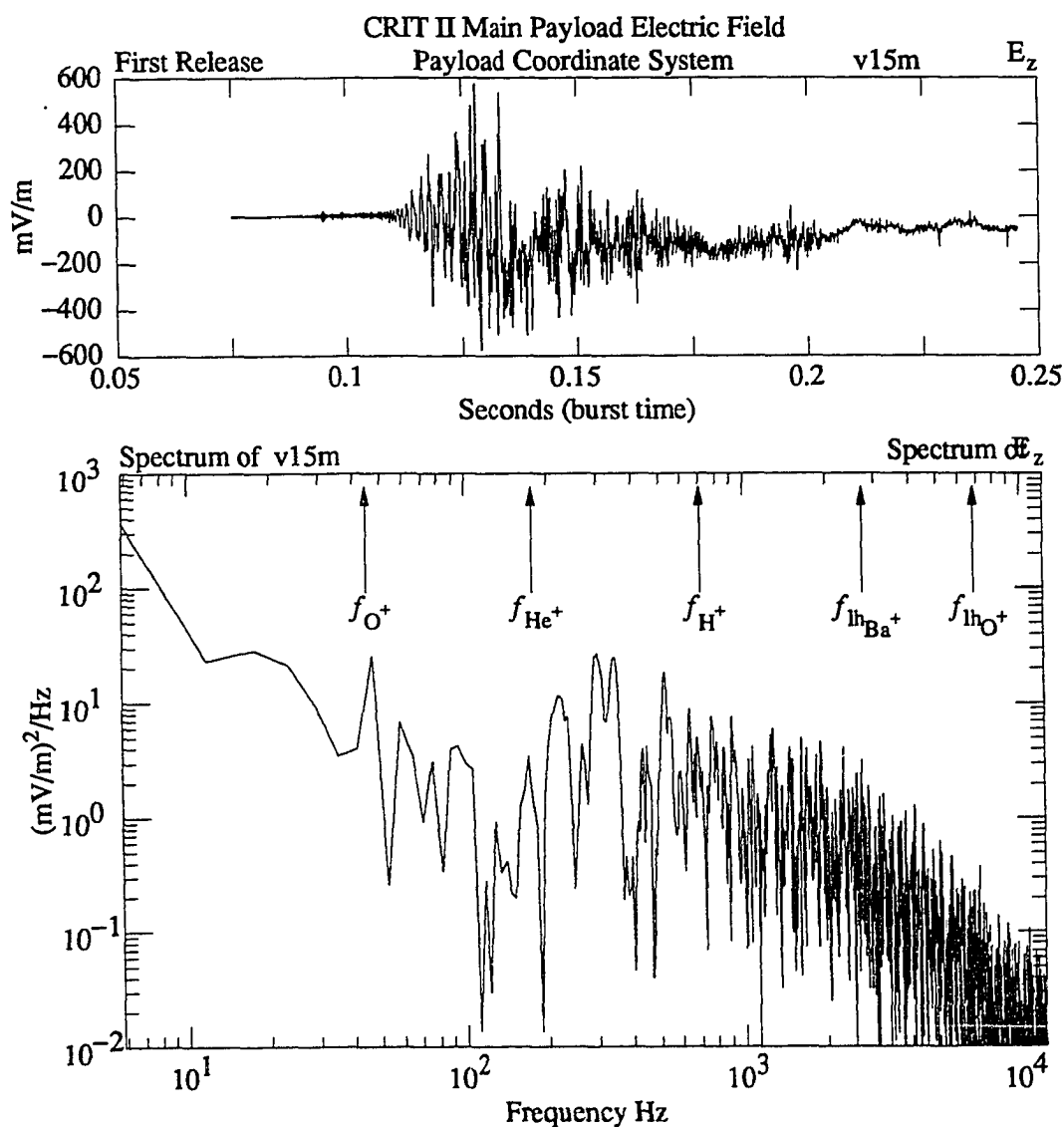


Figure 3.11: Frequency spectra of electric field data parallel to $\vec{V}_{Ba}$ during the first release and the associated time series from which the spectra was made (see Fig. 3.7).

3.1.4 Magnetic Field Data

Magnetic field deviations associated with the CRIT II first release are shown in the geomagnetic aligned coordinate system of Fig. 3.8 in Fig. 3.12. In the bottom panel the total magnitude of the field is calculated from the previous three measurements. With the passage of the ionization front, large (~ 200 nT) deviations were observed in each of the directions. These magnetic field perturbations cannot be the result of the displacement current in Amperes law alone, or in other words we can write

$$\nabla \times \vec{B} > \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} \quad (3.1)$$

To show this we estimate the electric field implied by the magnetic perturbations. Using the 500 m scale size of the barium stream, 200 nT for the magnetic deviations, and a short, but reasonable, time scale for $\vec{E}$ (≈ 1 ms), the required electric field is on the order of 35 kV/m. This is four orders of magnitude greater than any observed field. We therefore conclude that these magnetic field fluctuations are signatures of currents carried by charged particles such as the polarization current in an Alfvén wave or the current resulting from the motion of the newly created barium ions.

After the passage of the ionization front smaller perturbations, spaced about 0.2 s, the time for one barium ion gyration, are observed in the δb_x and δb_y directions while virtually no fluctuations were detected in the direction along the geomagnetic field. This periodicity, easily observed in the total magnitude (bottom panel), correlates with ions observed by the particle detectors (Fig. 3.4). These oscillations, lasting about 1.0 s, have been convincingly shown to result from the phase bunched gyration of a cloud of barium ions by *Brenning et al.* [1991b]. The physical idea behind the barium swarm model is that the barium ions will have some phase and

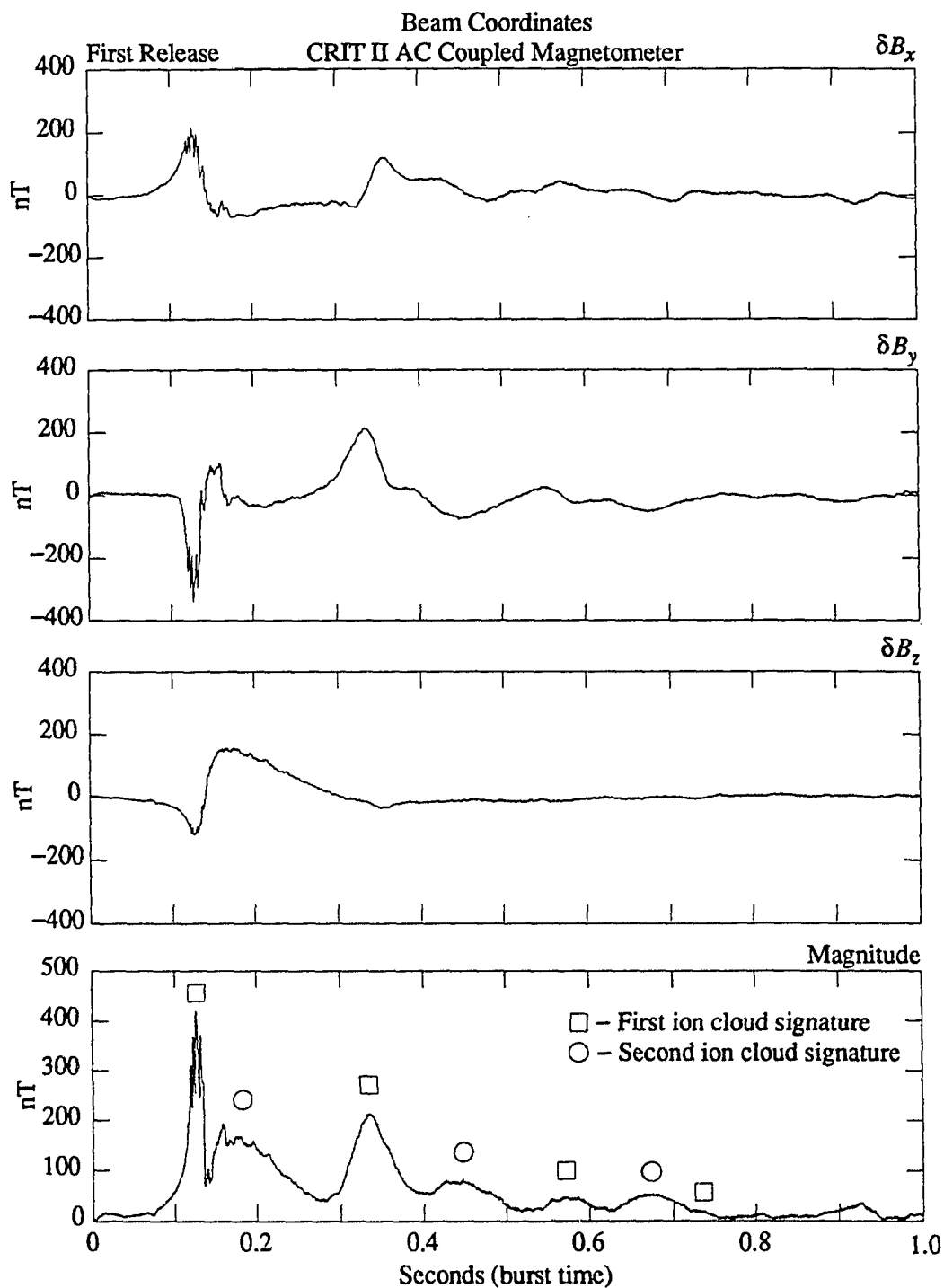


Figure 3.12: CRIT II, first release, fluctuations from the geomagnetic field.

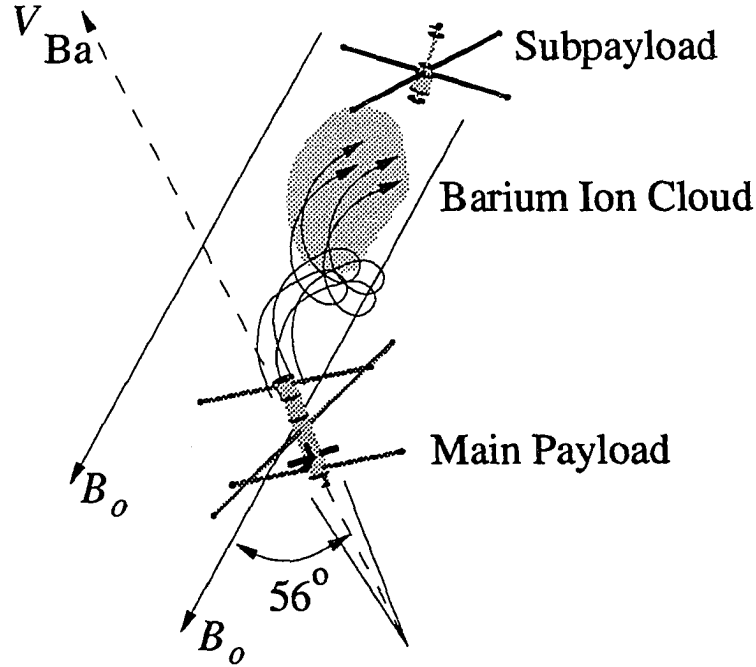


Figure 3.13: A sketch of the barium swarm motion along the magnetic field.

density correlation when they leave the barium stream and therefore collectively spiral up the field lines in a swarm as illustrated in Fig. 3.13. Two barium ion clouds were apparently created; one within the first part of the ionization front and the other towards the end of the front. Due to their large gyro radius (approximately 300 m) these phase-correlated ions constitute a cross-field ion current which is closed by geomagnetic aligned currents. In the barium swarm model this field aligned current is drawn by an electric field pattern which is maintained by the ion motion across $\vec{B}_0$ and which launches Alfvén waves in both directions along the magnetic field. The main result of *Brenning et al.* [1991b] is that the cloud motion is characterized by the K parameter (in MKS units)

$$K = \frac{\delta n e \mu_0 V_A l_{||}}{4 B_0} \quad (3.2)$$

where δn is the density in the barium swarm, V_A is the Alfvén velocity in the

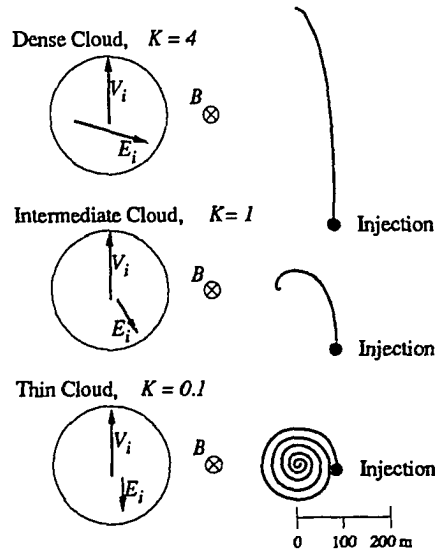


Figure 3.14: A sketch of the motion of three barium ion clouds of different density after *Brenning et al. [1991c]*.

ambient ionosphere, and $l_{||}$ is the extent of the ion cloud along $\vec{B}_0$. Fig. 3.14 taken from *Brenning et al. [1991c]* illustrates ion cloud motion for dense, intermediate and thin clouds, where the separation between these types is based on the K value of (3.2). For a dense cloud, $K \gg 1$, the ions continue “skidding” in the original direction of motion. The ambient plasma in the flux tube through the cloud is dragged with the motion, and the internal field in the cloud is close to the self-polarization field, $\vec{E} = -\vec{V}_i \times \vec{B}_0$ where $\vec{V}_i$ is the velocity of the cloud. For a cloud of intermediate density, $K = 1$, the ion motion is rapidly stopped, on the time scale of the ion gyro frequency. For a thin cloud $K \ll 1$ we have a collective spiral motion with decreasing amplitude. This is the case closest to the CRIT II observations. A detailed comparison of this theory with CRIT II measurements is presented by *Brenning et al. [1991c]*.

We now return to the magnetometer data during the passage of the ionization front. To gain further insight into the data we transform it into a coordinate system that is aligned with the barium stream, or more precisely, because of the problem

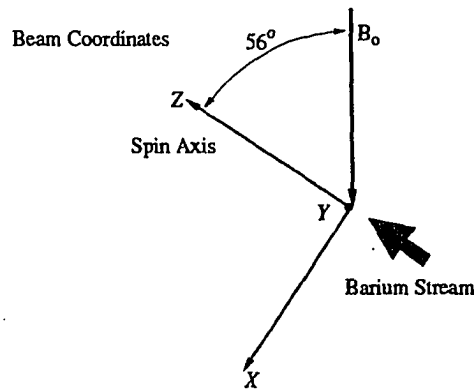


Figure 3.15: A coordinate system that is aligned with the barium beam.

with attitude control, into a coordinate system that is within 15° of the barium stream. This coordinate system is sketched in Fig. 3.15. The beam coordinate system is a rotation about the Y axis from the magnetic coordinates system such that the Z axis is aligned with the spin axis of the rocket. Data rotated into this coordinate system is presented in Fig. 3.16. The δB_x and δB_y are perturbation measurements of the magnetic field across the neutral stream, with the δB_y data panel measured along the $\vec{V}_{Ba} \times \vec{B}_0$ direction. The third panel, δB_z , presents the fluctuations parallel to the barium stream direction. The smaller perturbations riding on the first pulse seen primarily in the δB_y data indicate a highly variable current system within the ionization front which will be discussed in Chapter 5 along with the electric field variations on that same time scale. In the discussion for the remainder of this chapter, we will only consider the quasi-dc magnetic fluctuations during ionization front.

At the time of the flash (0.135 s) the δB_y magnetic perturbation rapidly changed direction from negative to positive indicating the presence of a particularly strong current. If we assume the observed magnetic perturbations result from a current system that is stationary in the frame of the barium stream then an estimate of

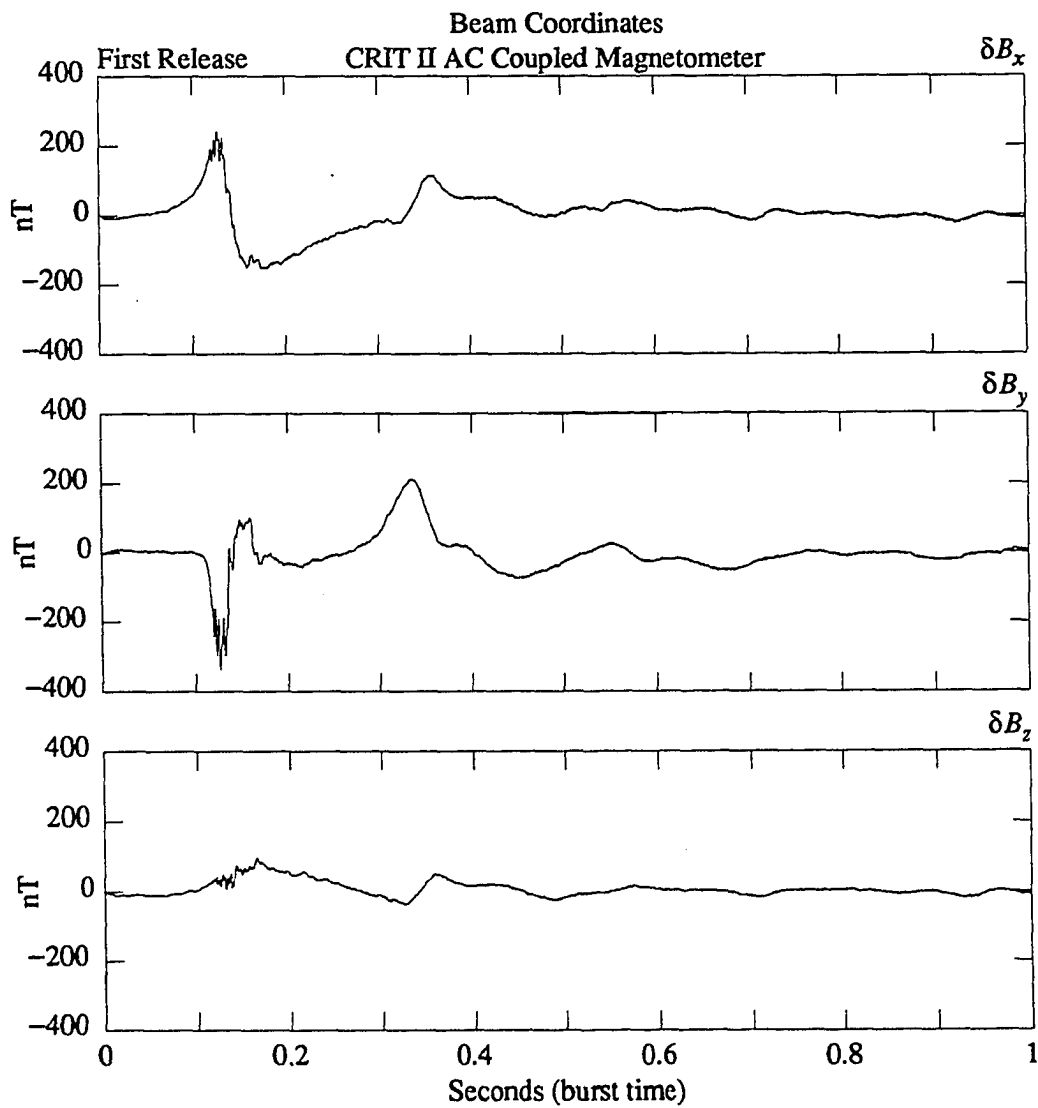


Figure 3.16: CRIT II, first release, fluctuations from the geomagnetic field in a barium stream aligned coordinate system (see Fig. 3.15.).

the currents can be made from Amperes law.

$$\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right)\hat{i} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right)\hat{j} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)\hat{k} = \mu_o(J_x\hat{i} + J_y\hat{j} + J_z\hat{k}) \quad (3.3)$$

The only spatial derivatives measured by the magnetometer are along the z direction as the current system sweeps past the payload. There are no measurements of spatial derivatives along x or y because a single magnetometer was used. This becomes an important point in this discussion. We therefore neglect the $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$ terms realizing we cannot determine the total current due to our limited data set. The $\frac{\partial}{\partial z}$ terms become $\frac{1}{\bar{V}_{Ba}}\frac{\partial}{\partial t}$ and so

$$-\frac{1}{\bar{V}_{Ba}}\frac{\partial B_y}{\partial t}\hat{i} + \frac{1}{\bar{V}_{Ba}}\frac{\partial B_x}{\partial t}\hat{j} = \mu_o(J_x\hat{i} + J_y\hat{j}) \quad (3.4)$$

Two points become apparent, first the δB_z channel should show no fluctuations from the passage of this type of current system and second, the magnetometer measurements are insensitive to currents aligned with the neutral stream. They will exist, but they will be undetected by the magnetometer in this model. The relative absence of fluctuations along the z direction can be seen in Fig. 3.16 by comparing the fluctuations of the δB_x and δB_y with the δB_z channel.

Because J_z currents are undetectable by the magnetometer, the measured J_x current could result from the projection of a field aligned current $J_{\parallel}$ as pictured in Fig. 3.17. Using (3.4) and δB_y from Fig. 3.16, the field aligned current that would be consistent with the positive derivative at 0.135 s is 11.5 mA/m². This current density is 10³ times larger than those typically measured in auroral arcs. Since currents associated with these changes in δB_y are so large, it seems only reasonable that they are field aligned. Currents in the $\bar{V}_{Ba} \times \bar{B}_o$ direction, or across the neutral stream, show up as derivatives in δB_x .

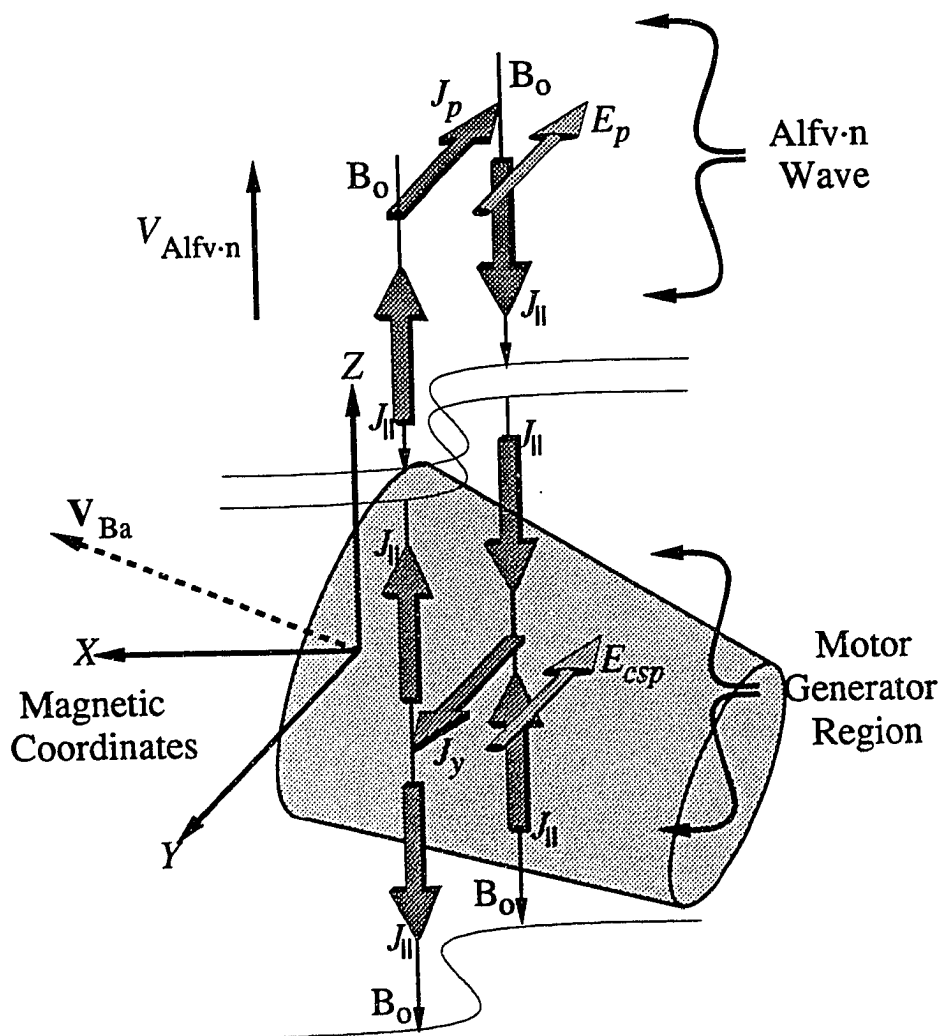


Figure 3.18: The cross stream current system which can exchange axial momentum of the barium stream with the ionosphere.

electric field, E_{csp} , remains in the same direction, the volume of the stream acts like a motor. Apparently the energy source for the generator, the production of new barium ions, shuts off and the polarization along the flux tube collapses, similar to a capacitor discharging. This discharge drives the large field aligned current previously discussed and turns the region within the stream into a motor. Using (3.4) the current associated with the positive derivative of δB_X is $\approx 0.4 \text{ mA/m}^2$ and the current associated with the negative derivative is $\approx 1.3 \text{ mA/m}^2$. The velocity change of a barium ion cloud, δv , due to each of these $\vec{J} \times \vec{B}_0$ forces can be estimated by,

$$\delta v = 1/\rho \int J_y B_0 dt \quad (3.5)$$

where ρ is the mass density of the barium ion cloud. If these $\vec{J}_y \times \vec{B}_0$ forces act on a barium ion cloud with twice the background plasma density then the first half of the ionization front will slow the cloud by $\approx 700 \text{ m/s}$ and the second half will accelerate ions by $\approx 7.0 \text{ km/s}$. Clearly the $\vec{J}_y \times \vec{B}_0$ force on the quasi-dc time scale cannot be responsible for significantly slowing newly produced ions from the neutral stream. The acceleration of barium ions in the second half of the front may provide the source of phase bunched barium ions for the second barium cloud which is easily observed in Fig. 3.4 and Fig. 3.16.

A second current system exists within the ionization front due to the newly created barium ions. This current system is related to the ionizing front model proposed by *Piel et al.* [1978] and is sketched in Fig. 3.19. The newly created barium ions are unmagnetized on the time scale of the ionization front, while the newly produced and ambient electrons are magnetized. The motion of barium ions perpendicular to $\vec{B}_0$ constitutes a current along the barium stream which is carried away by field aligned currents and closed through the polarization current of an Alfvén wave. Alternately, this field aligned current system can be explained

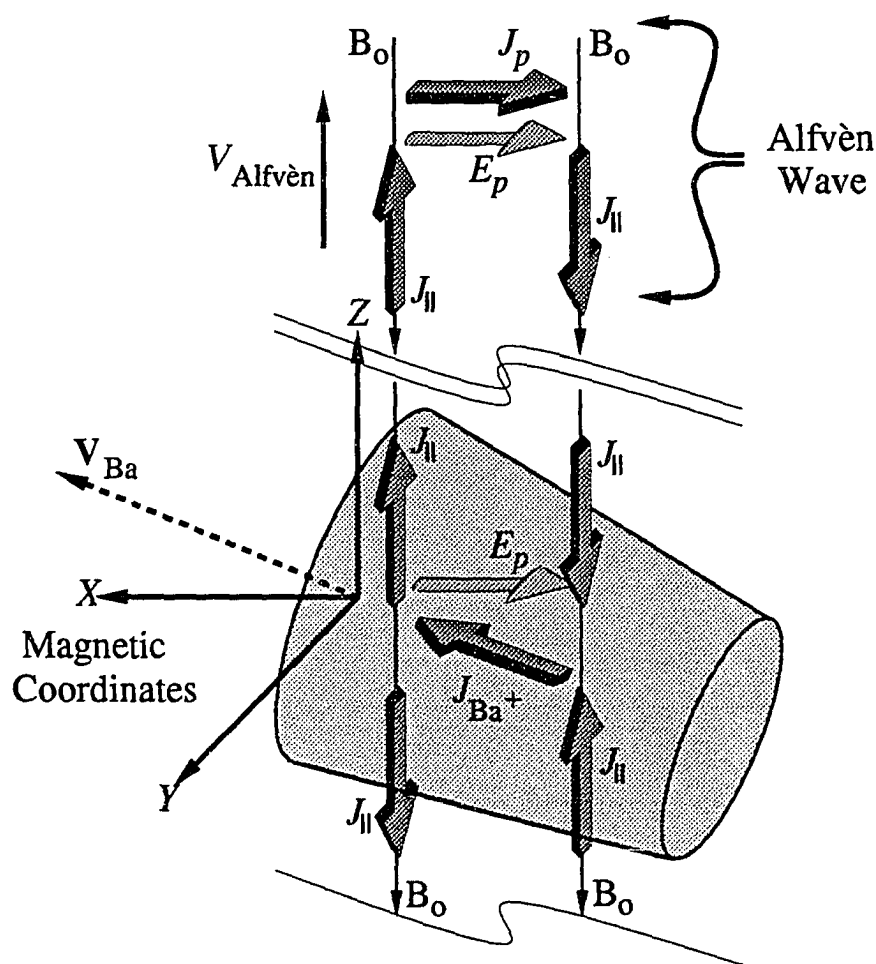


Figure 3.19: A sketch of currents required to maintain charge neutrality for an unmagnetized ion beam.

by considering an ion cloud displacing from its charge neutralizing electron cloud. Since the electrons are magnetized and cannot move with the ions, a positive space charge is created in the direction of the ions motion and a negative space charge at the back of the cloud. To maintain charge neutrality electrons are pulled along the geomagnetic field towards the front of the cloud and pushed away from the rear along the field. This positive space charge at the front and negative space charge at the rear sets up an internal electric field, E_p , such that a generator is formed. The resulting Hall current is in the same direction as the dynamo current of *Haerendel* [1982], J_y in Fig. 3.18, and suggests a relation between the two models.

In reality, both current systems exist at the same time and the actual fields are a superposition of the two models. The field aligned current is the sum of the divergence of the barium ion current and the Hall current, $j_{||} = J_y + j_{Ba+}$. We note, from a measurement perspective, that the field due to parallel currents shown in Fig. 3.19 will be more strongly detected at the main payload than those due to parallel currents carried at the edges of the barium cloud in Fig. 3.18. *Brenning et al.* [1991b] has shown that for steady state the dominant current system in a thin ion cloud is that of Fig. 3.19. This thin cloud will gyrate with the barium cyclotron period. This is the type of ion cloud that is observed after the ionization front. For thick ion clouds the current system of Fig. 3.18 dominates and the cross stream polarization electric field approaches $E_p = -\vec{V}_{Ba} \times \vec{B}_0$. A thick cloud will thus tend to skid across the magnetic field in the original direction of the ions, and is usually the type produced in magnetospheric and solar wind chemical releases relying on photoionization.

The differentiation between the current systems can be seen more clearly from CRIT II second release data of Fig. 3.20 in beam coordinates. Unlike the first burst data, the coordinate system is here almost perfectly aligned with the barium

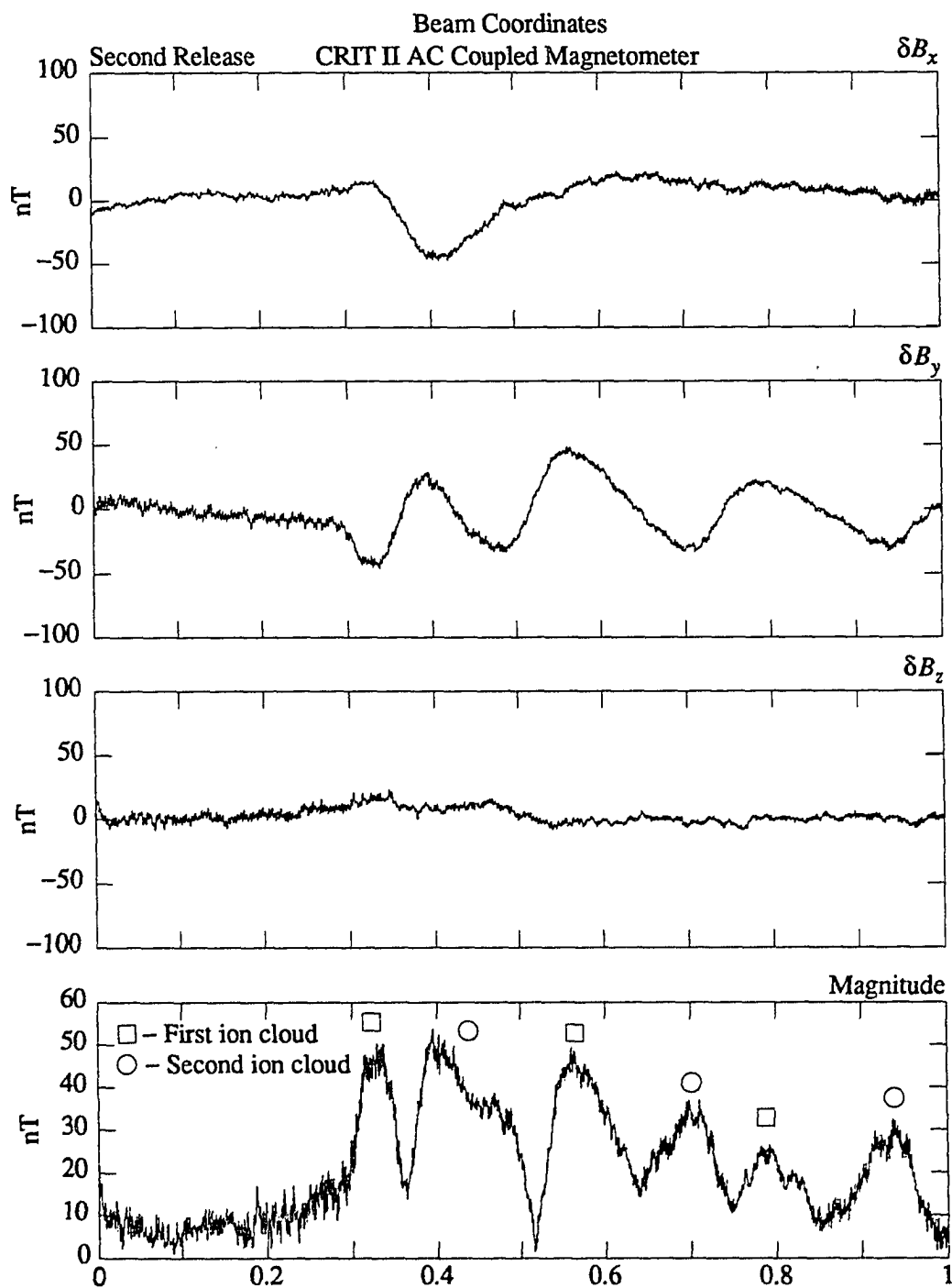


Figure 3.20: CRIT II, second release, fluctuations from the geomagnetic field.

stream. Signatures of the momentum exchanging cross stream currents (Fig. 3.18) are only seen during the passage of the ionization front as derivatives in δB_x . Once again a strong negative derivative just at the end of the ionization front indicates a reversal of the momentum exchanging current J_y . This turns the generator region into a motor which accelerates ions. The parallel currents (Fig. 3.19), δB_y derivatives, occur during the front. After the front, the δB fluctuations continue. The fluctuations are signatures of a gyrating barium explained by the barium swarm model of *Brenning et al.* [1991c].

3.2 Subpayload Results

Measurements of the electric and magnetic fields at the subpayload are presented in Fig. 3.21. The first two panels, v12m and v34m, show the electric field components perpendicular to the spin axis and the third panel, v15m, is a nonorthogonal measurement that has a major component parallel to the payload spin axis. The field component parallel to the spin axis can be calculated from v15m and v12m and is plotted in the fourth panel called v05. The spin axis of the subpayload was about 10° from the geomagnetic field, so qualitatively these measurements are in a geomagnetic aligned coordinate system. The orientation of the perpendicular components relative to the barium stream is unknown. Similar to main payload data, a major feature in the subpayload electric field data is a fluctuation with a period of 0.2 s. This is the mapping along the geomagnetic field of the electric field within the barium swarm. The perpendicular quasi-dc electric field in the ionization front at the main payload presumably maps up the geomagnetic field at the Alfvén speed to the subpayload where the field strength was 150 mV/m. In contrast to the electric field observations at the main payload, very little wave activity was observed

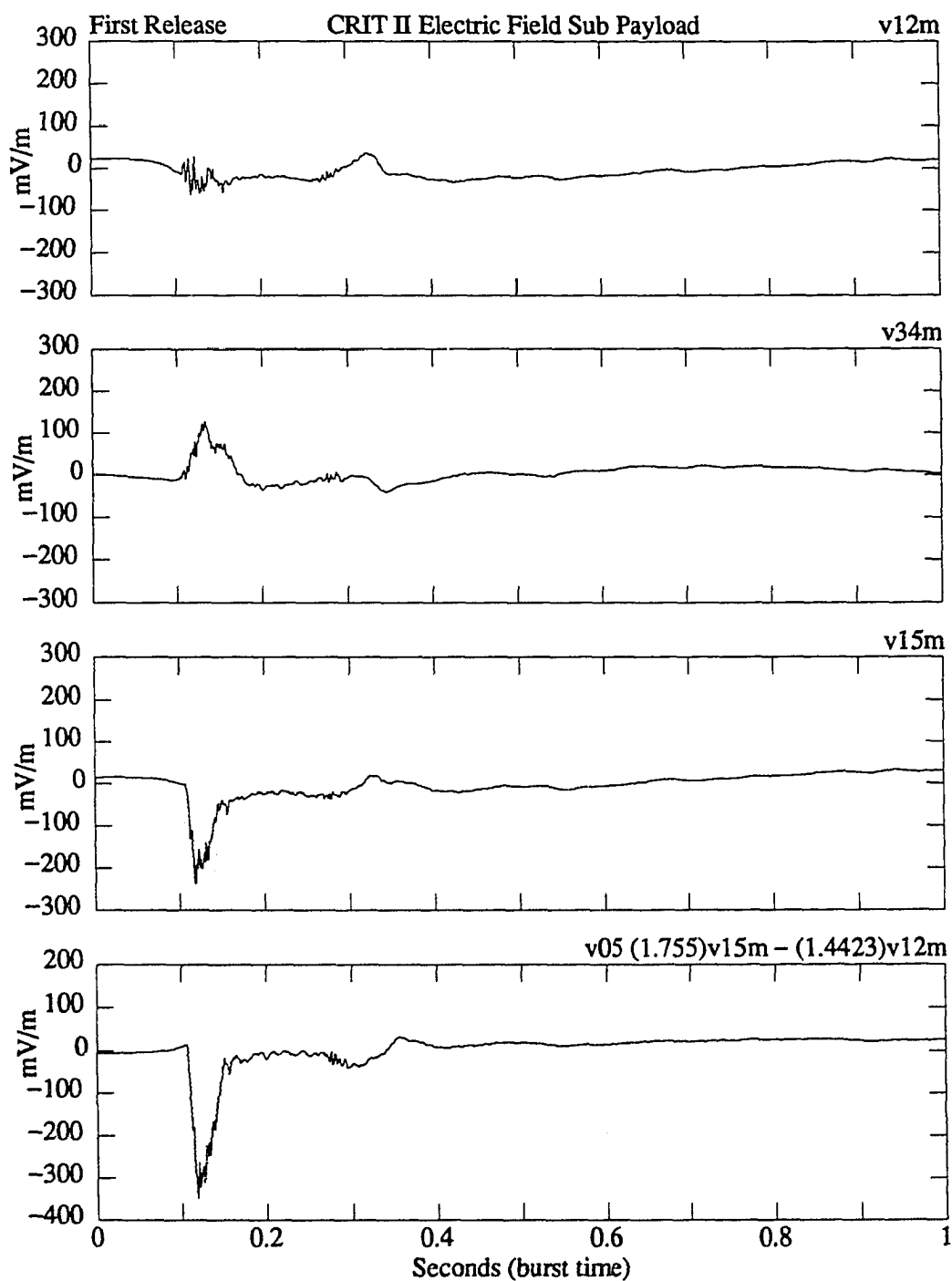


Figure 3.21: Electric field measurements at the CRIT II subpayload during the first release.

in the 300-900 Hz band. Either these waves have group velocities approximately perpendicular to $\vec{B}_0$, or these waves must have been damped in propagation along $\vec{B}_0$. The energy of these waves may have been channeled into heating the electron population as suggested by CIV theories.

The first pulse appears to have a significant geomagnetic parallel electric field component (v05, Fig. 3.21), 300 mV/m, which is correlated in time with the arrival of the ionization front at the main payload. However, this measurement appears to have been corrupted by an unbalanced collection of electrons by the electric field sensors and this result is discounted. Energetic electrons were detected at the subpayload by the electron charged particle analyzer during the ionization front. The most likely source of these heated electrons is the ionization front around the main payload. These 10 eV electrons escaping along the geomagnetic field constitute a field aligned current. This current of electrons is suspected to have been collected unevenly, relative to the other sensors, at the fifth electric field sensor. The current density calculated by Mr. Kan Liou of University of New Hampshire (*personal communication, 1991*) from the particle measurements is presented in Fig. 3.22. The subpayload body is between the source of the hot electrons and sensor #5 (Fig. 2.11), and the 10° tilt of the payload relative to the geomagnetic field is not enough to bring the sensor out of the payload's electron flux shadow which ends at ~13°. The sheath capacitance of the electric field sensors is about 13 pF so the potential of the sensors can indeed change on the observed time scales. There is an apparent contradiction since such a field aligned flux of electrons would substantially charge the payload, which, according to experimenters at University of New Hampshire is not observed. The electron current density of Fig. 3.22 is almost 80 times larger than the thermal oxygen ion current density (25 $\mu\text{A}/\text{m}^2$). Since the ion thermal current cannot balance the collected electron flux, a conductor would

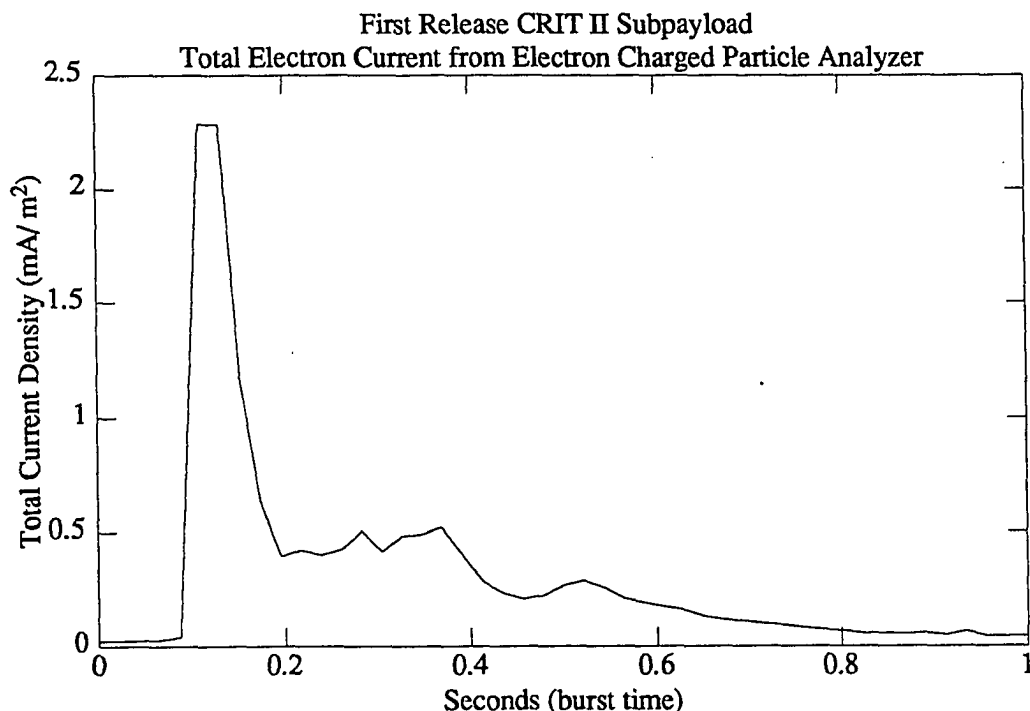


Figure 3.22: Geomagnetic aligned current density due to energetic electrons at the CRIT II subpayload during the first release.

charge significantly negative relative to the plasma until enough ions are collected or the electron flux is reduced by the potential (≈ 10 eV). This type of charging would have precluded obtaining any data from the electric field instrument unless the spacecraft also charged in the same sense. We are therefore forced to shade the data during the ionization front of the electric field measurements obtained with the fifth sensor. We return to this discussion below.

Fig. 3.23 shows the magnetic field observations made with a fluxgate magnetometer at the subpayload during the first release of CRIT II. The observations have been transformed into a non-rotating geomagnetic aligned coordinate system to separate the geomagnetic field, $\vec{B}_0$, from the fluctuations. The top two panels are the measurements perpendicular to $\vec{B}_0$ and the bottom panel contains the parallel measurement. Ideally the magnitude of $\vec{B}_0$ would be a dc offset in the bottom

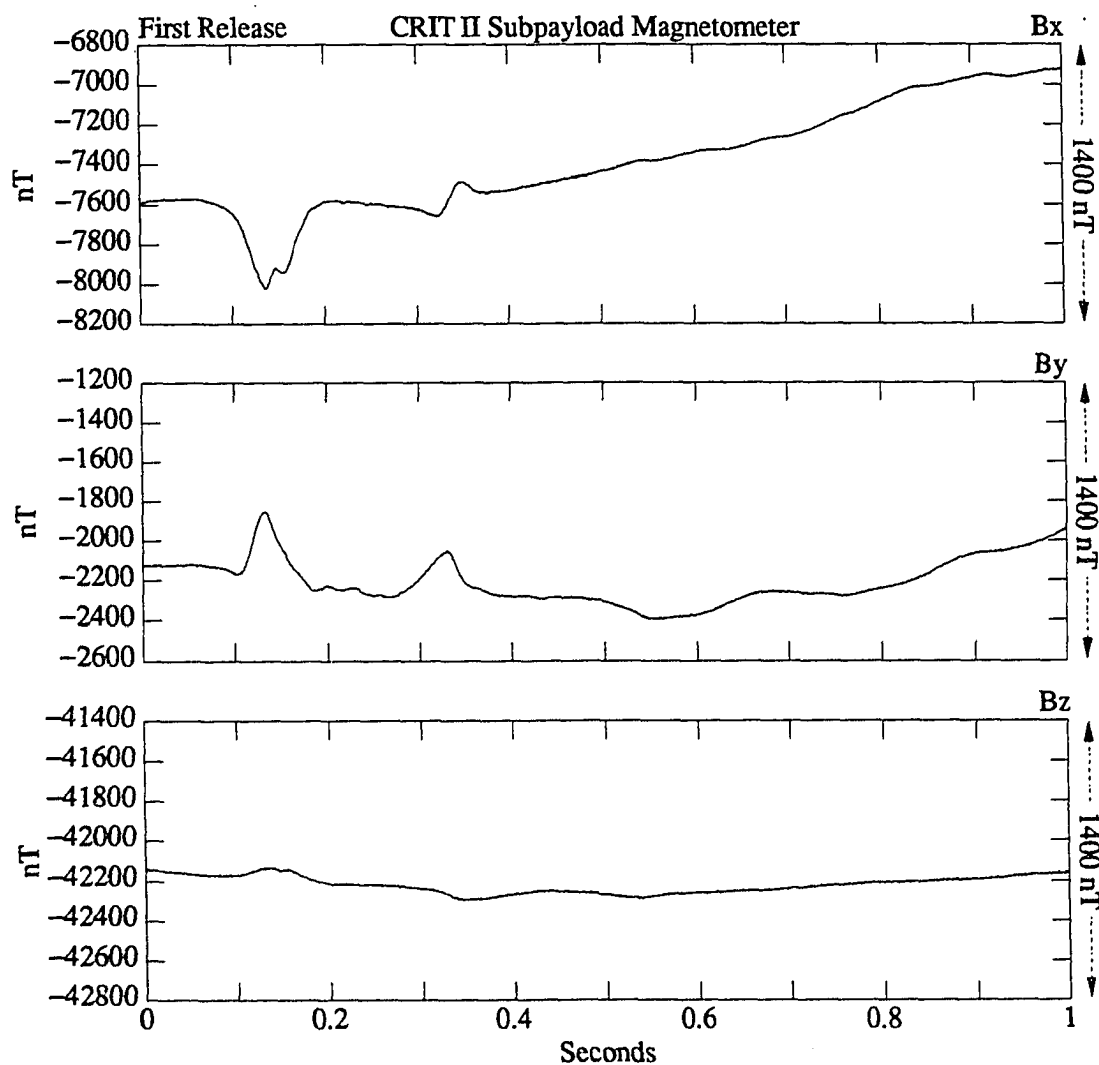


Figure 3.23: Electric field measurements at the CRIT II subpayload during the first release.

panel equal to the geomagnetic field strength and the top two panels would have no dc offsets, but errors in determining the subpayload attitude have resulted in both dc offsets and slow ≈ 5 s trends, remnants of the spin frequency. A strong 500 nT pulse, which corresponds to the first electric field pulse and the arrival of the ionization front at the main payload, was detected perpendicular to the geomagnetic field. The second magnetic pulse occurred at the same time as the second electric field pulse. As noted earlier, the momentum of the reacting neutrals is thought to be transported to the surrounding ionosphere via Alfvén waves. The subpayload experiment was designed to detect this type of phenomena. To verify the consistency of these electric and magnetic field observations with this theory we found the ratio of the perpendicular electric and perpendicular magnetic field fluctuations in MKS units to be ≈ 276 km/s in the first pulse and ≈ 233 km/s in the second pulse. Both are close to the local Alfvén speed of 300 km/s. To further verify this theory the directions of electric and magnetic field measurements need to be placed in the same coordinate system whose orientation relative to the main payload is known. The directions of the fields could then be tested against Fig. 1.3 to determine if momentum is actually being exchanged. The good agreement between the electric and magnetic field data even in the early time data suggests that the shadowing problem only affects the parallel electric field data. If the perpendicular electric fields are uncorrupted, as we believe, then the mapping of this field back to the region around the main payload suggests that the electric field in the ionization front is also 150 mV/m (see Fig. 3.21). Unfortunately the direction of this field is unknown, but if we assume the field is in the direction of E_{csp} then it would correspond to a slow down of the ions by about 3 km/s.

Another observation at the subpayload supporting the barium swarm model is that two and possibly three Alfvén wave packets were observed to originate from

the region of interaction around the main payload. These Alfvén wave pulses were separated in time by about the barium cyclotron period ($2\pi/\Omega_{\text{Ba}+}$). The ionization front, moving with the barium neutrals, would have moved out of contact with the field lines near the main payload. This means that pulses following the initial ionization front are associated with magnetized barium ions, created within the front, returning after their first and second gyro orbits.

3.3 Summary

For intercomparison purposes, in this chapter we have concentrated on a single second of data beginning at the time of the first barium release of CRIT II but have supplemented the discussion with data from the second release. Data from each instrument is presented for this period, including data from the subpayload. The observations during this period are conveniently divided into two parts. The first part is dominated by the physics occurring within an ionization front and the second part by the physics of the barium swarm [Brenning *et al.*, 1991*b*]. The barium swarm itself results from a process occurring within the ionization front. In one sense the late time field and particle data are akin to the ringing of a bell after it has been struck. The strike of the bell is analogous to the ionization front passage.

Within the ionization front significant plasma density increases, more than twice the background density, were measured with the arrival of the fastest neutral barium (15 - 9.7 km/s). A crude estimate of the ionization rate within the front, using these plasma density increases over the entire 0.1 s period of the front, gives $5 \times 10^6 \text{ cm}^{-3} \text{ s}^{-1}$, but there are indications the plasma was deposited during a shorter period of time, so the ionization rate was probably higher. The barium ions

produced during this short 0.1 s dominate the physics afterwards. Within the ionization front the particle measurements indicate barium ions were being scattered in energy and direction. Hot electrons were detected within the ionization front and for a short period after. The flux of hot electrons within the front was so great the detectors were saturated for brief periods. The peak ionization rate due to hot electrons was estimated by *Torbert et al.* [1991] to be over $10^6 \text{ cm}^{-1} \text{ s}^{-1}$.

The electric field measurements within the front have been examined in two frequency regimes, the quasi-dc fields, which vary on the barium cyclotron time scales and the wave fields. The quasi-dc electric field data were contaminated by effects of the neutral barium on the electric field sensors. Only an upper limit on the size of the fields during the ionization front can be deduced from the data. The subpayload field measurement is more reliable and suggests a field of 150 mV/m. This is too small a value to account for the scattering and slowing of barium ions observed by the particle detectors, implying that, the larger (400 mV/m) wave fields must contribute to this process. In addition, the upper limit on the quasi-dc fields is too small to allow the plasma in the ionization front to move with the neutral barium stream. The wave fields during the ionization front peak at frequencies well below the lower hybrid frequency. We will show in Chapter 5 that this is not a Doppler shifting artifact. CIV theories have relied heavily on modified two stream instabilities exciting lower hybrid waves, but these measurements suggest that other modes are being driven preferentially. All that is essential for CIV is the excitation of a wave mode which grows and can transfer energy to the electrons since a significant portion of the energy must end up in the electron population. The observed waves may fulfill these requirements.

The magnetic field fluctuations within the ionization front at the main payload were so large that they cannot be the result of a displacement current. Under

plausible assumptions we estimate currents along the magnetic field and currents across the barium stream perpendicular to the field. The cross field currents were only detected within the front, when momentum exchange was occurring. The parallel currents were larger, and in addition to closing the cross field currents, they indicated a flow of electrons neutralizing the space charge of the moving barium cloud. The subpayload positioned outside of the path of the ionization front, but in contact with it along the geomagnetic field, detected large electric and magnetic fluctuations consistent with an Alfvén wave. Several of these Alfvénic pulses were observed on the main and the subpayload with a time spacing equal to the barium cyclotron period. The absolute direction of the current system within the Alfvén wave is unknown, but it seems reasonable, in light of the evidence of momentum exchange, that part of the current system is the closure current for the cross field current at the main payload.

Chapter 4

The CRIT II Density Enhancements

The focus of this chapter is on the plasma density measurements made during both releases of the CRIT II experiment. In Section 3.1.1 these data were presented on a 1.0 second time scale. In this chapter the data will be examined on the much shorter time scale of the ionization front. This allows us to focus on events occurring within a 0.1 s period for the first release, and for the second release, on events occurring within a 0.2 s period. A distinguishing feature of the data is that a plasma density increase was detected within the ionization front. Possible explanations for these density enhancements will be considered, such as bunching of the ambient plasma by collisional 'snowplowing' of the neutral stream and productions of ions through collisional charge exchange and charge stripping. We will show that the first release density enhancement can possibly be explained by collisional ionization but that the second release density enhancement cannot. The discussion will conclude with an argument that both these plasma enhancements can be explained by a relatively inefficient CIV process. Although we do not present a definitive study of the data

because of inconsistent estimates of a relevant cross section in the literature, we do suggest that these plasma enhancements are signatures of CIV

4.1 Measurements

The high resolution plasma density measurements during the first burst are presented in Fig. 4.1. Within the ionization front a number of short period density enhancements and depletions appear to grow throughout the passage of the ionization front. To aid in determining the over all density enhancement two averaged curves have been over laid. The first curve is the data averaged by a factor of 30 and the second curve by a factor of 120. Clearly, despite the fluctuations, a net increase in the plasma density was detected during the first release. The second release data is presented in a similar format in Fig. 4.2

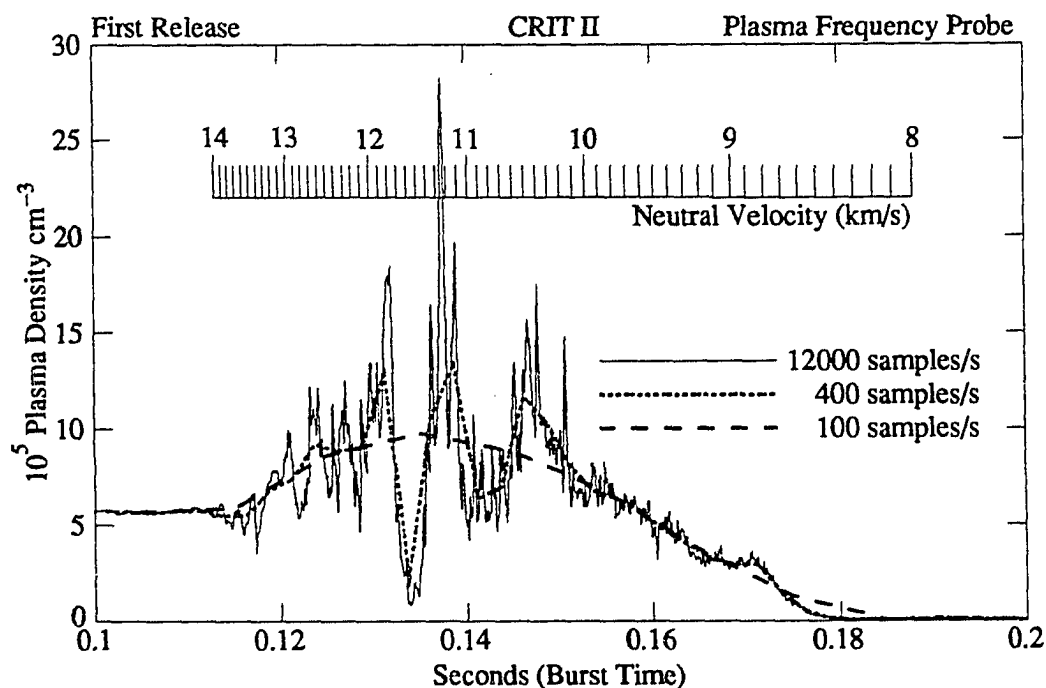


Figure 4.1: Plasma density measurements during the first release of CRIT II along with the velocity of barium neutrals.

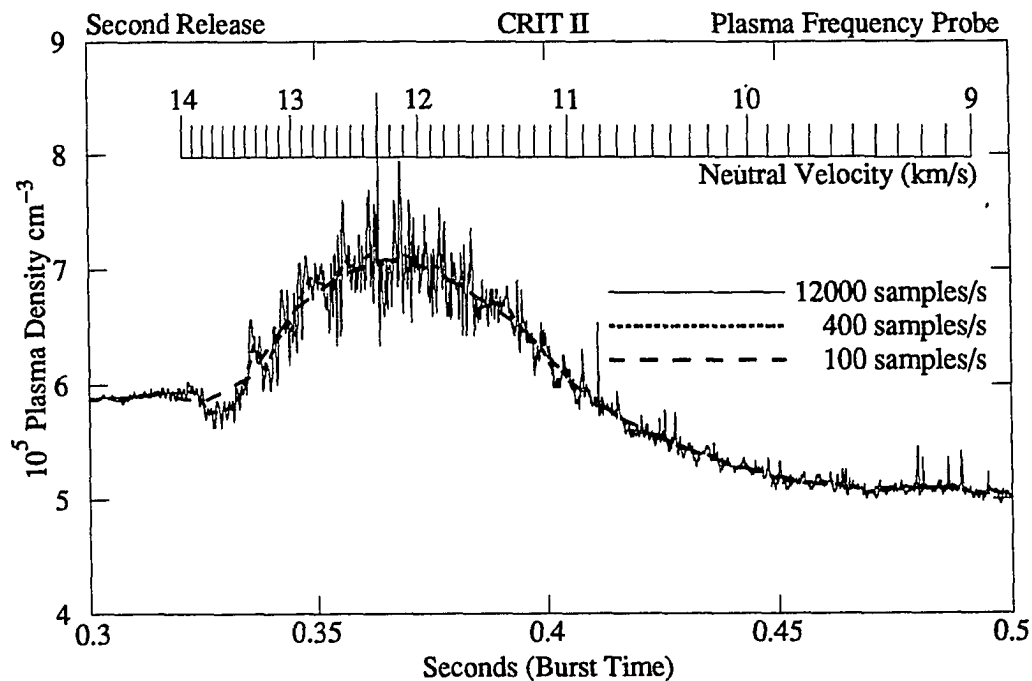


Figure 4.2: Plasma density measurements during the second release of CRIT II and velocity of barium neutrals.

Some comment on the velocity scale is in order. Inconsistencies between the neutral stream model and the observations have prompted a reevaluation of the release distances provided by NASA. The problem, simply stated, is that the distance of the main payload from the first release seems too great and the distance from main payload to the second release seems too small. The arrival time of the fastest neutral barium, traveling at 13.5 km/s according the neutral model, does not correlate well with the times of the observed density fluctuations. Since the neutral velocity was determined by ground based optical measurements of barium ions arriving in sunlight, we believe the 13.5 km/s cutoff for the fastest barium. The distance of the chemical release cannisters from the main payload was calculated from the time of flight and the total impulse *both* cannisters *together* had on the main payload. The total impulse was determined from the deviation of the main payload's trajectory by NASA, and equally divided between the two cannisters.

The separation velocity of each cannister, proportional to relative masses, was then calculated. There was not a separate measurement for each chemical cannister's ejection velocity. It is probable that the individual ejection impulses were not actually equal [*personal communication Phil Ward, NASA, 1991*]. The first cannister, the one creating the second release, appears from the main payload data to have been ejected faster than the second cannister. The neutral stream model 'fits' the first release data better if the release actually occurred 1.40 km away (rather than 1.58 km), and the second release data 'fits' the neutral stream model if it actually occurred 4.31 km away (rather than 4.88 km). These distances are consistent in that the sum of the individual impulses required equals the total impulse the main payload was measured to have received. These release distances will be used for the more detailed discussion of the ionization front and were used in plotting the neutral stream velocities in Fig. 4.1 and Fig. 4.2.

The general wave form is verified by a fixed bias Langmuir probe which gives further confidence in the observations. We consider the PFP data more reliable since vehicle potential changes do not affect the measurements.

In order to verify that the plasma enhancements are due to a strong plasma beam interaction, and quite possibly to the CIV process itself, we must rule out more classic explanations for the data. These competing mechanisms can be organized into two classes: (a) a mechanism in which existing oxygen plasma is merely pushed around by the neutral stream (e.g. the collisional snow plow effect), and (b) plasma production or modification mechanisms such as charge exchange and charge stripping. In Sections 4.2 and 4.3 we investigate these two types of processes and conclude that they cannot explain our data. In the final section we discuss the implications our data have for the physics of CIV.

4.2 Non-ionizing Sources of Plasma Structuring

4.2.1 Neutral Wind Forces

Since we measure the electron density we need to investigate whether the neutral wind forces of the barium stream, pushing ionospheric plasma, could be the cause of the density enhancements observed at the main payload during the releases. To partially answer this question we may refer to Fig. 2.5. The peak collision frequency of the neutral barium stream with the background O^+ plasma is plotted versus distance from release in this figure. If the collision frequency is greater than Ω_{O^+} , one would expect that the plasma could be pushed across the magnetic field. This would result in compression of the plasma and local density enhancements and depletions. This is clearly the case within the neutral stream for the first kilometer. After this distance, the collision frequency rapidly falls below Ω_{O^+} and the possibility of pushing plasma across the magnetic field ceases. The location of the main payload, for each of the four releases, was beyond the point where conventional wisdom would expect the plasma to be significantly affected by forces perpendicular to $\vec{B}_0$. Estimating the result of collisional forces along the geomagnetic field is not as straightforward. Comparisons of the ion-neutral collision frequency to the gyro frequency are not important, but the duration and magnitude of the collisional force relative to the inertia of the ions are.

Since the CRIT releases occurred at a nominal 45° to the geomagnetic field, there is a possibility of pushing oxygen ions along $\vec{B}_0$. This scenario is sketched in Fig. 4.3. The leading edge of the barium stream contacts the geomagnetic field below the main payload and the parallel component of the collisional force starts pushing background plasma ions along the field towards the main payload. As the leading edge progresses towards the payload, the density bump is pushed along the

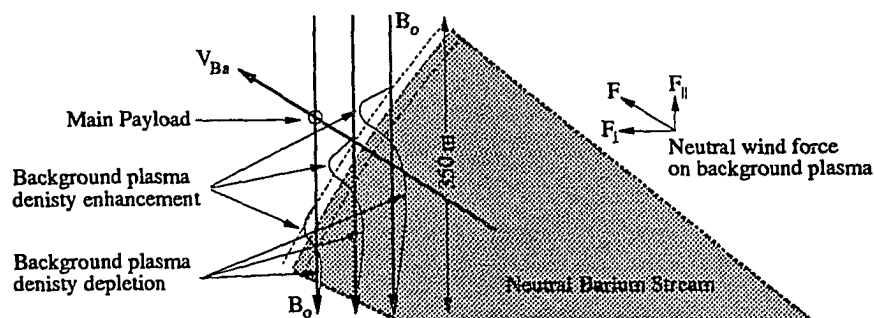


Figure 4.3: The neutral barium stream can push a density wave along the geomagnetic field at its leading edge.

field, growing as it goes. A depletion forms behind the local density enhancement. The barium stream acts like a snowplow by moving the background plasma within its path to the side, or in this case by sliding it up along the geomagnetic field. The magnitude of the enhancement and depletion observed at the main payload will depend on the specifics of the neutral stream forces. There are several complicating factors. First, the barium neutral density is Gaussian across the width of the stream. The half density maximum is 6.6° for fast barium, but the stream width increases for the slower neutrals [Wescott *et al.*, 1990]. This limits the region of plasma along the field that can be compressed before reaching the main payload to about 175 m for the first release of CRIT II. Second, the barium stream density peaks at velocities near 11 km/s and falls off rapidly with decreasing barium velocity. This results in a collisional force which is rather sharply peaked in time. Whether this rapidly changing force can compress enough plasma in 175 m is not obvious and has been the subject of a computer simulation to realistically take into account these complicating matters.

4.2.2 A Fluid Model for Collisional Forcing

Consider a neutral stream collisionally interacting with a magnetized plasma. Following the derivation of *Seyler and Providakes* [1987], we formulate a one-dimensional model where the magnetic field $\vec{B}_0$ is in the $\hat{z}$ direction, the neutral stream velocity u is in the $\hat{x}$ direction, and there are no variations in the $\hat{y}$ direction. This model will later be modified for the case of an unmagnetized plasma or for modeling motion along a magnetic field by setting $B = 0$ and making a few slight modifications. The plasma is warm, collisional, and quasineutral. Electrons are allowed to flow along the field to close the ion current while ions are not allowed to move along B for the perpendicular model. This quasineutrality condition is stated as

$$\nabla \cdot \vec{J} = 0 \quad (4.1)$$

Letting $n_e \approx n_i = n$, where the subscripts i and e stand for ions and electrons, the current is expressed as

$$\vec{J} = en[(v_{ix} - v_{ex})\hat{x} + (v_{iy} - v_{ey})\hat{y} + (v_{iz} - v_{ez})\hat{z}] \quad (4.2)$$

for this model, (4.1) and (4.2) reduce to

$$\frac{\partial}{\partial x}[n(v_{ix} - v_{ex})] - \frac{\partial}{\partial z}(nv_{ez}) = 0 \quad (4.3)$$

we relate the z derivative to the x derivative with a free parameter θ .

$$\frac{\partial}{\partial z} = \theta \frac{\partial}{\partial x} \quad (4.4)$$

and integrate to get

$$n(v_{ix} - v_{ex}) - \theta(nv_{ez}) = K \quad (4.5)$$

where K is a constant of integration representing the initial field aligned current, which is properly set to zero for this model. Now we turn to the electron momentum equation to find expressions for v_{ex} and v_{ez} in (4.5).

$$m_e n_e \left(\frac{\partial \vec{v}_e}{\partial t} + (\vec{v}_e \cdot \nabla) \vec{v}_e \right) = q_e n_e (\vec{E} + \vec{v}_e \times \vec{B}) - \nabla P_e - n_e m_e \nu_{e,n} (\vec{v}_e - \vec{u}_n) \quad (4.6)$$

Assuming $(\vec{v}_e \cdot \nabla) \vec{v}_e$ and the electron inertia term, $\partial \vec{v}_e / \partial t$, are small compared to the collision term $\nu_{e,n} \vec{v}_e$, the previous equation reduces to the following set of equations, one for each component $\hat{x}$, $\hat{y}$, and $\hat{z}$.

$$\begin{aligned} \hat{x} \quad \frac{e}{m_e} E_x &= -\Omega_e v_{ey} - \nu_{e,n} v_{ex} - \frac{\gamma_e k T_e}{m_e} \frac{1}{n} \frac{\partial n}{\partial x} \\ \hat{y} \quad 0 &= \Omega_e v_{ex} - \nu_{e,n} v_{ey} \\ \hat{z} \quad \frac{e}{m_e} E_z &= -\nu_{e,n} v_{ez} - \frac{\gamma_e k T_e}{m_e} \frac{1}{n} \frac{\partial n}{\partial z} \end{aligned} \quad (4.7)$$

where $\Omega_e = \frac{em_e}{B_0}$ is the electron gyro frequency. Since the neutral stream is directed in the $\hat{x}$ direction, the rate of electron-neutral collisions is not necessarily symmetric along and across the neutral stream. If the velocity of the electrons is much greater than this streaming motion, this asymmetry can be neglected. The thermal velocity of ionospheric electrons, about 170 km/s, is much greater than the velocity of the neutral barium beam, 10 km/s, so $\nu_{e\parallel} \approx \nu_{e\perp} = \nu_{e,n}$ is the combined collision frequency of electrons with neutrals and ions parallel and perpendicular to the neutral stream. This will not be the case for ion-neutral collisions and different collision frequencies will be required. We solve these equations for v_{ex} and v_{ez}

letting $\vec{E} = -\nabla\phi$ and $\alpha = 1 + \Omega_e^2/\nu_{e,n}^2$.

$$v_{ex} = \frac{1}{\alpha\nu_{e,n}} \left[\frac{e}{m_e} \frac{\partial\phi}{\partial x} - \frac{\gamma_e k T_e}{m_e} \frac{1}{n} \frac{\partial n}{\partial x} \right] \quad (4.8)$$

$$v_{ez} = \left[\frac{1}{\nu_{e,n}} \frac{e}{m_e} \frac{\partial\phi}{\partial z} - \frac{\gamma_e k T_e}{m_e} \frac{1}{n} \frac{\partial n}{\partial z} \right] \quad (4.9)$$

Substituting v_{ex} and v_{ey} from (4.8) and (4.9) into (4.5) and applying the conditions of (4.4) to (4.9) we arrive at

$$\frac{e}{m_i} \frac{\partial\phi}{\partial x} = \frac{m_e}{m_i} \left(\frac{\nu_{e,n}\alpha}{1 + \theta^2\alpha} \right) v_{ix} - \frac{\gamma_e k T_e}{m_i} \frac{1}{n} \frac{\partial n}{\partial x} \quad (4.10)$$

Now we turn to the momentum equation for ions.

$$m_i n_i \left(\frac{\partial \vec{v}_i}{\partial t} + (\vec{v}_i \cdot \nabla) \vec{v}_i \right) = q_i n_i (\vec{E} + \vec{v}_i \times \vec{B}) - \nabla P_i - n_i m_i \nu_{i,n} (\vec{v}_i - \vec{u}_n) \quad (4.11)$$

Applying the approximations previously stated, this reduces to the following set of equations for the $\hat{x}$ and $\hat{y}$ components.

$$\frac{\partial v_{ix}}{\partial t} + v_{ix} \frac{\partial v_{ix}}{\partial x} = \frac{e}{m_i} E_x + \Omega_i v_{iy} - \frac{\gamma_i k T_i}{m_i} \frac{1}{n} \frac{\partial n}{\partial x} - \nu_{||i,n} v_{ix} + \nu_{||i,n} u_{nx} \quad (4.12)$$

$$\frac{\partial v_{iy}}{\partial t} + v_{ix} \frac{\partial v_{iy}}{\partial x} = -\Omega_i v_{ix} - \nu_{\perp i,n} v_{iy} \quad (4.13)$$

using $\vec{E} = -\vec{\nabla}\phi$ and substituting (4.10) in (4.12), we arrive at the following equation for the acceleration of ions along the $\hat{x}$ direction as a function of the forcing neutral barium wind u_{nx} .

$$\frac{\partial v_{ix}}{\partial t} + v_{ix} \frac{\partial v_{ix}}{\partial x} = -\frac{m_e}{m_i} \left(\frac{\nu_{e,n}\alpha}{1 + \theta^2\alpha} \right) v_{ix} - \nu_{||i,n} v_{ix} - c_s \frac{1}{n} \frac{\partial n}{\partial x} + \Omega_i v_{iy} + \nu_{||i,n} u_{nx} \quad (4.14)$$

Table 4.1: Model parameters for the first release of CRIT II.

Parameter	Values	Comments
$\nu_{e,n}$	10^3s^{-1}	[Kelley, 1989] page 35
$\nu_{\perp i,n}$	0.5s^{-1}	[Kelley, 1989] page 35
$\nu_{\parallel i,n}$	$\approx 10 \text{s}^{-1}$	Changes with x and t , Fig. 2.5
Ω_e	$7.56 \times 10^6 \text{s}^{-1}$	$B_o = 0.43$ Gauss
Ω_i	$2.57 \times 10^2 \text{s}^{-1}$	for oxygen
α	5.71×10^7	unitless
m_e/m_i	5.4×10^{-4}	unitless
u_n	0 to $1.35 \times 10^4 \text{ m/s}$	large for large $\nu_{\parallel i,n}$

The constant $c_s = \sqrt{k(\gamma_e T_e + \gamma_i T_i)/m_i}$ is the velocity for ion acoustic waves. Equations (4.13), (4.14) and the continuity equation,

$$\frac{\partial n}{\partial t} + \frac{\partial}{\partial x}(n v_{ix}) = 0 \quad (4.15)$$

make up the one dimensional fluid model for stream forces perpendicular to B . The one dimensional model for stream forces parallel to B is recovered from (4.14) and (4.15) by letting $\Omega_i = 0$ and $\alpha/(1 + \theta^2 \alpha) = 1$. The parallel model is simpler than the perpendicular model since the gyro motion of the fluid does not have to be followed.

In Table 4.1 we have grouped parameters for our model appropriate to the first release of CRIT II. By comparing the magnitude of the terms their relative importance can be estimated. We particularly want to estimate the sensitivity of the simulation to the free parameter θ .

$$\left[\nu_{\parallel i,n} + \frac{m_e}{m_i} \left(\frac{\nu_{e,n} \alpha}{1 + \theta^2 \alpha} \right) \right] v_{ix} \approx \left[10 + \frac{0.54}{\theta^2} \right] v_{ix} \quad (4.16)$$

if $\theta = 0.73$ then the $\nu_{\parallel i,n}$ term is on the same order as the $1/\theta^2$ term and larger values for θ reduce the importance of the term. Smaller values of θ increase the drag

on the ion velocities, making them harder to move and thus reducing the density enhancement effects we are looking for.

To simulate equations (4.13), (4.14), and (4.15) we first normalize the time, dimensional, and amplitude scales. We choose to scale the equations by the ion cyclotron frequency, Ω_i .

$$t = \frac{\tau}{\Omega_i} \quad (4.17)$$

Distance will be scaled with a generalized wavelength λ that is determined by the real, physical length of the simulation divided by the number of grid points used.

$$x = \lambda r \quad (4.18)$$

Amplitudes, or energy, is scaled by $\gamma = \Omega\lambda$ to further simplify the equations.

$$\begin{aligned} v &= \gamma V \\ u &= \gamma U \end{aligned} \quad (4.19)$$

The normalized equations for the perpendicular model are given by

$$\frac{\partial n}{\partial \tau} = \frac{\partial}{\partial r}(nV_{ix}) \quad (4.20)$$

$$\frac{\partial V_{ix}}{\partial \tau} = -V_{ix} \frac{\partial V_{ix}}{\partial r} - \left[\frac{\nu_{\parallel i,n}}{\Omega_i} + \frac{m_e}{m_i} \left(\frac{\nu_{e,n}}{\Omega_i \theta^2} \right) \right] V_{ix} - \gamma^{-2} c_s \frac{1}{n} \frac{\partial n}{\partial r} + V_{iy} + \frac{\nu_{\parallel i,n}}{\Omega_i} U_{nx} \quad (4.21)$$

$$\frac{\partial V_{iy}}{\partial \tau} = -V_{ix} \frac{\partial V_{iy}}{\partial r} - V_{ix} - \frac{\nu_{\perp i,n}}{\Omega_i} V_{iy} \quad (4.22)$$

The normalized equations for the motion parallel to $\vec{B}_0$ are given by.

$$\frac{\partial n}{\partial \tau} = \frac{\partial}{\partial r}(nV_{iz}) \quad (4.23)$$

$$\frac{\partial V_{iz}}{\partial \tau} = -V_{iz} \frac{\partial V_{iz}}{\partial r} - \left[\frac{\nu_{||i,n}}{\Omega_i} + \frac{m_e}{m_i} \frac{\nu_{e,n}}{\Omega_i} \right] V_{iz} - \gamma^{-2} c_s \frac{1}{n} \frac{\partial n}{\partial r} + \frac{\nu_{||i,n}}{\Omega_i} U_{nz} \quad (4.24)$$

These coupled nonlinear equations have been solved using the Fourier pseudospectral method for computing the spatial derivatives. The time advance was accomplished by using a second order Runge-Kutta, and de-aliasing was incorporated by grid shifting. The results of the simulation are discussed in the following sections.

4.2.3 Simulation Results

The geometry of the CRIT II experiment was used for the simulation (Fig. A.2); the injection angle was chosen to be 57° and simulation results for distances of the first (1.58 km) and second (3.88 km) releases will be presented. The neutral stream model described in Section 2.2.1 was incorporated into the simulation. The component of the collisional force, $\nu_{||i,n}u$ (parallel to the barium stream), which is along the geomagnetic field is plotted in Fig. 4.4 for the CRIT II first release. Profiles of the parallel force are plotted for times during the passage of the ionization front. The solid lines have been labeled every 0.01 s after the release time and the dashed lines for time midway between solid profiles. The center of the barium beam is directed at the main payload located at the origin along the abscissa. Clearly the neutral stream force builds up from below the main payload along the geomagnetic field, pushing plasma towards it. The peak collisional force reaches the payload at 0.135 seconds after release time and compression of plasma along the field starts 35 ms before. The time evolution of the parallel force at the location of the main payload is presented in Fig. 4.5

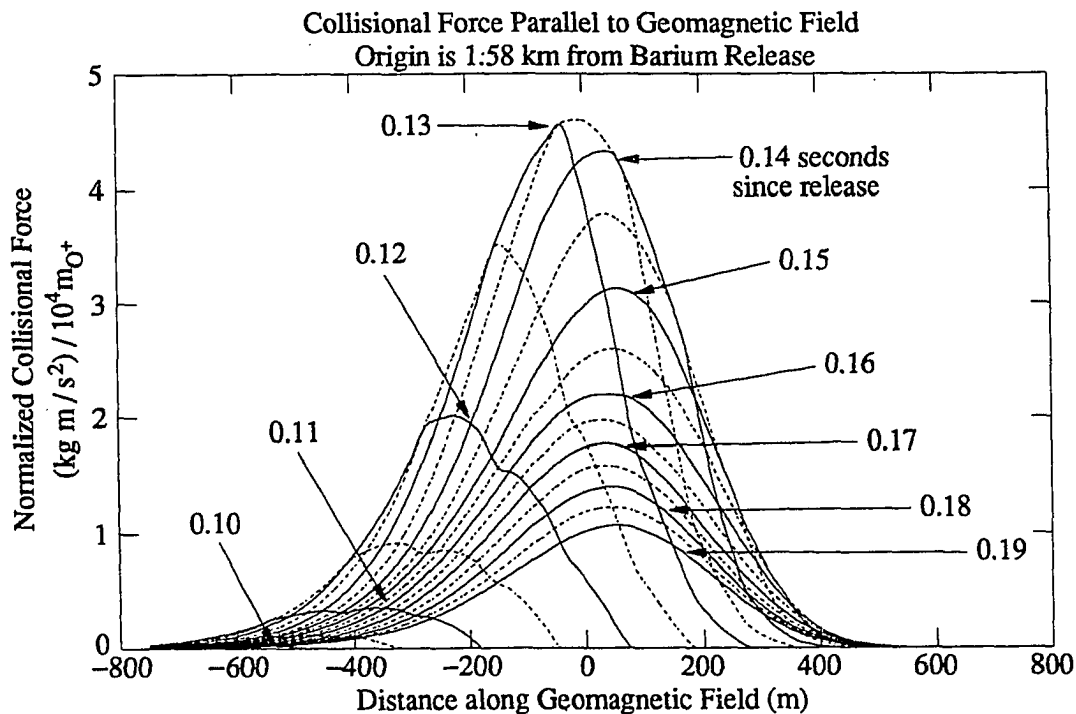


Figure 4.4: The collisional forces on oxygen ions along the geomagnetic field due to the neutral barium stream at times of the CRIT II first release ionization front. Dashed profiles have times between labeled solid profiles. The distance origin is located 1.58 km from the barium release point, and distance is measured along the geomagnetic field.

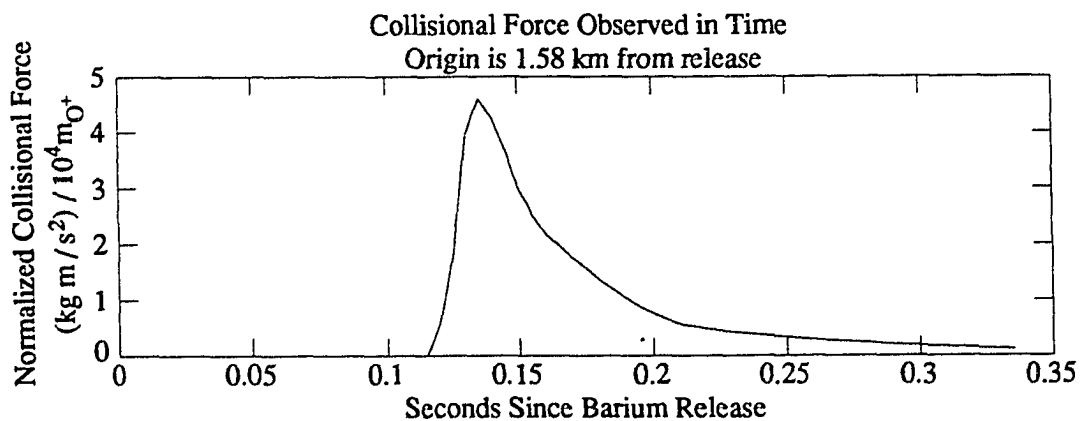


Figure 4.5: The local parallel collisional force on oxygen ions due to the neutral barium stream at times when the CRIT II first release ionization front reached the payload.

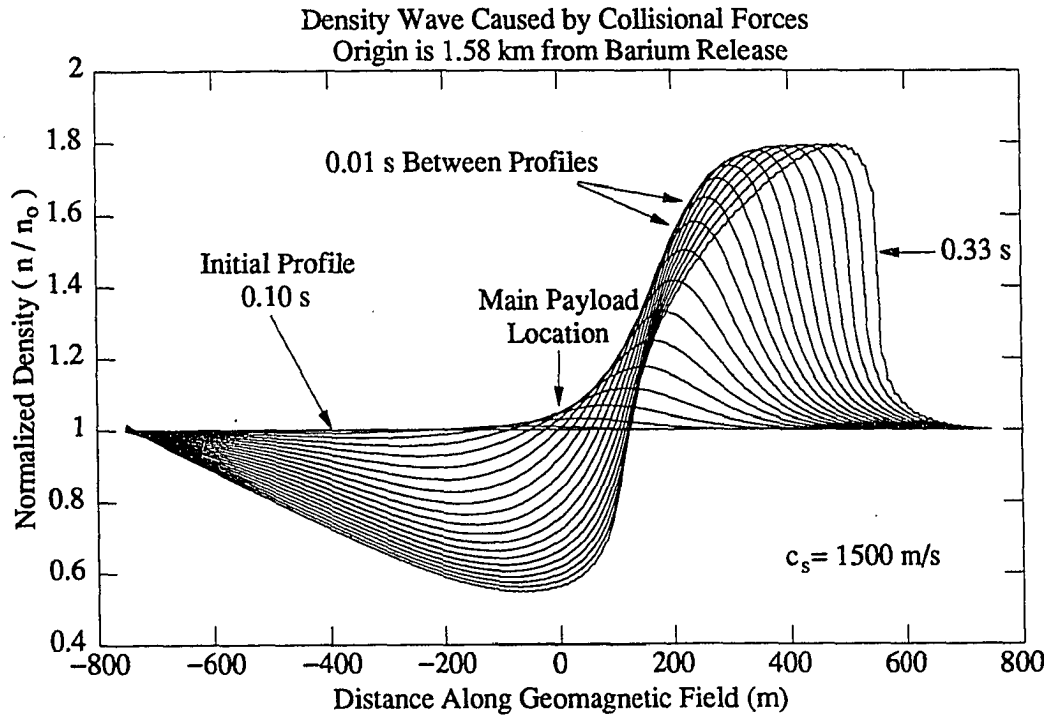


Figure 4.6: Simulated density profiles along the geomagnetic field at times of the CRIT II first release ionization front. The distance origin is located 1.58 km from the barium release point, and distance is measured along the geomagnetic field.

Parallel Model Results

The collisional force pushes the plasma within the barium stream along the geomagnetic field until it is out of the stream's path. This phenomenon can be seen in Fig. 4.6, which presents density profiles from the simulation for the first release. The density profiles in Fig. 4.6 are separated in time by .01 seconds and $c_s = \sqrt{k(T_e + 3T_i)/m_i} = 1500 \text{ m/s}$ is approximately equivalent to $T_e = T_i = 1000\text{K}$ for O^+ ions. The main payload location is at 0 along the abscissa of the graph. This is also where the center of the neutral stream is directed. Comparisons with Fig. 4.4 demonstrate that plasma is moved out of the neutral stream and a 'hole' develops along its path. At the distance of the payload from the first release, the plasma is forced out along the field faster than the ion acoustic velocity ($\approx 2500 \text{ m/s}$).

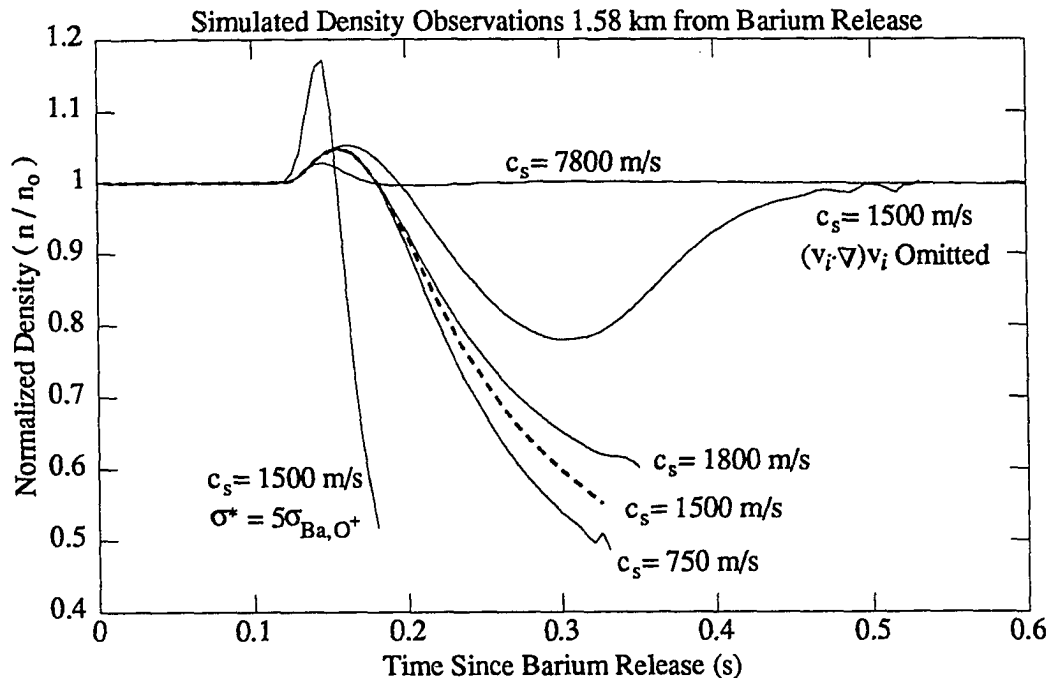


Figure 4.7: Simulated plasma density at the main payload affected by collisional forces along the geomagnetic field for the first release of CRIT II (dashed line). Plots for variations of c_s and σ , the neutral ion cross section, are also made.

This plasma forms a wave which, for these conditions, steepens and breaks after the last plotted profile or at about 0.38 s (Fig. 4.6). This condition is analogous to the atmospheric shock wave formed by a supersonic jet. The turbulent breaking of the wave will eventually lead to a plethora of shorter scale waves and the heating of electrons. The breaking of this large ion acoustic shock wave will occur long after the passage of the ionization front. Electrons heated in the turbulence of this breaking wave way may contribute to the extended, though minor, ionization of the neutral stream observed by the particle detectors (Fig. 3.3) occurring after the ionization front passage.

Simulated density fluctuations at the location of the main payload are presented in Fig. 4.7. Each of the graphs end at about the time the density wave breaks, and the equations cannot be simulated further. First, consider the density history at

the main payload which is displayed with a dashed line. The ion acoustic velocity, $c_s = 1500 \text{ m/s}$ is calculated for the ambient plasma, $T_e = T_i = 1000 \text{ K}$. A very small density enhancement, only about 5% of the background, is predicted to occur at the main payload due to collisional forces. Observations by the plasma frequency probe were more than twice the background density during the ionization front and peaks were over 4 times the background (Fig. 4.1). The depletion observed after the passage of the front by the PFP dropped off much faster than in Fig. 4.7 and was much deeper. The simulation was run with variations in temperature or c_s to estimate their effects. Choosing $c_s = 1800 \text{ m/s}$ represents a three fold increase in electron temperature, and $c_s = 7800 \text{ m/s}$ would be equivalent to an oxygen plasma composed of 10 eV electrons and 1000 K ions. For small variations of c_s , there are minor changes in the observed density enhancements and changes in the depth of the depletion. For $c_s = 7800 \text{ m/s}$ the depletion is much smaller. This can be attributed to the plasma's enhanced ability to diffuse back into the volume swept out by the barium stream. Also, with such a high acoustic velocity, the density wave does not break since the velocity of the expelled plasma is no longer supersonic.

Two other simulations were made using CRIT II first burst parameters with the results shown as well in Fig. 4.7. Because the cross section for momentum exchange ($2.307 \times 10^{-19} \text{ m}^2$) was approximated from atomic radii [Kittel, 1976] using the hard sphere approximation the sensitivity of the observed density enhancement to this parameter was explored. When the cross section was increased by a factor of five, the density enhancement at the payload was larger, but not large enough to explain the observations. For the other run the convective derivative nonlinearity, the $V_{iz} \frac{\partial V_{iz}}{\partial r}$ term of (4.24), was turned off to verify its importance. We conclude that the nonlinear steepening this term introduces is important and cannot be neglected in the formation of the hole.

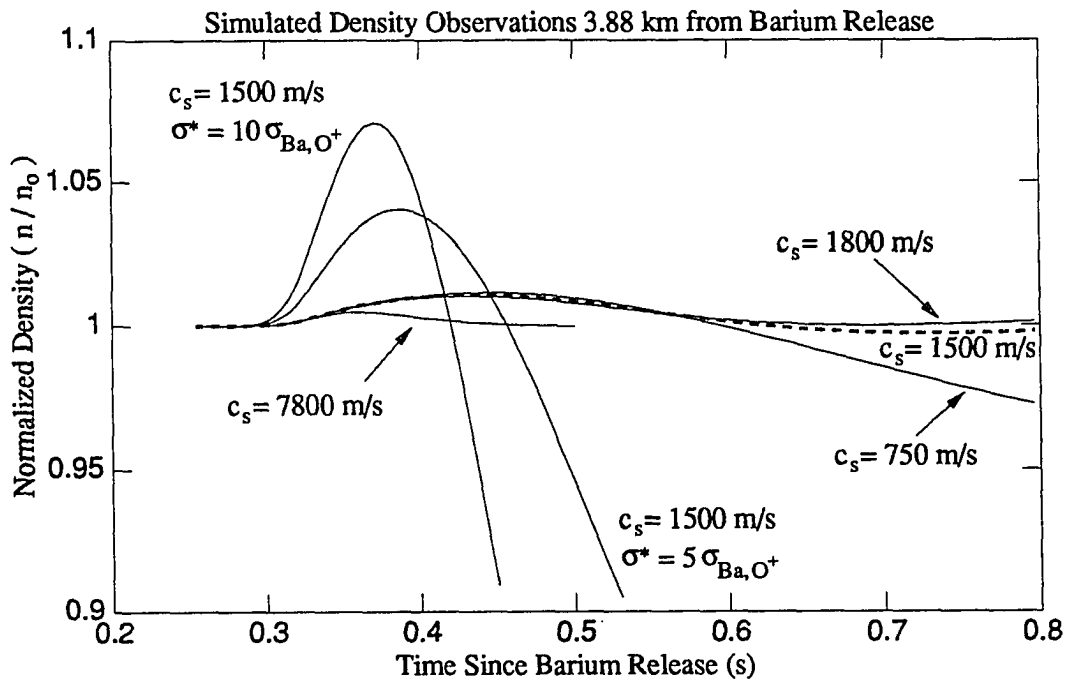


Figure 4.8: Simulated plasma density at the main payload affected by collisional forces along the geomagnetic field for the second release of CRIT II (dashed line). Plots for variations of c_s and σ , the neutral ion cross section, are also made.

Simulated density fluctuations at the location of the main payload for the second release of CRIT II are presented in Fig. 4.7. None of the various simulations produced breaking waves along the geomagnetic field since the collisional force is reduced by more than an order of magnitude. Instead a density ripple traveling along the field was produced. For $c_s = 7800$ m/s this wave reached the boundary of the simulation at 0.5 s, and that simulation had to be terminated. The cross section for momentum exchange of neutral barium with oxygen ions was increased by a factors of 5 and 10, but none of the runs produced density enhancements or depletions comparable to the observations.

Perpendicular Model Results

Fig. 4.9 presents the density perturbations driven by collisional forces of the neutral

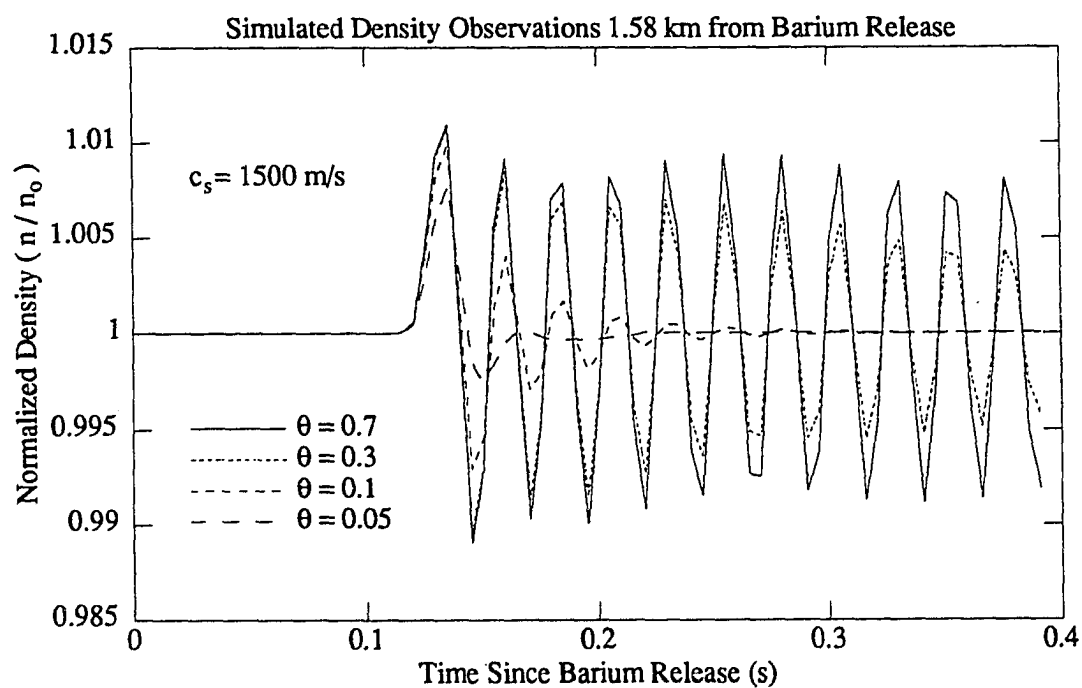


Figure 4.9: Simulated plasma density at the main payload affected by collisional forces perpendicular to the geomagnetic field for the second release of CRIT II. Plots for variations of the free parameter θ are made.

stream perpendicular to the geomagnetic field. Several runs of the model were made with various values for the free parameter θ . This parameter affects the damping of the density oscillations. Small values increasing the damping and values larger than 0.7 have almost no effect on the model. The oscillations in the simulated density history are at the oxygen cyclotron frequency. Oscillations at the oxygen cyclotron frequency were observed for the lowest energy ions by the particle detector looking nearly perpendicular to $\vec{B}_0$ (Fig. 3.4). This simulation provides a mechanism to explain these oscillations. Since the observed oscillations continue more than 0.5 s in the data, values of θ greater than 0.5 in the model reproduce the observations more accurately. Lastly, no large scale density enhancements were produced by the simulation. Collisional forces perpendicular to the geomagnetic field cannot explain observations of the PFP during the ionization front.

The results of both the parallel and perpendicular simulations have strongly suggested the density enhancements observed at the main payload of CRIT II could not be the results of ion-neutral collisional forces. The parallel model has shown plasma is pushed in required magnitudes as described in Fig. 4.3, but this snow plowed plasma enhancement is not significant at the location of the main payload. The density enhancements simulated with the parallel model are about 5% of what is observed by the plasma frequency probe. The density enhancements outside of the neutral stream along the field become the largest, reaching 80% of the background. The perpendicular model can be criticized as an incomplete representation of the physics since polarization of the ionosphere within the neutral stream cannot be simulated in a one dimensional model. We expect that inclusion of this effect would change the results, but not such that more plasma could be pushed across the magnetic field than what was demonstrated in the parallel model.

4.3 Plasma Deposition

4.3.1 Dynamical Effects on the Observations

We conclude from the previous section that the bulk of the density enhancements observed by the plasma frequency probe must result from the rapid ionization of barium. The next logical question is how much barium was ionized? The amount of barium deposited, or the ionization rate, cannot be simply calculated from the increase above the background level. There are several processes occurring which effectively change observed density relative to that deposited. The first is diffusion in a multi-component plasma, and the second results from the motion of barium ions relative to the region of deposition. These dynamical effects will be discussed before we begin a comparison of ionizing mechanisms.

Consider what will happen if a heavy barium plasma, about twice the background density, is quickly deposited in a magnetized oxygen plasma. This problem was studied by *Schunk and Szuszczewicz* [1991] in one dimension along the magnetic field, in two dimensions by *Ma and Schunk* [1990], and in three dimensions by *Ma and Schunk* [1991]. These simulations show that the oxygen ions within the barium ion cloud rapidly diffuse along the magnetic field. As the electrons associated with the newly produced barium ions try to diffuse along the magnetic field they pull the more mobile ions, oxygen, with them to maintain charge neutrality. This multi-species diffusion occurs more rapidly for the light ions, oxygen ions in our case, than ambipolar diffusion would for a single ion plasma, much as in the polar wind where hydrogen is accelerated to high velocity. The barium cloud is left intact on short time scales $\tau \approx \Omega_{Ba}^{-1}$. Quantitatively the report of these simulations and these experimental measurements cannot be readily compared, but qualitatively the simulations indicate oxygen ions will be pulled out rapidly enough to explain

aspects of the CRIT II observations.

The size of the barium ion cloud is comparable to the barium gyro radius for the CRIT II releases. This means the gyro motion of barium will also affect localized density measurements. If the cloud is deposited rapidly, $\tau < \Omega_{Ba}^{-1} = 200$ ms, then the gyro motion will completely remove the barium ions from the initial area of deposition. By the diffusion mechanism described above, the barium cloud will also act to reduce the local O^+ density during the period of deposition and this decrease will track the motion of the cloud with some delay. The neutral barium is injected at 57° to the geomagnetic field, so barium ions produced by collisional processes will also move from their region of creation along the geomagnetic field. Finally, the observed barium density will be affected by barium ions produced earlier, or up stream, moving into the region of detection. These barium ions will come from a region extending back along the neutral stream approximately $1/8$ to $1/4$ of a barium gyro radius. These effects qualitatively explain the data of Fig. 4.1 and Fig. 4.2.

Barium ions are deposited locally within about 25 ms or over $1/8$ of a barium gyro radius corresponding to the high velocity of the barium neutrals from which they came. Also during this time, barium ions created upstream can move into the measurement volume. Oxygen is forced out along $\vec{B}_0$ throughout this time and the following 25 ms. After about 25 ms, the newly created barium ions are moving more than $1/4$ of the way around their orbits and have slid along the geomagnetic field until locally (where they were created) they appear to rapidly disappear. Since the oxygen has been displaced a plasma depletion is observed behind the barium cloud. In addition there will be depleting effects by neutral wind forces along the magnetic field as shown in Fig. 4.7 and Fig. 4.8. Depletions resulting from the motion of a heavy ion cloud across the magnetic field were observed in the three-

dimensional simulations of *Ma and Schunk* [1991]. None of the simulations include the collisional forces of the neutral stream with the ambient plasma which was shown by the simulation in Section 4.2.1 to have significant effects on the depletion following the ionization front.

4.3.2 Collisional Ionization Processes

The two collisional ionization processes most relevant to the CRIT experiments, charge exchange and charge stripping, have been reviewed in Section 1.4. Both of these process can produce barium ions, but they have different effects on the local oxygen plasma. The charge exchange process trades one barium ion for one oxygen ion and the net plasma density remains constant. Charge stripping does not use up ambient ions, and therefore can lead to density enhancements. Barium ions produced by either process, n_{Ba^+} , can be approximately described by a non-linear, first order, differential equation, *i.e.*

$$\frac{dn_{\text{Ba}^+}}{dt} = (n_{\text{Ba stream}} - n_{\text{Ba}^+})(n_r - n_{\text{Ba}^+})v_{\text{Ba stream}}\sigma_r \quad (4.25)$$

where n_r is the initial density of the reacting species (oxygen or oxygen ions), $v_{\text{Ba stream}}$ is the velocity of the barium stream and σ_r is the cross section for the process. Equation (4.25) is the continuity equation neglecting the divergence term and including a production term. For this model we have assumed barium ions do not move and reverse reactions, those which convert barium ions to barium neutrals, can be neglected. The cross sections, σ_r , can be found in Table 1.3, the neutral stream model in Section 2.2.1, and the neutral oxygen density in Table 3.1. Equation (4.25) has been numerically integrated using a fourth-order Runge-Kutta method.

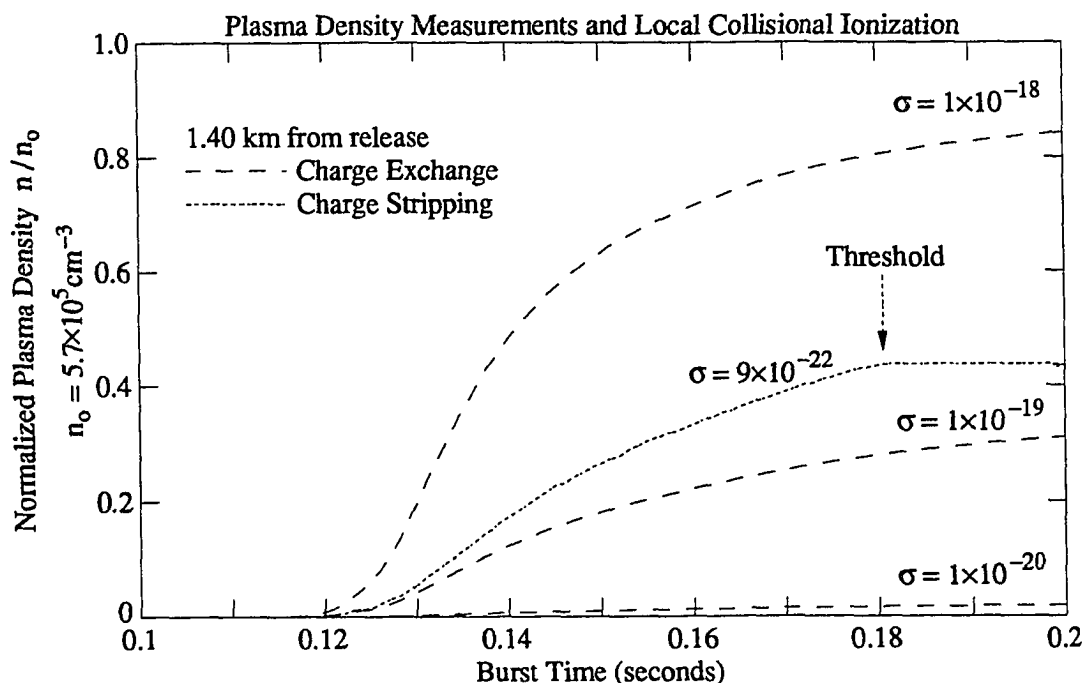


Figure 4.10: A comparison between CRIT II first release density measurements and production of barium ions by collisional process.

In Fig. 4.10 we compare PFP measurements for the first release which have been normalized to the background density with the predicted ionization of barium by collisional processes. Diffusion and ion motion effects are not included in the numerical integration of (4.25), therefore, the density depletion occurring after the ionization front is not reproduced. Clearly, without the addition of barium ions produced up stream, the rise time is much too slow, and the over all production is too little to explain the observations. This is because both of these processes require significant neutral barium density to become effective. If the larger cross section for charge exchange $\sigma = 10^{-18} \text{ m}^2$ is accurate (which we doubt) then about 80% of the ambient oxygen ions will be consumed by charge exchange. This process could help explain the observed depletion.

The effect of barium ions produced up stream that move with the neutral barium velocity can be studied in approximation with a one dimensional continuity

equation,

$$\frac{dn_{\text{Ba}^+}}{dt} + \frac{d}{dt}(n_{\text{Ba}^+} v_{\text{Ba stream}}) = n_{\text{Ba stream}} n_r v_{\text{Ba stream}} \sigma_r \quad (4.26)$$

where n_{Ba^+} , $v_{\text{Ba stream}}$, $n_{\text{Ba stream}}$, n_r , and σ_r are defined the same way they are for (4.25). The production term has been simplified so that extinction of n_r is not accounted for, therefore the production of barium ions will be over estimated. This is only a significant concern for oxygen ions when assuming the highest estimate of the charge exchange cross section ($\sigma = 10^{-18}$) where 80% of the ambient ions are consumed (Fig. 4.10). The simulation spatial grid starts 300 m up stream from the observation point. Barium ions further up stream than this approximate distance are not allowed to contribute to the local density since their gyro motion will have turned them out of the neutral stream. The depletion of barium ions due to the finite size of the barium cloud combined with their motion (along $\vec{B}_0$ and gyro motion) is not accounted for in (4.26), therefore, the hole will not be reproduced.

The predicted barium density for both charge stripping and charge exchange is plotted in Fig. 4.11 along with the measured plasma density for the first release. For this figure we are concerned with the first half of the simulation, or up to approximately 0.14 seconds. After this time loss mechanisms for barium ions, gyro or field aligned motion, will become significant and are not contained in the model. Qualitatively, the collisional processes deposit barium over the same time scale, $1/8$ of a gyro orbit, as is observed in the plasma density data. The rise time is very similar. Uncertainties in the charge exchange cross section make conclusions problematical about the production of barium ionization. Either all of the barium ions were created by this process, or none, depending on value of the cross section assumed. We tend to believe the lower estimates for the charge exchange cross section, but this point cannot be decided until measurements are made. We note in passing that a significant number of barium ions are created by charge strip-

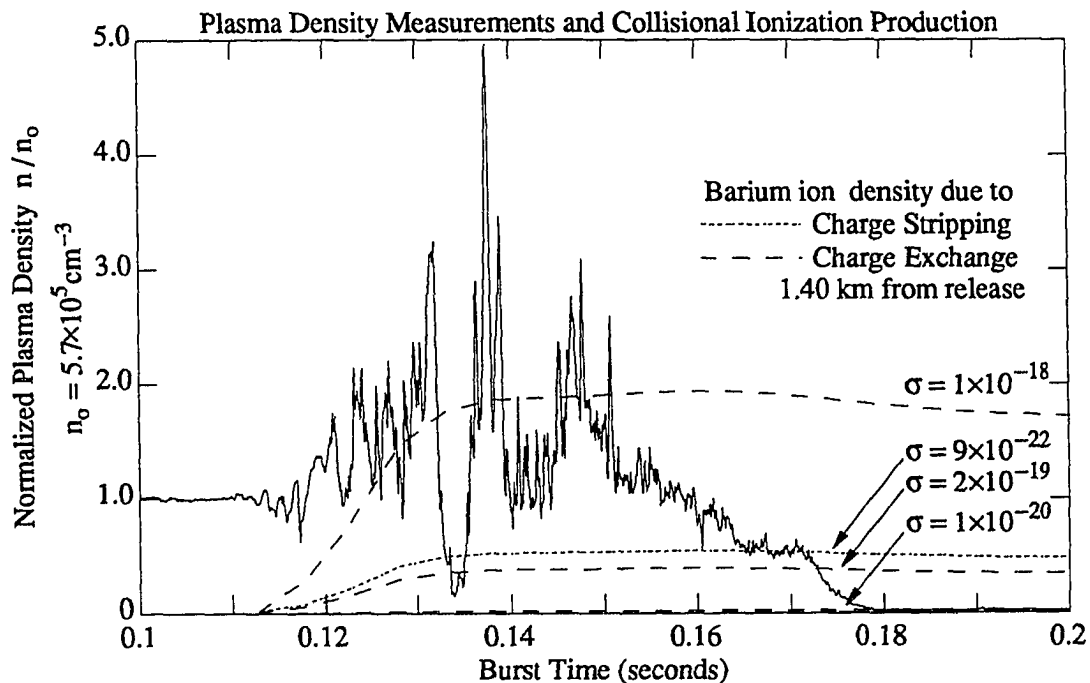


Figure 4.11: A comparison between CRIT II second release density measurements and production of barium ions by collisional process including motion of barium ions.

ping, and the CIV process may need this as a seeding mechanism. This is one possible explanation for the high variability of CIV experiments in space since the neutral oxygen density varies by two orders of magnitude at the altitudes of the experiments.

An estimate of the total plasma density can be made for the charge exchange process by adding the barium ion density of Fig. 4.11 to the oxygen density implied by Fig. 4.10. Since charge stripping does not use up oxygen ions, we may simply add the deposited barium ions of Fig. 4.11 to the background plasma density. Both processes overestimate the plasma density since loss of oxygen by diffusion along the geomagnetic field is ignored. The density hole is not recreated since gyro and field aligned motion of barium ions are also ignored. Comparison plots with measured plasma density are presented for both releases in Fig. 4.12 and Fig. 4.13. We

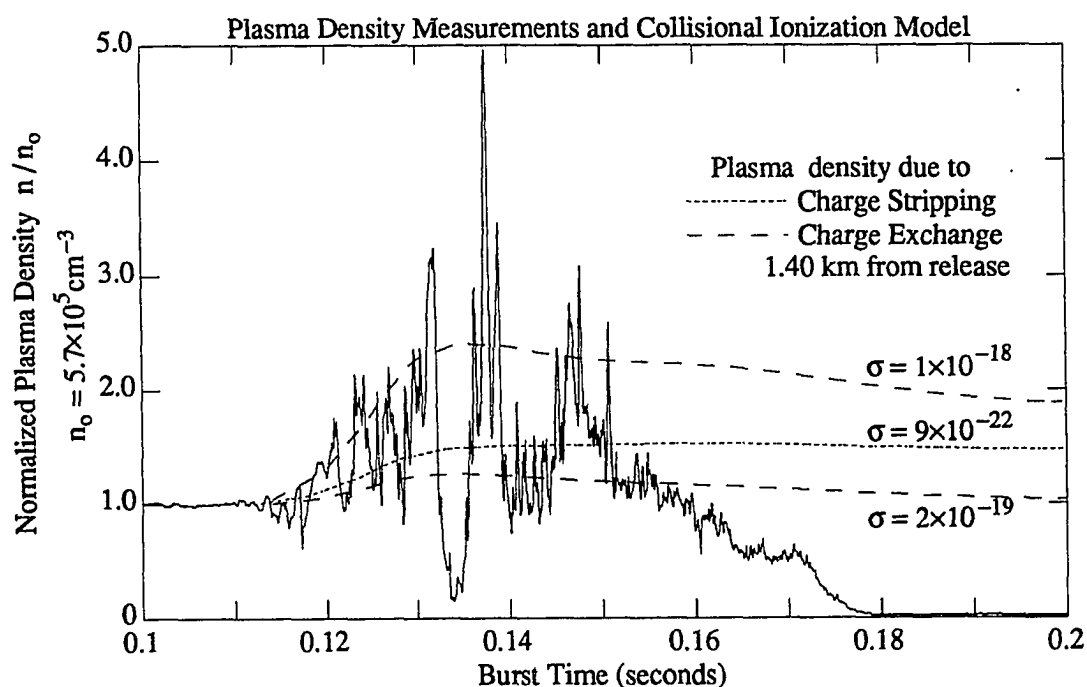


Figure 4.12: A comparison between CRIT II first release density measurements and calculated plasma density increases affected by collisional ionization of barium.

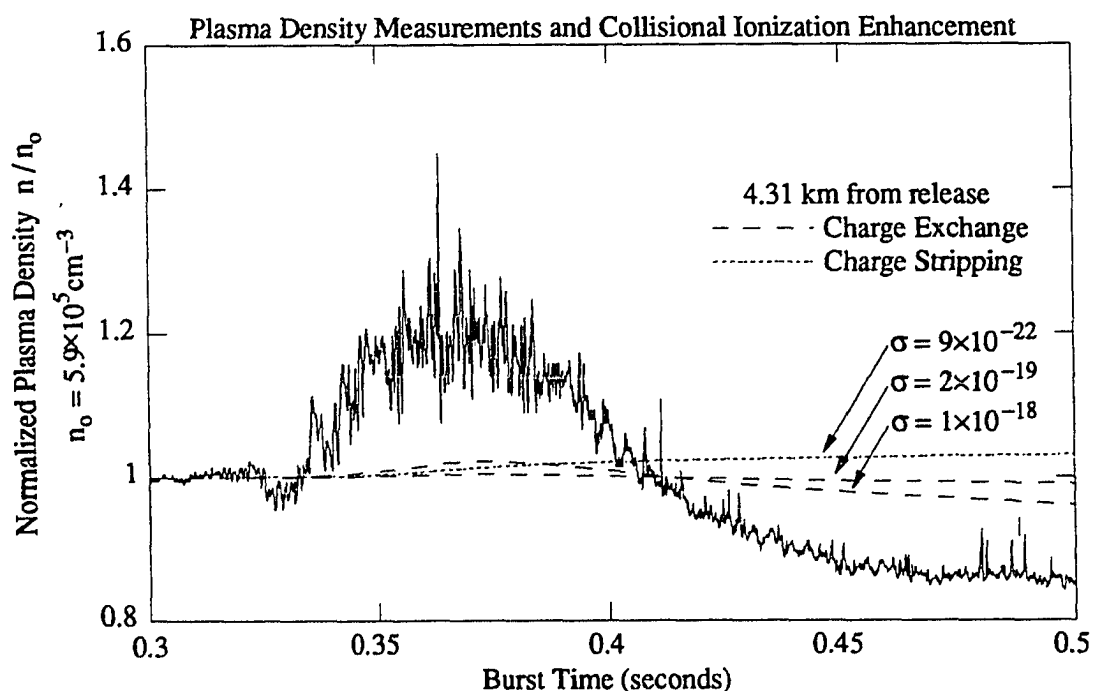


Figure 4.13: A comparison between CRIT II second release density measurements and calculated plasma density increases affected by collisional ionization of barium.

point out that the time span of Fig. 4.13 is twice that of Fig. 4.12 so the loss of barium, due to transport, is more apparent at the latter times in this figure. It is unsurprising, given the previous figures, that with the largest estimate of the charge exchange cross section the model matches the observed first release data, but it is very significant that it does not match the second release. This clearly indicates, for the second release, that some other process, such as CIV, contributed to the ionization of barium. Both charge stripping and charge exchange occur together so a combination of stripping and charge exchange cross section of about 2×10^{-19} could also reasonably explain the average first release data.

4.3.3 CIV Calculations

The CIV process does produce density enhancements and can much more easily explain the organization of plasma within the ionization front. In Section 1.2.1 we discussed the basics of conservation of energy and momentum that are required of the CIV mechanism. If the ionospheric mass entering into a CIV reaction is given by that in the volume swept out by two Alfvén waves traveling away from the interaction volume along the magnetic field (Fig. 1.3), then the ionization rate can be calculated from the conservation principals for a CIV reaction. The expression one arrives at for the ionization rate, $\dot{n}$, depends on two unknown constants, β and η and was given earlier as (1.9), but we repeat it here, *i.e.*

$$\dot{n} = 2\beta \frac{m_i}{m_n} \frac{n_o V_a}{l_{||}} \left(\eta \frac{v_n^2}{V_c^2} - 1 \right) \quad (4.27)$$

Recall that $\eta < 1$ is the efficiency with which the CIV mechanism uses the initial kinetic energy of neutral barium for ionization, $\beta > 1$ is a factor used to account for the mass contributed by plasma proximate to the flux tube along which the Alfvén

waves travel, v_n is the velocity of the neutral stream, the quantity $m_n n_o$ is the ambient mass density, $l_{||}$ is the extent of the neutral stream along the geomagnetic field, m_n is the mass of a neutral particle, V_a is the Alfvén speed, and V_c is the critical velocity for the neutral species. *Haerendel* [1982] took $\beta = 2$, which we will also use, realizing we have just swept a lot of physics under the rug in doing so. His reasoning was that the effective volume in an incompressible media for momentum transfer is twice the volume of injection. Using the parameters in yTable 3.1, (4.27) can be numerically integrated for various values of η to obtain the barium density as a function of time. The minimum value of η for which a CIV will occur is found by setting $\eta v_1^2/V_c^2 - 1 = 0$ and using the highest velocity for the neutral barium, 13.5 km/s, for the threshold. This gives a minimum η of 0.04; in other words, if η is less than 0.04 there is not enough energy in the neutral stream for a CIV reaction.

A comparison of first burst CRIT II density measurements and plasma production by CIV is presented in Fig. 4.14. The average efficiency required to make (4.27) match the data is relatively low ($0.05 < \eta < 0.06$) when compared with theoretical estimates of η , but several characteristics of the CIV process qualitatively agree with the measurements. First is the relatively explosive ionization occurring with the arrival of the fastest barium, and second is the shut off after a brief period $\tau \approx 1/8\Omega_{Ba}$. The shut off ($\dot{n} = 0$) occurs at the time when the slope of the model density curves go to zero. Since diffusion and gyro motion effects are not included in the numerical integration of (4.27) the curves do not reproduce the density depletion. The fastest neutral barium arrives in sufficient density, $n_{Ba} \approx 10^8 \text{ cm}^{-3}$ that the observed ionization can easily be generated. The sudden shutoff occurs when the barium velocity falls below the threshold velocity for CIV (1.2). These deposition characteristics are precisely those needed to explain the qualitative nature of the density measurements.

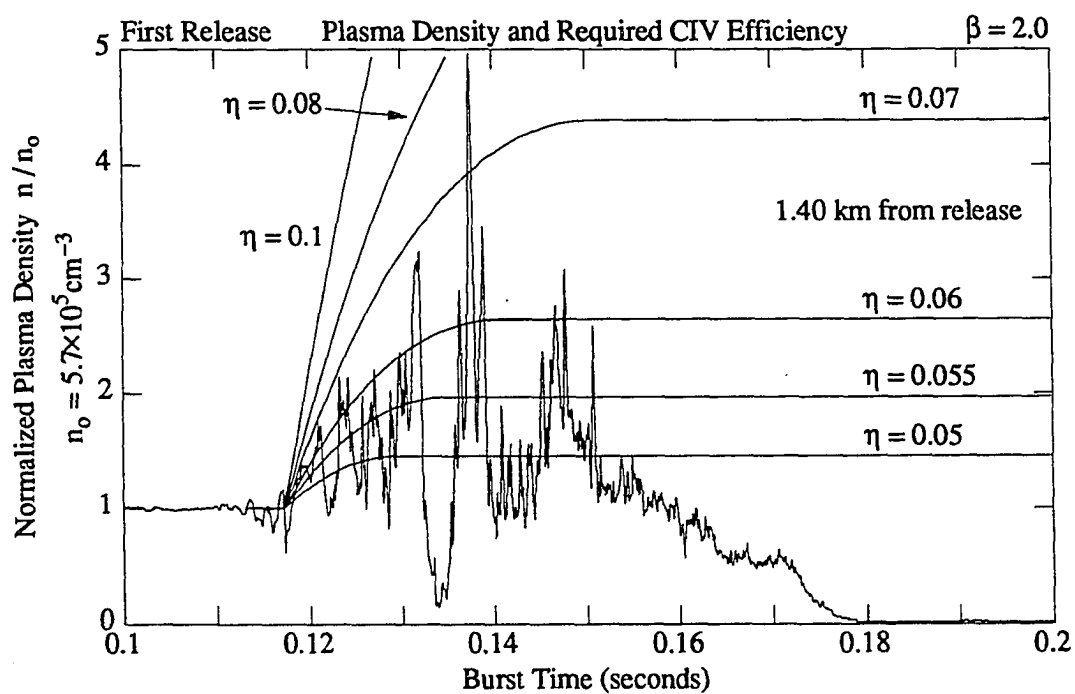


Figure 4.14: A comparison between CRIT II first release density measurements and production of barium ions by a mass loading limited CIV process occurring with efficiency η .

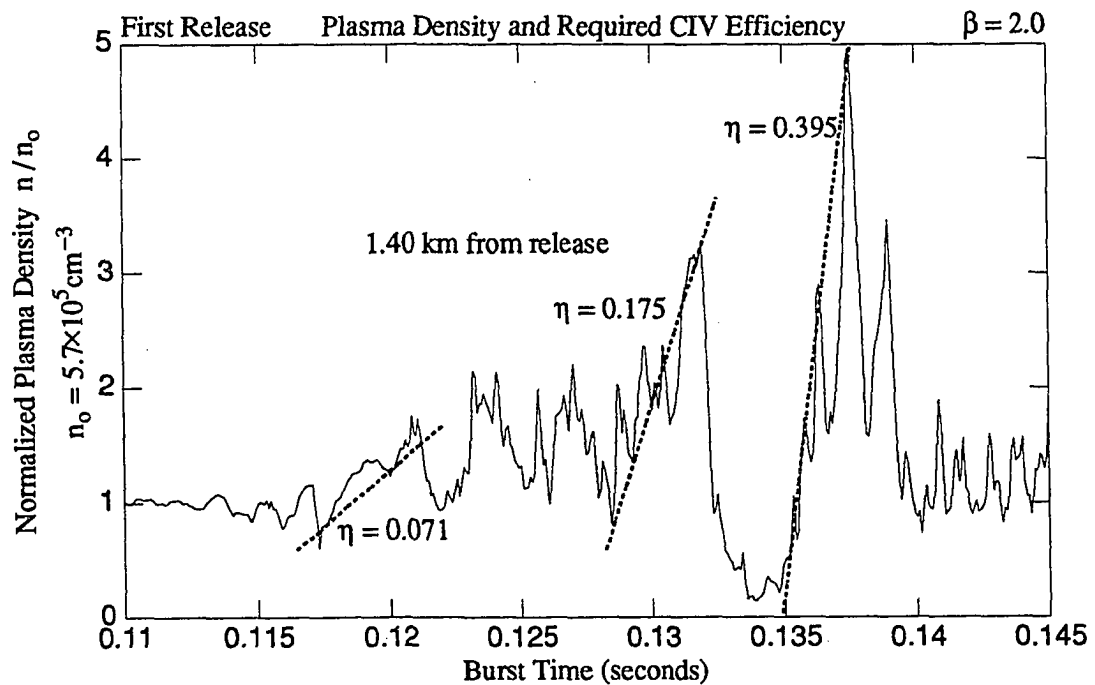


Figure 4.15: A comparison between the required efficiency of a mass loading limited CIV process and regions of density enhancement within the CRIT II first release ionization front.

The spiky features of the data are not explained by this average model. Spatially these features are only 15 to 20 m wide, if as we assume they are moving with the neutral velocity. They could result from an instability between the newly produced ions and the background plasma or be fundamental to the production process itself. In this former scenario, electric fields of an instability bunch ions into a repeating structure with about a 30 m wavelength. We note that bunching of the plasma by an instability cannot explain the density enhancements alone since integration of the data shows a net deposit of plasma within the period of this structure. Another possible explanation is that the CIV process is actually occurring in many short flashes. Each of these spikes would be the result of a CIV front. The efficiencies needed to produce plasma at the required rates are shown in Fig. 4.15. The required values of η are not unreasonable considering proposed

theoretical values which range from 0.25 to 0.67 [Galeev and Chabibrachmanov, 1983; Raadu, 1978; McBride *et al.*, 1972]. The CIV process requires field aligned currents to maintain the required momentum balance. If the background ionosphere cannot supply the required current some process might occur to shut off the current. This would be in contrast to just limiting the field aligned current at some maximum value. Once the current is shut off CIV cannot be sustained and rapid diffusion of oxygen brings the local plasma density down. Such a field aligned current would naturally be carried by electrons and the hypothesized mechanism which shuts down the current would be triggered by the electron velocity approaching some threshold. If the current is shut down, with the motion of electrons stopped, then the process which cut off the flow of electrons would cease to exist. The background plasma would once again be free to supply field aligned currents to a CIV process. This process would be repeated as long as the neutral stream velocity was above the required threshold for CIV.

A comparison of the CIV process for different values of η and the PFP measurements during the second release of CRIT II are presented in Fig. 4.16. Once again we see the qualitative agreement between the shape of the density curves for the CIV process and the actual measurements. The CIV process deposits barium and also shuts off rapidly enough to produce a localized cloud of barium ions that share nearly the same phase along their gyro orbits. The time span of Fig. 4.16 is twice that of Fig. 4.14 so spiky features appear more compressed. These features during the second release exist, but do not dominate as during the first release. The average efficiency of the second release is much reduced from the first release. If we assume the overall η is dropping linearly with distance from the release point (for lack of a better model) then the CIV process will only occur within ≈ 6 km from the release! This is an extremely significant result since the majority of CIV

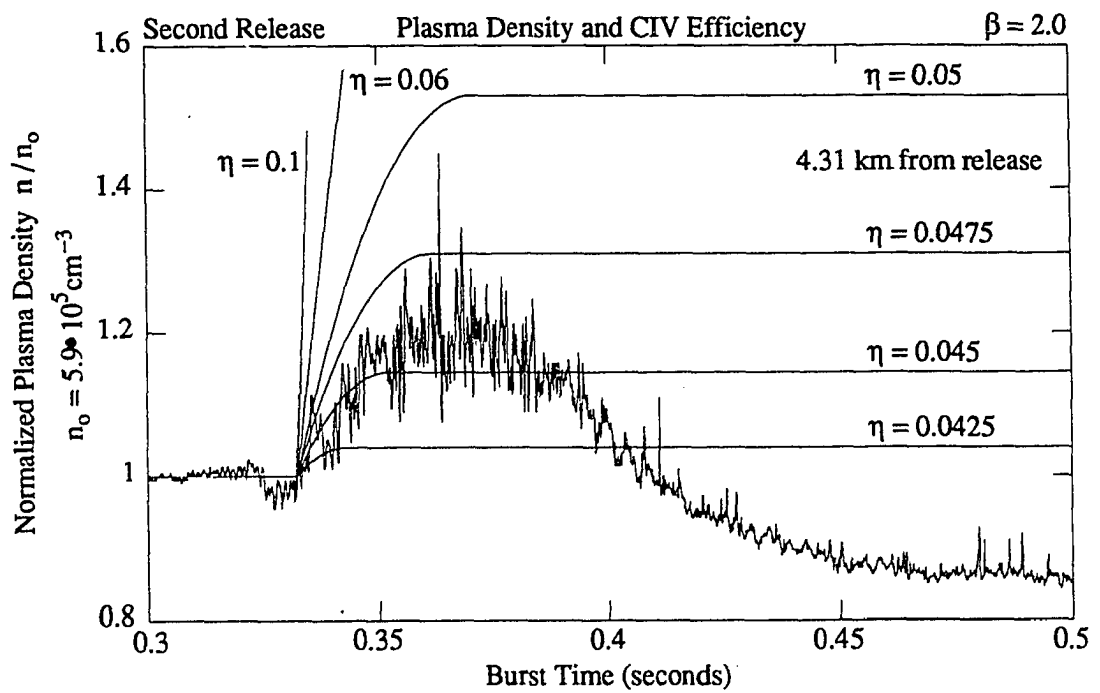


Figure 4.16: A comparison between CRIT II second release density measurements and production of barium ions by a mass loading limited CIV process occurring with efficiency η .

experiments are analyzed from the ground with optical instrumentation. These instruments are limited in their resolution such that ionization occurring within this small distance is essentially undetectable. To summarize, we would not expect the ground based measurements to detect the enhanced ionization observed by the *in situ* instrumentation during the CRIT II experiment. More important, the density enhancements measured *in situ* appear to result from a CIV process.

4.4 Summary

In this chapter we have concentrated on the plasma density measurements during the passage of the ionization front. Broadly, two features are observed in both of the CRIT II releases. The first feature is a density enhancement accompanying the fastest neutral barium (13.5 to about 10.0 km/s). The second feature is a plasma density depletion which occurs after the enhancement. The density enhancement is observed over a time of about $\frac{1}{5}\Omega_{Ba^+}^{-1}$ which implies that its width along the neutral stream is on the order of a barium gyro radius. This enhancement is of central interest since one of the objectives of the CRIT experiments was to create barium ions through a CIV reaction. We have therefore focused on processes which might produce a density enhancement.

Three processes have been examined as candidate explanations for the observed density data structure. These methods are 1) collisional forcing by the neutral stream, 2) collisional ionization processes, and 3) the critical ionization velocity (CIV) phenomena. Through a process of elimination we converged on CIV as the only consistent explanation for the observed density enhancement during the second release. The first method of producing density structures was through collisional forces of the dense neutral barium stream acting on the ambient ionosphere. At the

location of the payloads, the collisional forces were too small to push magnetized plasma across the neutral stream, and the upper limit on the electric fields observed within the density enhancement precluded any self polarization field that would let the plasma $\vec{E} \times \vec{B}_0$ drift across the magnetic field at the required velocity (see Chapter 3). From this we conclude the the density enhancement is created around (within about 300 m) the main payload. The other possibility is that plasma was being snowplowed by the neutral stream along the geomagnetic field. We found that within the neutral barium stream the density enhancements produced by this effect were much too small to explain the data even if we varied the relevant parameters to increase the effectiveness of the process. We did find that this snowplow effect would contribute to a density depletion at times in good agreement with the data and propose it as a contributing factor to its development.

The second candidate for producing a density enhancement, collisional ionization, ionizes part of the neutral barium stream either through charge exchange with oxygen ions or charge stripping against oxygen neutrals. There are several factors which make these calculation problematical, and primary among them is the uncertainty of the charge exchange cross section. The other factors are the motion of the barium ions after they have been produced, and the reaction of the background plasma to the newly produced barium ions. We have endeavored, with an approximate model for the motion of newly produced barium ions, to simulate the density observations with these ionization processes. Depending on the value used for the charge exchange cross section, the first release density enhancement can be completely explained without resorting to CIV. This is not the case for the second release where neither charge exchange nor charge stripping can explain the data and CIV is the only explanation. Finally, we note that a rather low value for η (efficiency of the CIV process) is needed to explain the data. The avalanche like

ionization that would produce barium ions in such quantities that the CIV process would be unmistakable conclusion did not occur.

Chapter 5

Magnetic and Electric Fields in the Ionization Front

Events occurring within the ionization front are the least understood phenomena of the CRIT data set. Yet, these events are a key to understanding if a CIV reaction was initiated during the releases. In this chapter we will present the reader with a portfolio of measurements and related calculations from the CRIT II ionization front. No consistent, all-embracing theory exists to guide the presentation of the data. Instead, we will examine aspects of the data that are possible signatures of CIV, that are related to the various current theories for the CIV microprocess, and that show the data set is internally consistent. The focus will be on the magnetic and electric field data within the ionization front from the first release of CRIT II. In particular, we will focus on the shorter time and spatial variations of the data since we have already discussed the larger scale phenomena in Chapter 3. Intense currents are implied by the magnetic field oscillations and we will explore their relationship with the electric fields and with the plasma density fluctuations that are at the same scale. We note with some frustration that the density, magnetic, and

Beam Coordinates

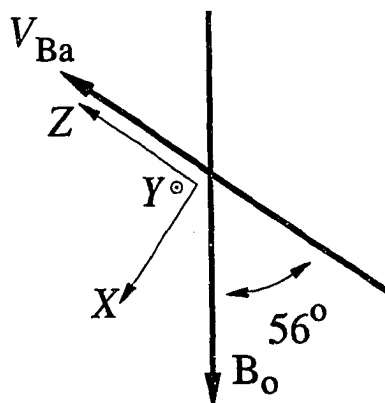


Figure 5.1: Beam coordinate system for first release, CRIT II main payload.

electric field perturbations within the heart of the neutral stream are all strongly correlated, but the exact relationship is not clear. Many of the current CIV theories predict that lower hybrid waves are the mechanism for heating electrons to ionizing energies. Since waves are thought to be excited through a modified two stream instability, we explore the commonly used dispersion relation and compare it to our data, concluding that the observed waves are not fully described by it.

5.1 Magnetic Fields

An overview of the magnetic field measurements has been presented in Chapter 3. In particular, we assumed a model current system that moved with the barium ions and related the magnetic field observations to our model. Briefly, in the beam coordinate system (Fig. 5.1), the δB_x fluctuations are associated with field aligned currents and were observed during the ionization front and each time a magnetized swarm of phase bunched barium ions passed the payload. The field aligned currents are presumed to close above and below the release height through the polarization current within an Alfvén wave propagating along $\vec{B}_0$. The δB_y fluctuations are

associated with currents across the barium stream and are important because they can slow or accelerate the barium ions through a $\vec{j}_y \times \vec{B}_0$ force. This current was only observed during the front and was not observed in the resulting barium swarms. The field aligned currents close two current systems, the cross field currents just discussed (Fig. 3.18) and barium stream aligned currents (Fig. 3.19) which are not detectable by the magnetometer. These barium stream aligned currents result from the motion of newly created barium ions which, on the time scale of the ionization front, can be treated as unmagnetized.

In this chapter, we focus on the magnetic field measurements within the ionization front and the current system observed within it. We will concentrate on a 0.1 s period for the first release and a 0.2 s period for the second release.

5.1.1 Overview

Magnetometer data have been transformed to a beam coordinate system (Fig. 5.1) and are presented in Fig. 5.2 for the first release and in Fig. 5.3 for the second release. In the longer time scale magnetic field data of Chapter 3, we assumed that the fluctuations result from a current system moving parallel to the neutral stream. Under this assumption we showed that no fluctuations are expected in the δB_z channel as long as the $\hat{z}$ -axis of the magnetometer is aligned with the direction of motion of the current system. Examination of the data in Fig. 5.2 and Fig. 5.3 shows that this is indeed the case in beam aligned coordinates, particularly for the second release where the $\hat{z}$ -axis and the beam were quite close to parallel. For the high frequency fluctuations which are the subject of this chapter, the same observation holds in Fig. 5.2, that is, the δB_z variations are smaller than δB_x and δB_y . There are almost no high frequency variations during the second release, but this is well explained by the poor frequency response of the magnetometer and the

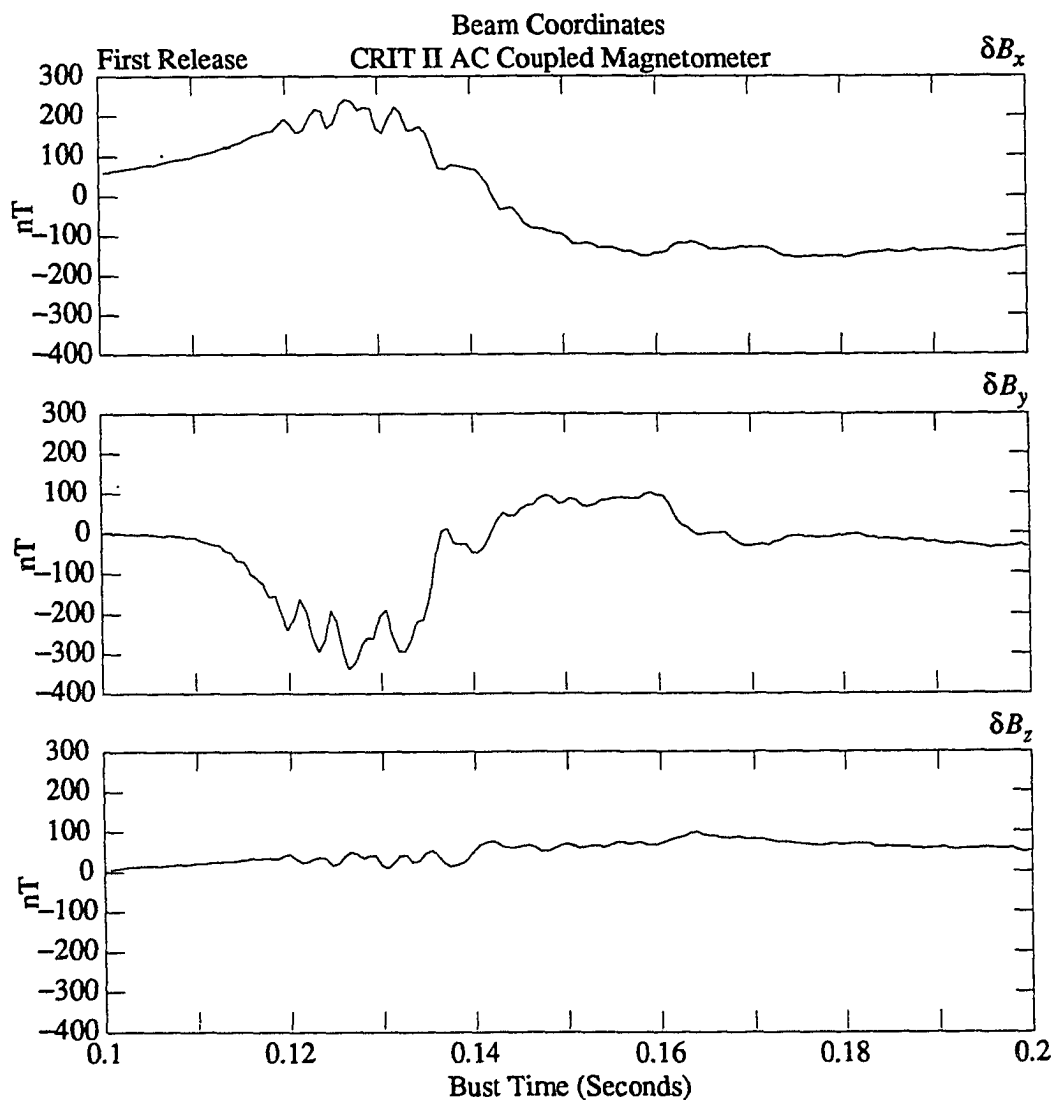


Figure 5.2: CRIT II first release ac-coupled magnetometer data rotated into a neutral barium stream aligned coordinate system.

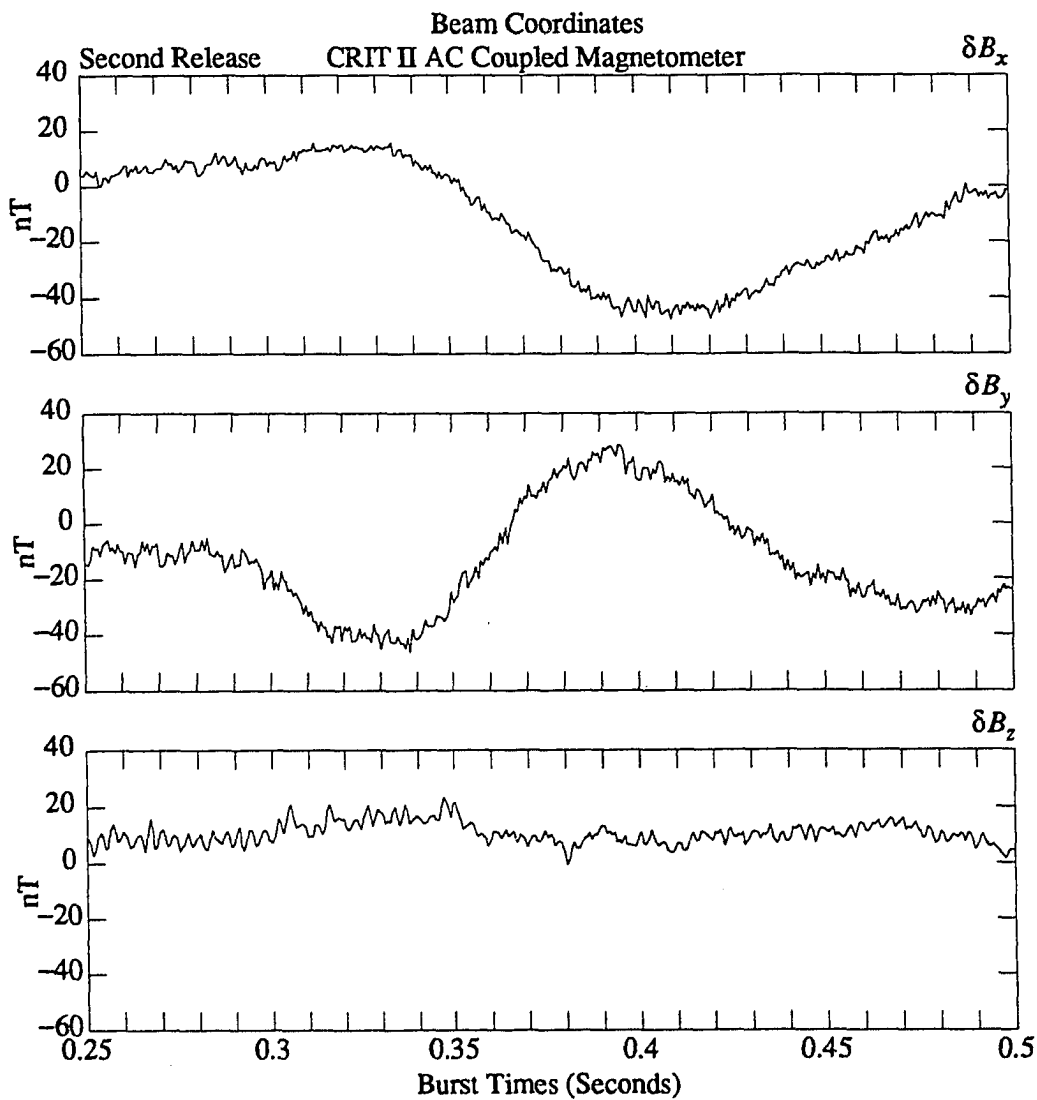


Figure 5.3: CRIT II second release ac-coupled magnetometer data rotated into a neutral barium stream aligned coordinate system.

clear detection of electric field oscillations near 1 kHz.

Another way to view this result is to consider the high frequency fluctuations to be a wave-like phenomena. Then Maxwell's equation $\nabla \cdot \vec{B} = 0$ reduces to $\vec{k} \cdot \delta \vec{B} = 0$ and from Fig. 5.2 we conclude that the angle between $\vec{k}$ and the rocket $\hat{z}$ -axis is less than 10° . This is consistent with the notion that the waves are traveling parallel to the barium beam and with the interferometric observations of the electric fields described later in this chapter which also show that $\vec{k}$ is parallel to the barium beam.

There are distinct similarities between the first and second release data as well as some important differences. Fluctuations in the observations can be conveniently divided into short and long time scales. The longer time scale, approximately 50 ms ($0.25\tau_{Ba^+}$), is observed in first release (Fig. 5.2) and second release (Fig. 5.3) data in both δB_x and δB_y channels, albeit with about an order of magnitude smaller signal for the second release. Magnetic and electric field signatures on this longer time scale were also observed at the subpayload about 1.7 km away and are thought to be the signatures of an Alfvén wave. The short time scale fluctuations (~ 3 ms) were not observed at the subpayload. The subpayload detectors could have observed such short time scale phenomena if it had been present, however, and therefore they must be a highly damped mode outside of the barium stream.

The most distinct difference between the first and second release data is the absence of short time scale fluctuations within the second release data. However the high frequency limit on the magnetometer was only 600 Hz. As noted above we believe the second release magnetic fluctuations would have been around 1 kHz and were thus strongly filtered. For example, the four pole filter in the magnetometer would reduce the amplitude of 1 kHz fluctuations by a factor of 100, placing them in the noise. We consider 1 kHz because the first release fluctuations (≈ 350 Hz)

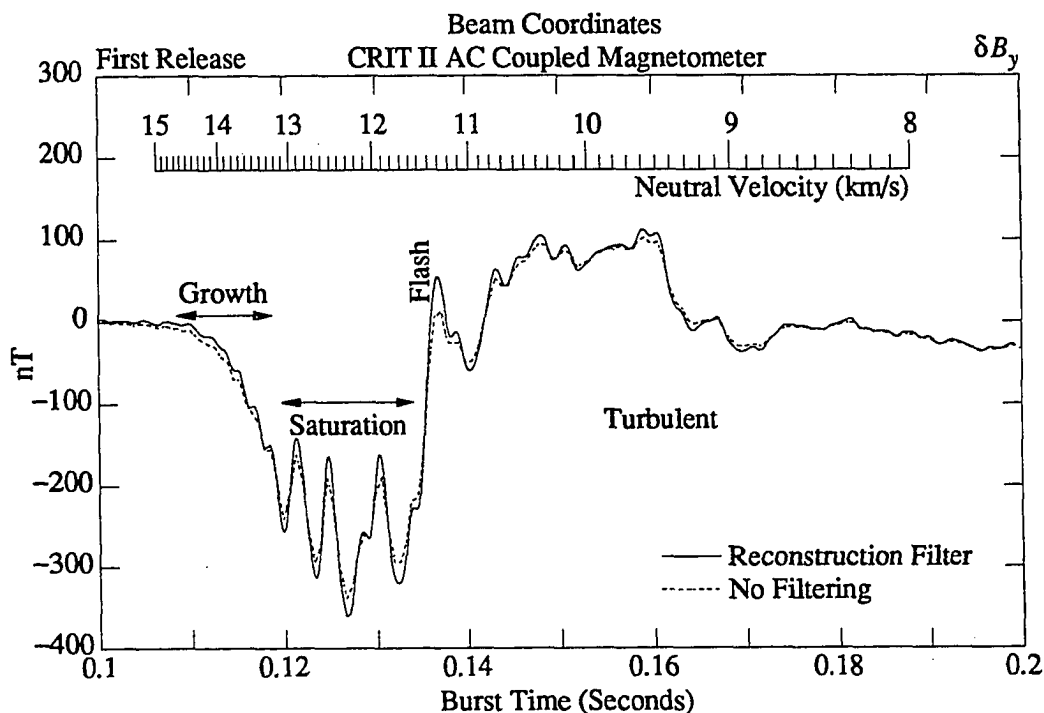


Figure 5.4: Data from the δB_y channel of the magnetometer during the passage of the ionization front of the CRIT II first release.

were correlated with first release electric field oscillations, and the corresponding second release electric field oscillations were around 1 kHz. The major differences between the measurement locations for the two releases were the neutral barium density and the resulting ionization, which was much lower for the second release and is probably related to the frequency of the fluctuations.

To aid in the discussion of the first release magnetic field measurements and the short time scale fluctuations, we present the annotated Fig. 5.4. A scale for the neutral stream velocity is included in this plot of the δB_y data and several regions have been noted. We refer to these regions as growth, saturation, and turbulent because of the corresponding features of the electric field data. These same regions will be discussed in Section 5.2 in conjunction with the electric field data. The term flash refers to the peak of the photometer observations. The short time scale

fluctuations, with frequencies around 350 Hz, are close enough to the sampling frequency of the instrument, 1.5 kHz, that each peak is composed of only about 2 or 3 samples. A reconstruction filter implementing bandlimited interpolation has been used to 'fill in' between the original points in a way that is consistent with the 600 Hz bandlimit of the data [Oppenheim and Willsky, 1983]. Both the processed and original data are presented in Fig. 5.4 for comparison.

Through the growth region the δB_y perturbation grows increasingly more negative. More importantly the slope of the signal steadily becomes more negative. Since this slope is directly related to the field aligned current, more demands are made on the ambient plasma to deliver current through this period. The saturation zone is co-located with the region when the δB_y data start to strongly oscillate. The currents implied by the fluctuations during the saturation period time are much larger than those associated with the longer time scale fluctuations. The flash region separates the saturation zone from the turbulent region. At the time of the flash the largest current density is implied by the positive derivative of the magnetic field data. We note that almost all of the instruments, photometer, PFP, particle detectors, etc, on the main payload detected this flash period as being uniquely active. Throughout the turbulent region magnetic fluctuations are measured, but they are much weaker than those of the growth and saturation regions.

5.1.2 Currents

The relationship between the currents and the magnetic field observations was previously derived in Chapter 3 and is reproduced here.

$$-\frac{1}{\bar{V}_{Ba}} \frac{\partial B_y}{\partial t} \hat{i} + \frac{1}{\bar{V}_{Ba}} \frac{\partial B_x}{\partial t} \hat{j} = \mu_o (J_x \hat{i} + J_y \hat{j}) \quad (5.1)$$

Before applying (5.1) to the data of Fig. 5.2 and Fig. 5.3 to calculate currents, an estimate for $\vec{V}_{Ba}$ is required. The correct velocity to use is that of the current system, which is unmeasured. Conventional wisdom would suggest this current system is moving with the slowed barium ions, but this velocity is not well defined by any measurements. A lower bound on the current can be found by using the velocity of the neutral stream. Later in this chapter we analyze the data for coincident electric field structures to determine phase velocity. If this measured phase velocity for electric field structures is used, which is slower than the neutral stream, then the currents would be 1.8 to 2.4 times larger. We have chosen to use the neutral barium stream velocity to calculate currents since it has a well defined value.

Field Aligned Currents

The estimated field aligned currents are presented in Fig. 5.5. In addition the growth, saturation, and turbulent regions are also noted and a scale for the neutral barium velocity has been included. The oscillating currents implied even by this lower-limit estimate are large, 10^3 times larger than those measured in an active auroral arc. The maximum field aligned current that one might expect the ionosphere could supply to the ionization front, assuming no accelerating geomagnetic aligned electric fields, is given when all of the electrons are moving with their thermal velocity. This current density, J_{th} , is 10 mA/m^2 ($T_e = 1000 \text{ K}$) and is noted on Fig. 5.5 by two dashed lines. Clearly the field aligned currents are of this order of magnitude. This may imply a geomagnetically aligned electric field may be required to create such large currents, especially since this estimate is a lower limit.

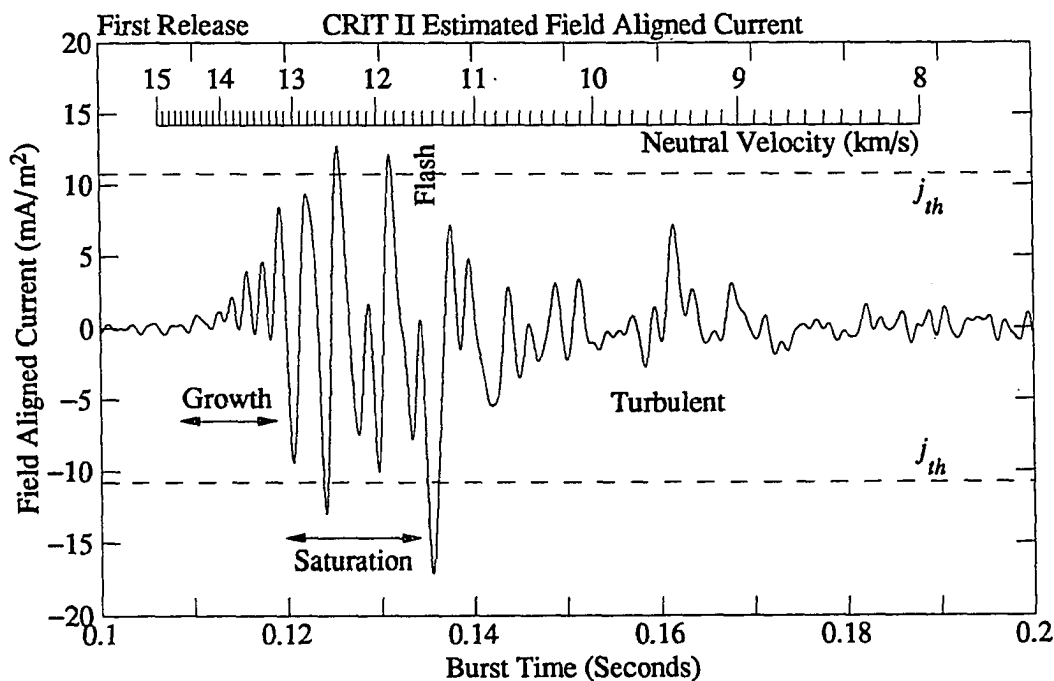


Figure 5.5: A lower limit estimate of the field aligned currents within the ionization front of the first release of CRIT II.

Momentum Exchange Currents

A lower limit estimate of the currents across the barium stream is presented in Fig. 5.6. These currents are perpendicular to the geomagnetic stream and are most likely carried by the electron Hall drift. To sustain such an electron current a polarizing electric field is required perpendicular to the geomagnetic field and in the $\hat{x}$ direction. Later in this Chapter we will show that such a field exists.

5.2 Electric Fields

In this section, we concentrate on the fluctuating electric fields within the ionization front with frequencies over about 100 Hz. The quasi-dc measurements, fluctuations with frequencies less than this, were corrupted during the ionization front passage as discussed in Chapter 3. The electric field data was transmitted over two different

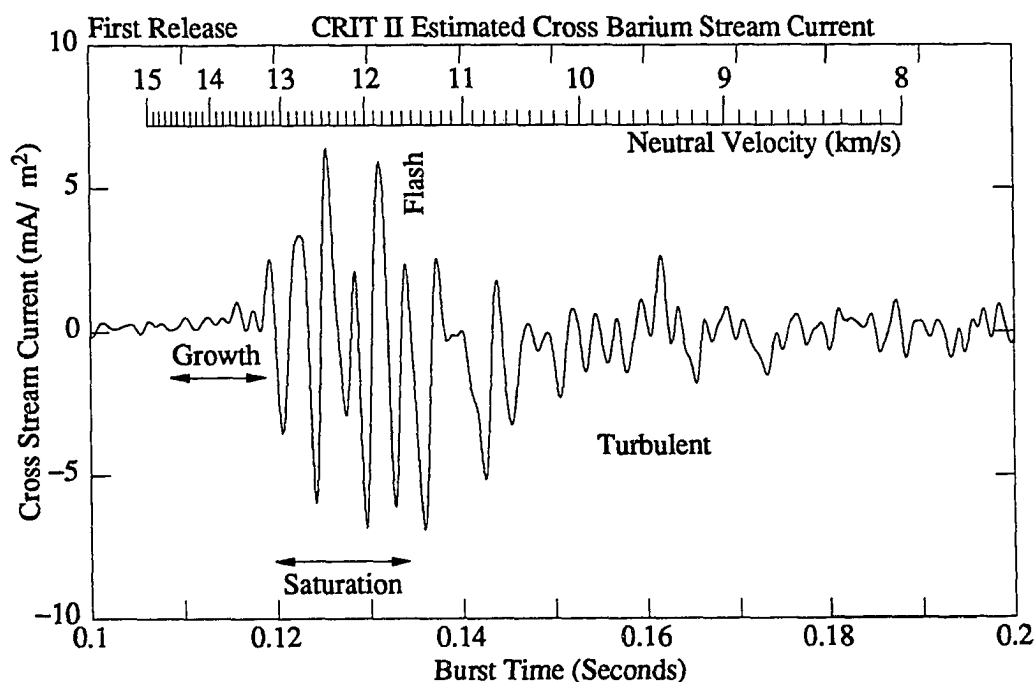


Figure 5.6: A lower limit estimate of the currents across the barium stream during the CRIT II, first release, ionization front. Currents could be 1.4 to 1.6 times larger.

telemetry links. The best data came through the PCM system since the data were digitized on the payload and no additional noise was introduced by the telemetry system. Data transmitted through the FM-FM telemetry system appear to have been corrupted, more than expected, by malfunctioning components in the payload. Therefore, the majority of the data analysis has been done using three primary electric field measurements from the PCM system, two orthogonal measurements across the neutral stream and one measurement along the neutral stream direction. The assumption, in making this statement, is that the neutral stream was aligned with the spin axis of the payload. NASA provided positions and attitudes for the main payload which, combined with the optical observations of the University of Alaska for the release directions, indicate the spin axis was actually misaligned by 15° for the first release and 8° for the second. There has been no systematic

evaluation of the margin of error in these numbers, but experience gained during the analysis process leads us to assume a standard error of about 5° .

Data from these three primary PCM channels, which have been filtered to remove the corrupted quasi-dc components, are presented in Fig. 5.7 for the first release and in Fig. 5.8 for the second. The δE_x and δE_y channels are measurements across the neutral stream and the δE_z channel is a measurement along the neutral stream. A high pass filter with flat phase response has been used to remove the quasi-dc fields. These figures presents the data on the same time scale as the density data discussed in Chapter 4 and the magnetometer data in Section 5.1. Later, in this Chapter we will bring all three sets of measurements during the ionization front together for comparison in the same plots, but first we discuss just the electric field data.

5.2.1 Overview

The electric fields within the ionization front of the first release developed through several distinct phases, whereas the development of the fields during the second release was not as distinct. In this section, we will use the electric field data measured along the direction of the neutral stream to discuss the time evolution of the electric field within the front. These data are presented in Fig. 5.9. We have also labeled a scale with the neutral stream velocity, and inset a plot of the instantaneous frequency of the oscillations for a short period of time. Velocities in the scale greater than that of the fastest neutral barium, 13.5 km/s, have been shaded. The electric field data can be divided into the same four distinct periods, namely 1) the precursor waves, 2) the growth phase, 3) the saturated waves, and 4) the turbulent region. These regions have been marked on Fig. 5.9 and will be treated separately. The electric field during the second release appears to have

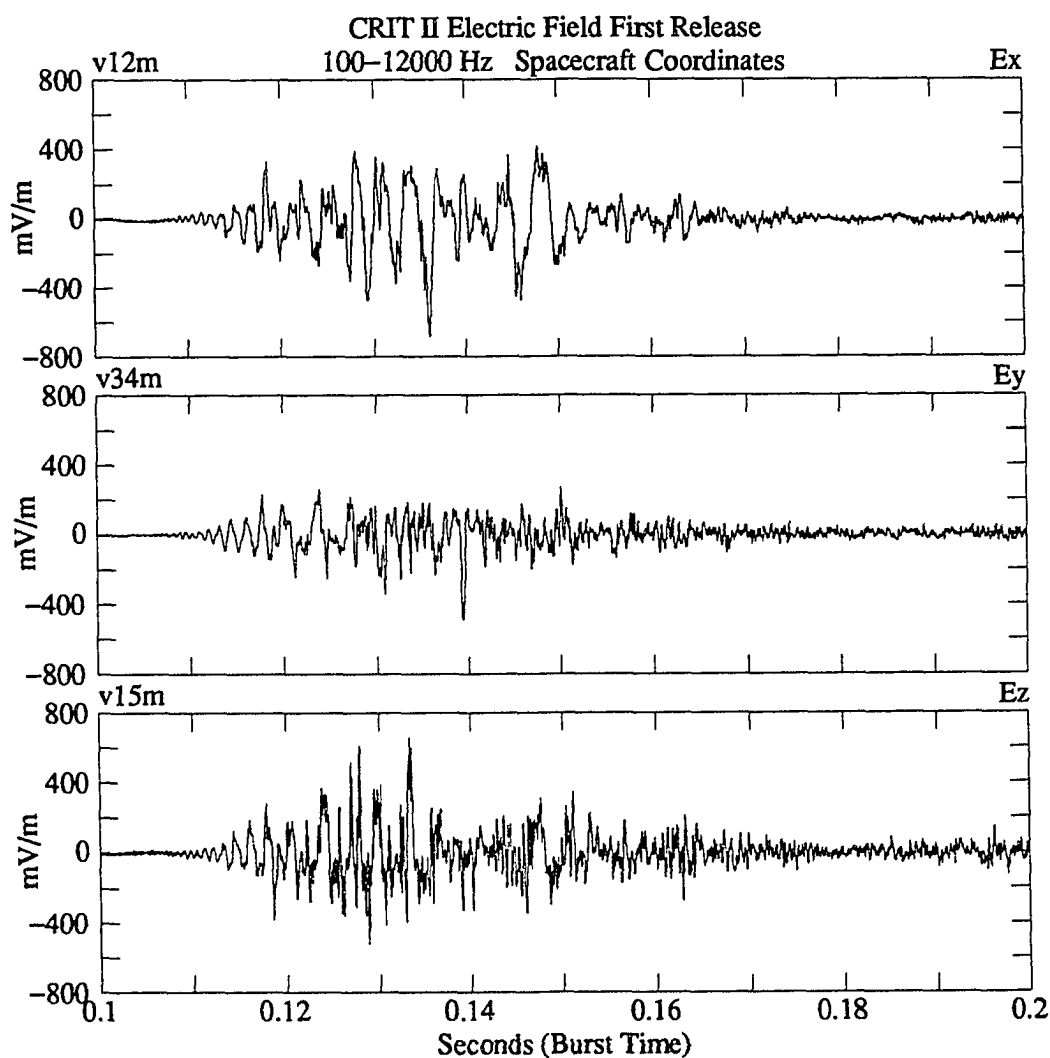


Figure 5.7: Electric field data from first release of CRIT II. Data has been high pass filtered at 100 Hz.

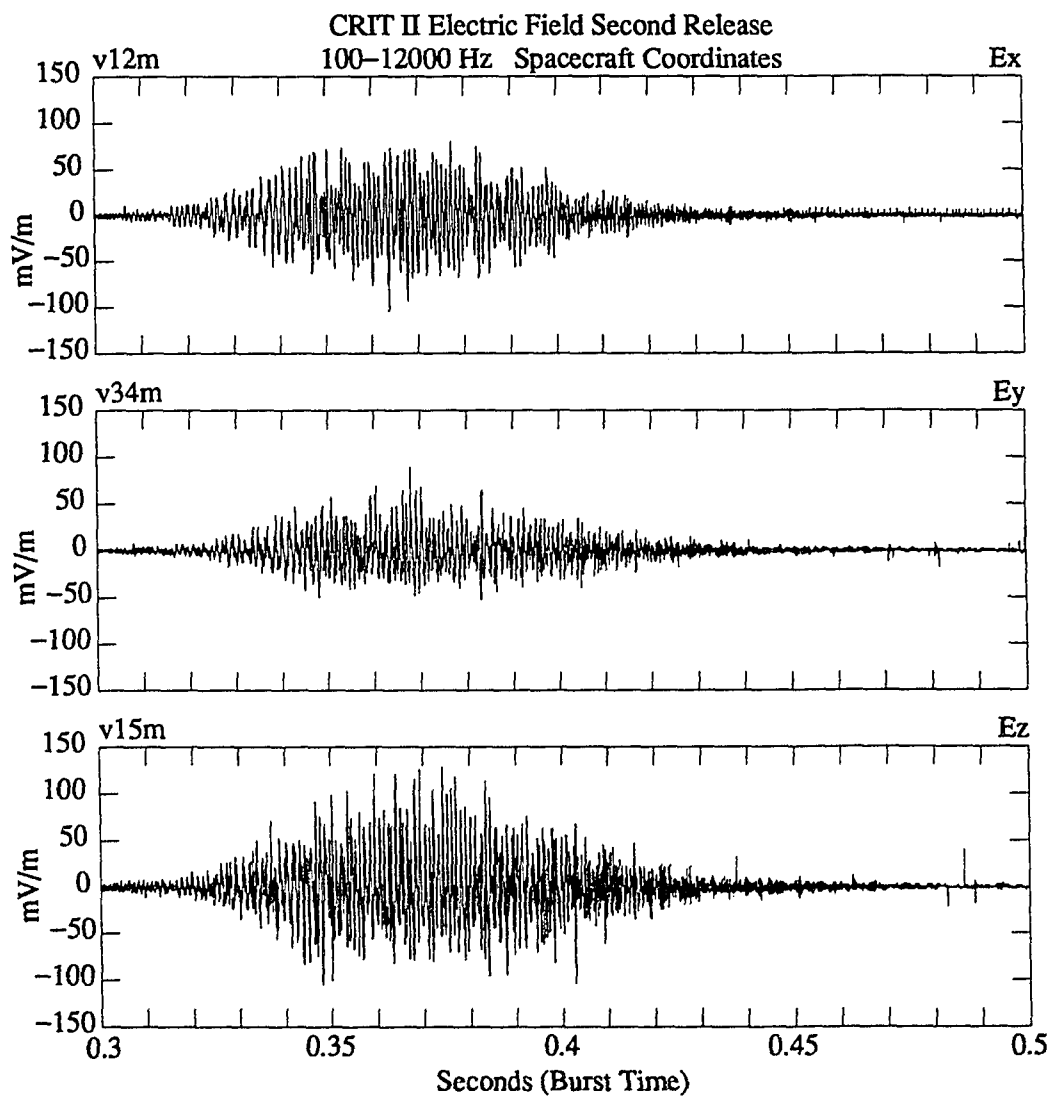


Figure 5.8: Electric field data from second release of CRIT II. Data has been high pass filtered at 100 Hz.

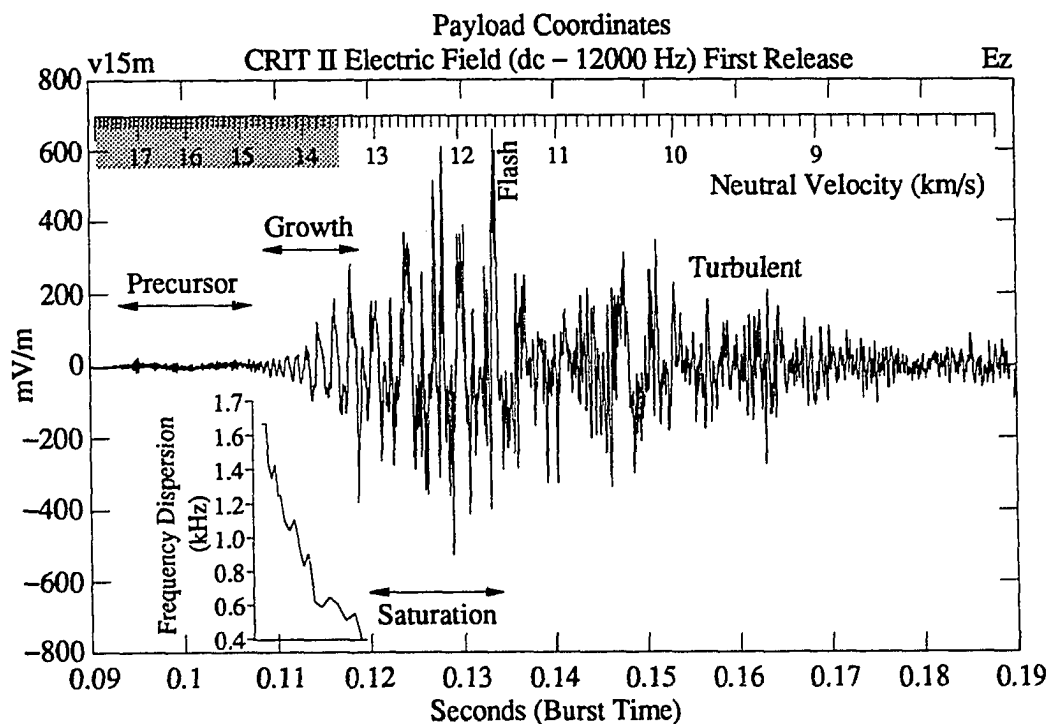


Figure 5.9: CRIT II first release electric field data along the direction of the neutral stream. The velocity of the neutrals is noted at the top, and the frequency dispersion during the growth period is plotted and inset.

passed through only the first two stages of this progression. A likely explanation is that those stages are directly linked to the ionization rate within the front.

Precursor Phase

The precursors waves are barely visible on this plot and given this name because they arrive at the main payload before the barium neutrals. The frequency band of these waves is around 7 kHz or just above the Oxygen lower hybrid frequency. The measured amplitude of these bursts of lower hybrid waves peaks at about 20 mV/m, which implies at least 24 mV/m fields perpendicular to $\vec{B}_0$. This value is actually a lower limit since for electrostatic waves if the electric field sensor spacing is close to the wavelength a spatial filtering effect reduces the measurements. The fact that the modulation envelope for these waves is different for the different channels, and thus different sensor spacing (Fig. 5.7), implies that the wave length of these oscillations is indeed on the order of the detector boom lengths or in the range of 1 to 2 m. The group velocity of lower hybrid waves is along the magnetic field and typically greater than 100 km/s for ionospheric conditions, much faster than the neutral stream. These waves are propagating ahead of the Barium neutrals since they arrive before the stream. A possible scenario is sketched in Fig. 5.10, but we caution that there is no experimental evidence for the propagation direction implied in the figure. The observations are also consistent with both Condor [Kelley *et al.*, 1986] and CRIT I [Kelley *et al.*, 1991] which both observed precursor waves near the lower hybrid frequency of the ambient plasma.

Growth Phase

The next feature in the electric field measurements is a stage of apparent growth starting at about 0.108 s burst time and continuing to about 0.118 s. The beginning

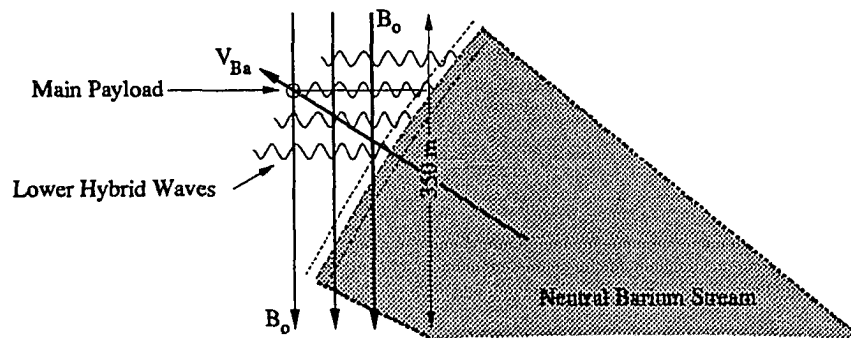


Figure 5.10: Lower hybrid waves, which have group velocities greater than the velocity of the neutral barium stream, can propagate ahead of the barium stream. These waves will be observed at the main payload before the arrival of neutral barium.

of this period would correspond to the arrival of 14.7 km/s particles from the release and at the end with 13.2 km/s particles. We note that the fastest barium is expected to be produced at 13.5 km/s and the density of barium with velocities greater than this would be very small. A thin coating of grease was applied to the surface of the barium metal in the shape charges to protect it from moisture. This hydrocarbon film would probably have been accelerated to this velocity range by the release (*personal communication, E. M. Wescott, 1989*). Another source of particles might be collisionally accelerated ionospheric neutral oxygen which could form a tenuous oxygen ion beam by charge exchanging with ambient oxygen ions. In short, we can hypothesize a number of possible, but unmeasured processes that could destabilize waves and make them grow during this period.

There are several distinguishing features of these waves: a) they decrease in frequency with time, b) they are narrowly banded in frequency ($\delta f/f \approx 0.1$), and c) their amplitude changes with about a 0.3 ms time constant ($\gamma \approx 0.1\omega_o$, where ω_o is the observed average wave frequency and γ is the growth rate). The waves start out at about 1.6 kHz and then drop in frequency throughout the growth phase to about 300 Hz. This frequency dispersion is plotted and inset below the growth

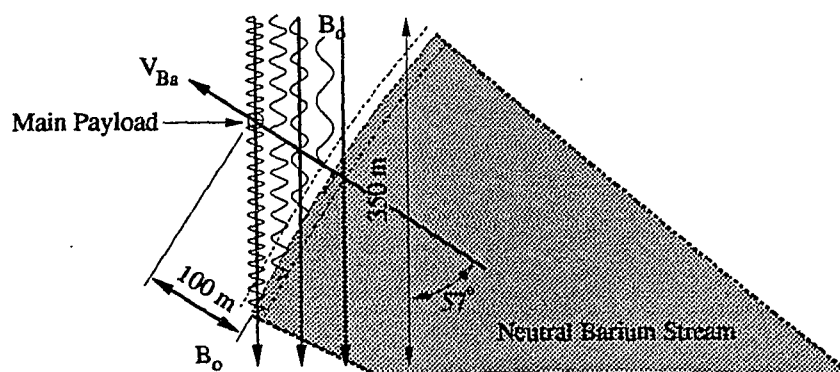


Figure 5.11: Growth region electric fields are probably a mapping along the geomagnetic field of the electric field structures within the barium stream. The waves have been exaggerated in this sketch.

region in Fig. 5.9. The time base of this inset plot is matched with the electric field time scale so a direct comparison of the changing wave frequency can be made. We note that this frequency dispersion cannot be explained as a simple Doppler shift of a constant wavelength structure moving with the velocity of the neutral barium stream. For example, if we assume the phase velocity of these waves is that of the neutral stream, or equivalently that the dispersion relation is given by

$$\omega = kv_{\text{neutrals}} \quad (5.2)$$

then the wavelengths, parallel to the neutral stream, would range from 10 to 40 m throughout this growth period.

A possible explanation for the source of these growth waves, which ride ahead of the neutral barium, is sketched in Fig. 5.11. The barium stream is injected at 57° to the geomagnetic field. This results in contact of the barium stream with the flux tube threading the the main payload before the neutral barium arrives at the main payload. The electric fields within the barium stream might map along the geomagnetic field and therefore be observed at the main payload before the

barium. The frequency of the waves in the sketch have been exaggerated to show how higher frequency data originate from the edges of the barium stream where the density is lower.

The width of the barium stream, measured to the half maximum of the neutral density, is approximately 32 m. If we assume the electric field structure maps along the geomagnetic field with a speed on the order of the Alfvén velocity (300 km/s) then the fields will be observed at the main payload 0.6 ms after contact with the flux tube. The waves will be detected some 7.2 ms before the neutral barium arrives at the payload since the neutral velocity (13.5 km/s) is much slower than the Alfvén velocity. If we use the width of the neutral stream when it is down to 36% of maximum, two sigma, then the waves will be detected some 8.6 ms before the barium stream. This is very close to the roughly 10 ms width of the growth phase, but latter in this chapter we show that the electric field structures are traveling slower than the neutrals, at about 10 km/s. This value results in almost a perfect match for the onset of the growth phase.

In essence, data in the growth region are a sampling of the electric fields radially across the neutral stream. The neutral barium density is falling off exponentially, given by (2.5), and the frequency shift through the growth phase results from either shorter scale lengths, or higher frequency waves being generated within the lower density portions of the neutral stream. This is in general agreement with the second release data where the neutral stream density was lower and higher frequencies are observed. Finally, we note that the energetic electron particle detectors measured on average an isotropic population of electrons around 20 eV through this phase indicating a preheating of the electrons, possibly by these waves.

Saturation Phase

The waves of the growth period blend into the saturation region which begins at approximately 0.118 s burst time. The division between the growth region and the saturation region is not distinct, which is consistent with the model presented for the growth phase. The saturation region is characterized by squared off waves with a period of ≈ 300 Hz which appear to be clamped by some limiting process, hence the name for this region. This region begins simultaneously with the arrival of 13.5 km/s neutral barium, and throughout the saturation region isotropic populations of energetic electrons were detected with peak energies at 10 eV and 100 eV as shown in Fig. 3.5. The saturated waves correspond with the ionization of barium and the deposition region of the density data as discussed in Section 4.3.3. Lastly, during this period the magnetometer detected the signatures of the oscillating current system at the same frequency.

A potential drop across the saturation waves can be estimated if we assume the electric field structure is moving with the neutral stream. The width of the saturation features in Fig. 5.9 is ≈ 20 m and the wave amplitudes are about 300 mV/m. A simple integration of the field over a square wave implies a potential able to directly accelerate electrons to about 6 eV. This energy range is consistent with the energetic electron particle measurements during this period which observed electrons heated in a band around 10 eV.

Turbulent Phase

The last region identified begins just after the flash was observed by the photometer. At this same time the electron particle detectors were saturated briefly by fluxes of hot electrons. No information is available about the energies or fluxes other than that they were greater than the instrument threshold. The magnetometer measured

a strong current sheet and the density probe indicated a localized depletion. The neutral model predicts the arrival of the bulk of the barium at this time. After the flash the electric field drastically changed character. Throughout the growth and saturation region a coherent signal could be observed, but this coherent signal is lost during the turbulent period, hence the name. The analysis of the density measurements in Chapter 4 shows that the production of barium is largely shut off throughout the turbulent region. We believe this production of barium ions is the origin of the difference between the turbulent region and the growth and saturation regions.

5.2.2 Interferometry

Payload Configuration

The electric field dipole antennas on the main payload were configured such that some measurement pairs could be used as interferometers. With appropriate assumptions, interferometric techniques can be used to obtain some information on the phase velocity of the structures being observed [Briggs *et al.*, 1950]. This type of analysis adds significantly to the characterization of the waves within the ionization front.

Consider, for example, an electric field pattern moving with the barium beam towards the main payload as illustrated in Fig. 5.12. Electric field components of this structure will first be observed at the leading antenna, a measurement between sensor 1 and sensor 2, and later by the trailing antenna, a measurement between sensor 5 and sensor 6. Measurements between sensor pairs 1-5 and 2-6, 2-4 and 3-5 provide additional baselines. The phase centers for these electric field measurements are located midway between the sensors. The electric field component observed is that projected along the vector between the sensors. Assuming that the structure

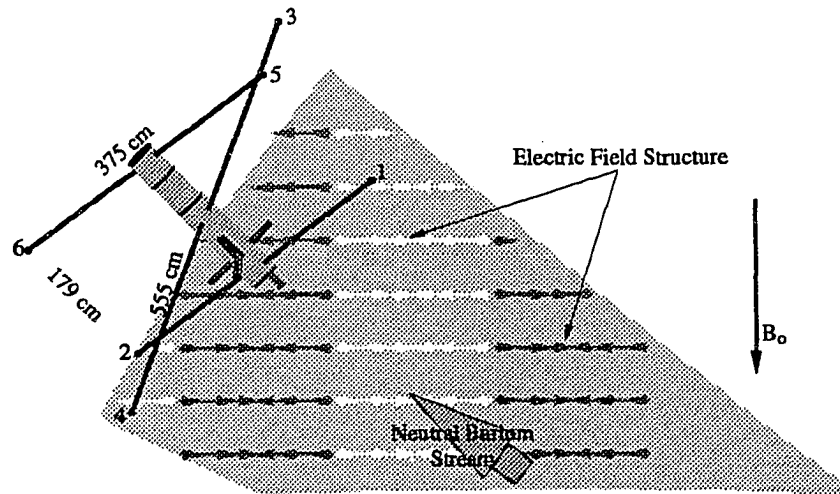


Figure 5.12: A diagram of an electric field structure within the neutral stream passing the CRIT II main payload. Different probe pairs observe the electric field structure at different times.

is not significantly changing with time during its transit, a stationarity assumption, the observed time delay and separation distance of the phase centers can be used to obtain a phase velocity between the phase centers. The observed time delay, τ , can be related to the wave vector, $\vec{k}$, of the impinging electric field structure.

$$\tau = \frac{\vec{k} \cdot \vec{d}}{\omega} \quad (5.3)$$

where ω is the observed wave frequency and $\vec{d}$ is the vector distance between the phase centers of the measurements. If the time delay is measured for three different baselines (three linearly independent $\vec{d}$) then (5.3) can be written as a matrix equation of the form $\vec{\tilde{d}} \cdot \vec{k} = \omega \vec{\tau}$. The wave vector can be calculated from the

measurements by inverting the matrix $\vec{d}$.

$$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \omega \begin{bmatrix} d_{1x} & d_{1y} & d_{1z} \\ d_{2x} & d_{2y} & d_{2z} \\ d_{3x} & d_{3y} & d_{3z} \end{bmatrix}^{-1} \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} \quad (5.4)$$

A listing of the electric field measurements made between different probe pairs and their associated bandwidth, sampling rates, and telemetry systems is given in Appendix C. There are several complicating problems in carrying out the analysis described. The first problem is that not all of the needed channels were transmitted on the same telemetry link. Approximately half of the channels needed for such an analysis were transmitted through the FM-FM telemetry system and the time base between the PCM and FM-FM telemetry systems is only accurate to approximately 1 ms. The time base problem can be resolved by correlating the v34 m channels which were transmitted on both telemetry channels. A second problem is that the quality of the data transmitted through the FM-FM telemetry system is poor compared to the PCM system. This can be easily observed by comparing the PCM data of Fig. 5.7 to the FM-FM data of Fig. 5.13. Other problems which conspire to make this analysis difficult are different sampling rates for the channels and different bandwidths of the measurements.

Correlations

The time delay between measurements has been calculated by making cross correlations in the time domain. This method works well if the signals are sufficiently coherent which will be the case for the growth phase and throughout most of the saturation region. This kind of analysis is not expected to work well for the turbulent region where the electric field structures are not coherent. We chose this

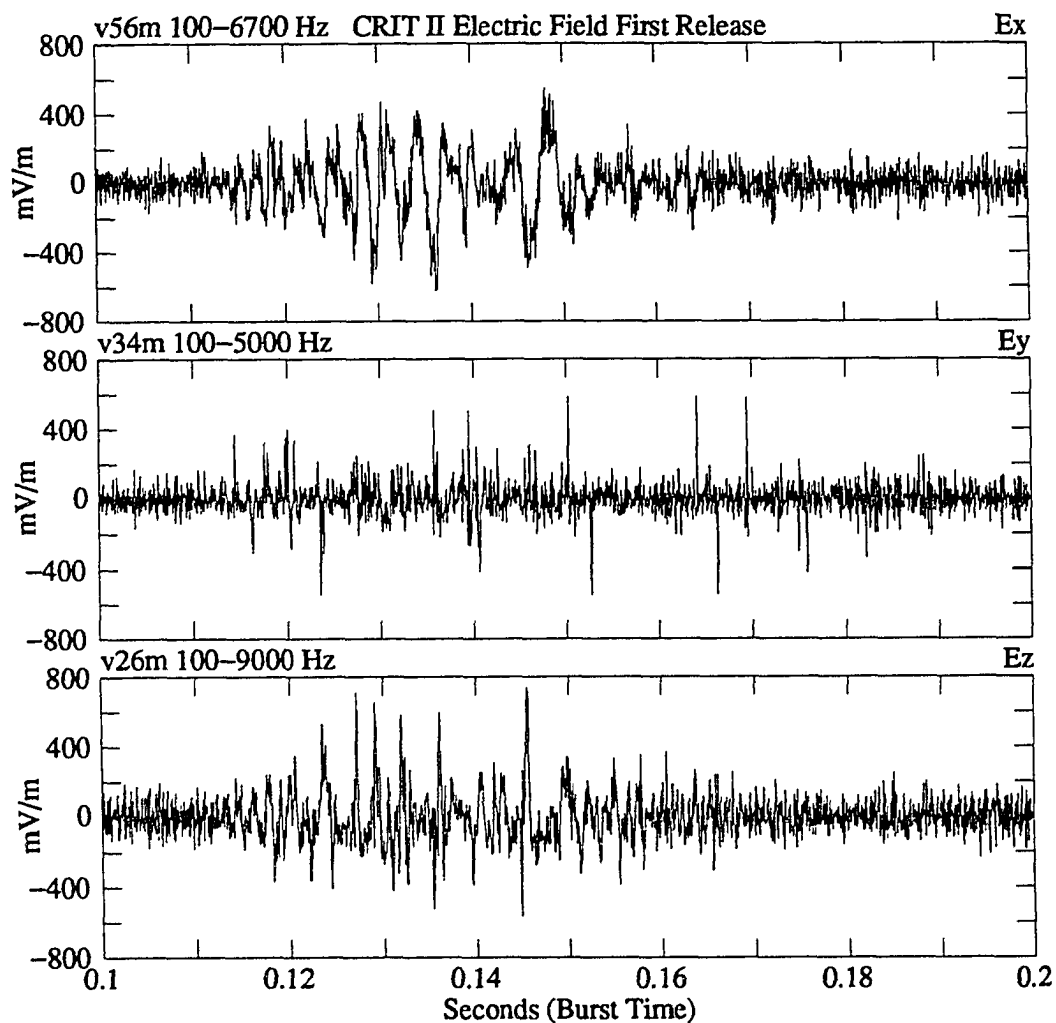


Figure 5.13: Electric field data from first release of CRIT II transmitted through the FM-FM telemetry system. Data have been high pass filtered at 100 Hz.

method so that variations in the correlation time can be followed through the data, and because the event to be studied is not suitable (too few data points) for frequency domain techniques. The normalized correlation function for two time series, $g(t)$ and $f(t)$ is given by.

$$\text{corr}_{g,f}(\tau) = \frac{\langle g(t) \cdot f(t - \tau) \rangle}{[\langle g^2(t) \rangle \cdot \langle f^2(t - \tau) \rangle]^{1/2}} \quad (5.5)$$

where the $\langle \rangle$ notation denotes the expectation, which, for our analysis, is a summation over t . The range of the normalized correlation function is 0 to 1 where 1 indicates the series are identical [Papoulis, 1984]. For the standard analysis, correlations were made out of 80 data points (3.3 ms) and the previous 40 data points were overlapped as we passed through the data.

An example of the correlation function for E_x measured between sensors 1-2 (Fig. 5.12) and E_x measured between sensors 5-6 is presented in Fig. 5.14. The time series can be found in the top panels of Fig. 5.7 and Fig. 5.13. We have analyzed the data differently for this figure to bring out features of the correlation of our data. Correlations were made from 120 data points (5 ms of data) and the entire correlation function is presented on the left of Fig. 5.14. Each correlation has been self normalized by its peak value which is plotted on the right in Fig. 5.14. The correlation function begins to develop a localized peak after about 0.11 s burst time which is at the beginning of the growth period (see Fig. 5.9). The correlation time delay of this peak is about 0.2 ms. The observed phase velocity, given the 1.79 m separation of the measurement phase centers, is therefore approximately 8.9 km/s. This peak in each correlation exists until about 0.165 seconds burst time, which is into the turbulent regime. The correlation function becomes random at approximately the same time the density depletion is observed and after the turbulent

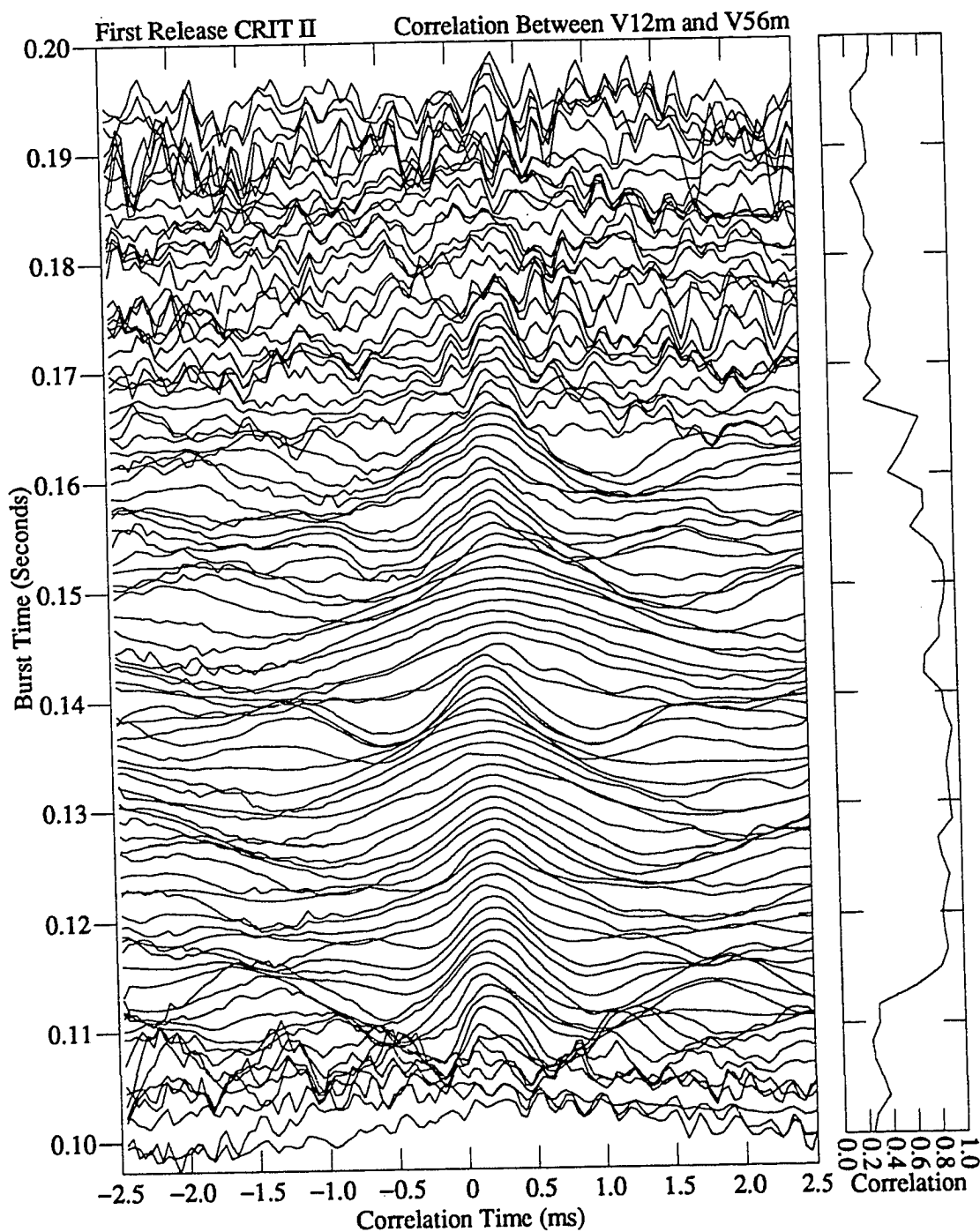


Figure 5.14: Correlation between two E_x measurements spaced along the path of the neutral stream. Each correlation has been self normalized to its peak value.

region discussed above. This suggests the coherence length does not become small compared to the baseline between the measurements until the turbulent region. Since significant electric fields are still observed after 0.165 seconds, this suggests turbulent waves are generated by the passage of the ionization front.

Electrostatic $\vec{k}$ Vector

To determine a wave vector, $\vec{k}$, from the observations, the correlation times between three pairs of measurements must be made. The channels which are available for this type of analysis are v12 m and v56 m, v15 m and v26 m, and v24 m and v35 m. The last pair of channels, v24 m and v35 m, were both transmitted via the PCM telemetry channel, but v24 m was only sampled at 750 Hz. This reduces the resolution of τ measured along this baseline relative to the other channels. The peak correlation and associated correlation time verses burst time are presented for these three pairs of channels in Fig. 5.15. The correlation (dashed line) and correlation time (solid line) from Fig. 5.14 are shown in the top panel of Fig. 5.15. The period of interest, where the correlations are the largest, has been shaded. Clearly the correlation times are reasonably well defined from 0.115 through 0.135 s or through the growth and saturation phases of the electric field. The correlation time is almost constant along the neutral stream (top panel). In the center panel v15 m is correlated against v26 m. This measurement is almost directly across the neutral stream and corresponds to electric field structures moving across the neutral stream. These structures slow from 22 km/s to 5 km/s during the period from 0.12 to 0.13 s.

The wave vectors of the structures during the shaded period of Fig. 5.15 have been calculated by (5.4) and are presented in Fig. 5.16. The components of $\vec{k}$ have been transformed into a coordinate system aligned with the geomagnetic field and

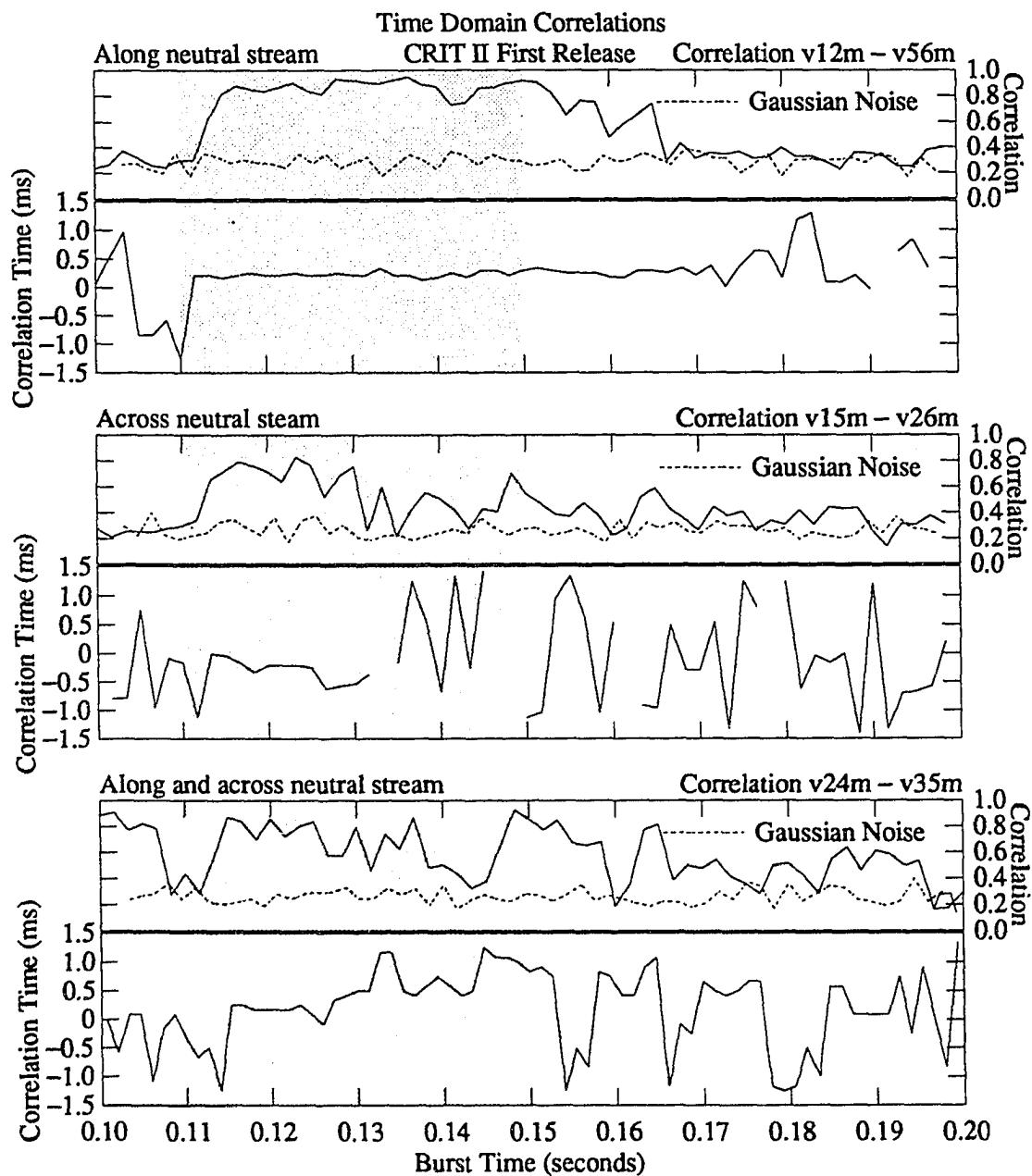


Figure 5.15: Peak Correlation and correlation time measured along three linearly independent baselines during the first release of CRIT II.

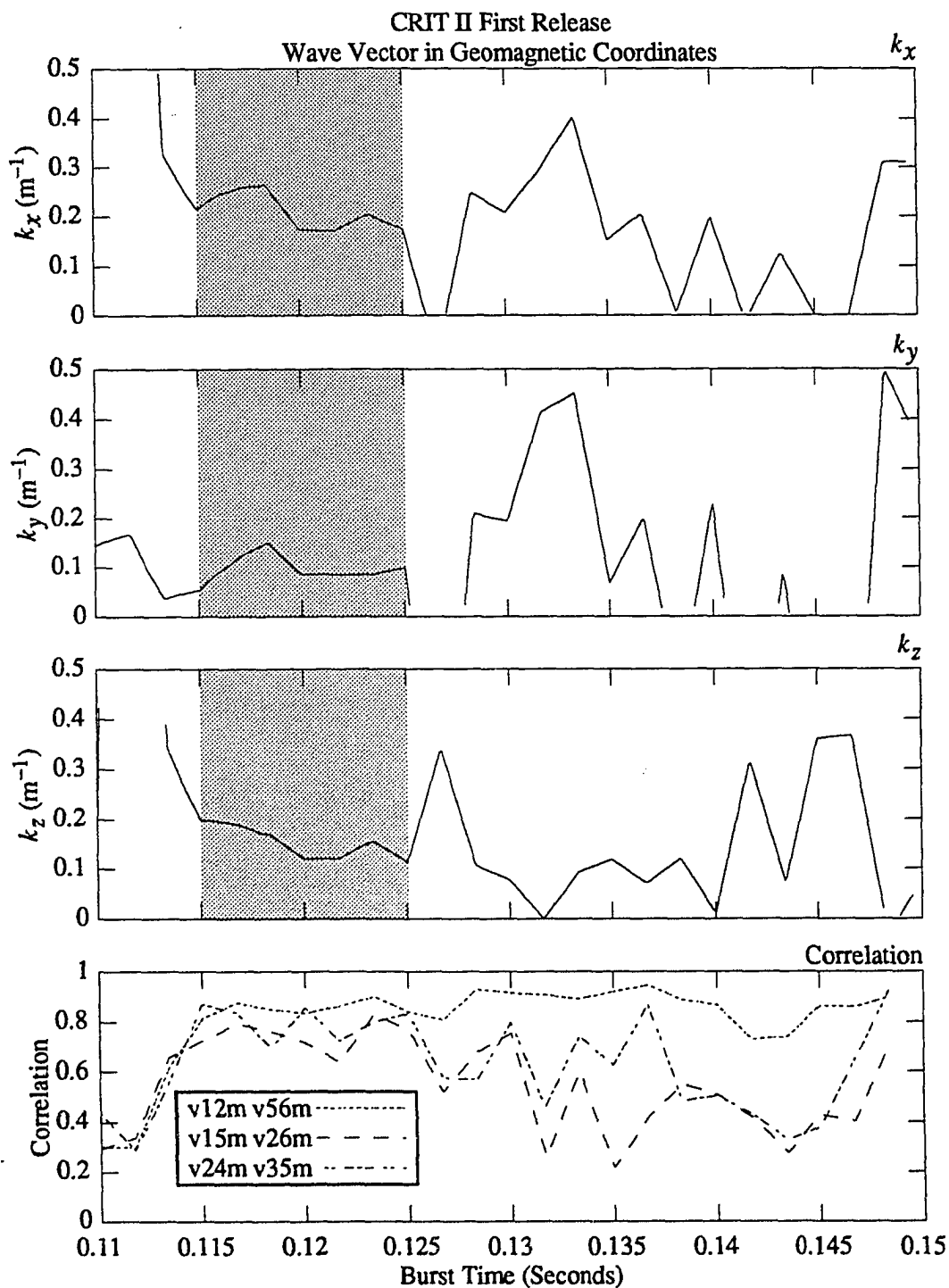


Figure 5.16: Components of the wave vector for the electric field structures expressed in a geomagnetic aligned coordinate system.

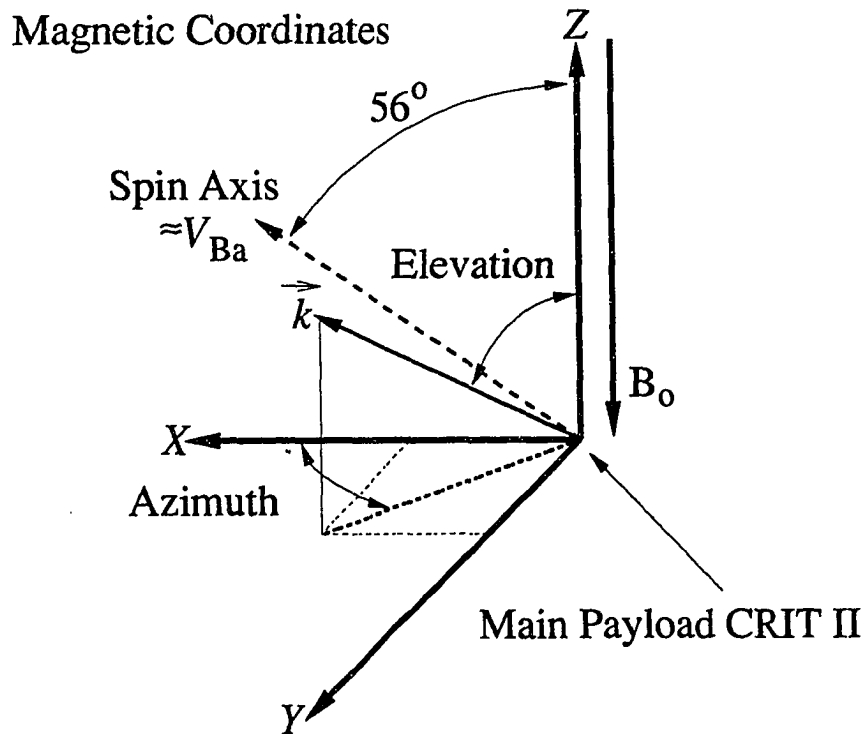


Figure 5.17: Definition of geomagnetic aligned coordinate systems and the azimuth and elevation for $\vec{k}$.

the direction of the barium stream. This is the same magnetic coordinate system used throughout this study and is sketched again for reference in Fig. 5.17. The electric field structures have large values of k_z during the growth and saturation regimes; in fact k_z is nearly as large as k_x . To simplify the results of this analysis and present physical implications we have plotted the azimuth and elevation of $\vec{k}$ in Fig. 5.18 and the associated wave length. The azimuth and elevation angles are defined in Fig. 5.17. The elevation of $\vec{k}$ during the saturation period, approximately 55° , is almost precisely the elevation of the barium neutral beam, $\approx 57^\circ$. Even the azimuth for these waves allows them to be aligned with the neutral stream given the uncertainty of the neutral stream direction relative to the magnetic coordinate system. The wave length of these structures appears to go through several stages. Initially, during the growth period (0.110 to 0.115 s) the structures are approxi-

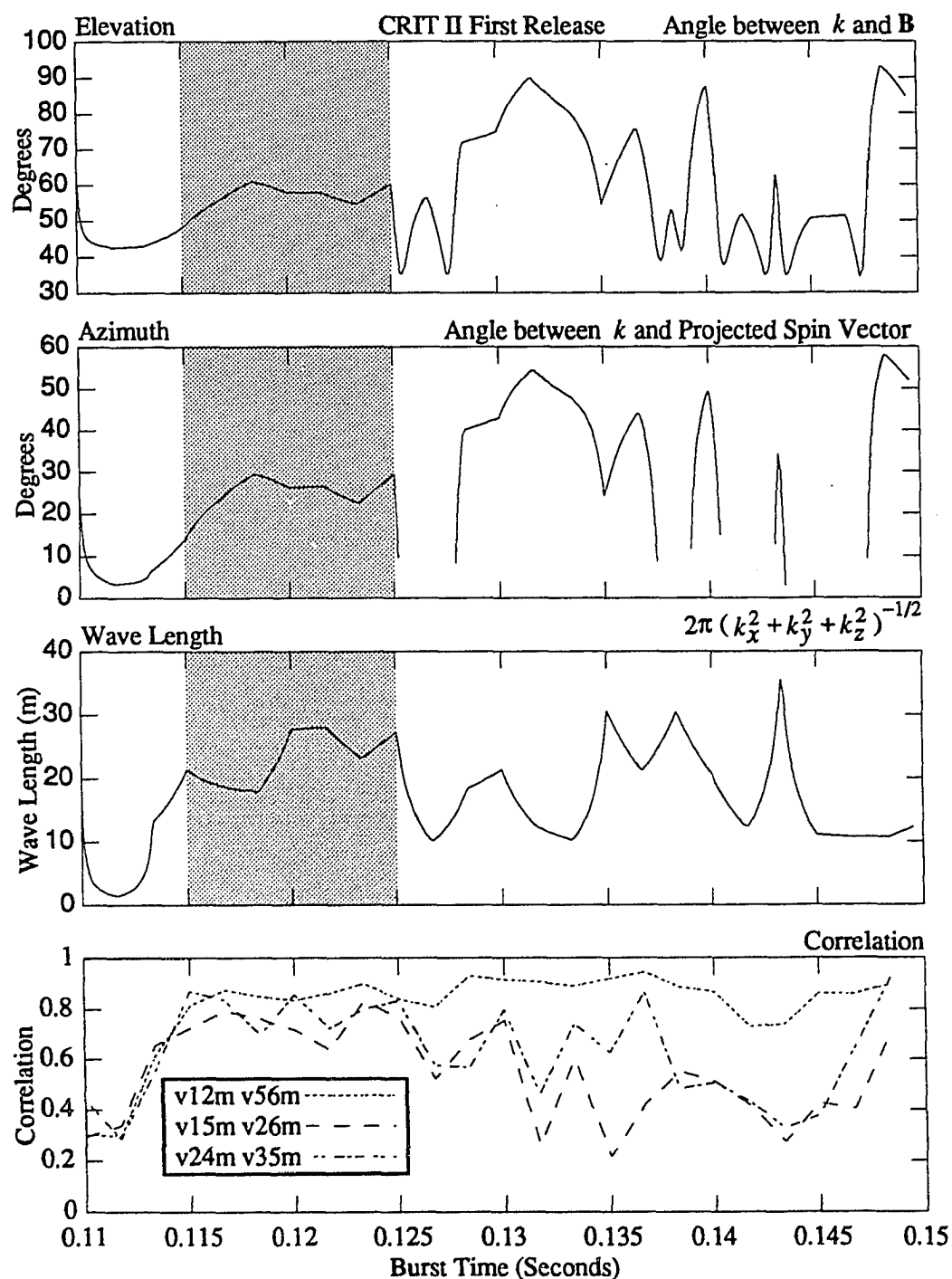


Figure 5.18: The azimuth and elevation of the wave vector, $\vec{k}$, and the wavelength of the waves in the first release ionization front of CRIT II.

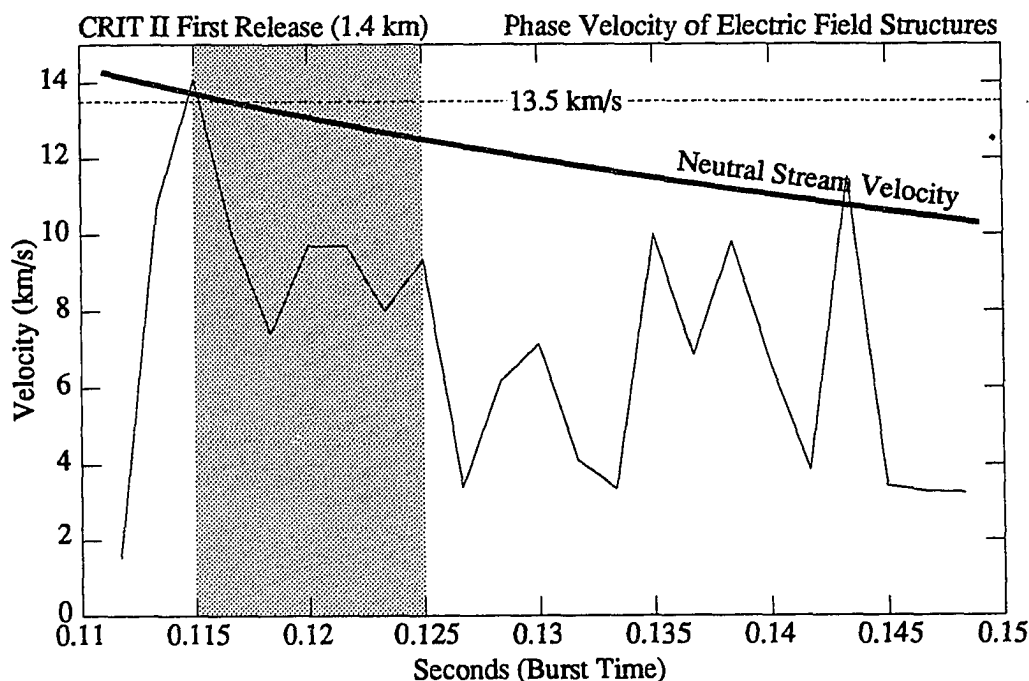


Figure 5.19: The phase velocity of the electric field structures within the ionization front, $\omega/|\vec{k}|$ for the first release of CRIT II.

mately 10 to 15 m. During the saturation period the wave length stretches out to approximately 30 m and then drops back to 15 m. At the time of the flash observed by the photometer (0.133 s) the waves seem to jump back to approximately 30 m. For comparison the collisionless skin depth for this plasma, c/ω_p , is 7.1 m.

This interferometric analysis has shown that the electric field structures within the saturation region are moving approximately in the direction of the neutral barium stream, at 57° to $\vec{B}_0$. In Fig. 5.19 we plot both $\omega/|\vec{k}|$, or the phase velocity of these electric field structures and the neutral stream velocity. The electric field structures are moving slower than the neutral barium, but this is expected since the waves should be associated with the motion of the barium ions. These barium ions will be slowed since they provide the free energy for the growth of the electric field structures. If the barium ions are slowed as suggested then they have given up about 50% of their original kinetic energy. This would be enough energy to

maintain a CIV type reaction given sufficient efficiency in the rest of the CIV microprocess.

5.2.3 Electric Field Data in Magnetic Coordinates.

Transformation of the electric field measurements into another coordinate system is not a straightforward process and can only be achieved by using the results of the interferometric analysis. Since the measurement phase centers of our detectors are not aligned, different components of the electric field have small phase shifts relative to each other. These phase shifts make possible the interferometric analysis, but must be removed before the data can be transformed to a different coordinate system. Effectively, the phase centers for three linearly independent measurements must be shifted, through analysis of the data, so they coincide. This is achieved by time shifting the data appropriately. The time shifts can be worked from the interferometric results in Fig. 5.15. Briefly, the electric field data obtained from the measurement between sensors 1 and 2 is delayed by half of the correlation time measured along the stream (top panel of Fig. 5.15), and the electric field data obtained for the measurement between sensors 1 and 5 is delayed by half of the correlation time measured across the stream (center panel of Fig. 5.15). The third electric field component, the measurement between sensors 3 and 4, is not shifted since the other two measurements phase centers have been aligned with its phase center through this process.

The most useful coordinate system to transform the electric field into is the magnetic coordinate system previously defined in Fig. 5.17 and used throughout this thesis. Electric field data which have been transformed to this geomagnetically aligned coordinate system are presented in Fig. 5.20. The E_x direction is perpendicular to the geomagnetic field in the approximate direction of the barium stream

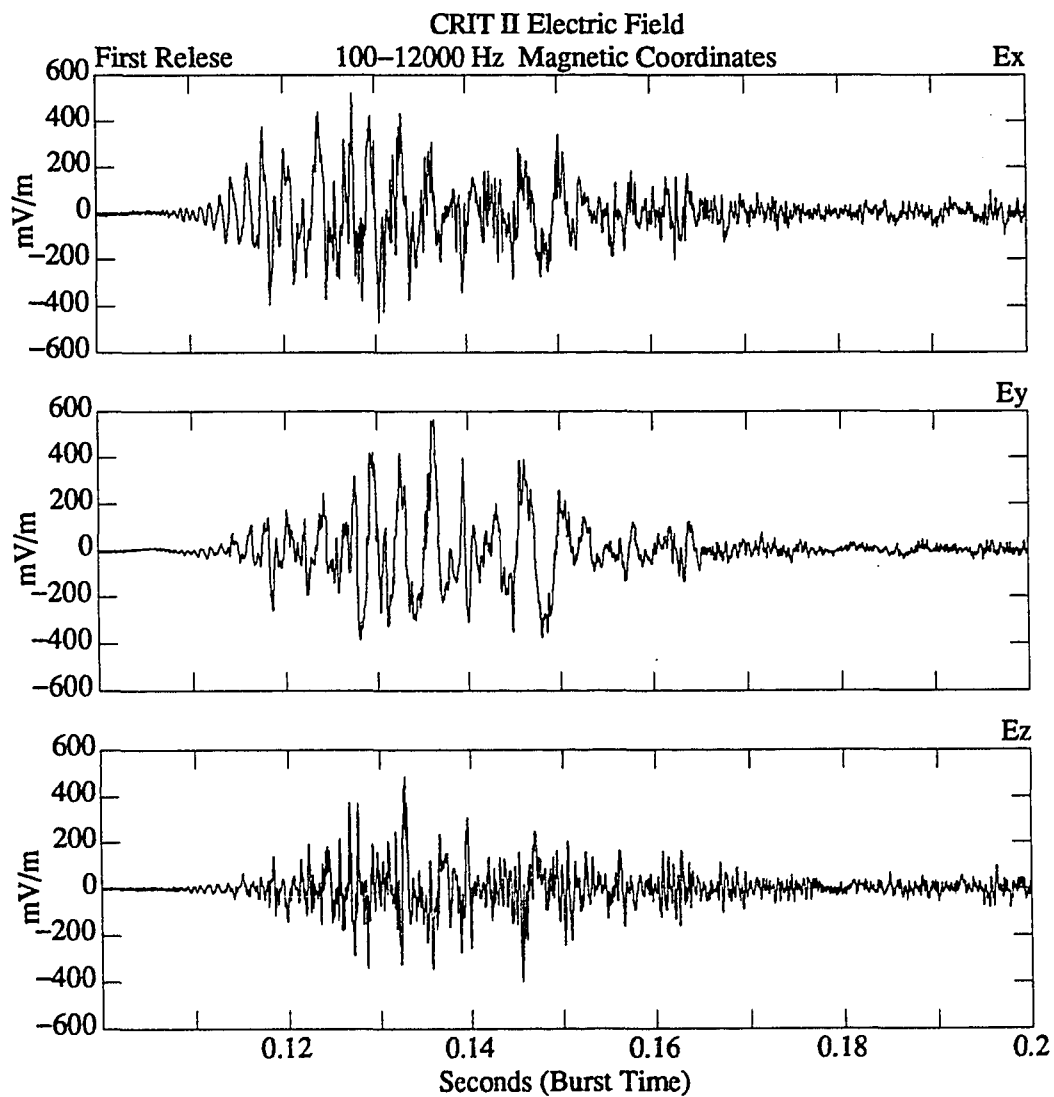


Figure 5.20: Electric field data in geomagnetic aligned coordinates from the first release of CRIT II.

and E_z is anti-parallel to the geomagnetic field. The results of the rotations are not as clean as hoped for, but they do suggest that the electric fields of these waves are mainly perpendicular to the magnetic field and aligned in the direction of the neutral stream projected perpendicular to $\vec{B}_0$.

An interesting result of this analysis and the interferometric analysis is the relationship between $\vec{E}$ and $\vec{k}$. The angle between these two vectors is large $\sim 57^\circ$ implying that this wave mode is a hybrid between transverse (electromagnetic, $\vec{k} \perp \vec{E}$) and longitudinal (electrostatic, $\vec{k} \parallel \vec{E}$) wave modes. Generally, the transverse modes are much faster than the longitudinal wave modes and it is not obvious what type of transverse mode would be slow enough to form this hybrid. From an intuitive argument, the observed wave mode makes sense. Firstly the waves traveling with the ions will be able to take energy out of the ions over a longer period of time and will grow larger than those waves which do not. Therefore, it is reasonable to expect the wave $\vec{k}$ to align with the ion motion. Secondly the high conductivity along the geomagnetic field makes it impossible to maintain an electric field along $\vec{B}_0$. Therefore, the parallel wave field of a purely electrostatic wave is effectively shorted out, and probably results in large field aligned currents. Thirdly the alignment of the electric field in the direction the barium stream, but perpendicular to $\vec{B}_0$, is the most efficient direction for slowing barium ions or for taking energy from the barium stream. Finally, there is almost no component of $\delta\vec{B}$ parallel to the beam. From $\nabla \cdot \vec{B} = 0$ it is clear that the electromagnetic component of this process travels parallel to the barium beam.

Rather than seeking slow electromagnetic modes, we argue that even the wave observations correspond to a moving current system in the beam frame. Not surprisingly these are phase locked with the electric fields of the primary wave which are perpendicular to $\vec{B}_0$. That is we suggest that an electrostatic wave has a phase

velocity equal to the projection of the beam velocity perpendicular to $\vec{B}_0$. This wave is of large amplitude ($\vec{k} \cdot \vec{E} \gg k_B T_e / e$) and may affect the local plasma density in several ways. The plasma will be organized by the wave itself as in any electrostatic wave. Second, the associated parallel electric field undoubtedly heats the electrons. These hot electrons may ionize neutral barium on the same spatial scales. The phase relationship between $\delta \vec{E}$ and δn could be quite complicated. Indeed we find that the same spatial scales exist in δn and $\delta \vec{E}$ but that the phase relationship varies. A linear theory of electrostatic waves precludes a $\delta \vec{B}$ but given the large localized barium densities and perpendicular electric fields it is hardly surprising that we experimentally detect strong $\delta \vec{B}$!

5.3 Ionization Front Overview

In this section, we will bring together the *in situ* density, magnetic field, and electric field measurements of the previous sections and from Chapter 4. We make this comparison during the ionization front of the the first release of CRIT II by plotting data from the plasma frequency probe, the ac-coupled magnetometer, and the electric field detectors in the same figures. All of the data have been similarly filtered so that the highest frequency component is 600 Hz, which is the limit of the magnetic field measurements. In addition, we have removed the phase delays in the magnetometer data that were introduced by the telemetry filter. This four pole filter introduced a very significant phase lag ($\approx 90^\circ$) at the primary frequency observed in the data (≈ 300 Hz) and 180° near the cutoff frequency (600 Hz). The phase delay of both the PFP and the electric field data is negligible for frequencies under 600 Hz.

5.3.1 Comparisons of Density, J_z , and E_x Data

In Fig. 5.21 we present the plasma density, the J_z or field aligned current, and the E_x component of the electric field in magnetic coordinates. First, consider the relationship between the current and the electric field. Lines have been drawn between corresponding maxima and minima which clearly show that J_z is 180° out of phase with E_x throughout the ionization front. The one exception occurs at the time of the flash (approximately 0.135 s) observed by the photometer when there is a 90° phase difference. The phase of the density fluctuations relative to the field aligned current changes through the ionization front. At first the current is in phase with the density structures. This occurs during the growth phase of the electric field and before the bulk of the barium arrives. After this period, the phase switches to a 180° difference through the saturation regime. After the flash and a localized density depletion the two signals are in phase again.

The interferometric analysis of Section 5.2.2 showed that the wave vector for the electric field was aligned with the barium stream and that $\vec{E}$ was largely perpendicular to $\vec{B}_0$. If the E_x field were a component of an electrostatic wave, $\vec{k} \parallel \vec{E}$, then a large geomagnetic aligned component of the electric field would exist, and this field, by observation, would be 180° degrees out of phase with the field aligned current. The 180° phase differences between j_z and the proposed E_z imply that such an electrostatic wave would be acting as a source of power or as a generator. In reality, the high conductivity along $\vec{B}_0$ essentially shorts out the generator field and E_z is actually much smaller than the field required to make $\vec{E} \parallel \vec{k}$. We can estimate a magnitude for E_z by balancing the energy produced by this generator, $j_z E_z$, against the energy lost by the barium ions through slowing, since they are the ultimate source of energy for the waves. The energy density in the newly created ions, assuming n_o ions are produced in 0.02s, is about $18 \mu\text{Jm}^{-3}$. If 50% of this en-

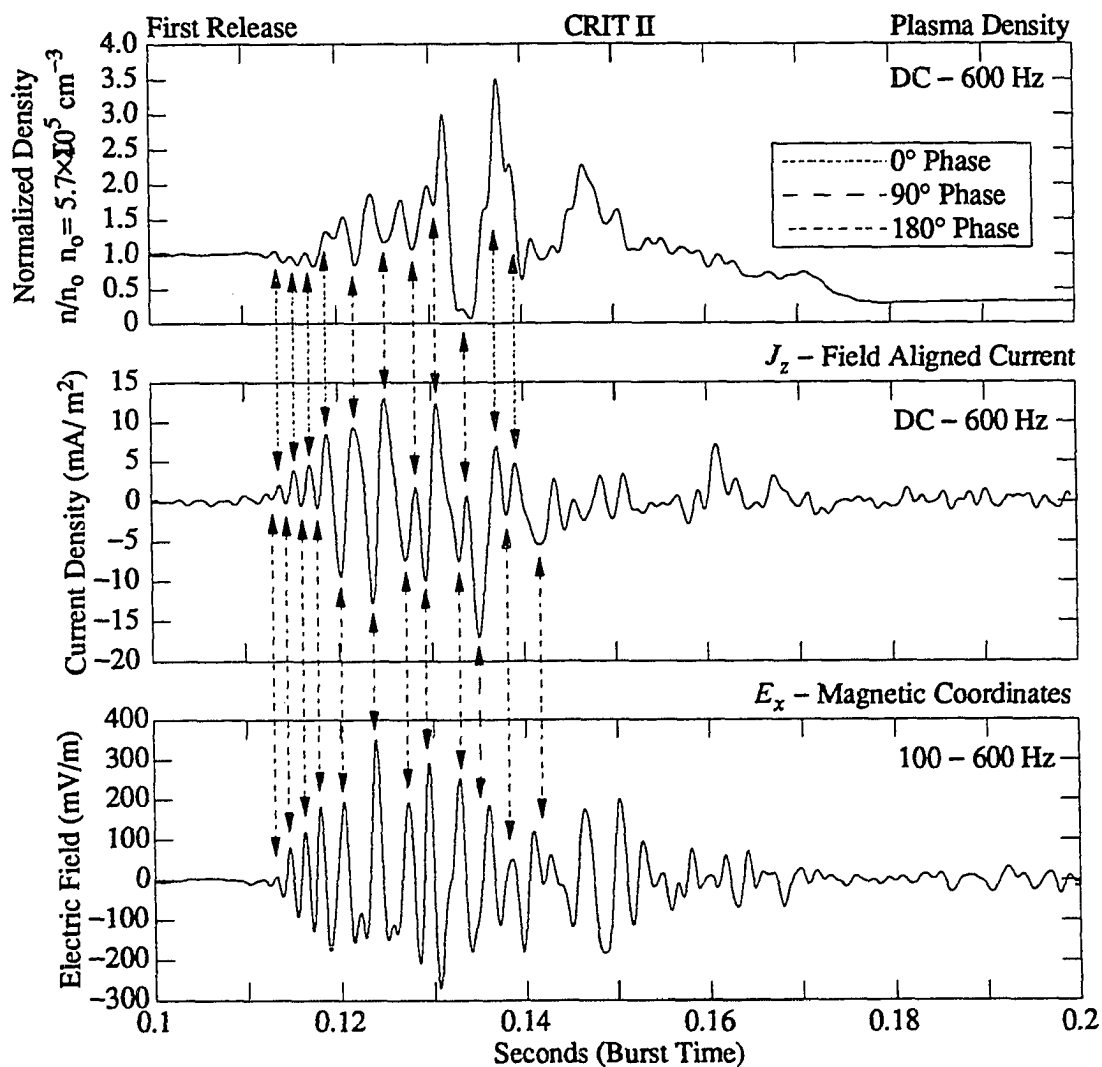


Figure 5.21: A comparison between plasma density and J_z and E_x in geomagnetic aligned coordinates. Data have been similarly filtered.

ergy is dissipated, which we estimate from Fig. 5.19, by driving the generator waves over this period of time, then the required parallel electric field, E_z , is ≈ 40 mv/m. This value may seem large, but it is clearly consistent with the rotated E_z electric field of Fig. 5.20 and only requires the total field to be 6° off perpendicular to $\vec{B}_0$.

5.3.2 Comparisons of Density, J_y , and E_y Data

In Fig. 5.22 we present the plasma density, the J_y or cross stream current, and the E_y component of the electric field in magnetic coordinates. Consider the relationship between the current and the electric field. Once again lines have been drawn between corresponding maxima and minima which show that J_y is $\sim 90^\circ$ out of phase with E_y until 0.135 s, or the time of the flash. After this period the phase relationship between J_y and E_y seems to vary from being in phase to being 180° out of phase. The phase of the density fluctuations relative to the cross stream current varies through the ionization front and no clear pattern can be deciphered.

Since J_y and E_y are approximately 90° out of phase, very little power, if any, is dissipated by these cross stream oscillations. An analogy is the relationship between $\vec{E}$ and $\vec{J}$ in a capacitor. The energy is stored in the internal electric fields and given back within each cycle of the wave.

5.4 Lower Hybrid Waves

Lower hybrid waves are thought to play a crucial role in heating electrons in the CIV microprocess [Goertz *et al.*, 1990; Galeev and Chabibrachmanov, 1983; Raadu, 1978]. For the conditions of the CRIT II experiment, the lower hybrid frequency would be 6.9 kHz for a pure oxygen plasma and 2.4 khz for a barium plasma. Because of the low frequencies we have detected the question naturally arises as to whether the

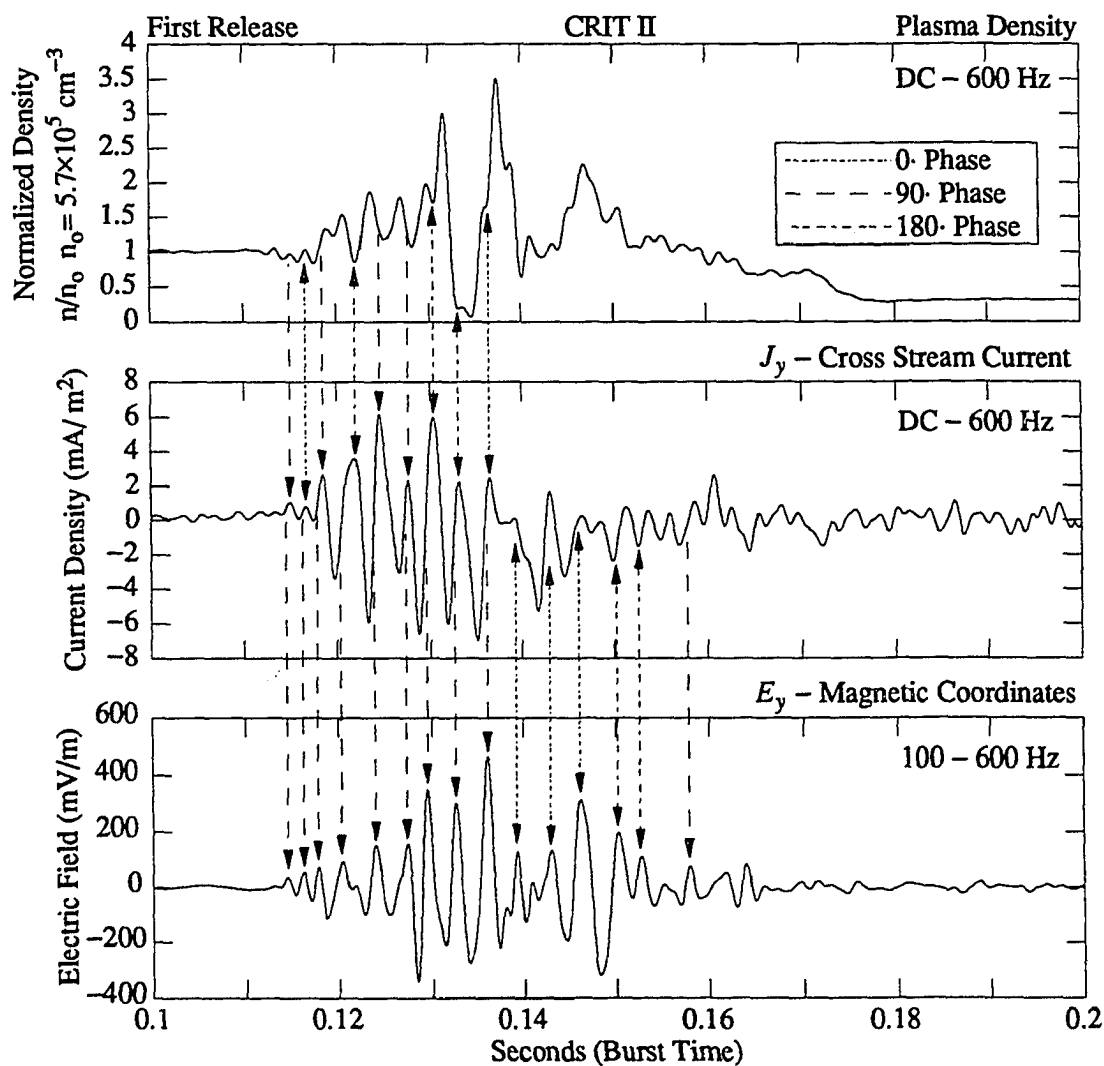


Figure 5.22: A comparison between plasma density and J_y and E_y in geomagnetic aligned coordinates. Data have been similarly filtered.

waves observed during the passage of the ionization front are actually lower hybrid waves generated by the streaming barium ions or some other mode. In a simplified way out of the dilemma, one might suggest that lower hybrid waves are Doppler shifted by the barium beam to the frequency range observed, but, as we shall see this simple explanation does not hold up to analysis. Another experimental result, that $\vec{k}$ is not parallel to $\vec{E}$, also seems to rule out the standard lower hybrid mode and other pure electrostatic modes where $\vec{k} \parallel \vec{E}$ and both are nearly perpendicular to $\vec{B}_0$. To illustrate the problems with existing theory, we will compare the results of our electric field analysis with the standard dispersion relation used in CIV theories as an initial postulate. Since more modes are described in this dispersion relation than just lower hybrid waves, we gain some insight into observed waves by examining these other modes which are possibly related to the observed waves.

5.4.1 Dispersion Relation

We begin by presenting the fluid dispersion relation for a cold oxygen plasma with a collisional drag, ν_e , on the electrons that is destabilized by an unmagnetized barium ion beam.

$$\begin{aligned}
 1 = & \frac{\omega_{pe}^2(\omega + i\nu_e)}{\omega[(\omega + i\nu_e)^2 - \Omega_e^2]} \left[1 - \frac{\Omega_e^2}{(\omega + i\nu_e)^2} \cos^2 \theta \right] \\
 & + \frac{(1 - \alpha)\omega_{pO+}^2}{(\omega^2 - \Omega_{O+}^2)} \left[1 - \frac{\Omega_{O+}^2}{\omega^2} \cos^2 \theta \right] \\
 & + \frac{\alpha\omega_{pBa+}^2}{(\omega - kV_o \cos \phi)^2}
 \end{aligned} \tag{5.6}$$

In this equation $\omega_{pe,pO+,pBa+}$ are the plasma frequencies for electrons, oxygen ions, and barium ions respectively, and where $\Omega_{e,O+}$ are the gyro frequencies for electrons and oxygen ions. The quantity α is the fraction of barium ion density to electron

density ($\alpha = n_{\text{Ba}^+}/n_e$), θ is the angle between the wave vector, $k = |\vec{k}|$, and the magnetic field, and ϕ is the angle between the wave vector and the barium stream. The velocity of the barium beam is given by V_o and the fluctuating quantities vary as $\exp i(\vec{k} \cdot \vec{r} - \omega t)$. Implicit in the dispersion relation of (5.6) is the assumptions that $\vec{k} \parallel \vec{E}$ and $\delta \vec{B}$. Therefore, we do not expect this dispersion relation to fully describe the observed waves within the ionization front. Equation (5.6) is written in the reference frame of the background plasma which is essentially identical to the reference frame of the main payload. All Doppler shifting of waves relative to the main payload reference frame are included. Equation (5.6) is written in the form of susceptibilities for, electrons (first term), oxygen ions (second term), and the barium beam (third term).

The standard dispersion relation [Papadopoulos, 1989] used when considering lower hybrid waves in CIV can be found from (5.6) by neglecting collisions ($\nu_e = 0$) and assuming Ω_{O^+} and Ω_{Ba^+} are small compared to $\omega_{p\text{O}^+}$, $\omega_{p\text{Ba}^+}$, and ω which in turn are small compared to ω_{pe} and Ω_e .

$$\frac{\alpha \omega_{lh, \text{Ba}^+}}{(\omega - k V_o \cos \phi)^2} + \frac{\omega_{lh, \text{O}^+}}{\omega^2} \left[\frac{m_{\text{O}^+}}{m_e} \cos^2 \theta + (1 - \alpha) \right] = 1 \quad (5.7)$$

All quantities have been previously described for (5.6) except for the lower hybrid frequency for a single component plasma of density n which is given by

$$\omega_{lh, \text{O}^+, \text{Ba}^+} = \frac{\Omega_e^2 \omega_{p\text{O}^+, p\text{Ba}^+}^2}{\Omega_e^2 + \omega_{pe}^2} \quad (5.8)$$

where $\omega_{p\text{O}^+, p\text{Ba}^+, pe}$ is the plasma frequency for ions and electrons, respectively. We will first examine this simpler dispersion relation and then return to the more complex dispersion relation (5.6).

This dispersion relation (5.7) is fourth order in ω , and the roots of (5.7) can

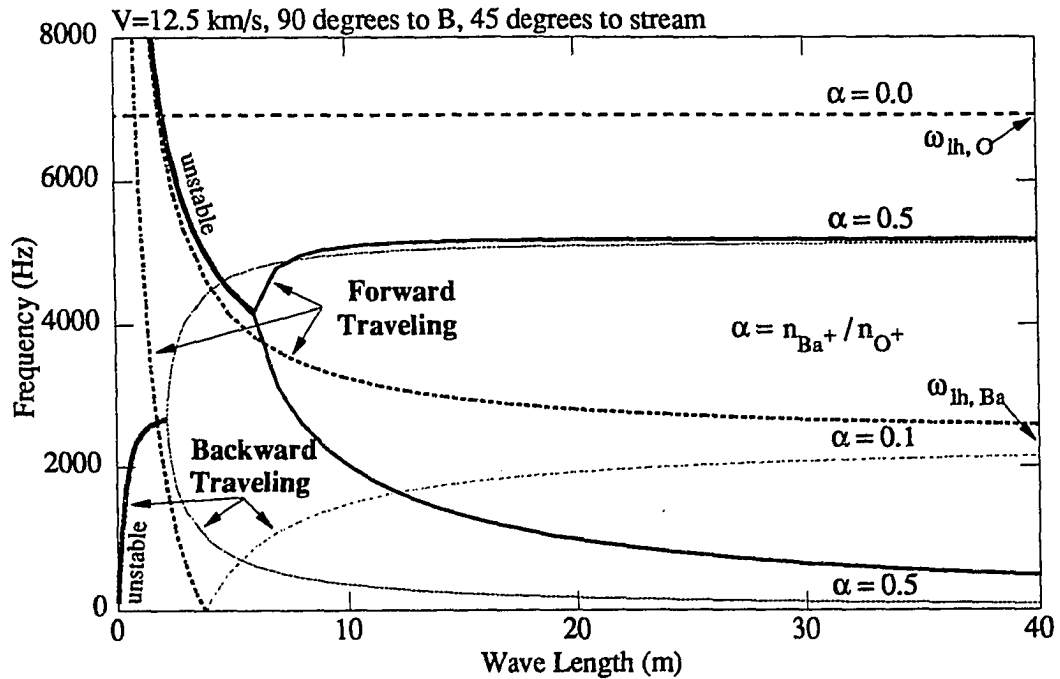


Figure 5.23: Roots of the dispersion relation for a magnetized barium beam in a cold, magnetized oxygen plasma.

be real or complex. The real roots describe characteristic frequencies for waves of the plasma. Since the coefficients of (5.7) are real, complex roots will occur in complex conjugate pairs describing both a growing and a damped wave at the same frequency. Negative values for ω describe backward traveling waves; i.e. waves traveling in the opposite direction to the barium beam. At most two unstable wave modes can be described by the dispersion relation, while up to four real frequencies can exist for a given k . The roots of (5.7) are plotted in Fig. 5.23 for no barium beam ($\alpha = 0$), a 12.5 km/s barium beam with half the oxygen density ($\alpha = 0.5$), and a 12 km/s barium beam with no oxygen ($\alpha = 1$). The angle between the wave vector, $\vec{k}$, and the barium stream is 45° and $\vec{k}$ is perpendicular to $\vec{B}_0$. For $\alpha = 1$ or $\alpha = 0.0$ equation (5.7) degenerates to a second order equation in ω that has real roots for $\vec{k} \perp \vec{B}_0$. When $\alpha = 0$ both a forward and a backward traveling wave at the oxygen lower hybrid frequency are found and neither wave is Doppler shifted,

since both waves are in the same reference frame as the payload. For $\alpha = 1$, again two waves exist that have frequencies at the barium lower hybrid frequency but in the reference frame of the streaming barium ions. Since this reference frame is moving with respect to the payload, the barium lower hybrid waves are Doppler shifted. The backward traveling wave is Doppler shifted such that at $\lambda \approx 4$ m the wave appears to stand still. For shorter wavelengths, this backward traveling wave becomes a forward traveling wave in the payload reference frame. For $\alpha = 0.5$ and for short wave lengths ($\lambda < 3$ m) two destabilized wave modes exist. These are shown in Fig. 5.23 by heavy lines. The forward traveling unstable waves have frequencies greater than 4 kHz, much too high to explain our observations. This branch has been studied extensively using kinetic theory by *McBride et al.* [1972] and *Scales* [1989] and is typically the lower hybrid mode referred to by CIV authors. The observed waves are not explained by this particular lower hybrid branch of the dispersion relation. The other unstable branch of backwards traveling waves have frequencies below 3 kHz and could be candidates to explain the observations, except the wave lengths are too short. The waves during the growth and saturation phases have frequencies between 1.6 kHz and 300 Hz, in agreement with this branch, but the observed wave lengths (~ 25 m) are too long. For wave lengths greater than about 8 m there are four real roots to the dispersion relation of (5.7). These four characteristic frequencies describe a pair waves with frequencies near the lower hybrid frequency for a barium-oxygen plasma and a pair of lower frequency waves. The lower frequency stable branches are the most likely candidates for our waves in this dispersion relation. In the next two sections we explore both the unstable, short wave length, low-frequency modes and the stable, longer wave length, low-frequency modes in greater detail.

5.4.2 Unstable Low-Frequency Modes

The characteristics of the unstable low-frequency mode verses wavelength are presented in Fig. 5.24 for different values of α . From top to bottom Fig. 5.24 presents plots of the frequency, growth rate, phase velocity, and the group velocity. If the frequency of the waves observed during either the saturation or growth phase are compared to this mode then the implied wave lengths are less than 100 cm. This result presents a problem from an observational stand point. Since the sensor spacings of the electric field experiment are longer than this (375 cm, 179 cm, 555 cm, etc), we would expect strong spatial filtering of the data for each of the measurements. This would make the electric field strengths much higher than those reported which are already quite large. In addition, the phase and group velocity of these waves is less than 200 m/s. The interferometric analysis of the data indicates the observed waves have much higher velocities and much longer wavelengths. Furthermore it is difficult to see how such slow waves could interact well with the beam. We therefore conclude that this mode is not related to the observations.

5.4.3 Low-Frequency Long-Wavelength Modes

There are two stable, low-frequency branches of the dispersion relation (5.7) for the parameters of the CRIT II experiment. One mode describes a forward traveling wave and the other mode describes backward traveling wave. Characteristics of these two modes are presented in Fig. 5.25. The frequency of these two modes versus wave length is presented in the top panel for variations of the barium ion beam density. Phase and group velocities are presented in the second and third panels. The angle of the barium stream with respect to the geomagnetic field is 57° for these plots, the barium stream velocity is 12.5 km/s, and we have considered a wave mode with $\vec{k}$ parallel to the beam. We have also plotted the results of the

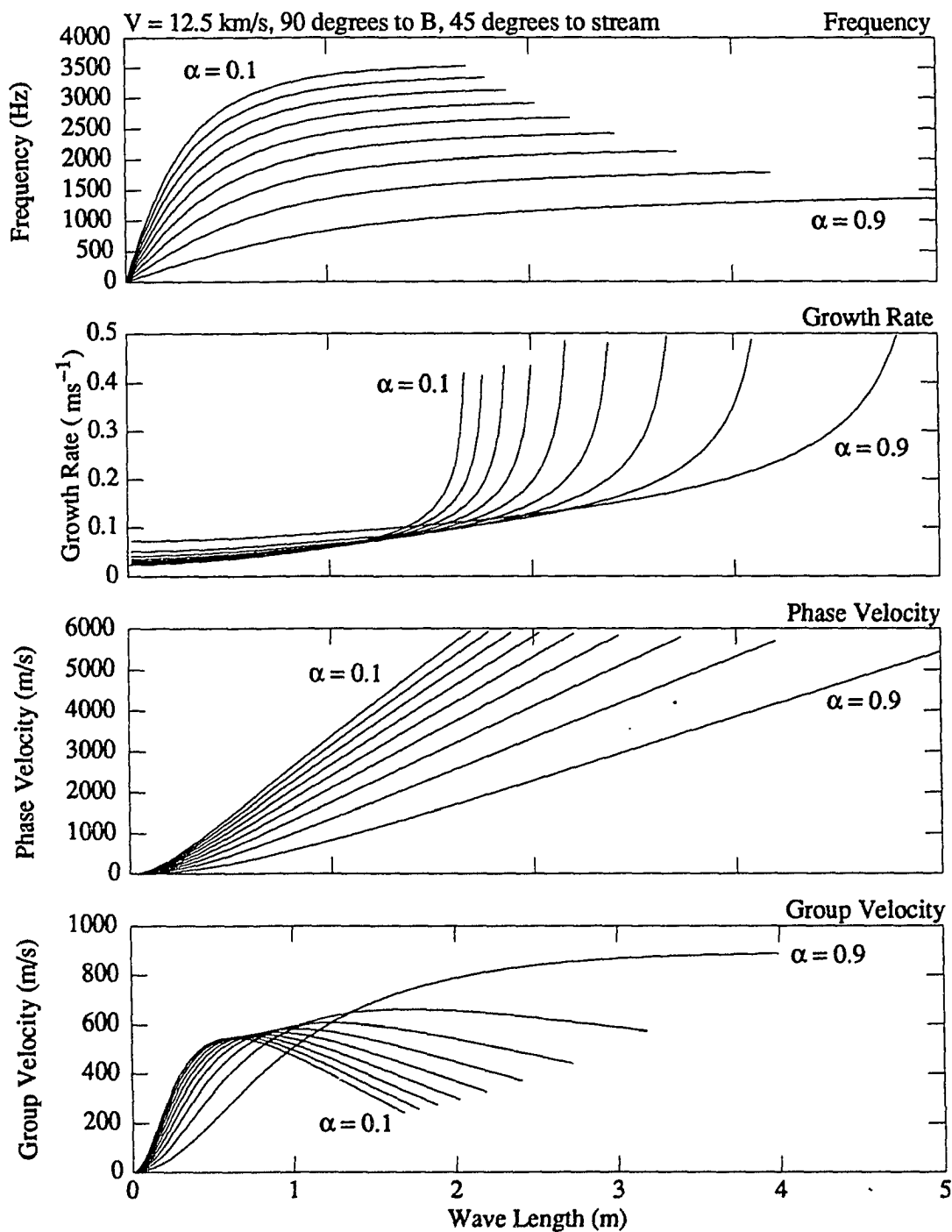


Figure 5.24: Unstable low frequency roots of the dispersion relation for an magnetized barium beam in a cold, magnetized oxygen plasma. The frequency, growth rate, phase velocity, and group velocity for different barium ion densities are plotted.

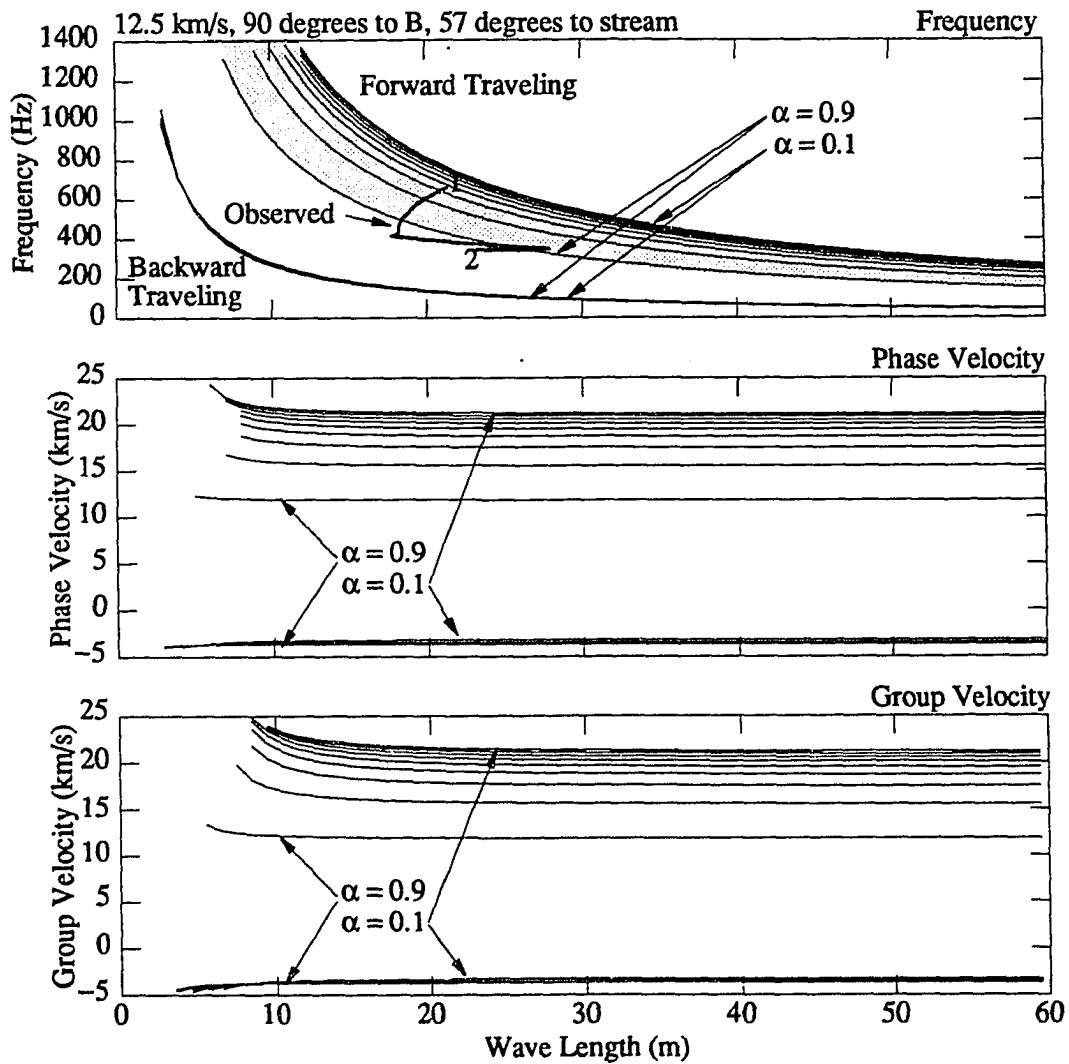


Figure 5.25: Stable low frequency roots of the dispersion relation for an magnetized barium beam in a cold, magnetized oxygen plasma. The frequency, phase velocity, and group velocity for different barium ion densities are plotted.

interferometry analysis in the same figure. The observed data start at 1 and end at 2 corresponding to the shaded period of time 0.115-0.125 s in Fig. 5.18. If we study electrostatic modes with phase velocities perpendicular to $\vec{B}_0$ (which might have a projected velocity in resonance with the beam) the results are similar.

The observed waves are forward traveling, as determined by the interferometry analysis of Section 5.2.2 and reside close to the forward traveling wave dispersion curves (shaded region of Fig. 5.25) in frequency-wavelength space. We propose that the observed waves are related to this mode since there are many features in agreement with the observations. The measured waves are created by a barium beam that is increasing in barium density with time. This will result, for relatively constant wavelength, in a dispersal towards lower frequencies according to the dispersion relation. The observed data are in general agreement with this effect, although there is no explanation for why the waves tend towards constant wavelength. The phase or group velocity of this mode is slightly greater, but within a factor of two of what is observed (Fig. 5.19). This mode is stable in the collisionless plasma model of (5.7), but if we add collisions (see (5.6)) then this mode is destabilized. This long-wavelength low-frequency mode can be described as a negative energy mode that is destabilized when a mechanism for energy dissipation exists.

To show that this mode is unstable when ν_e is included we start with (5.6) and derive a simplified form of the dispersion relation. The frequency of this mode is very near $\omega_0 = kV_0 \cos \phi$ or the so called beam mode since the waves are in resonance with the barium ion beam velocity. Therefore, the dominate terms in (5.6) for $\theta < 90^\circ$ are the parallel (to $\vec{B}_0$) electron and the barium beam susceptibilities.

$$1 = \frac{\omega_{pe}^2 \cos^2 \theta}{\omega(\omega + i\nu_e)} + \frac{\alpha\omega_{pBa}^2}{(\omega - \omega_0)^2} \quad (5.9)$$

Once again, $\omega_{pe,pBa+}$ are the plasma frequencies for electrons and barium ions, α is the fraction of barium ions, and θ is the angle between the wave vector $\vec{k}$ and $\vec{B}_0$. We assume a frequency $\omega = \omega_o + i\delta$ where $\omega_o \gg \delta$ and substitute into (5.9). Solving for δ we get

$$\delta = \pm \sqrt{\frac{\alpha m_e}{m_{Ba+}} \frac{\omega_o}{\cos \theta} \left(1 + i \frac{\nu_e}{\omega_o}\right)^{1/2}} \quad (5.10)$$

where we have approximated that $\omega_{pe}^2 \cos^2 \theta \gg \omega_o(\omega_o + i\nu_e)$. If we make the additional assumption that the electrons are very collisional on the time scale of the wave ($\nu_e > \omega_o$) then we can write that $\omega = \omega_o - \gamma + i\gamma$ where the growth rate γ is defined as

$$\gamma = \cos^{-1} \theta \sqrt{\frac{\alpha \omega_o \nu_e}{2} \frac{m_e}{m_{Ba+}}} \quad (5.11)$$

For CRIT II parameters, the growth rate is in the range of $\gamma \approx 0.002\omega_o$ for $\theta \approx 45^\circ$, but as θ approaches 90° the growth rate increases. For modes with wave vectors close to perpendicular the reduced dispersion relation given by (5.9) is invalid and the whole dispersion relation of (5.6) must be solved. Numerical solutions show that growth rates as large as $\gamma \approx 0.085\omega_o$ for $\theta \simeq 90^\circ$ exist and that they are insensitive to the collision frequency ν_e . This is approximately the growth rate of the waves observed during the growth phase of the first release as described in Section 5.2.1. Furthermore a high collision frequency model implies that the wave electric field should be in phase with the electron velocity, a result in agreement with the data in Fig. 5.21.

There remains the fundamental difference between the data and this model concerning the direction of the wave vector $\vec{k}$. The interferometric analysis lead to the conclusion that the $\vec{k}$ of the waves were aligned with the neutral beam and hence at 57° to the magnetic field while the observations show that $\vec{E}$ is perpendicular to $\vec{B}_0$ and that a strong fluctuating magnetic field exists which also has $\vec{k}$ parallel to

the beam. The dispersion relations we have considered only describe electrostatic waves ($\vec{k} \parallel \vec{E}$) and to more accurately describe the observations an electromagnetic correction to the dispersion relation must be made. Nevertheless, we have been highly successful at matching the characteristics of the observed waves with this electrostatic model suggesting that the electrostatic mode of this mixed mode wave dominates the waves characteristics. A more detailed analysis is called for that includes the electromagnetic corrections to further explain the CRIT II observations. Also we need to argue that some effective collision frequency exists in order to raise $\nu_e > \omega_o$ since classical collisions are such that $\nu_e \approx 10^3 \approx 0.5\omega_o$. In this regard we note that the collision process could include the ionizing effects of e^- on barium, not just wave particle effects.

5.5 Summary

In this chapter we have reviewed the data during the ionization front which is also the period when barium ionization is observed and when CIV could be active. Strongly correlated electric and magnetic field fluctuations were observed during the front. In addition, density perturbations were observed on the same time scales, but the phase relationship with the key parameters are not systematic throughout the front.

A rotation of the electric field into a geomagnetic aligned coordinate system showed that these electric field fluctuations are nearly perpendicular to $\vec{B}_o$ and in the plane containing $\vec{B}_o$ and the beam velocity. The electric fields in these waves are strong enough to slow and scatter the newly created barium ions which is implied to occur by the energetic particle detector data. An interferometric analysis of the electric field structures showed that they were traveling slightly slower than

neutral barium stream. If we make the very reasonable assumption that they are traveling with the bulk of the barium ion motion then the neutrals must have given up about 50% of their kinetic energy. Presumably this energy has gone into these electric and magnetic field oscillations and from there into heating of electrons.

The corresponding magnetic field oscillations were nearly perpendicular to the barium beam which was about 57° from $\vec{B}_0$. They can be interpreted as a current system moving with the barium beam that is driven by the electric fields. Large currents are implied by the magnetic field oscillations that are along $\vec{B}_0$ and are 180° out of phase with the primary wave electric field. If this primary wave field is about 5° off of perpendicular to $\vec{B}_0$ then the power transfer implied by $J_{\parallel}E_{\parallel}$ is in agreement with the implied slowing of the barium ions.

An analysis of the fluid dispersion relationship for an unmagnetized barium beam in a cold, collisional plasma has revealed two points. First, the mode which is commonly used in CIV theories when determining the energy transfer of the neutral stream to electrons, the lower hybrid mode, is not the dominate wave mode observed *in situ* during the CRIT II releases. Second, there exists a mode which is resonate with the beam velocity and is only destabilized in the presence of an energy dissipation mechanism, electron collisions, that agrees approximately in frequency, wavelength, and growth rate with the observed waves of the ionization front. This wave mode is purely electrostatic ($\vec{k} \parallel \vec{E}$), but the observations clearly point out that the observed waves are not purely electrostatic since the angle between $\vec{k}$ and $\vec{E}$ is about 57° . The resulting $\vec{k}$ vector as revealed by interferometry and from $\nabla \cdot \vec{B} = 0$ is parallel to the stream. We therefore have proposed a hybrid electromagnetic-electrostatic wave whose primary characteristics, frequency, wavelength, and growth rate are dominated by the electrostatic component. In our model the barium beam generates a large amplitude wave much like a boat on water. The wave phase

velocity perpendicular to $\vec{B}_0$ locks in with the stream and creates corresponding nearly field aligned structures in the barium plasma. These structures are so intense and localized that net currents exist perpendicular and parallel to $\vec{B}_0$ which create fluctuating magnetic fields which must be transverse to the source velocity, that is, to the beam.

With respect to CIV, this proposed wave mode is consistent with the requirements of the CIV microprocess. Considering energy balance, all that is required for is a plasma collective effect that will take energy out of a beam of newly created ions and that will heat electrons. The observations and the proposed wave mode are consistent with this requirement, but there is much that is still unknown about this mode. The efficiency of this whole process has not been determined except that we believe about 50% of the initial energy of the barium ions is removed by this wave mode. The amount of energy received by the electrons has not yet been determined from the particle measurements and may be impossible to estimate given that the detector saturated. Finally, the observations are most like the homogeneous model of [Raadu, 1978] where the ions are slowed by an unspecified force and the instability is driven by the ions. In our observations and model the wave electric fields slow the ions, and the instability, which heats the electrons parallel to $\vec{B}_0$ is also driven by streaming newborn ions.

Chapter 6

Summary and Suggested Research Directions

'It seems a shame,' the Walrus said,
 'To play them such a trick.
After we've brought them out so far,
 And made them trot so quick!'
The carpenter said nothing but
 'The butter's spread too thick!'
—*Lewis Carroll, The Walrus and the Carpenter, Through the Looking-
Glass and What Alice Found There*

The critical ionization velocity effect may play a major role in the formation, shaping, and distribution of matter in the universe. The basic theory as proposed by Hannes Alfvén back in the 1950s is based on sound principals of energy balance; if there is sufficient kinetic energy in the directed motion of a neutral cloud for the entire cloud to ionize then nature may find a way to release the kinetic energy for ionization. The magnetized plasma through which the neutral cloud passes plays the part of a catalyst, promoting a reaction that would not otherwise occur. For the past 35 years the question of whether a magnetized plasma can act as such a catalyst for the ionization of a streaming neutral cloud has been investigated. Laboratory

results have confirmed CIV and now the investigation of CIV has returned to the original setting in which it was proposed, space. In this thesis we report on the most complete set of diagnostic experiments conducted to date in an environment which is potentially conducive to CIV in space

6.1 Summary

In this thesis we have presented data, analysis, and reasonable theory for a set of experiments named CRIT I & II (for CRITICAL Ionization Tests). The experiments were carried out using very similar payloads launched on two sounding rockets in 1986 and 1989, respectively. These were two of the most complex sounding rocket payloads launched up to those times. The object was to detect CIV in space and to interpret the data in the light of existing theories for the processes. The data from the second rocket, CRIT II, was much richer, and was therefore the focus of this thesis. In this summary we will review evidence for a CIV signature in our data and comment on the contributions and relevance of this thesis research.

The premise of CIV is that it will create new plasma, therefore the first and primary signature of CIV that we hoped to detect was copious amounts of ionized barium. The prevailing expectation was that CIV would be the unmistakable conclusion given the amount of ionization that might occur. *In situ* density probes and ground based optical observations were therefore planned to detect the barium ions. The second signature which we hoped to detect were hot electrons that would have energies sufficient to ionize barium since these hot electrons are the predicted result of the CIV process. Energetic electron particle detectors were included in the *in situ* instrumentation to detect these signatures. The third signature was a slowing and scattering of the newly created barium ions. This would indicate that

the energy of the neutral stream was being tapped in the first step of the CIV microprocess. Therefore, energetic ion particle detectors were part of the *in situ* diagnostics. The published theories of CIV to date have proposed that lower hybrid waves should be observed, since through these waves energy is transported out of the newborn ions and deposited into the thermal motion of electrons. Therefore the fourth signature, which we were led to expect from these theories, was lower hybrid waves, and the electric field experiment was well suited for detecting these waves. Another signature would be evidence of momentum coupling between the newly produced ions and the ambient ionosphere. Therefore, a subpayload was positioned outside of the barium stream in the ambient plasma to detect the response of the ambient plasma. Finally, sensitive magnetometers were included on the payloads to detect the expected current systems.

6.1.1 Ionization of Barium

We did not observe new plasma being produced in such quantities that CIV was clearly the only possible explanation, but we did see a sizable density enhancement. In this thesis we showed that this density enhancement could not have been produced near the explosion and then transported across the geomagnetic field to the observation point. In fact the observed internal electric field were so small that virtually no transport across the geomagnetic field would be expected. We also showed that the collisional force of the neutral barium stream acting on the background ionosphere could not plow the plasma together and thereby create the observed density enhancement. Finally, we attempted to sort out how much of the density enhancement resulted from CIV and how much resulted from expected collisional ionization, but such an analysis is problematical. The first problem is that the required cross sections for collisional ionization have not been measured

and estimates vary by two orders of magnitude. Another problem is properly accounting for transport of the barium ions along the geomagnetic field due to their injection velocity and across $\vec{B}_0$ due to gyro motion. We presented a set of approximate calculations based on a one dimensional simulation showing that the density enhancement observed during the first release can be completely explained by collisional ionization processes but only if the largest estimates of cross sections are used, which is probably unreasonable. More important, the density enhancement observed during the second release cannot be explained by collisional ionization, even with the largest estimates for the cross sections. This second release data may be the first unambiguous detection of CIV in space. It seems more likely that the smaller values for the cross section are correct and that some fraction of the density enhancement during the first release was also due to ionization by hot electrons, or CIV. There is evidence that CIV occurred in several 'flashes', but that this high ionization rate was not sustained in the first release. In summary, we are fairly certain that some portion of the density enhancements is due to the ionization of barium by CIV but accurate measurements of cross sections for classical ionization are needed and more detailed simulations of plasma motion are required to clarify this assertion.

6.1.2 Hot Electrons

Hot electrons were detected during both releases of CRIT II. The analysis of the data is not yet complete, but preliminary work by researchers at University of New Hampshire suggests there are sufficient hot electrons to explain the density enhancements by electron impact ionization of the neutral barium stream (*personal communication, Roy Torbert, 1991*). Definitive results are not to be expected from the electron particle detectors for two reasons. First, the detectors were saturated

by large fluxes of electrons such that at crucial times the data are corrupted. Second, the time resolution of the detectors is rather marginal for diagnosing the rapidly changing event. In this thesis we briefly showed that the potential across the short time scale electric field structures within the ionization front are large enough to accelerate electrons to the observed energies. Further quantitative analysis of the electron particle data is called for before definitive statements about hot CIV electrons and ionization rates can be made from the perspective of the particle data. This is clearly beyond the scope of the present thesis.

6.1.3 Slowing and Scattering of Barium Ions

There are several signatures in the ion particle data that indicate barium ions are scattered during the passage of the ionization front. One signature is associated with the clouds of barium ions that are periodically observed by the detectors at intervals separated by the barium gyro period and that have pitch angles nearly perpendicular to $\vec{B}_0$. Since the injection angle for newly created barium ions was 57° , significant scattering must have occurred. A second signature is that during the passage of the ionization front few ions are observed with trajectories aligned with the neutral barium stream but many are observed at other pitch angles. This is in contrast to the period of time after the ionization front when barium ions are observed with trajectories aligned with the barium beam and few are observed to be scattered. Both of these signatures imply that energy is being taken out of the newly created barium ions, which is essential for CIV to occur. Little has been done in the way of a quantitative analysis of the ion particle data and more analysis is needed. Again, like the energetic electron particle data, the time resolution is marginal for studying in detail the events of greatest interest within the ionization front.

In this thesis we have shown that the electric field data are consistent with the slowing of barium ions, and that the barium ions have given up approximately half of their kinetic energy. This assertion is based on the premise that the electric field structures are moving with the bulk of the barium ions. We showed through an interferometric analysis that these electric field structures were traveling slower than the neutral barium stream.

6.1.4 Waves in the Ionization Front

The dominant waves that were observed within the ionization fronts were not the classical lower hybrid mode. This is perhaps the most significant contribution of this thesis to the understanding of CIV in space since it was not the expected observation. The simplest description of the observed waves is that they are in resonance with the barium ion beam ($\omega = \vec{k} \cdot \vec{V}_{\text{beam}}$). The analysis of both the electric field data and the magnetic perturbation data indicated that the wave vector of the observed mode was aligned with the neutral stream and was thus at an angle of 57° to $\vec{B}_0$. The analysis also showed that the electric field within the waves was approximately perpendicular to $\vec{B}_0$, indicating that the waves were not a purely electrostatic mode. By hypothesizing an effective electron collision frequency in excess of classical values, we have found that a previously stable branch of the dispersion relation solution becomes unstable. This “low frequency” lower hybrid mode seems to explain many of our results and perhaps some laboratory results as well.

Although the observed waves were not the classical lower hybrid waves predicted by existing theory, this does not mean CIV could not have occurred. All that is required of the waves is that they take energy out of the newly created barium ions and transport it into a population of hot electrons. The data strongly suggest that

this is exactly what is occurring within the ionization front. Previous theoretical calculations for the efficiency of the CIV process relied on the growth and saturation of lower hybrid waves, but now these calculations need to be made using the observed wave mode. This type of analysis may be a key to explaining the relatively low yield of the CRIT experiments. In this thesis we have just scratched the surface of a theoretical understanding of the observed waves and more analysis is called for so that efficiency calculations for the CIV process can be made.

6.1.5 Momentum Coupling

The momentum of the newly created barium ions must be absorbed by the ambient plasma if the energy of the ions is to be tapped. A theorized mechanism for momentum coupling is a current system within the ionization front that is closed in the ambient plasma. The momentum of the new ions is thought to be transmitted to the ambient plasma by the $\vec{J} \times \vec{B}_0$ forces acting on such a current system. Part of such a current system in the ambient ionosphere was detected by the subpayload in the form of an Alfvén wave, but the polarization of the wave relative to the ionization front is currently unknown. Therefore, we are uncertain of the direction of the polarization current and the acceleration of the ambient plasma in the flux tube threading the ionization front. The orientation of the Alfvén wave relative to the ionization front can be determined given the available information and should be investigated in future research. Currents within the ionization front were detected in the correct direction to slow the newly created ions, but on two different time scales. The longer time scale currents are on the same time scale as the Alfvén wave front detected by the subpayload, but they appear to be too small to significantly slow any new born barium ions. The shorter time scale currents do not appear to close in the ambient ionosphere but are of approximately the correct magnitude.

We are still uncertain about this issue of momentum coupling within the context of the CRIT experiments.

6.2 Future Research Directions

There are many unanswered questions about the CRIT experiments and about CIV in space. Several of these questions have been alluded to in the previous section. For example: What is the source of the small scale structuring within the first release density enhancement? What percent of the density enhancements can be attributed to ionization of barium by CIV? How efficient are electrons heated by the newly proposed wave mode and what parameters determine the wavelength of maximum growth for the proposed wave mode? Why were the CRIT experiments relatively inefficient in terms of CIV? Given time and resources, some of these questions can be explored with the current data set.

In any future CIV experiments much more attention should be paid to details associated with determining the exact deployment of each section of the payload. A design requirement should be that the distance between each payload piece, instrumentation or chemical release, is to be known to within about 10 m. This is not the case with the CRIT experiments and has caused some ambiguities in the data analysis.

A second design requirement should be planned knowledge of the orientation of the instrumented payloads to within about 0.1° . This might be achieved through the use of star sensors instead of relying on attitude gyros, which seem prone to failure and lead to little confidence in the roll phase of the payload. A more accurate attitude determination would allow much more precise rotations of the various data. For instance, the small geomagnetically aligned electric field component could be

separated from the much larger perpendicular components.

Serious thought should be given to a payload, instrumented with electric field detectors, that is rapidly spinning and coneing and is within the path of the neutral stream. The dc offsets in the electric field data could be identified removed with the aid of such motion. This can be accomplished because the coneing and spinning payload effectively samples the dc offsets in three dimensional space whereas a payload that is only spinning samples the instrument offsets in the spin plane or two dimensional space. This would allow quasi-dc electric field measurements to be obtained in the ionization front since the offsets due to changes in contact potential could be removed from the data.

Some simple improvements to the instrumentation would yield a much better data set. For instance, a much better electric field interferometer should be included since this aspect of the CRIT II analysis has proven to be particularly useful in diagnosing the ionization front. Careful attention should be paid to the timing between measurements and the sampling rates channels making up the interferometer. The bandwidth of the dc magnetometer should be raised and possibly supplemented with a search coil magnetometer since so much useful data is available in the magnetic field signatures of CIV experiments. One might even consider making interferometric measurements of the magnetic field perturbations, which seem to be locked into the moving ion beam frame. Finally, it may prove feasible to include a second plasma frequency probe to make interferometric measurements of the density perturbations. In short, although CRIT was very successful, many improvements could be made in the instrumentation and execution of future tests for CIV in space.

Appendix A

Positional Data and Projections

A.1 CRIT I Deployment Results

The CRIT I rocket was launched from Wallops Island, VA May 13 1986 at 7h 40m 0.0s UT. Post flight analysis of the CRIT I flight revealed that friction in the gimbal bearings combined with a long slow motion of the rocket on the pad before launch caused the gyro to lose reference; that is, the gyro did not freely turn in its bearings, but turned with them. Also, during the ascent of the CRIT I vehicle, the gyro additionally slewed due to vibration. All combined, the errors added to about a 23° slip of the reference direction, and since the entire motion of the payload was referenced to this gyro, the experiment was deployed differently than planned. This problem was initially noticed during the comparison of attitude with radar measurements of the location of the main and subpayloads. This was later confirmed by examining the supposed attitude during the boost phase of the rocket. Using the methods previously described, the attitude of the main payload was eventually determined. Once the attitude of the main payload was known, the resulting deployment of the experiment could be calculated. Fig. A.1 summarizes

the positions of the payloads at the times of both releases. The locations of the payload have been projected in two perpendicular views. Data used to create Fig. A.1 are tabulated in this Appendix. The top projection in Fig. A.1 is parallel to the geomagnetic field and contains the vector between the main payload and the burst. From this projection we can see the injection angle was 47.6° for the first burst and 47.8° for the second. The first burst occurred 1.986 km and the second 4.344 km away from the main payload. The subpayload was not on the same field line as the main payload, and misplaced from its intended position. This is clearly observed in the lower projection in Fig. A.1. The lower projection is perpendicular to the local geomagnetic field and viewed as if looking down the magnetic field line. The field line threading the subpayload was clearly located outside the bulk of the neutral barium jet.

On the main payload, all booms and doors successfully deployed. The situation for the subpayload was not so fortunate since only two (#2 & #3) of the four booms tripped micro-switches signifying complete deployment. At least one of the booms, #1, hung up during deployment. An estimate of the booms resulting position, made by comparing $\mathbf{V} \times \vec{\mathbf{B}}_0$ measurements between the boom sets, placed the boom 66° up from the intended spin plane. Because the ends of these booms were heavily weighted the resulting moments of inertia were significantly different than planned. The payload was no longer symmetric and the spin axis of the subpayload was shifted significantly with respect to the body axis of the payload.

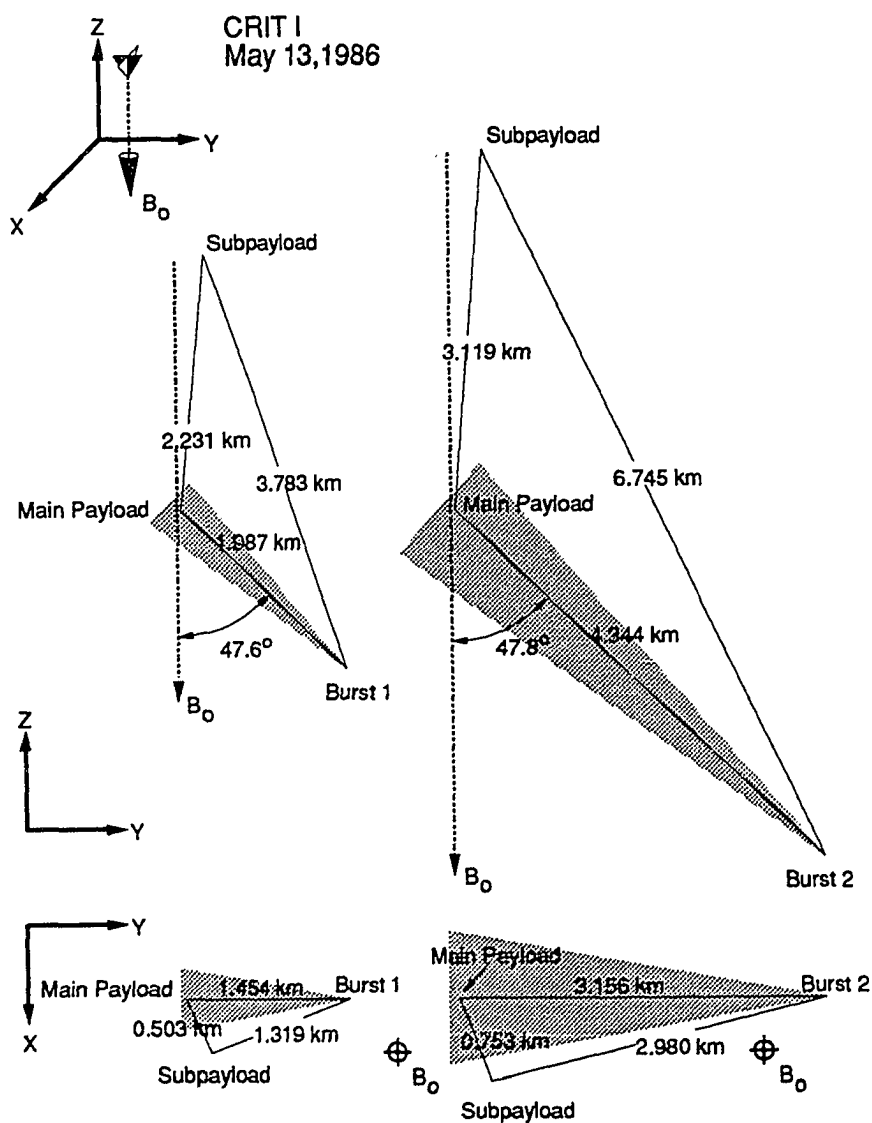


Figure A.1: Projection of locations, distances, and angles onto the Y-Z and X-Y planes for both barium releases during the CRIT I experiment. The coordinate system in which the projections are made is defined such that the Z direction is antiparallel to the geomagnetic field, the origin is located at the main payload, and the chemical cannister is located in the Y-Z plane. The figure is drawn to scale.

A.1.1 Release One of CRIT I

Table A.1: General information about CRIT I launch and first release

CRIT I Launch		
Date	May 13, 1986	(day 133)
Time	7:40:0.0	UT
Launch Site Location		
Geodetic latitude	37.83	degrees
Geodetic longitude	-75.49	degrees
Gyro Reference or Gyro Uncage		
Azimuth	173.0	degrees
Elevation	90.0	degrees
Attitude of Main Payload Spin Axis		
Pitch	-23.6	degrees
Yaw	-9.2	degrees
Attitude of Neutral Beam at Main Payload		
Azimuth	14.7	degrees
Elevation	64.2	degrees

Table A.2: The location of the payloads in geodetic coordinates at the time of the first release in the CRIT I experiment.

Positions of payload in Geodetic Coordinates				
Name	Latitude	Longitude	Altitude	Units
Main	37.605189	-73.209345	400768.00	Deg, Deg, Meters
Sub	37.596284	-73.212532	402766.00	Deg, Deg, Meters
Burst	37.598297	-73.211309	398965.00	Deg, Deg, Meters

Table A.3: Some vectors of interest expressed in a Up, East, and North coordinate system located at the main payload at the time of the first chemical release in the CRIT I experiment. Angles between these various vectors are computed in the second table.

At the location of the Main payload					
Name	<i>X Up</i>	<i>Y East</i>	<i>Z North</i>	Magnitude	Units
Earth Field	-0.4043	-0.0328	0.1694	0.4396	Gauss
Ba Beam	0.9003	0.1104	0.4210	1.0000	Unitless
Spin Axis	0.9063	0.0927	0.4124	1.0000	Unitless
Main_Burst	-1800.5500	-184.2720	-819.4518	1986.8160	Meters
Main_Sub	2001.0981	-299.1954	-1043.7289	2276.6821	Meters
Burst_Sub	3801.6481	-114.9233	-224.2771	3809.9916	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	132.39	47.61
Earth Field	Spin Axis	132.96	47.04
Earth Field	Main_Burst	47.04	132.96
Earth Field	Main_Sub	167.24	12.76
Earth Field	Burst_Sub	159.75	20.25
Ba Beam	Spin Axis	1.18	178.82
Ba Beam	Main_Burst	178.82	1.18
Ba Beam	Main_Sub	54.28	125.72
Ba Beam	Burst_Sub	29.51	150.49
Spin Axis	Main_Burst	180.00	0.00
Spin Axis	Main_Sub	53.47	126.53
Spin Axis	Burst_Sub	28.69	151.31
Main_Burst	Main_Sub	126.53	53.47
Main_Burst	Burst_Sub	151.30	28.70
Main_Sub	Burst_Sub	24.77	155.23

Table A.4: Projection of some vectors of interest projected onto a plane perpendicular to the geomagnetic field. The projection of the vectors are expressed in a geomagnetically aligned coordinate system located at the main payload at the time of the first chemical release in the CRIT I experiment. The coordinate system is defined such that the X direction is perpendicular to both the main_burst vector and the geomagnetic field. The Z direction is antiparallel to the geomagnetic field. Angles between these various vector projections are computed in the second table.

At the location of the Main payload					
Name	X	Y	Z	Magnitude	Units
Earth Field	0.0000	0.0000	0.0000	0.0000	Gauss
Ba Beam	-0.0179	-0.7384	0.0000	0.7386	Unitless
Spin Axis	0.0000	-0.7318	0.0000	0.7318	Unitless
Main_Burst	0.0000	1453.9212	0.0000	1453.9212	Meters
Main_Sub	454.3218	215.9504	0.0000	503.0337	Meters
Burst_Sub	454.3218	-1237.9708	0.0000	1318.7039	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	85.83	94.17
Earth Field	Spin Axis	85.93	94.07
Earth Field	Main_Burst	94.07	85.93
Earth Field	Main_Sub	95.40	84.60
Earth Field	Burst_Sub	87.57	92.43
Ba Beam	Spin Axis	1.39	178.61
Ba Beam	Main_Burst	178.61	1.39
Ba Beam	Main_Sub	116.82	63.18
Ba Beam	Burst_Sub	21.55	158.45
Spin Axis	Main_Burst	180.00	0.00
Spin Axis	Main_Sub	115.42	64.58
Spin Axis	Burst_Sub	20.15	159.85
Main_Burst	Main_Sub	64.58	115.42
Main_Burst	Burst_Sub	159.85	20.15
Main_Sub	Burst_Sub	95.27	84.73

Table A.5: Projections of some vectors of interest onto a plane containing both the earth field and main_burst vector. The projection of the vectors are expressed in a geomagnetically aligned coordinate system located at the main payload at the time of the first chemical release in the CRIT I experiment. The coordinate system is defined such that the X direction is perpendicular to both the main_burst vector and the geomagnetic field. The Z direction is antiparallel to the geomagnetic field. Angles between these various vector projections are computed in the second table.

At the location of the Main payload					
Name	X	Y	Z	Magnitude	Units
Earth Field	0.0000	0.0000	-0.4396	0.4396	Gauss
Ba Beam	0.0000	-0.7384	0.6741	0.9998	Unitless
Spin Axis	0.0000	-0.7318	0.6815	1.0000	Unitless
Main_Burst	0.0000	1453.9212	-1354.0867	1986.8160	Meters
Main_Sub	0.0000	215.9504	2220.4140	2230.8907	Meters
Burst_Sub	0.0000	-1237.9708	3574.5008	3782.8068	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	132.39	47.61
Earth Field	Spin Axis	132.96	47.04
Earth Field	Main_Burst	47.04	132.96
Earth Field	Main_Sub	174.45	5.55
Earth Field	Burst_Sub	160.90	19.10
Ba Beam	Spin Axis	0.57	179.43
Ba Beam	Main_Burst	179.43	0.57
Ba Beam	Main_Sub	53.16	126.84
Ba Beam	Burst_Sub	28.50	151.50
Spin Axis	Main_Burst	180.00	0.00
Spin Axis	Main_Sub	52.59	127.41
Spin Axis	Burst_Sub	27.93	152.07
Main_Burst	Main_Sub	127.41	52.59
Main_Burst	Burst_Sub	152.07	27.93
Main_Sub	Burst_Sub	24.66	155.34

A.1.2 Release Two of CRIT I

Table A.6: General information about CRIT I launch and Second release

CRIT I Launch		
Date	May 13, 1986	(day 133)
Time	7:40:0.0	UT
Launch Site Location		
Geodetic latitude	37.83	degrees
Geodetic longitude	-75.49	degrees
Gyro Reference or Gyro Uncage		
Azimuth	173.0	degrees
Elevation	90.0	degrees
Attitude of Main Payload Spin Axis		
Pitch	-23.0	degrees
Yaw	-8.9	degrees
Attitude of Neutral Beam at Main Payload		
Azimuth	14.0	degrees
Elevation	64.2	degrees

Table A.7: The location of the payloads in geodetic coordinates at the time of the second release in the CRIT I experiment.

Positions of payload in Geodetic Coordinates				
Name	Latitude	Longitude	Altitude	Units
Main	37.518549	-72.551546	377020.00	Deg, Deg, Meters
Sub	37.506216	-72.556576	379832.00	Deg, Deg, Meters
Burst	37.503734	-72.555361	373057.00	Deg, Deg, Meters

Table A.8: Some vectors of interest expressed in a Up, East, and North coordinate system located at the main payload at the time of the second chemical release in the CRIT I experiment. Angles between these various vectors are computed in the second table.

At the location of the Main payload					
Name	X Up	Y East	Z North	Magnitude	Units
Earth Field	-0.4073	-0.0346	0.1716	0.4433	Gauss
Ba Beam	0.9003	0.1053	0.4223	1.0000	Unitless
Spin Axis	0.9111	0.0822	0.4040	1.0000	Unitless
Main_Burst	-3957.8631	-357.0266	-1755.0089	4344.2150	Meters
Main_Sub	2816.2369	-471.1884	-1440.4309	3198.1323	Meters
Burst_Sub	6774.0999	-114.1618	314.5780	6782.3611	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	132.21	47.79
Earth Field	Spin Axis	133.40	46.60
Earth Field	Main_Burst	46.60	133.40
Earth Field	Main_Sub	166.37	13.63
Earth Field	Burst_Sub	153.94	26.06
Ba Beam	Spin Axis	1.80	178.20
Ba Beam	Main_Burst	178.20	1.80
Ba Beam	Main_Sub	54.05	125.95
Ba Beam	Burst_Sub	23.50	156.50
Spin Axis	Main_Burst	180.00	0.00
Spin Axis	Main_Sub	52.54	127.46
Spin Axis	Burst_Sub	21.98	158.02
Main_Burst	Main_Sub	127.46	52.54
Main_Burst	Burst_Sub	158.02	21.98
Main_Sub	Burst_Sub	30.56	149.44

Table A.9: Projection of some vectors of interest projected onto a plane perpendicular to the geomagnetic field. The projection of the vectors are expressed in a geomagnetically aligned coordinate system located at the main payload at the time of the second chemical release in the CRIT I experiment. The coordinate system is defined such that the X direction is perpendicular to both the main_burst vector and the geomagnetic field. The Z direction is antiparallel to the geomagnetic field. Angles between these various vector projections are computed in the second table.

At the location of the Main payload					
Name	X	Y	Z	Magnitude	Units
Earth Field	0.0000	0.0000	0.0000	0.0000	Gauss
Ba Beam	-0.0238	-0.7403	0.0000	0.7406	Unitless
Spin Axis	0.0000	-0.7266	0.0000	0.7266	Unitless
Main_Burst	0.0000	3156.5321	0.0000	3156.5321	Meters
Main_Sub	706.5532	261.8953	0.0000	753.5295	Meters
Burst_Sub	706.5532	-2894.6367	0.0000	2979.6207	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	93.55	86.45
Earth Field	Spin Axis	93.62	86.38
Earth Field	Main_Burst	86.38	93.62
Earth Field	Main_Sub	90.67	89.33
Earth Field	Burst_Sub	94.00	86.00
Ba Beam	Spin Axis	1.85	178.15
Ba Beam	Main_Burst	178.15	1.85
Ba Beam	Main_Sub	112.18	67.82
Ba Beam	Burst_Sub	15.56	164.44
Spin Axis	Main_Burst	180.00	0.00
Spin Axis	Main_Sub	110.34	69.66
Spin Axis	Burst_Sub	13.72	166.28
Main_Burst	Main_Sub	69.66	110.34
Main_Burst	Burst_Sub	166.28	13.72
Main_Sub	Burst_Sub	96.62	83.38

Table A.10: Projections of some vectors of interest onto a plane containing both the earth field and main_burst vector. The projection of the vectors are expressed in a geomagnetically aligned coordinate system located at the main payload at the time of the second chemical release in the CRIT I experiment. The coordinate system is defined such that the X direction is perpendicular to both the main_burst vector and the geomagnetic field. The Z direction is antiparallel to the geomagnetic field. Angles between these various vector projections are computed in the second table.

At the location of the Main payload					
Name	X	Y	Z	Magnitude	Units
Earth Field	0.0000	0.0000	-0.4433	0.4433	Gauss
Ba Beam	0.0000	-0.7403	0.6719	0.9997	Unitless
Spin Axis	0.0000	-0.7266	0.6871	1.0000	Unitless
Main_Burst	0.0000	3156.5321	-2984.7126	4344.2150	Meters
Main_Sub	0.0000	261.8953	3108.0932	3119.1076	Meters
Burst_Sub	0.0000	-2894.6367	6092.8058	6745.4581	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	132.23	47.77
Earth Field	Spin Axis	133.40	46.60
Earth Field	Main_Burst	46.60	133.40
Earth Field	Main_Sub	175.18	4.82
Earth Field	Burst_Sub	154.59	25.41
Ba Beam	Spin Axis	1.17	178.83
Ba Beam	Main_Burst	178.83	1.17
Ba Beam	Main_Sub	52.59	127.41
Ba Beam	Burst_Sub	22.36	157.64
Spin Axis	Main_Burst	180.00	0.00
Spin Axis	Main_Sub	51.42	128.58
Spin Axis	Burst_Sub	21.19	158.81
Main_Burst	Main_Sub	128.58	51.42
Main_Burst	Burst_Sub	158.81	21.19
Main_Sub	Burst_Sub	30.23	149.77

A.2 CRIT II Deployment Results

The CRIT II rocket was launched from Wallops Island, VA May 4 1989 at 1h 42m 10.0s UT. The gyro on the CRIT II rocket seemed to have functioned properly, but incorrect software for the attitude control system caused the chemical canisters to be deployed along different vectors than planned. To verify the accuracy of the CRIT II gyro a new method was used to independently determine the vehicle attitude. This method made use of the $\mathbf{V} \times \mathbf{\bar{B}}_0$ measurement from the electric field instruments, the velocity vector of the payload from radar tracking, and the magnetometer to determine attitude. Since this information agreed with the gyro attitude solution we felt confident the gyro problems during the CRIT I flight did not occur during the CRIT II flight. Fig. A.2, analogous to Fig. A.1, summarizes the positions of the payloads at the times of the releases. Data used to create Fig. A.2 are also tabulated in this Appendix. The top projection in Fig. A.2 contains the vector between the main payload and the burst. From this projection we can see the injection angle was 57.0° for the first burst and 44.5° for the second. The first burst occurred 1.584 km and the second 3.880 km away from the main payload. The subpayload was correctly deployed and on virtually the same field line as the main payload during the first release. The location of the subpayload during the second release could not be controlled as ideally due to constraints of orbital mechanics. The second projection in Fig. A.2 is perpendicular to the local geomagnetic field and viewed as if looking down the magnetic field line. The field line threading the subpayload was clearly located inside the bulk of the neutral barium jet in both cases.

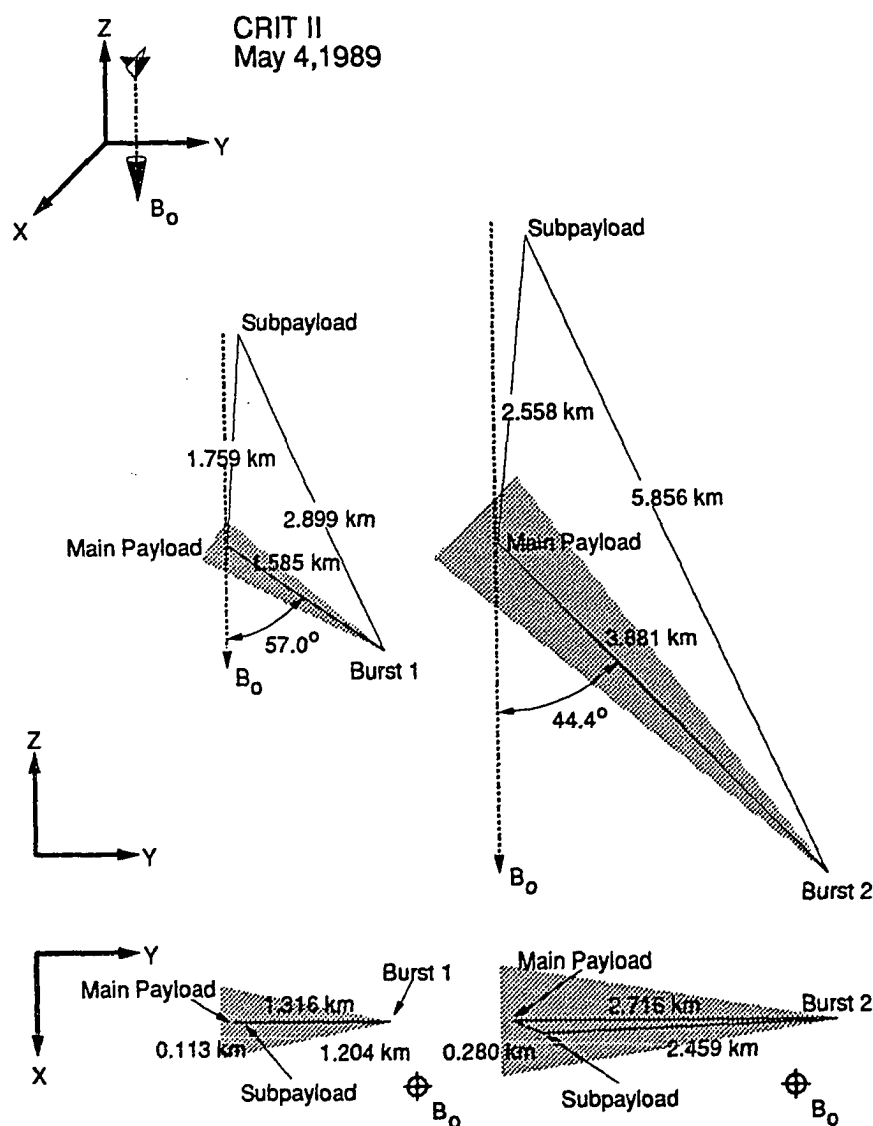


Figure A.2: Projection of locations, distances, and angles onto the Y-Z and X-Y planes for both barium releases during the CRIT II experiment. The coordinate system in which the projections are made is defined such that the Z direction is antiparallel to the geomagnetic field, the origin is located at the main payload, and the chemical cannister is located in the Y-Z plane. The figure is drawn to scale.

A.2.1 Release One of CRIT II

Table A.11: General information about CRIT II launch and first release

CRIT II Launch		
Date	May 4,1989	(day 124)
Time	1:42:10.0	UT
Launch Site Location		
Geodetic latitude	37.83	degrees
Geodetic longitude	-75.49	degrees
Gyro Reference or Gyro Uncage		
Azimuth	104.0	degrees
Elevation	90.0	degrees
Attitude of Main Payload Spin Axis		
Pitch	55.5	degrees
Yaw	15.5	degrees
Attitude of Neutral Beam at Main Payload		
Azimuth	119.10	degrees
Elevation	19.25	degrees

Table A.12: The location of the payloads in geodetic coordinates at the time of the first release in the Crit II experiment.

Positions of payload in Geodetic Coordinates				
Name	Latitude	Longitude	Altitude	Units
Main	37.156851	-73.875899	423983.00	Deg, Deg, Meters
Sub	37.151336	-73.875606	425616.00	Deg, Deg, Meters
Burst	37.163782	-73.889173	423475.00	Deg, Deg, Meters

Table A.13: Some vectors of interest expressed in a Up, East, and North coordinate system located at the main payload at the time of the first chemical release in the Crit II experiment. Angles between these various vectors are computed in the second table.

At the location of the Main payload					
Name	<i>X Up</i>	<i>Y East</i>	<i>Z North</i>	Magnitude	Units
Earth Field	-0.3954	-0.0316	0.1698	0.4315	Gauss
Ba Beam	0.3297	0.8249	-0.4591	1.0000	Unitless
Spin Axis	0.5667	0.6858	-0.4568	1.0000	Unitless
Main_Burst	-510.6457	-1257.1632	818.6938	1584.7643	Meters
Main_Sub	1634.9289	27.7629	-647.1286	1758.5615	Meters
Burst_Sub	2145.5746	1284.9261	-1465.8224	2898.8206	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	122.90	57.10
Earth Field	Spin Axis	138.53	41.47
Earth Field	Main_Burst	56.17	123.83
Earth Field	Main_Sub	176.32	3.68
Earth Field	Burst_Sub	155.47	24.53
Ba Beam	Spin Axis	15.80	164.20
Ba Beam	Main_Burst	176.22	3.78
Ba Beam	Main_Sub	60.76	119.24
Ba Beam	Burst_Sub	32.66	147.34
Spin Axis	Main_Burst	164.27	15.73
Spin Axis	Main_Sub	45.11	134.89
Spin Axis	Burst_Sub	17.38	162.62
Main_Burst	Main_Sub	120.15	59.85
Main_Burst	Burst_Sub	148.36	31.64
Main_Sub	Burst_Sub	28.21	151.79

Table A.14: Projection of some vectors of interest projected onto a plane perpendicular to the geomagnetic field. The projection of the vectors are expressed in a geomagnetically aligned coordinate system located at the main payload at the time of the first chemical release in the Crit II experiment. The coordinate system is defined such that the X direction is perpendicular to both the main_burst vector and the geomagnetic field. The Z direction is antiparallel to the geomagnetic field. Angles between these various vector projections are computed in the second table.

At the location of the Main payload					
Name	X	Y	Z	Magnitude	Units
Earth Field	0.0000	0.0000	0.0000	0.0000	Gauss
Ba Beam	0.0643	-0.8371	0.0000	0.8396	Unitless
Spin Axis	0.0865	-0.6566	0.0000	0.6623	Unitless
Main_Burst	0.0000	1316.4992	0.0000	1316.4992	Meters
Main_Sub	3.4295	112.9185	0.0000	112.9705	Meters
Burst_Sub	3.4295	-1203.5808	0.0000	1203.5856	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	93.20	86.80
Earth Field	Spin Axis	93.07	86.93
Earth Field	Main_Burst	86.63	93.37
Earth Field	Main_Sub	86.57	93.43
Earth Field	Burst_Sub	93.36	86.64
Ba Beam	Spin Axis	3.11	176.89
Ba Beam	Main_Burst	175.61	4.39
Ba Beam	Main_Sub	173.87	6.13
Ba Beam	Burst_Sub	4.23	175.77
Spin Axis	Main_Burst	172.50	7.50
Spin Axis	Main_Sub	170.76	9.24
Spin Axis	Burst_Sub	7.34	172.66
Main_Burst	Main_Sub	1.74	178.26
Main_Burst	Burst_Sub	179.84	0.16
Main_Sub	Burst_Sub	178.10	1.90

Table A.15: Projections of some vectors of interest onto a plane containing both the earth field and main_burst vector. The projection of the vectors are expressed in a geomagnetically aligned coordinate system located at the main payload at the time of the first chemical release in the Crit II experiment. The coordinate system is defined such that the X direction is perpendicular to both the main_burst vector and the geomagnetic field. The Z direction is antiparallel to the geomagnetic field. Angles between these various vector projections are computed in the second table.

At the location of the Main payload					
Name	X	Y	Z	Magnitude	Units
Earth Field	0.0000	0.0000	-0.4315	0.4315	Gauss
Ba Beam	0.0000	-0.8371	0.5432	0.9979	Unitless
Spin Axis	0.0000	-0.6566	0.7493	0.9963	Unitless
Main_Burst	0.0000	1316.4992	-882.2174	1584.7643	Meters
Main_Sub	0.0000	112.9185	1754.9292	1758.5582	Meters
Burst_Sub	0.0000	-1203.5808	2637.1466	2898.8185	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	122.98	57.02
Earth Field	Spin Axis	138.77	41.23
Earth Field	Main_Burst	56.17	123.83
Earth Field	Main_Sub	176.32	3.68
Earth Field	Burst_Sub	155.47	24.53
Ba Beam	Spin Axis	15.79	164.21
Ba Beam	Main_Burst	179.15	0.85
Ba Beam	Main_Sub	60.70	119.30
Ba Beam	Burst_Sub	32.49	147.51
Spin Axis	Main_Burst	165.06	14.94
Spin Axis	Main_Sub	44.91	135.09
Spin Axis	Burst_Sub	16.70	163.30
Main_Burst	Main_Sub	120.15	59.85
Main_Burst	Burst_Sub	148.36	31.64
Main_Sub	Burst_Sub	28.21	151.79

A.2.2 Release Two of CRIT II

Table A.16: General information about CRIT II launch and second release

CRIT II Launch		
Date	May 4, 1989	(day 124)
Time	1:42:10.0	UT
Launch Site Location		
Geodetic latitude	37.84	degrees
Geodetic longitude	-75.49	degrees
Gyro Reference or Gyro Uncage		
Azimuth	104.0	degrees
Elevation	90.0	degrees
Attitude of Main Payload Spin Axis		
Pitch	55.5	degrees
Yaw	15.5	degrees
Attitude of Neutral Beam at Main Payload		
Azimuth	113.10	degrees
Elevation	35.40	degrees

Table A.17: The location of the payloads in geodetic coordinates at the time of the second release in the CRIT II experiment.

Positions of payload in Geodetic Coordinates				
Name	Latitude	Longitude	Altitude	Units
Main	36.911980	-73.400607	409718.00	Deg, Deg, Meters
Sub	36.904348	-73.401145	412114.00	Deg, Deg, Meters
Burst	36.923244	-73.431184	407507.00	Deg, Deg, Meters

Table A.18: Some vectors of interest expressed in a Up, East, and North coordinate system located at the main payload at the time of the second chemical release in the CRIT II experiment. Angles between these various vectors are computed in the second table.

At the location of the Main payload					
Name	<i>X Up</i>	<i>Y East</i>	<i>Z North</i>	Magnitude	Units
Earth Field	-0.3958	-0.0328	0.1718	0.4328	Gauss
Ba Beam	0.5793	0.7498	-0.3198	1.0000	Unitless
Spin Axis	0.5731	0.6797	-0.4577	1.0000	Unitless
Main_Burst	-2215.7540	-2898.2522	1322.6712	3880.5787	Meters
Main_Sub	2398.6448	-51.0417	-893.2491	2560.0774	Meters
Burst_Sub	4614.3988	2847.2105	-2215.9203	5857.4386	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	135.54	44.46
Earth Field	Spin Axis	139.25	40.75
Earth Field	Main_Burst	44.42	135.58
Earth Field	Main_Sub	173.72	6.28
Earth Field	Burst_Sub	155.18	24.82
Ba Beam	Spin Axis	8.88	171.12
Ba Beam	Main_Burst	178.69	1.31
Ba Beam	Main_Sub	50.25	129.75
Ba Beam	Burst_Sub	19.65	160.35
Spin Axis	Main_Burst	172.27	7.73
Spin Axis	Main_Sub	46.91	133.09
Spin Axis	Burst_Sub	17.24	162.76
Main_Burst	Main_Sub	129.72	50.28
Main_Burst	Burst_Sub	160.36	19.64
Main_Sub	Burst_Sub	30.64	149.36

Table A.19: Projection of some vectors of interest projected onto a plane perpendicular to the geomagnetic field. The projection of the vectors are expressed in a geomagnetically aligned coordinate system located at the main payload at the time of the second chemical release in the CRIT II experiment. The coordinate system is defined such that the X direction is perpendicular to both the main_burst vector and the geomagnetic field. The Z direction is antiparallel to the geomagnetic field. Angles between these various vector projections are computed in the second table.

At the location of the Main payload					
Name	X	Y	Z	Magnitude	Units
Earth Field	0.0000	0.0000	0.0000	0.0000	Gauss
Ba Beam	0.0228	-0.7001	0.0000	0.7004	Unitless
Spin Axis	-0.1142	-0.6427	0.0000	0.6528	Unitless
Main_Burst	0.0000	2715.9635	0.0000	2715.9635	Meters
Main_Sub	105.0287	259.5410	0.0000	279.9867	Meters
Burst_Sub	105.0287	-2456.4226	0.0000	2458.6669	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	96.02	83.98
Earth Field	Spin Axis	100.57	79.43
Earth Field	Main_Burst	83.25	96.75
Earth Field	Main_Sub	75.28	104.72
Earth Field	Burst_Sub	95.79	84.21
Ba Beam	Spin Axis	11.94	168.06
Ba Beam	Main_Burst	178.14	1.86
Ba Beam	Main_Sub	156.10	23.90
Ba Beam	Burst_Sub	0.58	179.42
Spin Axis	Main_Burst	169.93	10.07
Spin Axis	Main_Sub	168.04	11.96
Spin Axis	Burst_Sub	12.52	167.48
Main_Burst	Main_Sub	22.03	157.97
Main_Burst	Burst_Sub	177.55	2.45
Main_Sub	Burst_Sub	155.52	24.48

Table A.20: Projections of some vectors of interest onto a plane containing both the earth field and main_burst vector. The projection of the vectors are expressed in a geomagnetically aligned coordinate system located at the main payload at the time of the second chemical release in the CRIT II experiment. The coordinate system is defined such that the X direction is perpendicular to both the main_burst vector and the geomagnetic field. The Z direction is antiparallel to the geomagnetic field. Angles between these various vector projections are computed in the second table.

At the location of the Main payload					
Name	X	Y	Z	Magnitude	Units
Earth Field	0.0000	0.0000	-0.4328	0.4328	Gauss
Ba Beam	0.0000	-0.7001	0.7137	0.9997	Unitless
Spin Axis	0.0000	-0.6427	0.7575	0.9935	Unitless
Main_Burst	0.0000	2715.9635	-2771.7202	3880.5787	Meters
Main_Sub	0.0000	259.5410	2544.7207	2557.9220	Meters
Burst_Sub	0.0000	-2456.4226	5316.4409	5856.4969	Meters

Angles Between Vectors		θ	$180 - \theta$
Earth Field	Ba Beam	135.55	44.45
Earth Field	Spin Axis	139.69	40.31
Earth Field	Main_Burst	44.42	135.58
Earth Field	Main_Sub	174.18	5.82
Earth Field	Burst_Sub	155.20	24.80
Ba Beam	Spin Axis	4.14	175.86
Ba Beam	Main_Burst	179.97	0.03
Ba Beam	Main_Sub	50.27	129.73
Ba Beam	Burst_Sub	19.65	160.35
Spin Axis	Main_Burst	175.89	4.11
Spin Axis	Main_Sub	46.14	133.86
Spin Axis	Burst_Sub	15.51	164.49
Main_Burst	Main_Sub	129.76	50.24
Main_Burst	Burst_Sub	160.38	19.62
Main_Sub	Burst_Sub	30.62	149.38

Appendix B

Data Catalog

This appendix contains a survey of electric field, magnetic field and plasma density measurements from both chemical releases of CRIT I and CRIT II. Each figure presents 1.0 second of data, and the horizontal width of all the plots is the same. The figures were originally produced on a 300 dpi printer so only about 1500 data points can unambiguously be represented on the page. Because of this basic limitation some the data was filtered before plotting to present a more accurate representation of what can be displayed and to make them more aesthetically pleasing. The starting time of each plot is referenced to the burst time, or time the command was sent to detonate the chemical release. Due to delays in decoding the detonate command and of the associated electronics, the actual release occurs about 13 ms later. The time base between various channels on the same payload and between the main and subpayloads is accurate to at least 1 ms or one printer point in these figures. The time base between some channels on same payload is better than 1 μ s.

B.1 CRIT I Main Payload

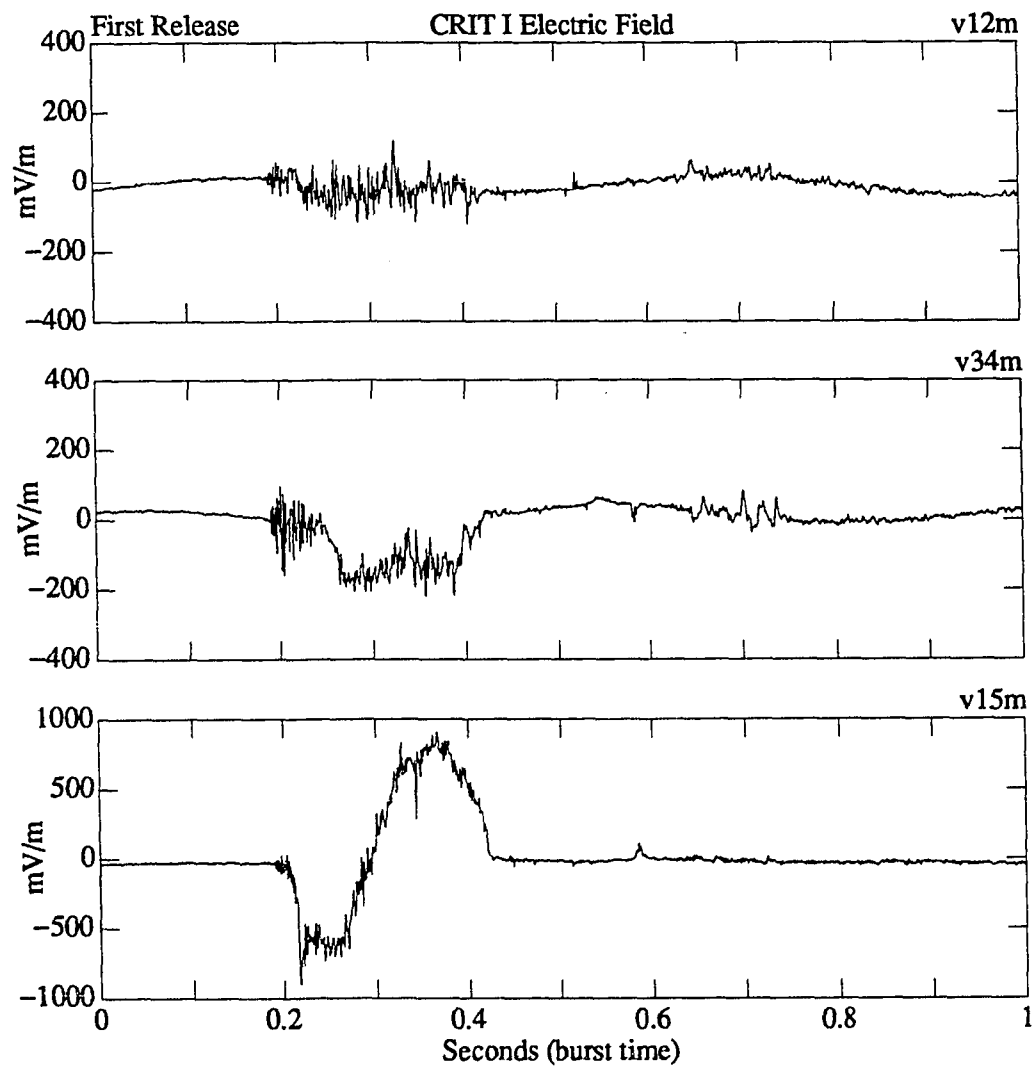


Figure B.1: Electric field measurements made at the main payload during the first chemical release of CRIT I. This data from the FM-FM telemetry link has been averaged, reducing the sampling rate to 1 kHz. No attempt has been made to remove the $V \times B_0$ induced electric field, despin the data, or remove DC offsets.

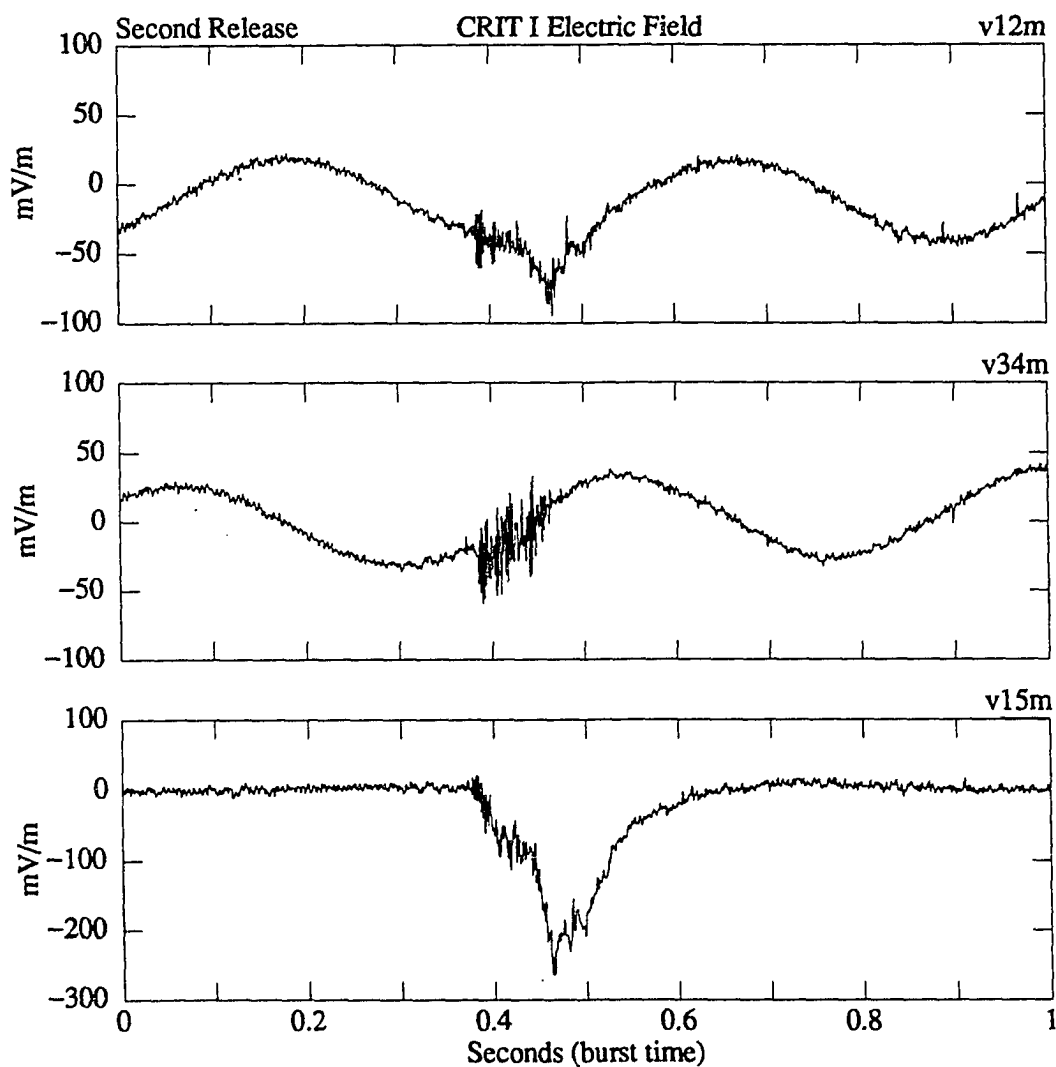


Figure B.2: Electric field measurements made at the main payload during the second chemical release of CRIT I. This data from the FM-FM telemetry link has been averaged, reducing the sampling rate to 1 kHz. No attempt has been made to remove the $V \times B_0$ induced electric field, despise the data, or remove DC offsets.

B.2 CRIT I Subpayload

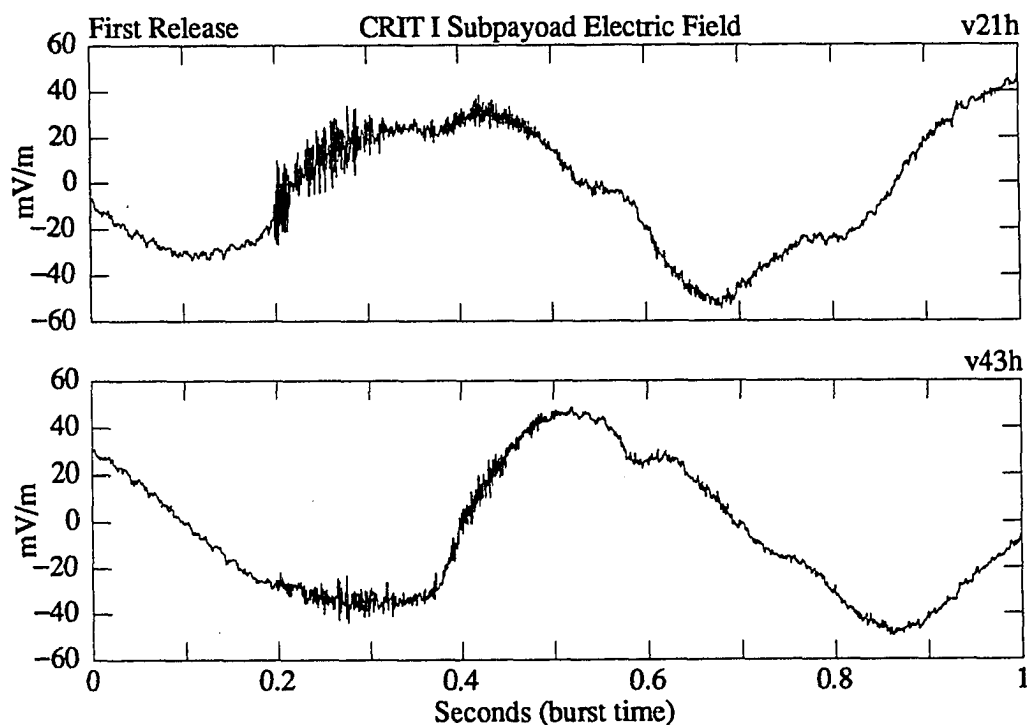


Figure B.3: Electric field measurements made at the subpayload during the first chemical release of CRIT I. This data from the FM-FM telemetry link has been averaged, reducing the sampling rate to 1 kHz. No attempt has been made to remove the $V \times B_0$ induced electric field, despise the data, or remove DC offsets. The amplitude of the $V \times B_0$ field as measured by v21h (top panel) was matched to v34h data (bottom panel) in an attempt to correct for the partial deployment of boom #1.

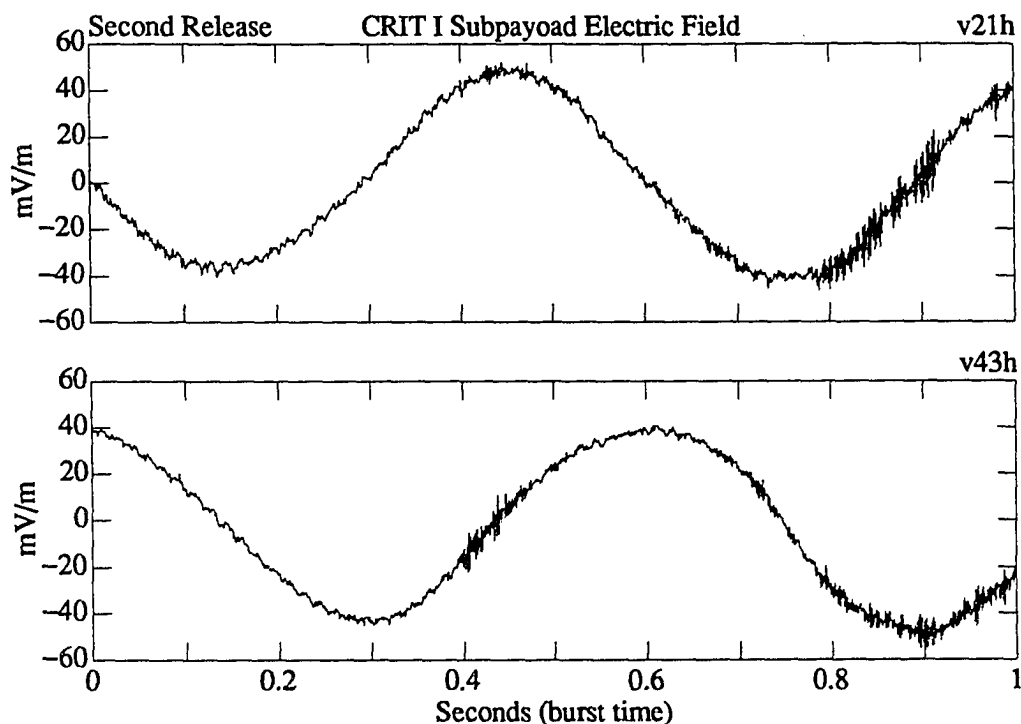


Figure B.4: Electric field measurements made at the subpayload during the second chemical release of CRIT I. This data from the FM-FM telemetry link has been averaged, reducing the sampling rate to 1 kHz. No attempt has been made to remove the $V \times B_0$ induced electric field, despin the data, or remove DC offsets. The amplitude of the $V \times B_0$ field as measured by v21h (top panel) was matched to v34h data (bottom panel) in an attempt to correct for the partial deployment of boom #1.

B.3 CRIT II Main Payload

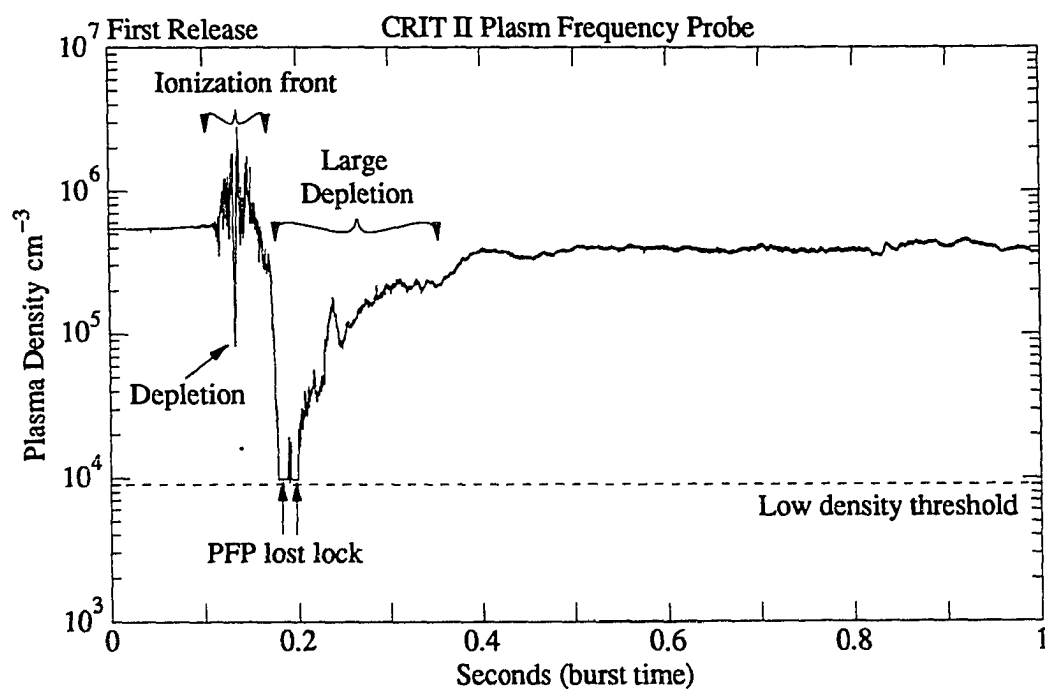


Figure B.5: Plasma density measurements made at the main payload by the plasma frequency probe during the first chemical release of CRIT II. This data, from the PCM telemetry link, is sampled at 12 kHz. The plasma density dropped below the low density threshold of the instrument for a short period of time.

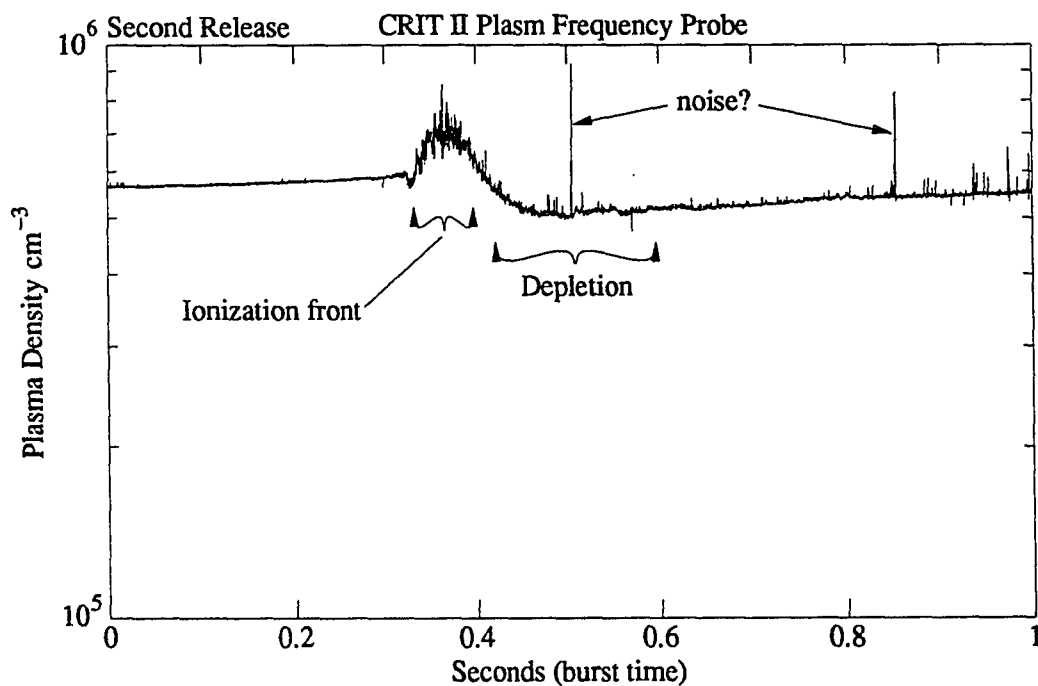


Figure B.6: Plasma density measurements made at the main payload by the plasma frequency probe during the second chemical release of CRIT II. This data, from the PCM telemetry link, is sampled at 12 kHz. Some of the large spikes are probably not physical and result from noise in the telemetry system.

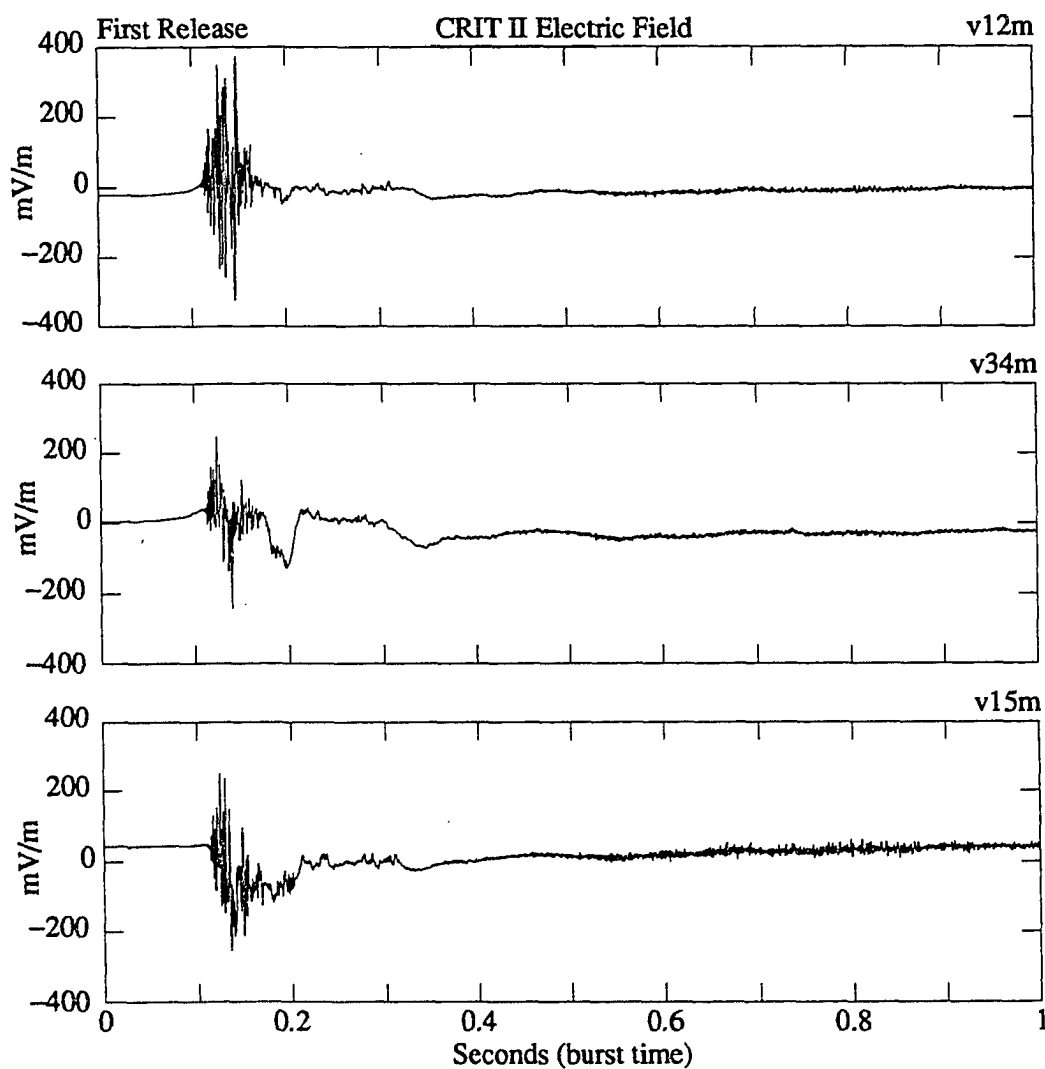


Figure B.7: Electric field measurements made at the main payload during the first chemical release of CRIT II. This data, from the burst memory PCM telemetry link, has been averaged, reducing the sampling rate to 1.5 kHz. No attempt has been made to remove the $V \times B_0$ induced electric field, despin the data, or remove DC offsets.

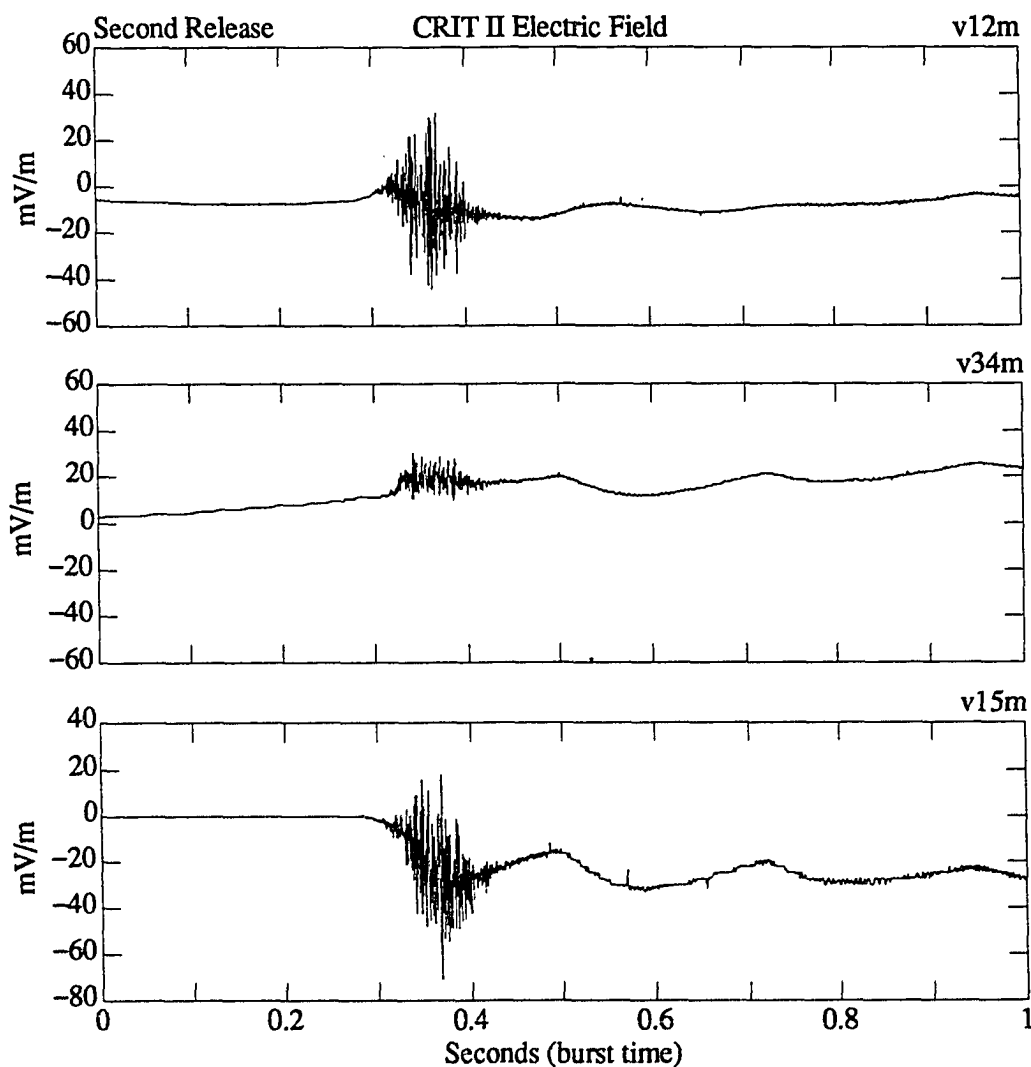


Figure B.8: Electric field measurements made at the main payload during the second chemical release of CRIT I. This data, from the burst memory PCM telemetry link, has been averaged, reducing the sampling rate to 1.5 kHz. No attempt has been made to remove the $V \times B_o$ induced electric field, despin the data, or remove DC offsets.

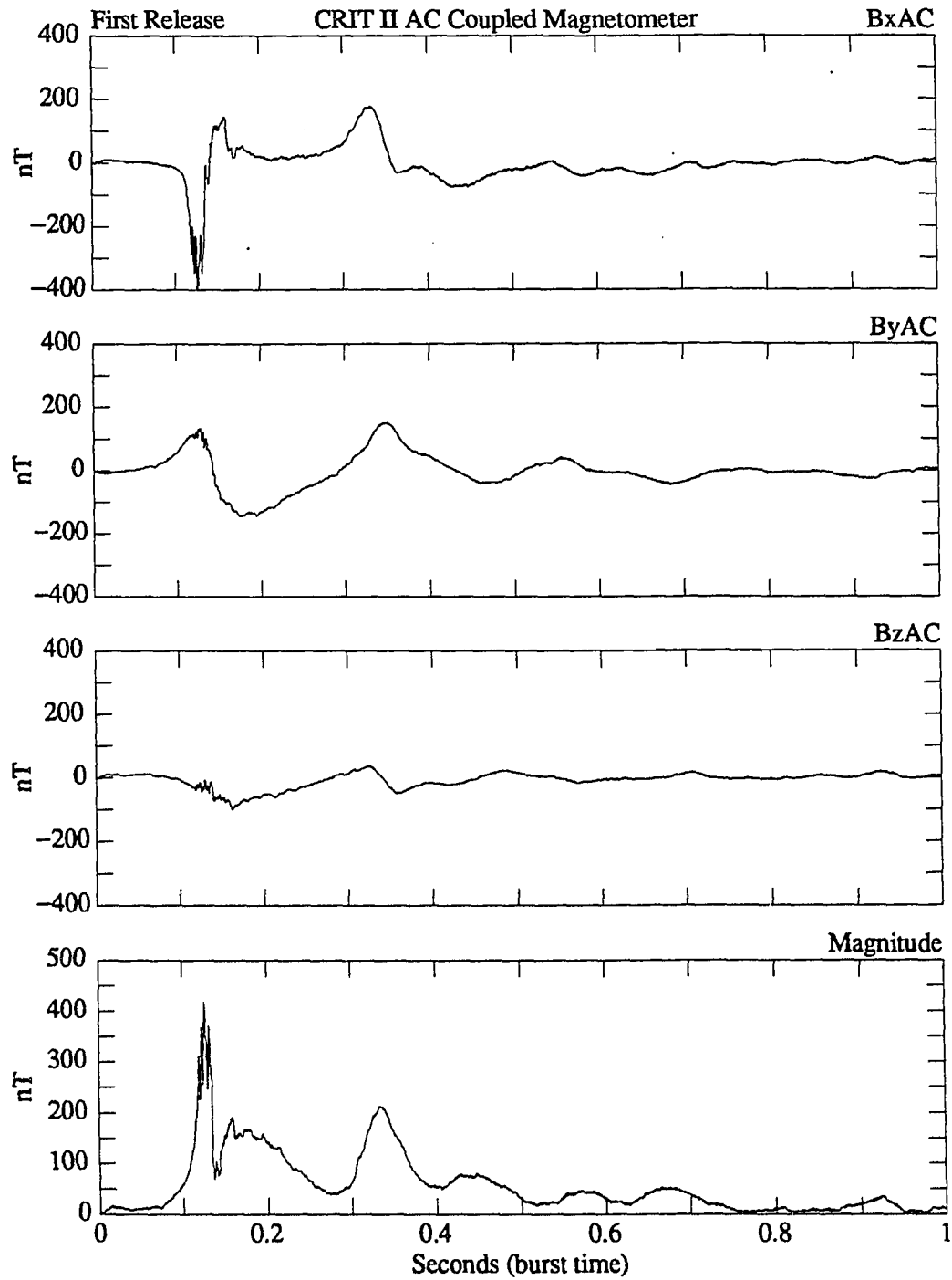


Figure B.9: Magnetic perturbation measurements made at the main payload during the first chemical release of CRIT I. This data, from the burst memory PCM telemetry link, is sampled at 1.5 KHz. The residual offsets were removed by subtracting data aligned in spin phase previous to the release, but no attempt has been made to despin the data. The magnitude, $\sqrt{B_{xAC}^2 + B_{yAC}^2 + B_{zAC}^2}$, is presented in the bottom panel.

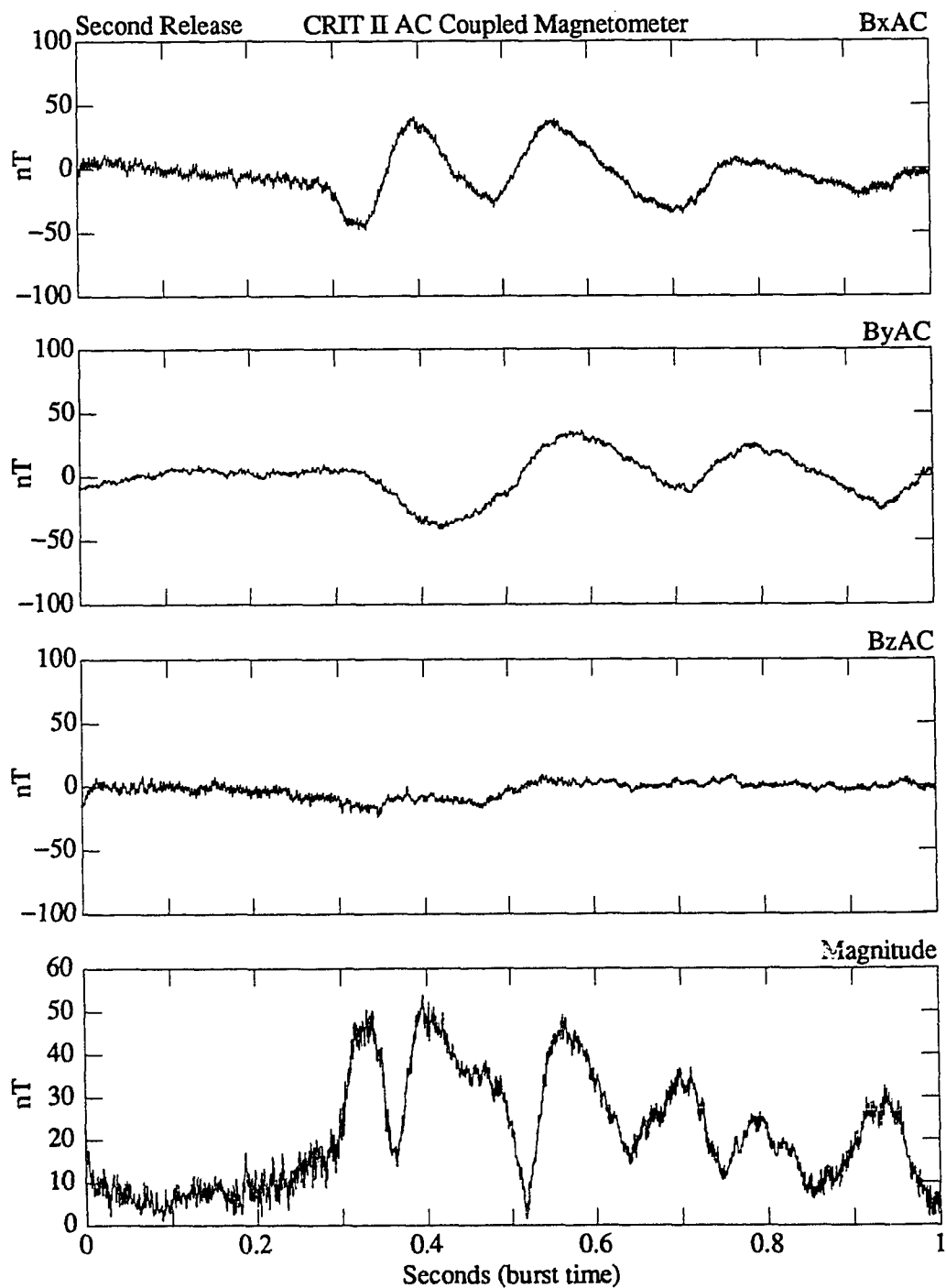


Figure B.10: Magnetic perturbation measurements made at the main payload during the second chemical release of CRIT I. This data, from the burst memory PCM telemetry link, is sampled at 1.5 KHz. The residual offsets were removed by subtracting data aligned in spin phase previous to the release, but no attempt has been made to despun the data. The magnitude, $\sqrt{B_{xAC}^2 + B_{yAC}^2 + B_{zAC}^2}$, is presented in the bottom panel.

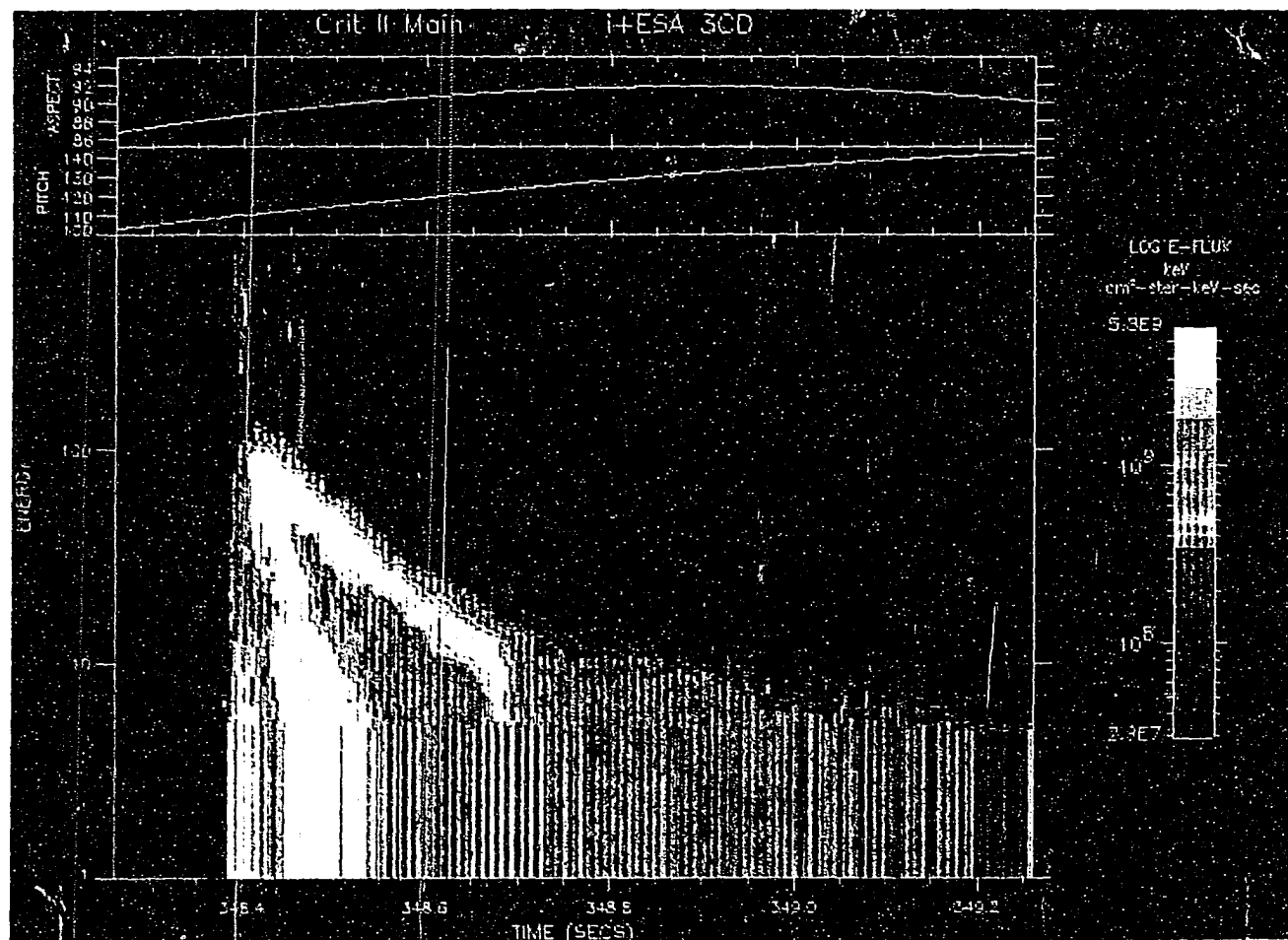


Figure B.11: A gray-scale version of Fig. 3.3. Time-energy spectrogram of ions at the CRIT II main payload during the first release. The look direction is back towards the release and nearly along the neutral stream direction.

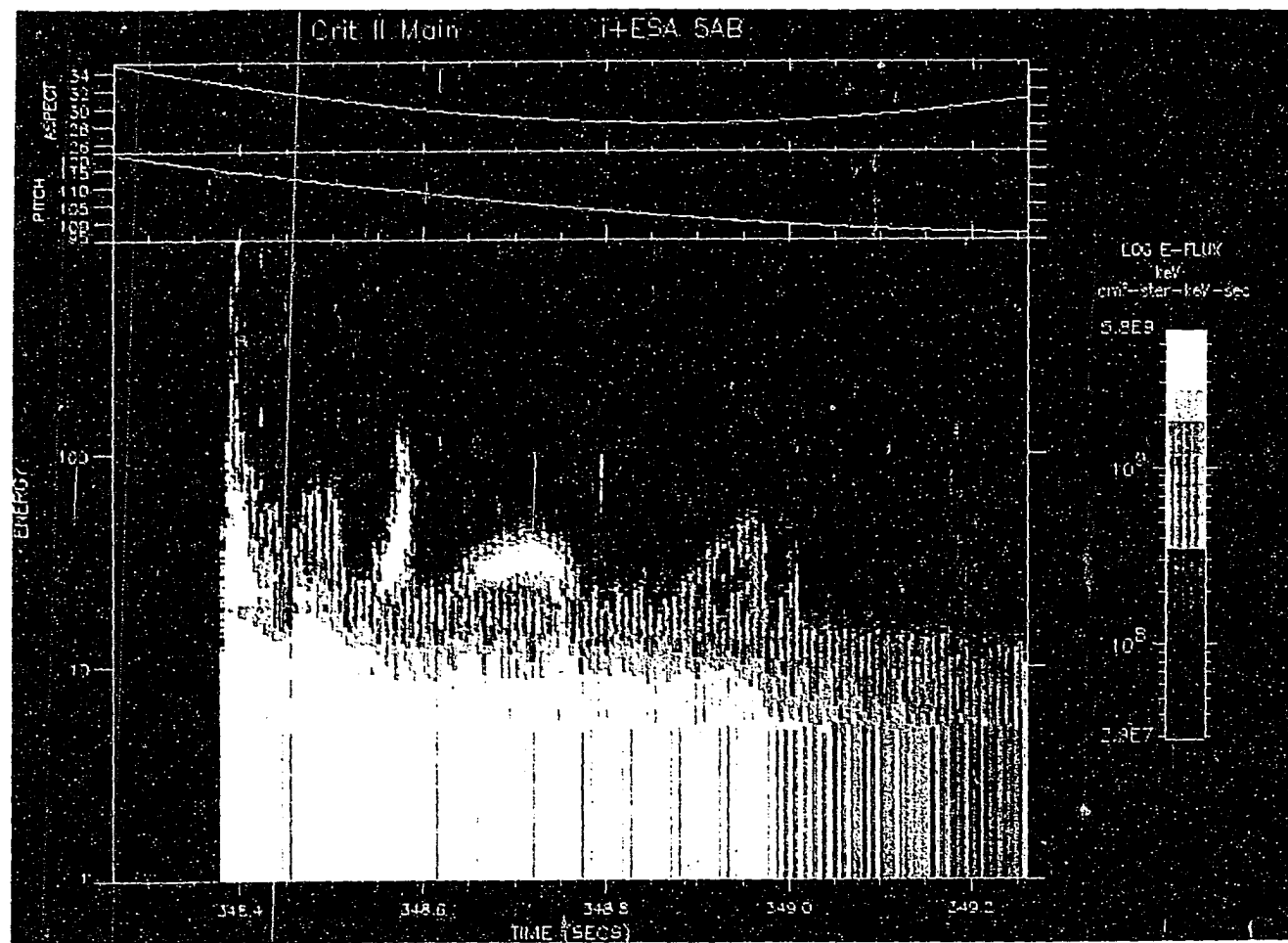


Figure B.12: A gray-scale version of Fig. 3.4. Time-energy spectrogram of ions at the CRIT II main payload during the first release. The look direction is nearly perpendicular to the geomagnetic field.

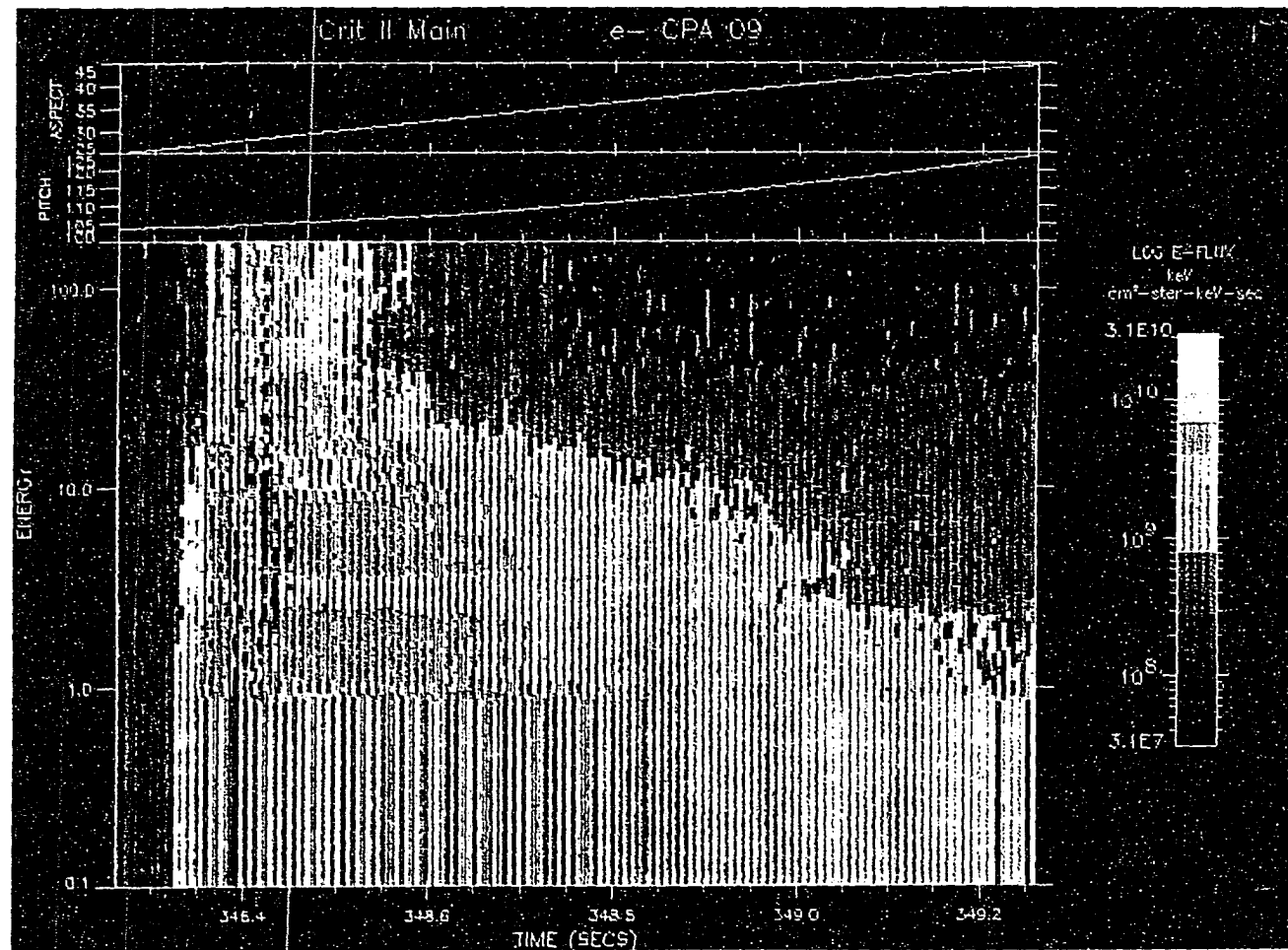


Figure B.13: A gray-scale version of Fig. 3.5. Time-energy spectrogram of electrons at the CRIT II main payload during the first release.

B.4 CRIT II Subpayload

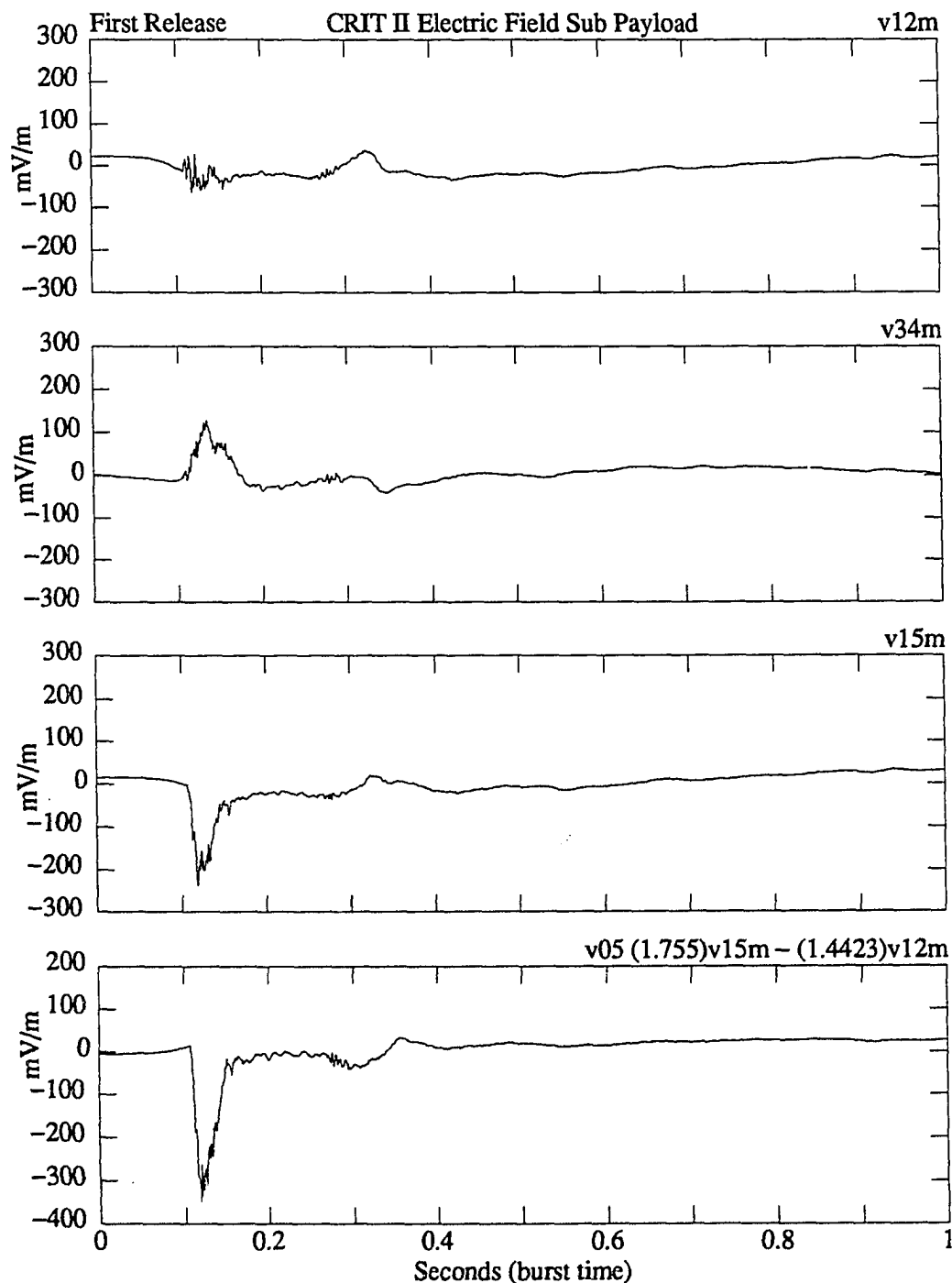


Figure B.14: Electric field measurements made at the subpayload during the first chemical release of CRIT II. This data, from the PCM telemetry link, is sampling at 1515 Hz. No attempt has been made to remove the $V \times B_0$ induced electric field, despin the data, but DC offsets have been carefully removed from all three channels. The bottom panel data, v05, is the electric field perpendicular to v12m and v34m as calculated from v15m and v12m.

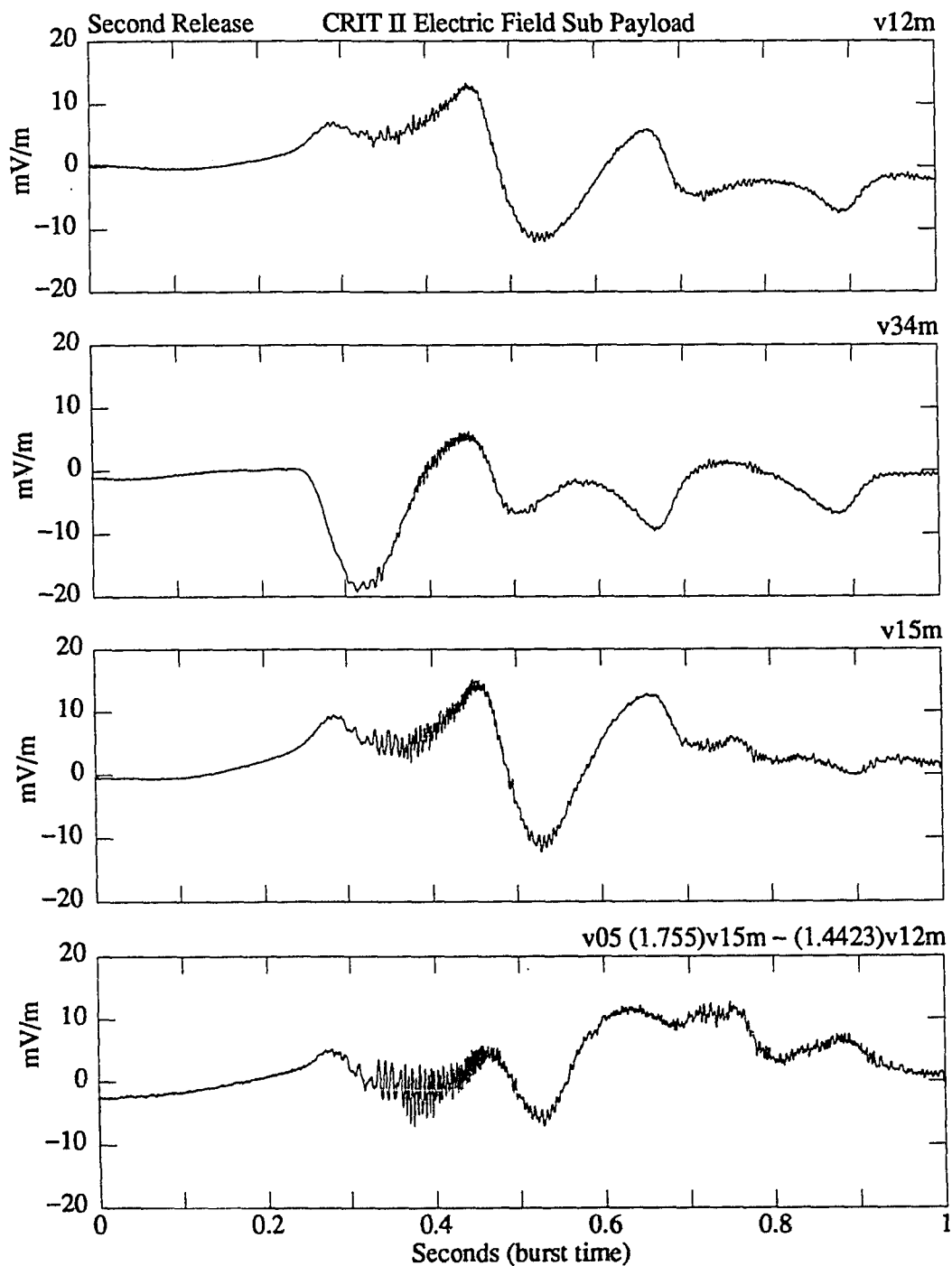


Figure B.15: Electric field measurements made at the subpayload during the second chemical release of CRIT II. This data, from the PCM telemetry link, is sampling at 1515 Hz. No attempt has been made to remove the $V \times B_0$ induced electric field, despin the data, but DC offsets have been carefully removed from all three channels. The bottom panel data, v05, is the electric field perpendicular to v12m and v34m as calculated from v15m and v12m.

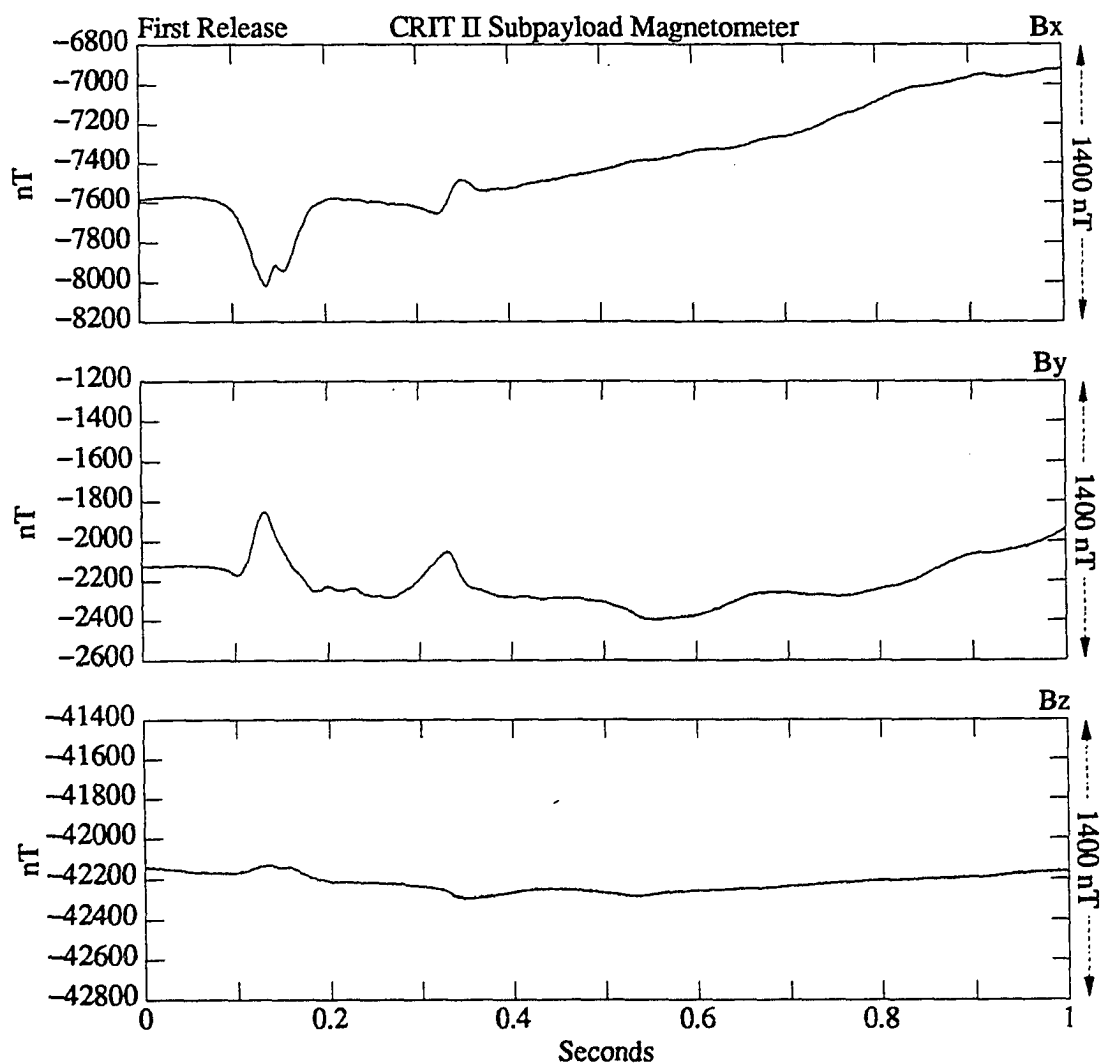


Figure B.16: Magnetic field measurements made at the subpayload during the first chemical release of CRIT I. This data from the PCM telemetry link was sampled at 1515 Hz. An imperfect attempt has been made to rotate the data into a magnetic aligned coordinate system so small perturbations can be observed. These perturbations are observed, but large DC offsets remain.

Appendix C

Instrumentation and Telemetry Lists

The Cornell telemetry lists from both CRIT I and CRIT II are reproduced here for reference. These lists contain information about what electric field measurements were made and how they were filtered. The telemetry lists are of two kinds, FM-FM telemetry lists and PCM telemetry lists. In addition, a few other instruments, such as magnetometers or swept frequency analyzer, are listed.

C.1 CRIT I Main Payload

Table C.1: Cornell University telemetry list and measurement information for FM-FM telemetry link to CRIT I main payload electric field experiment.

Data Function	IRIG#	Center (Khz)	Experiment Lower	Filters (Hz) Upper	Gain	MI
HF21	25	560	30	30000	8-16384	1.4
δB	24	400	180	7800	1000-2048	2.5
HF43	23	300	30	12000	8-16384	2.0
HF51	22	225	30	12000	8-16384	1.5
V12M	21	165	0	9000	1	1.3
V34M	20	124	0	9000	1	1.0
V15M	19	93	0	6700	1	1.0
V56M	18	70	0	5000	1	1.0
V26M	17	52.5	0	3800	1	1.0
V21H	16	40	0	1500	5	2.0
V43H	15	30	0	1500	5	1.5
AGC	14	22	0	1600	-	1.0
V51H	13	14.5	0	1000	5	1.0
V1S	12	10.5	0	390	0.94	2.0
Bx	5	1.3	0	-	-	-
By	4	0.96	0	-	-	-

C.2 CRIT I Subpayload

Table C.2: Cornell University telemetry list and measurement information for FM-FM telemetry link to CRIT I subpayload electric field experiment.

Data Function	IRIG#	Center (Khz)	Experiment Lower	Filters (Hz) Upper	Gain	MI
HF12	23	300	30	21000	64	1
HF34	22	225	30	16000	64	1
V12M	21	165	0	12000	1	1
V34M	20	124	0	9000	1	1
$\delta N_i/N_i$	19	93	30	6700	40	1
V21H	18	70	0	5000	5	1
V43M	17	52.5	0	3800	5	1
N_i	16	40	0	2900	1	1
V1S	15	30	0	2100	0.94	1
Mag-X	11	7.35	0	-	-	1

Table C.3: Cornell University telemetry list and measurement information for PCM telemetry link to CRIT I subpayload electric field experiment.

Data Function	Word ID	Samples #/Sec	Experiment Lower	Filters (Hz) Upper	Gain
V12M'	A6	208	0	100	1
V21H'	A3	208	0	100	5
V34M'	A4	208	0	100	1
N'_i	A5	208	0	100	1
Bx'	A7	208	0	100	5
By'	A8	208	0	100	5

C.3 CRIT II Main Payload

Table C.4: Cornell University telemetry list and measurement information for FM-FM telemetry link to CRIT II main payload electric field experiment.

Data Function	IRIG#	Center (Khz)	Experiment Filters (Hz)		Gain	MI
			Lower	Upper		
HF12	25	560	30	30000	100	1.4
HF15	24	400	30	12000	100	2.5
HF34	23	300	30	12000	100	1.9
HF25	22	225	30	12000	100	1.4
HF16	21	165	30	9000	100	1.4
V26M	20	124	0	9000	1	1
V56M	19	93	0	6700	1	1
V34M	18	70	0	5000	1	1
SFA	17	52.5	0	3800	1	-
V23M	16	40	0	1500	1	2
V12H	15	30	0	1500	5	1.5
V34H	14	22	0	1500	5	1.1
V15H	13	14.5	0	1000	5	1.0
V1S	12	10.5	0	390	0.94	2.0

Table C.5: Cornell University telemetry list and measurement information for PCM burst mode telemetry link to CRIT II main payload electric field experiment.

Data Function	Word ID	Samples #/Sec	Experiment Filters (Hz)		Gain
			Lower	Upper	
V12M'	FAN0	24000	0	12000	0.5
V34M'	FAN1	24000	0	12000	0.5
V15M'	FAN6	24000	0	12000	0.5
V35M'	FAN5	12000	0	6000	1
V46M'	FAN10	1500	0	750	1
V24M'	FAN11	1500	0	750	1
δB_x	AN5	1500	0.8	600	30
δB_y	AN12	1500	0.8	600	30
δB_z	AN25	1500	0.1	600	30

C.4 CRIT II Subpayload

Table C.6: Cornell University telemetry list and measurement information for FM-FM telemetry link to CRIT II subpayload electric field experiment.

Data Function	IRIG#	Center (Khz)	Experiment Lower	Filters (Hz) Upper	Gain	MI
HF12	25	560	30	12000	64	3.5
HF34	24	400	30	12000	64	2.5
HF15	23	300	30	12000	64	1.9
V12H	22	225	0	12000	5	1.4
V34H	21	165	0	12000	5	1
V15H	20	124	0	9000	5	1
$\delta N_i/N_i$	19	93	30	6700	1	1
TRADAT	18	70	-	-	-	-
V12M	17	52.5	0	3800	1	1
V34M	16	40	0	2900	1	1
N_i	15	30	0	2100	1	1
V1S	14	22	0	1650	0.94	1

Table C.7: Cornell University telemetry list and measurement information for PCM telemetry link to CRIT II subpayload electric field experiment.

Data Function	Word ID	Samples #/Sec	Experiment Lower	Filters (Hz) Upper	Gain	Notes
V12M'	A1	757.57	0	350	1	
V12H'	A2	1515.15	0	700	5	
V34M'	A5	757.57	0	350	1	
V34H'	A4	1515.15	0	700	5	
N_i'	A6	757.57	0	350	1	
V15M'	A3	1515.15	0	700	1	
V12M''	RIT4H,4L	1515.15	0.8	700	1	16 bit
V34M''	RIT5H,5L	1515.15	0.8	700	1	16 bit
V15M''	RIT6H,6L	1515.15	0.1	700	2	16 bit
N_i''	RIT7H,7L	1515.15	0.1	700	1	16 bit

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